

MERCHANTMEN-AT-ARMS

DAVID · W · BONE

ILLUSTRATED · BY · MUIRHEAD · BONE

MERCHANTMEN-AT-ARMS

DAVID · W · BONE

ILLUSTRATED · BY · MUIRHEAD · BONE

The Project Gutenberg eBook of Merchantmen-at-arms : the British merchants' service in the war

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Merchantmen-at-arms : the British merchants' service in the war

Author: David W. Bone

Illustrator: Sir Muirhead Bone

Release date: April 11, 2010 [eBook #31953]

Most recently updated: January 6, 2021

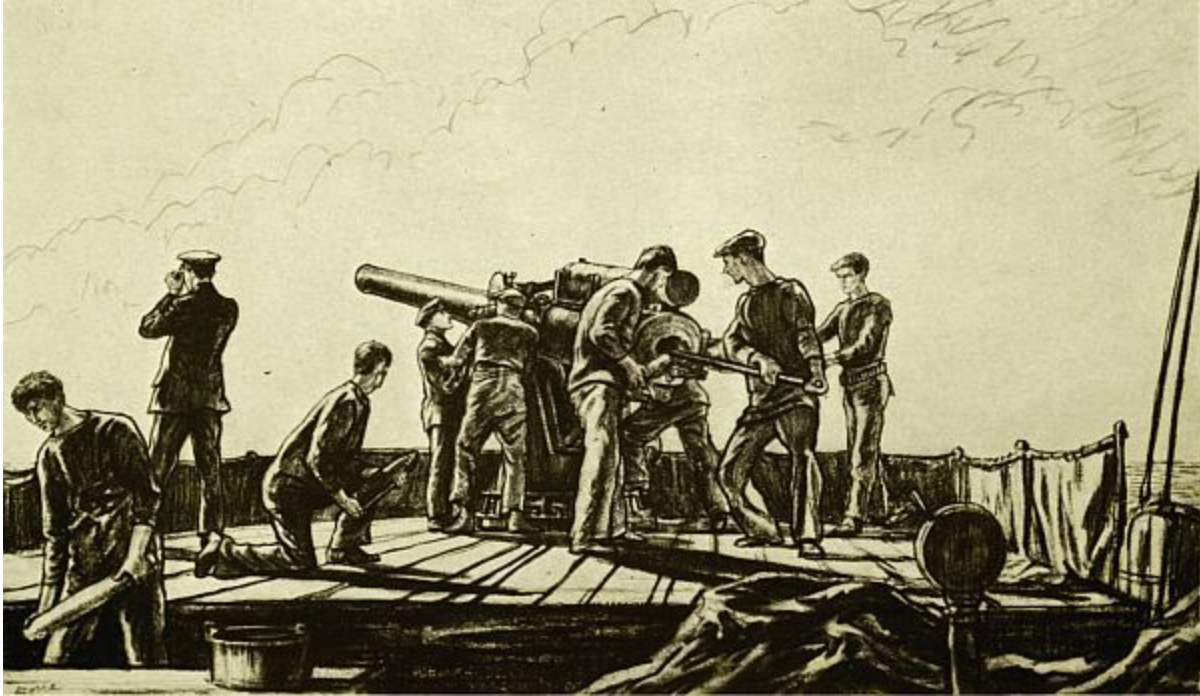
Language: English

Other information and formats: www.gutenberg.org/ebooks/31953

Credits: Produced by Tor Martin Kristiansen and the Online Distributed Proofreading Team at <https://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive) In memory of Thomas A. Noster, American Merchant Marine, from June 29, 1942-August 15, 1945.

*** START OF THE PROJECT GUTENBERG EBOOK
MERCHANTMEN-AT-ARMS : THE BRITISH MERCHANTS' SERVICE
IN THE WAR ***

Merchantmen-at-Arms



Frontispiece

MERCHANTMEN AT GUN PRACTICE

Merchantmen-at-Arms

**THE BRITISH MERCHANTS' SERVICE IN
THE WAR**

BY

DAVID W. BONE
DRAWINGS BY
MUIRHEAD BONE



LONDON: CHATTO & WINDUS
1919

All rights reserved

TO

ALGERNON C. F. HENDERSON

AS REPRESENTING A SYMPATHETIC AND UNDERSTANDING
GOVERNANCE IN AN IMPORTANT SECTION
OF THE BRITISH MERCHANTS'
SEA SERVICE

CONTENTS

PART I

	PAGE
I THE MERCHANTS' SERVICE	
OUR FOUNDATION	<u>3</u>
THE STRUCTURE	<u>14</u>
II OUR RELATIONS WITH THE NAVY	
JOINING FORCES	<u>21</u>
AT SEA	<u>26</u>
OUR WAR STAFF	<u>30</u>
III THE LONGSHORE VIEW	<u>44</u>
IV CONNECTION WITH THE STATE	
TRINITY HOUSE, OUR ALMA MATER	<u>53</u>
THE BOARD OF TRADE	<u>61</u>
V MANNING	<u>67</u>

PART II

VI THE COASTAL SERVICES	
THE HOME TRADE	<u>77</u>
PILOTS	<u>87</u>
LIGHTSHIPS	<u>91</u>
VII 'THE PRICE O' FISH'	<u>97</u>
VIII THE RATE OF EXCHANGE	<u>103</u>
IX INDEPENDENT SAILINGS	<u>110</u>
X BATTLEDORE AND SHUTTLECOCK	<u>116</u>
XI ON SIGNALS AND WIRELESS	<u>120</u>
XII TRANSPORT SERVICES	<u>125</u>
INTERLUDE	<u>132</u>
'THE MAN-O'-WAR'S 'ER 'USBAND'	<u>134</u>
XIII THE SALVAGE SECTION	

THE TIDEMASTERS	<u>141</u>
A DAY ON THE SHOALS	<u>147</u>
THE DRY DOCK	<u>156</u>
XIV ON CAMOUFLAGE—AND SHIPS' NAMES	<u>163</u>
XV FLAGS AND BROTHERHOOD OF THE SEA	<u>169</u>

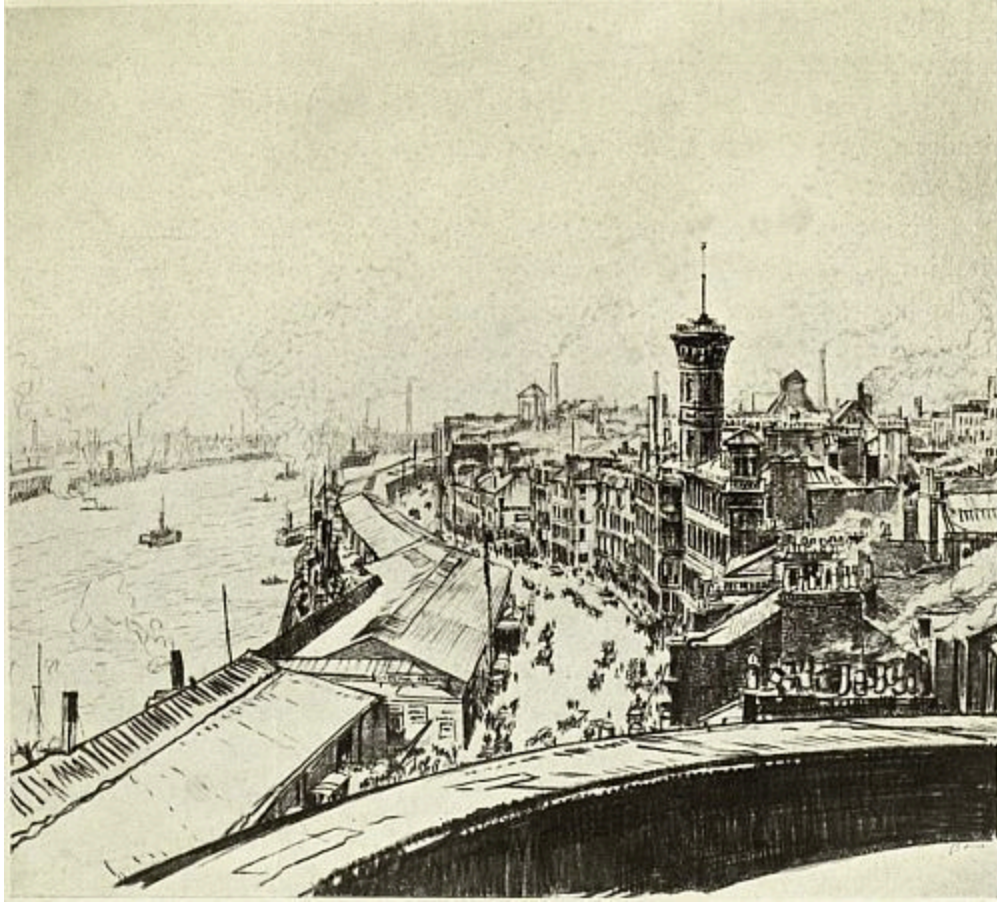
PART III

XVI THE CONVOY SYSTEM	<u>177</u>
XVII OUTWARD BOUND	<u>184</u>
XVIII RENDEZVOUS	<u>190</u>
XIX CONFERENCE	<u>198</u>
XX THE SAILING	
FOG, AND THE TURN OF THE TIDE	<u>205</u>
'IN EXECUTION OF PREVIOUS ORDERS'	<u>212</u>
XXI THE NORTH RIVER	<u>217</u>
XXII HOMEWARDS	
THE ARGONAUTS	<u>224</u>
ON OCEAN PASSAGE	<u>230</u>
'ONE LIGHT ON ALL FACES'	<u>236</u>
XXIII 'DELIVERING THE GOODS'	<u>244</u>
XXIV CONCLUSION: 'M N'	<u>252</u>
APPENDIX	<u>255</u>
INDEX	<u>257</u>

ILLUSTRATIONS

	PAGE
MERCHANTMEN AT GUN PRACTICE	<i>Frontispiece</i>
THE CLYDE FROM THE TOWER OF THE CLYDE TRUST BUILDINGS	xi
GRAVESEND: A MERCHANTMAN OUTWARD BOUND	3
THE BRIDGE OF A MERCHANTMAN	7
THE OLD AND THE NEW: THE <i>MARGARET</i> OF DUBLIN AND R.M.S. <i>TUSCANIA</i>	15
IN A MERCHANTMAN—BOMB-THROWER PRACTICE	21
A BRITISH SUBMARINE DETAILED FOR INSTRUCTION OF MERCHANT OFFICERS	31
THE D.A.M.S. GUNWHARF AT GLASGOW	33
INSTRUCTIONAL ANTI-SUBMARINE COURSE FOR MERCHANT OFFICERS AT GLASGOW	39
THE LOSS OF A LINER	44
THE MERSEY FROM THE LIVER BUILDINGS, LIVERPOOL	49
THE MASTER OF THE GULL LIGHTSHIP WRITING THE LOG	53
AT GRAVESEND: PILOTS AWAITING AN INWARD-BOUND CONVOY	59
TRANSPORTS LEAVING SOUTHAMPTON ON THE NIGHT PASSAGE TO FRANCE	67
LIVERPOOL: MERCHANTMEN SIGNING ON FOR OVERSEA VOYAGES	69
THE RULER OF PILOTS AT DEAL	77
A HEAVILY ARMED COASTING BARGE	83
THE LAMPMAN OF THE GULL LIGHTSHIP	93
MINESWEEPERS GOING OUT	97
SOUTHAMPTON WATER	103
'OUT-BOATS' IN A MERCHANTMAN	105
FIREMEN STANDING BY TO RELIEVE THE WATCH	111
QUEEN'S DOCK, GLASGOW	116
THE BRIDGE-BOY REPAIRING FLAGS	121
A TRANSPORT EMBARKING TROOPS FOR FRANCE	125

TRANSPORTS IN SOUTHAMPTON DOCKS	<u>129</u>
THE <i>LEVIATHAN</i> DOCKING AT LIVERPOOL	<u>135</u>
SALVAGE VESSELS OFF YARMOUTH, ISLE OF WIGHT	<u>141</u>
IN A SALVAGE VESSEL: OVERHAULING THE INSULATION OF THE POWER LEADS	<u>145</u>
A TORPEDOED MERCHANTMAN ON THE SHOALS: SALVAGE OFFICERS MAKING A SURVEY	<u>151</u>
A TORPEDOED SHIP IN DRY DOCK	<u>157</u>
DAZZLE	<u>163</u>
AN APPRENTICE IN THE MERCHANTS' SERVICE	<u>171</u>
A STANDARD SHIP AT SEA	<u>177</u>
BUILDING A STANDARD SHIP	<u>179</u>
THE THAMES ESTUARY IN WAR-TIME	<u>184</u>
DROPPING THE PILOT	<u>187</u>
EXAMINATION SERVICE PATROL BOARDING AN INCOMING STEAMER	<u>190</u>
DAWN: CONVOY PREPARING TO PUT TO SEA	<u>193</u>
EVENING: PLYMOUTH HOE	<u>198</u>
A CONVOY CONFERENCE	<u>201</u>
THE OLD HARBOUR, PLYMOUTH	<u>205</u>
CONVOY SAILING FROM PLYMOUTH SOUND	<u>207</u>
INWARD BOUND	<u>217</u>
A TRANSPORT LOADING	<u>219</u>
A CONVOY IN THE ATLANTIC	<u>224</u>
THE BOWS OF THE <i>KASHMIR</i> DAMAGED BY COLLISION	<u>227</u>
THE MAYFLOWER QUAY, THE BARBICAN, PLYMOUTH	<u>233</u>
EVENING: THE MERSEY FROM THE LANDING-STAGE	<u>241</u>
THE STEERSMAN	<u>243</u>
THE WORK OF A TORPEDO	<u>244</u>
TRANSPORTS DISCHARGING IN LIVERPOOL DOCKS	<u>245</u>
TROOP TRANSPORTS DISEMBARKING AT THE LANDING-STAGE, LIVERPOOL	<u>249</u>
'M N'	<u>252</u>



THE CLYDE FROM THE TOWER OF THE CLYDE TRUST BUILDINGS

INTRODUCTION

WRITTEN largely between the shipping crisis of 1917 and the surrender of German undersea arms at Harwich on November 20, 1918, this book is an effort to record a seaman's impressions of the trial through which the Merchants' Service has come in the war.

It is necessarily halting and incomplete. The extent of the subject is perhaps beyond the safe traverse of a mariner's dead reckoning. Policies of governmental control and of the economics of our management do not come within the scope of the book except as text to the diary of seafaring. Out at sea it is not easy to keep the right proportions in forming an opinion of measures devised on a grand scale, and of the operation of which we see only a small part. Our slender thread of communication with longshore happenings is often broken, and understanding is warped by conjecture.

In pride of his ancient trade, the seaman may perceive an importance and vital instrumentality in the ships and their voyages that may not be so evident to the landsman. By this is the mariner constantly impressed: that, without the merchant's enterprise on the sea—the adventure of his finance, his ships, his gear, his men—the armed and enlisted resources of the State could not have prevailed in averting disaster and defeat.

The unique experiences of individual seamen—the trials of seafaring under less favourable circumstances than was the writer's good fortune—the complaints and grievances of our internal affairs—are but lightly sketched. Many brother seamen may feel that the harassing and often despairing case of the average tramp steamer has not adequately been dealt with; that—in "Outward Bound," as an instance—the writer presents a tranquil and idyllic picture which cannot be accepted as typical. The bitter hardship of proceeding on a voyage under war conditions, with the same small crew that was found inadequate in peace-time, is hardly suggested; the extent of the work to be overtaken is perhaps camouflaged in that description of setting out. Reality would more frequently show a vessel being hurried out of dock on the top of the tide, putting to sea into heavy weather, with the

hatchways open over hasty stowage, and all the litter of a week's harbour disroutine standing to be cleared by a raw and semi-mutinous crew.

Criticism on these grounds is just: but it was ever the seaman's custom to dismiss heavy weather—when it was past and gone—and recall only the fine days of smooth sailing. If the hard times of our strain and labouring are not wholly over, at least we have fallen in with a more favouring wind from the land. Conditions in the Merchants' Service are vastly improved since Germany challenged our right to pass freely on our lawful occasions. Relations between the owner and the seamen are less strained. Remuneration for sea-service is now more adequate. The sullen atmosphere of harsh treatment on the one hand, and grudging service on the other, has been cleared away by the hurricane threat to our common interests.

Throughout the book there are some few extracts—all indicated by quotation marks—from the works of modern authors. The writer wishes to acknowledge their use and to mention the following: "Trinity House," by Walter H. Mayo; "The Sea," by F. Whympers; "The Merchant Seamen in War," by L. Cope Cornford; "Fleets behind the Fleet," by W. Macneile Dixon; "North Sea Fishers and Fighters" and "Fishermen in Wartime," both by Walter Wood; the pages of the *Nautical Magazine*.

The grateful thanks of writer and artist are tendered to Rear-Admiral Sir Douglas Brownrigg, Chief Naval Censor, and to Lord Beaverbrook and Mr. Arnold Bennett, of the Ministry of Information, for facilities and kindly assistance in preparation of the work. The writer's indebtedness to his Owners for encouragement and for generous leave of absence (without which the book could not have been written) is especially acknowledged.

Mr. Muirhead Bone's drawings reproduced in this book were executed during the war for the Ministry of Information with the co-operation of the Admiralty. They are now in the possession of the Imperial War Museum. With the exception of the illustrations on pages [44](#), [224](#), and [252](#), these drawings were made on the spot.

DAVID W. BONE

PART I



GRAVESEND: A MERCHANTMAN OUTWARD BOUND

I

THE MERCHANTS' SERVICE

OUR FOUNDATION

ALTHOUGH sea-interest of to-day finds an expression somewhat trite and familiar, the spell of the ships and the romance of voyaging drew an instant and wondering recognition from the older chroniclers. With a sure sense of right emphasis, yet observing an austere simplicity, they preserved for us an eloquent and adequate impression of the vital power of the ships. One outstanding fact remains constantly impressed in their records—that our island gates are set fast on the limits of tide-mark, leaving no way out but by passage of the misty sea-line; there is no gangway to a foreign field other than the planking of our vessels.

Grandeur of the fleets, the might of sea-ordnance, the intense dramatic decision of a landing, stand out in the great pieces the early writers and painters designed. Brave kingly figures wind in and out against the predominant background of rude hulls and rigging and weathered sails. The outline of the ships and the ungainly figures of the mariners are definitely placed to impel our thoughts to the distant sea-marches.

Happily for us, the passengers of early days included clerks and learned men on their pilgrimages, else we had known but little of bygone ship life. With interest narrowed by bounds of the bulwarks, they noted and recorded a worthy description. In the mystery of unknown seas, as in detail of the sea-tackle and the forms and usages of the ship, they penned a perfect register: down to the tunnage of the butts, we know the ships—to the 'goun of faldying' and the extent of their lodemanage, we recognize the men.

At later date we come on the seaman and his ships recorded and portrayed with a loving enthusiasm. Richard Hakluyt—"with great charges and infinite cares, after many watchings, toiles and travels, and wearying

out" of his weak body—sets out for us a wonderful chronicle of the shipping to his day. He grew familiarly acquainted with the chiefest 'Captaines,' the greatest merchants, and the best mariners of our nation, and acquired at first hand somewhat more than common knowledge of the sea. He saw not only the waving banners of sea-warriors and the magnificence of their martial encounters, but lauded victory in far voyages, the opening to commerce of distant lands, the hardihood of the Merchant Venturers. He realized the value of the seaman to the nation, not alone to fight battles on the sea, but as skilful navigators to further trade and intercourse. He was not ignorant "that shippes are to litle purpose without skillfull Sea-men; and since Sea-men are not bred up to perfection of skill in much lesse time than in the time of two prentiships; and since no kinde of men of any profession in the commonwealth passe their yeres in so great and continuall hazard of life; and since of so many, so few grow to gray heires; how needful it is that . . . these ought to have a better education, than hitherto they have had."

His matchless patience and care and exactitude were only equalled by his pride in the doings of the seamen and the merchants. With a joyful humility he exults in the hoisting of our banners in the Caspian Sea—not as robber marauders, but as peaceful traders under licence and ambassade—at the station of an English Ligier in the stately porch of the Grand Signior at Constantinople, at consulates at Tripolis and Aleppo, in Babylon and Balsara—"and which is more, at English Shippes coming to anker in the mighty river of Plate." In script and tabulation he glories in the tale of the ships, and sets out the names and stations of humble merchant supercargoes with the same meticulous care as the rank and titles of the Captain-General of the Armada.

Alas! There was none to set a similarly gifted hand to the further course of his lone furrow. Purchas tried, but there was no great love of his subject-matter to spread a glamour on the pages. Perhaps the magnitude of the task, ever growing and gathering, and the minute and unwearying succession of Hakluyt's "Navigations and Traffiques," discouraged and deterred less ardent followers. Of voyages and expeditions and discoveries there are volumes enough, but few such intimate records as "the Oathe ministered to the servants of the Muscovie company," or the instructions given by the Merchant Adventurers unto Richard Gibbs, William Biggatt, and John

Backhouse, masters of their ships, have been written since Hakluyt turned his last page.

As outposts to our field, roving bands on a frontier that rises and falls with the tide, the seamen were ever the first to apprehend the mutterings of war. With but little needed to set spark to the torch, they came in to foreign seaport or littoral with a fine confidence in their ships and arms. Truculent perhaps, and overbearing in their pride of long voyaging over a mysterious and threatening sea, they were hardly the ambassadors to aid settlement of a dispute by frank goodwill and prudence. Sailing outwith the confines of ordered government, their lawless outlook and freebooting found a ready rejoinder in restraint of trade and arbitrary imprisonment. Long wars had their seed in tavern brawls, enforcement "to stoope gallant [lower topsail] and vaile their bonets" for a puissant king or queen, brought a reckoning of strife and bloodshed.

Although military sea-captains, the glory of their victories, the worthiness of their ships and appurtenances, figure largely on the pages of subsequent sea-history, not a great deal has been written of the sailor captains and their mates and crews. Later chroniclers were concerned that their subjects should be grand and combatant: there was little room in their text for trading ventures, or for such humble recitals as the tale and values of hogshead or caisse or bale. A line of demarcation was slowly but inevitably ruling a division of our sea-forces. The service of the ships, devoted indifferently to sea-warfare or oversea trading—as the nation might be at war or peace—was in process of adjustment to meet the demands of a new sea-attack. The vessels were no longer merely floating platforms from which a military leader could direct a plan of rude assault and engage the arms of his soldiery, leaving to the masters and seamen the duty of handling the way of the ship. A new aristocracy had arisen from the decks who saw, in the pull of their sails, a weapon more powerful than shock ordnance, and resented the dictation of landsmen on their own sea-province. Sea-warfare had become a contest, more of seamanship and manœuvre, less of stunning impact and a weight of military arms.

In division of the ships and their service, it may quite properly be claimed that the Merchants' Service remained the parent trunk from which the new Navy—a gallant growing limb—drew sap and sustenance, perhaps,

in turn, improving the growth of the grand old tree. Certainly their service was an offshoot, for, since Henry VIII ordered laying of the first especial war keel, the sea-battles to the present day have been largely joined by the ships and men and furniture of the merchants, carrying on in the historic traditional manner of a fight when there was fighting to be done, a return to trade and enterprise when the great sea-roads were cleared to commerce. Stout old Sir John Hawkins, Frobisher, Drake, Davis, Amadas, and Barlow were merchant masters, shrewd at a venture, in intervals of, and combination with, their deeds of arms. Only a small proportion of State ships were in issue with the merchants' men to scourge the great Armada from our shores. Perhaps the existence of such a vast reserve in ships and men delayed the progress of purely naval construction. Only with the coming of steam was the line drawn sharply and definitely—the branch outgrowing the interlock of the parent stem.

With partial severance and division of the ships, the seamen—who had been for so long of one breed, laying down sail-needle and caulking-iron to serve ordnance and hand-cutlass or boarding-pike—had reached a parting of the ways, and become naval or mercantile as their habits lay. The State war vessels, built and manned and maintained for strictly military uses, increased in strength and numbers. Their officers and crews developed a new seamanship and discipline that had little counterpart on the commercial vessels. For a time the two services sailed, if not in company, within sight and hail of one another. On occasion they joined to effect glorious issues, but, with the last broadside of war, courses were set that quickly swerved the fleets apart.

Longer terms of peace gave opportunity for development on lines that were as poles apart. The Naval Service perfected and exercised their engines of war, and drilled and seasoned their men to automaton-like subservience to their plans. A broadening to democratic freedom, quickened by familiar intercourse with other nationals, had effect with the merchantmen in rousing a reluctance to a resort to arms; they desired but a free continuance of trading relations. Although differing in their operations and ideals, both services were striving to enhance the sea-power of the nation. Thomas Cavendish, Middleton, Monson, Hudson, and Baffin—merchant masters—explored the unknown and extended a field for mercantile ventures, but that field could have been but indifferently

maintained if naval power had not been advanced to protect the merchantmen in their voyaging.



THE BRIDGE OF A MERCHANTMAN

As their separation developed, relations grew the more distant between the seamen. While certainly protecting the traders from any foreign interference, the new Navy did little to effect a community of interest with their sea-fellows. Prejudices and distrust grew up. State jealousies and trade monopolies formed a confusion of interests and made for strained relations between the merchants and the naval chancelleries on shore. At sea, the arbitrary exercise of authority by the King's officers was opposed by revolutionary instincts for a free sea on the part of the merchants' seamen. Forcible impressment to naval service was the worst that could befall the traders' men. For want of energy or ability to carry through the drudgery of early sea-training, the naval officers took toll of the practised commercial seamen as they came in from sea. Bitter hardship set wedge to the cleavage. After long and perilous voyaging, absent from a home port for perhaps two or three years, the homeward-bound sailor had little chance of being allowed a term of liberty on shore—a brief landward turn to dissolve the

salt casing of his bones. Within sound of his own church bells, in sight of the windmills and the fields and the home dwelling he had longed for, he was haled to hard and rigorous sea-service on vessels of war. The records of the East India Company have frequent references to this cruel exercise of naval tyranny.

"On Thursday morning the Directors received the agreeable news of the safe arrival of the *Devonshire*, Captain Prince, from Bengal. . . . Her men have all been impressed by the Men-of-War in the Downs, and other hands were put on board to bring her up to her moorings in the River."

". . . On Sunday morning the Purser of the *William*, Captain Petre, arrived in town, who brought advice of the said ship in the Downs, richly laden, on Account of the Turkey Company: the Ships of War in the Downs impressed all her men, and put others on board to bring her up."

"Notwithstanding the Report spread about, fourteen days ago, that no more sailors would be impressed out of the homeward-bound ships, several ships that arrived last week had all their men taken from them in the Downs."

Serving by turns, as his agility to dodge the gangs was rated, on King's ship for a turn, then hauling bowline on a free vessel; forced and hunted and impressed, the shipmen had perhaps sorry records to offer the historian, then busy with the enthralling chronicles of fleet engagements and veiling with glamour the toll of battles. Perhaps it was, after all, the better course to preserve a silence on the traders' doings and leave to romantic conjecture a continuance of Hakluyt's patient story.

Since the date of naval offgrowth, the chronicles have not often turned on our commercial path. Lone voyages and encounters with the sea and storm are minor enterprises to the sack of cities and the clash of arms at sea. Unlike the Naval Service, we merchants' men hold few recorded titles to our keystone in the national fabric. The deeds and documents may exist, but they are lost to us and forgotten in the files of musty ledgers. The fruits of our efforts stand in the balances of commercial structure, and are perhaps more enduring than a roll of record. But, if we are insistent in our search, we may borrow from the naval charters, and read that not all the glory of our sea-history lies with the thunder of broadsides and the impact of a close

boarding. Engagement with the elements—a contest with powers more cruel and implacable than keen steel—efforts to further able navigation, the standard of our seamanship—drew notable recruits to the humbler sea-life. The small crews and less lavish gear on the freighters brought the essentials of the sea-trade to each individual of the ship's company. Idlers and landsmen learned quickly and bitterly that their only claim to existence on a merchant's ship lay in a rapid acquisition of a skill in seamanship. The lessons and the threats and enforcements did not come wholly from their superiors, to whose tyranny they might expose a sullen obstinance, and gain, perhaps, a measure of sympathy from their rude sea-fellows. Then—as later, in the keen sailing days of our clipper ships—their hardest taskmasters were foremast hands, watchmates, the men they lived with and ate with and worked with—bitter critics, unpersuadable, who saw only menace and a threat to their own safety in the shipping of a man who could not do man's work. On the decks and about the spars of a merchant vessel, each man of the few seamen carried two lives—his own and a shipmate's—in his ability to 'hand, reef, and steer.' There was no place on board for a 'waister,' a 'swabber,' longshoreman, or sea labourer. Every man had quickly to prove his ability: the unrelenting sea gave time for few essays.

Fertility of resource, dexterity to serve at all duties, skill at handling ship and canvas, were the results of sea-ship training. In the merchantmen great opportunities offered for advancement in all branches of the seaman's art. Long voyaging was better exercise for a progression in navigation than the daily pilotage of the war vessels. Blake, in his early days as a merchant supercargo, learnt his seafaring on rough trading voyages, and his training could not have been other than sound to persist, through twenty years shore-dwelling as a merchant at Bridgwater, until he was called from his counting-house to command our naval forces. Dampier was a tarry foremast hand in his day: whatever we may judge of his conduct, we can have nothing but admiration for his seamanship. Ill-equipped and short-handed, racked by sea-sores and scurvy, his expeditions were unparalleled as a triumph of merchant sea-skill. James Cook learned his trade on the grimy hull of an east-coast collier—to this day we are working on charts of his masterly surveys.

In later years the merit of the trading vessels as sterling sea-schools was equally plain. During intervals of combatant service, or as prelude to a

naval career, training on the merchants' ships was eagerly sought by ardent naval seamen who saw the value of its resource in practical seamanship, in navigation, and weather knowledge. Great captains did not disdain the measure of the instruction. They sent their heirs to sea in trading vessels to draw an essence in practice from their sea-cunning. Hardy, Foley, and Berry had borne a hand at the sheets and braces, and had steered a lading of goods abroad, before they came to high command of the King's ships. Who knows what actions in the victories of Copenhagen, the Nile, and Trafalgar (hinged on the cast of the winds) were governed by Nelson's early sea-lessons, under Master John Rathbone, on the decks of a West India merchantman?

For long after, relations and interchange between the two Services were not so intimate. Until coming of the Great War, with a mutual appreciation, we had little in common. Our friend and peacemaker—the influence of seafaring under square sail—languished a while, then died. In steam-power, with its growth of development and intricacy of application, we found no worthy successor to present as good an office. In the long span of a hundred years of sea-peace we grew apart. The gulf between the two great Services widened to a breach that only the rigours of a world-conflict could reconcile.

As though exhausted by the indefinite sea-campaign of 1812, the Royal Navy lay on their oars and saw their commercial sea-fellows forge ahead on a course that revolutionized sea-transport and sea-warfare alike. The Lords of the Admiralty would listen to no deprecation of their gallant old wooden walls: steam propulsion was laughed at. To the Merchants' Service they left the risk and the responsibility of venturing afar in the rude new ships. In this wise, to us fell the honour of leading the State service to a new order of seafaring. Iron hulls and steam propulsion came first under our hands. It was not long before our new command of the sea was noted. Somewhat grudgingly, the conservative sea-mandarins were brought to a knowledge that their torpor was fatal. The Navy stirred and lost little time in traversing the leeway. They progressed on a path of experiment and probation suited to their needs, striving to construct mightier vessels and to forge new and greater arms. Exploring every avenue in their quest for aid and material, every byway for furtherance of their aims, they drew strange road-fellows within their ranks, new workmen to the sea. The engines of their adoption called for crafty hands to serve and adjust them. Steam we knew in our time

and could understand, but auxiliary mechanics outgrew the limits of our comprehension; naval practice became a science outwith the bounds of our sea-lore, a new trade, whose only likeness to ours lay in its service on the same wide sea.

Parted from the need to draw arms, secure in the knowledge of adequate naval protection, the Merchants' Service developed their ships and tackle in the ways of a free world trade. By shrewd engagement and industry in the counting-house, diligence and forethought in the building-yards, keen sailing and efficiency on the sea, the structure of our maritime supremacy was built up and maintained. Monopolies and hindering trade reservations and restrictions barred the way, but yielded to the spirit of our progress. Vested interests in seas and continents had to be fought and conquered, and there was room and scope for lingering combative instincts in the keen competition that arose for the world's carrying trade. Other nations came on the free seas, secure in the peace our arms had wrought, and entered the lists against us. The challenge to our seafaring we met by skill and hardihood—keener and more polished arms than the weapons of our seafathers. The coming of competitors spurred us to sea-deeds in the handling of our ships and cargoes, dispatch in the ports, and activity in the yards, that brought acknowledged victory to our flag. Every sense and thought that was in us was used to further our supremacy. The craft and workmanship of the builders and enterprise of the merchants provided us with the most beautiful of man's creations on the sea—the square-rigged sailing ship of the nineteenth century. With pride we sailed her. We, too, brought science to our calling; rude, perhaps, and not readily defined save by a long, hard pupilage. Not less than the calibre of the new naval ordnance was the measure of our sail spread, not inferior to ironclad hulls the speed and beauty of our clippers—we paralleled the roads of their strategy by the masterly handling of a cloud in sail. With a regularity and precision as noted as our naval sea-brothers' advance in gunfire, we served the trade and the mails, and spread the flood of emigration to the rise and glory of the Empire.

With the decline of square sail, a new way of seafaring opened to us. In the first of our steam pioneering, we took our yards and canvas with us, as good part of our sea-kit; a safe provision, as we thought, against the inevitable failure we looked for in the new navigation. We were

conservatively jealous of our gallant top hamper, and scorned the promise of a power that only dimly as yet we understood. But—the promise held. In a few years we became converts to the new order, in which we found a greater security, a more definite reliance, than in the angles of our sail plane. There was no longer a need for our precious 'stand by,' and we unrigged the wind tackle and accepted our new shipmate, the marine engineer, as a worthy brother seaman. It was not only the spars and the cordage and the sails we put ashore. With all the gallant litter we unloaded, condemned to the junk-heap, went a part of our seamanship as closely woven to the canvas as the seams our hands had sewn.

In steam practice, new problems required to be studied and resolved; challenges to our vaunted sea-lore came up that called for radical revision of older methods and ideas. Changes, as wide and drastic as the evolutions of a decade in sail, were presented in a swift succession of as many days. With eyes now turned from aloft to ahead, we retyped our seamanship to meet the altered conditions of the veer in our outlook. Unhelped, if unhindered, in our efforts, we adapted our calling to the sudden and revolutionary innovations in construction and power of the new ships. We grew sensible of gaps in our knowledge, of voids in education that our earlier handicraft had not revealed. Severed, by press of our sea-work, from the facilities for study that now offered advancement to the landsman, we sought in alert and constant practice a substitute for technical instruction. By step and stride and canter we jockeyed each new starter from the shipyards, and studied their paces and behaviour on the vexed testing courses of the open sea. If our methods were rude in trial, they settled to efficiency in service. We paced in step with the rapid developments of the shipwright's art, the not less active contrivance of the engineers. We kept no man waiting for a sea-controller to his new and untried machine: there was no whistling for a pilot on the grounds of our reaches. From oversea dredger and frail harbour tug to the magnitude of an *Aquitania*, we were ever ready to board her on the launching ways and steer her to the limits of her draught.

A Hakluyt of the day would have a full measure for his enthusiasm in the shear of our keels on every sea, the flutter of our flags to all the winds. By virtue of worthy vessels and good seamanship, the Red Ensign was devoted to a world service; by good guardianship and commercial rectitude the

Merchants' Service held charge of the world's wealth in transport—the burden of the ships. All nations put trust in us for sea-carriage. The Spanish onion-grower on the slopes of Valencia, the Java sugar merchants, the breeders of Plata, looked to their harbours for sight of our hulls to load their products. Greek boatmen took payment for their cases on a scrap of dingy paper; the tide-labourers of the world demanded no earnest of their fees ere setting to work—our flag was their guarantor. The incoming of our ships brought throng to the quay-sides of far seaports; the outgoing sent the prospering merchants to the bank counters, to draw value from our skill in navigation, our integrity, and sea-care.

THE STRUCTURE

THE avalanche of war found us, if unprepared, not unready. The Merchants' Service was in the most efficient state of all its long story. Bounteous harvests had set a tide of prosperity to all parts of the world. Trade had reached the summit of a register in volume and account. The transport of the world's goods was busied as never before. With every outward stern wash went a full lading of our manufactures—a bulk of coal, a mass of wrought steel; foam at the bows—returning, brought exchange in food and raw materials, grist to the mills of our toiling artisans—a further provision for continuation of our trading. There were no idle keels swinging the tides in harbour for want of profitable employment; no seamen lounging on the dockside streets awaiting a 'sight' to sign-on for a voyage. Bulk of cargoes exceeded the tonnage of the ships, and the riverside shipyards resounded to the busy clamour of new construction. Advanced systems of propulsion had emerged from tentative stages, were fully tried and proved, and owners were adding to their fleets the latest and largest vessels that art of shipwrights and skill of the engineers could supply. We were well built and well found and well employed in all respects, not unready for any part that called us to sea.

On such a stage the gage was thrown. Right on the heels of the courier with challenge accepted, went the ships laden with a new and precious cargo—our gallant men-at-arms. Before a shot of ours was fired, the first

blow in the conflict was swung by passage of the ships: throughout the length of it, only by the sea-lanes could the shock be maintained.

Viewing the numbers and tonnage of the ships, the roll and character of the seamen, we were not uneasy for the sea-front. With the most powerful war fleet in the world boarding on the coasts of the enemy, we had little to fear. The transports and war-service vessels could be adequately safeguarded: the peaceful traders on their lawful occasions could trust in international law of the civilized seas, on which no destruction may be effected without cause, prefaced by examination. Of raiders and detached war units there might be some apprehension, but the White Ensign was abroad and watchful—it was impossible that the shafts of the enemy could reach us on the sea. For a time we set out on our voyages and returned without interference.



THE OLD AND THE NEW
THE MARGARET OF DUBLIN AND R.M.S. TUSCANIA

Anon, an amazing circumstance shocked our blythe assurance. In a new warfare, by traverse of a route we thought was barred, the impossible became a stern reality! While able, by power of their ships and skill and gallantry of the men, to keep the surface naval forces of the enemy doomed to ignoble harbour watch, the mightiest war fleet the seas had ever carried was impotent wholly to protect us! Our Achilles heel was exposed to merciless under-water attack, to a new weapon, deadly in precision and difficult to counter or evade. Throwing to the winds all shreds of honour and conscionable restraint, all vestiges of a sea-respect for non-combatants

and neutrals, the pacts and bounds of international law—the humane sea-usages that spared women and children and stricken wounded—the decivilized German set up the banners of a stark piracy, an ocean anarchy, to whose lieutenants the sea-wolves of an earlier age were but feeble enervated weaklings.

Piracy, gloried in and undisguised, faced us. Well and definite! We had known piracy in the long years of our sea-history: we had dealt with their trade to a full settlement at yard-arm or gallows. The course of our seafaring was not to be arrested by even the deep roots and deadly poison of this not unknown sea-growth: we had scaled the foul barnacles and cut the rank weeds before in the course of sea-development. If our ways had become peaceful in the long years of unchallenged trading, our habits were never less than combatant throughout a life of struggle with storm and tide. Not while we had a ship and a man to the helm would we be driven from the sea; our hard-won heritage was not to be delivered under threat or operation of even the most surpassing frightfulness. Jealousy for our seafaring, for our name as sailors, forbade that we should skulk in harbour or linger behind the nets and booms. Our work, our livelihood, our proud sea-trade, our honour was on the open sea. Our pride was this—that, in our action, we would be followed by the seafarers of the world. It was for no idle vaunt we boasted our supremacy at sea. If we could take first place of the world's seamen in time of peace, our station was to lead in war. We put out to sea—the neutrals followed. Had we held to port, German orders would have halted the sea-traffic of the world. With no shield but our seamanship, no weapon but the keenness of our eyes, no power of defence or assault other than the swing of a ready helm, we met the pirates on the sea, with little pretension in victory and no whining in defeat.

Challenged to stand and submit, the *Vosges* answered with a cant of the helm and hoist of her flag, and stood on her way under a merciless hail of shot. Unarmed, outsped, there was little prospect of escape—only, in an obstinate sea-pride, lay acceptance of the challenge. With decks littered by wreckage and wounded, bridge swept by shrapnel, water making through her torn hull, there was no thought to lay-to and droop the flag in surrender. When, at length, the ensign was shot away, there were men enough to hoist another. In hours their agony was measured, until, in despair of completing his foul work, the enemy gave up the contest. Reeking of the combat, the

Vosges foundered under her wounds. The sea took her from her gallant crew, but they had not given up the ship—their flag still fluttered at the peak as she went down. *Anglo-Californian* fought a grim, silent fight for four hours, matching the intensity of the German gunfire by the dogged quality of her mute defiance. *Palm Branch* turned away from galling fire at short range, double-banked the press in the stokehold, and cut and turned on her course to confuse the ranges. Her stern was shattered by shell, the lifeboats blown away; the apprentice at the wheel stood to his job with blood running in his eyes. Fire broke out and added a new terror to the situation. There was no flinching. Through it all the engines turned steadily, driven to their utmost speed by the engineers and firemen. A one-sided affair—a floating hell for seamen to stand by, helpless, and take a frightful gruelling! But they stood to it, and came to port.

If, under new and treacherous blows, our hearts beat the faster, there was little pause, no stoppage, in the steady coursing of our sea-arteries. We fought the menace with the same spirit our old sea-fathers knew. Undeterred by the ghastly handicap against us—the galling fetters of a policy that kept us unarmed, we pitted our brains and seamanship against the murderous mechanics of the enemy. To the new under-water attack there were few adequate counter-measures in the records of our old seafaring. We revised the standard manual, drew text from old games, shield from the cuttlefish, models for our sweeps from discarded sea-tackle. Special devices, new plans, stern services were called for; we devised, we specialized—our readiness was never more instant. Out of our strength we built up a new Service. Instruction and equipment came from the Royal Navy, but the men were ours. In the throes of our exertions the Merchants' Service repeated a tradition. The stout aged tree shot forth another worthy limb—a second Navy—not less ardent or resourceful than the first offshoot, now grown to be our guardian.

Our branches twined and interlocked in service of a joint endeavour. Under the fierce blast of war we swayed and weighed together in shield of our ancient foundation. Within our ranks we had cunning fishers, keen, resolute sea-fighters of the banks, to whom the coming of a strange mechanical devil-fish offered a new zest to the chase, a famous netting. Enrolled to Special Service, they engaged the enemy at his doorstep and patrolled the areas of his outset. Undaunted by the odds, deterred by no risk

or threat, they ranged and searched the sea-channels and cleared the lanes for our safe passage. To detect, to warn, to meet and counter-charge the submarine in his depths, to safeguard the narrow seas from hazard of the mines, was all in the day's work of the *Temporary* R.N.R.

Throughout all the enrolments, the divisions, the changes, and the training for new and special duties, there was no easing of the engines: we effected our adjustments and allotments under a full head of steam. All that the enemy could do could not prevent the steady reinforcement of our arms, the passage of our men, the transport of our trade. The long lines of our sea-communications remained unbroken, despite our losses and the grim spectre of the raft and the open boat. It could not be otherwise—and Britain stand. There could be no halt in the sea-traffic. Only from abroad could we draw supplies to raise the new leaguer of our island garrison; only by way of the sea could we retain and renew our strength.

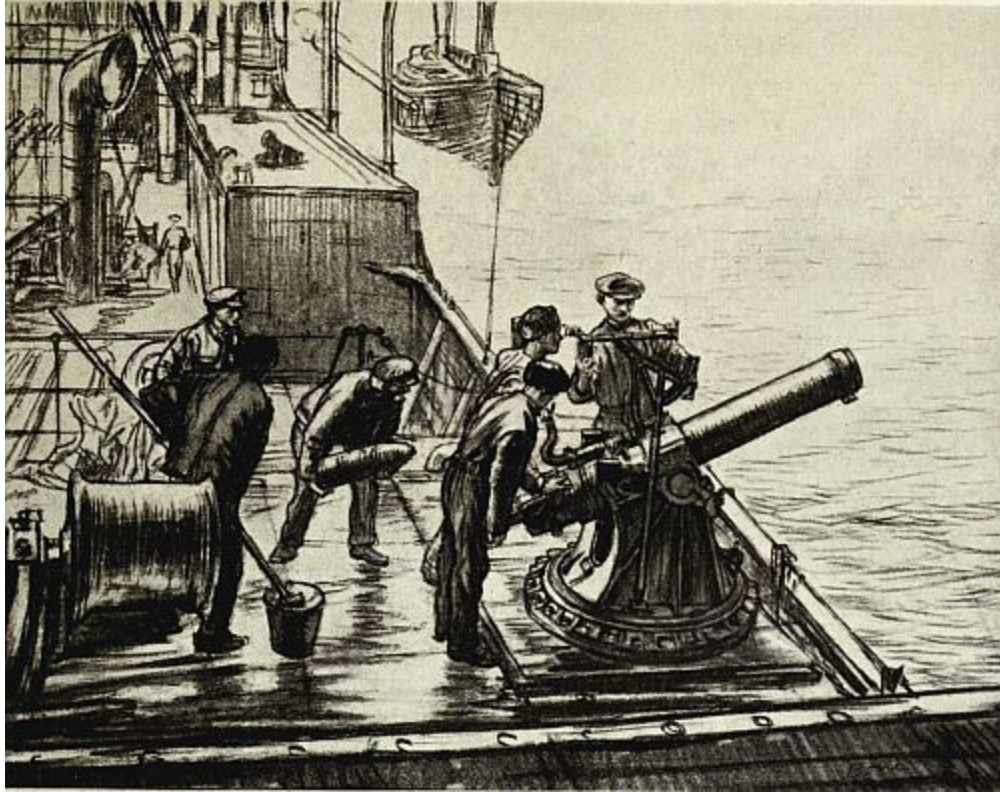
In time the intolerable shackles of inactive resistance were struck from our hands. Somewhat tardily we were supplied with weapons of defence and instructed in their use and maintainance. We went to school again, under tutelage of the Naval Service, and drew a helpful assistance from the tale of their courses since we had parted company. We were heartened by the new spirit of co-operation with the fighting service. Ungrudgingly they lent experts to direct our movement. They turned a stream of their inventive talent in the ways of gear and apparatus to protect our ships. They shipped our ordnance, and supplied skilled gunners to leaven our rude crews. More, they helped to strip the veneer of convention that hampered us—our devotion to standard practice in rules and lights and equipment. We learned our lessons. Even though the peaceful years had lessened our fighting spring, we had lost no aptitude for service of the guns in defence of our rights, nor for measure to deceive or evade. Armed and alert, we returned to the sea, confident in the discard of a weight in our handicap. We could strike back, and with no feeble blow—as the pirates soon learned.

There were scores to settle. *Palm Branch*, belying her tranquil name, took a payment in full for her shattered stern and the blood running in the steersman's eyes. Keen eyes sighted a periscope in time. The helm was put over and the white track raced across the stern, missing by feet. Baffled in under-water attack, the enemy hove up from his depths to open surface fire.

He never had opportunity. If look-out was good, gun action was as quick and ready in *Palm Branch*. Her first shot struck the conning-tower, the second drove home on the submarine, which sank. While all eyes were focused on the settling wash and spreading scum of oil, a new challenge came and was as speedily accepted. A shell, fired by a second submarine at long range, passed over the steamer. Slewing round to a new target, the gunners kept up a steady return, shot for shot. The submarine dropped farther astern, fearing the probe of a bracket: he angled his course to bring both his guns in action. Two pieces against the steamer's one! At that, he fared no better. Firing continuously, eighty rounds in less than an hour, he registered not one hit.

At length *Palm Branch's* steady, methodical search for the range had effect. Her gunners capped the day's fine shooting by a direct hit on the submarine's after-gun, shattering the piece. At evens again—the U-boat ceased fire and drew off, possibly under threat of British patrols approaching at full speed, more probably for the good and sufficient reason that he had had enough.

Not all our contests were as happily decided. If—shirking the issue of the guns, with no zest for a square fight—the German went to his depths, he had still the deadly torpedo to enforce a toll. The toll we paid and are paying, but there is no stoppage in the round by which the nation is fed and her arms served. The burden is heavy and our losses great, but we have not failed. We dare not fail.



IN A MERCHANTMAN—BOMB-THROWER PRACTICE

II

OUR RELATIONS WITH THE NAVY

JOINING FORCES

AFTER an interval of a hundred years, we are come to work together again, banded, as in the days of the Armada, to keep the seas against a ruthless challenger. In view of a new blood-bond between us, it is difficult to write coldly of the causes that have kept us apart. Only by preface of an affirmation can it be made possible. Through all our differences, prejudices, envies—perhaps jealousies—there ran at least one clear unsullied thread—our admiration for the Navy, our glory in its strength and power, our belief in its matchless efficiency.

We seamen, naval or mercantile, are a stout unmovable breed. Tenacity to our convictions is deeply rooted. The narrow trends of shipboard life give licence to a conservatism that out-Herods Herod in intensity, unreason—in utter sophistry. We extend this atmosphere to our relationships, to the associations with the beach, with other sea-services, with other ships—to the absurd pretensions of the other watch. "A sailorman afore a landsman, an' a shipmate afore all," may be a useful creed, but it engenders a contentious outlook, an intolerance difficult to reconcile. In the fo'c'sle, the upholding of a 'last ship' may lead to a broken nose; aft, the officers may quarrel, wordily, over the grades of their service; ashore, the captain may only reserve his confidences for a peer of his tonnage; over all, the distance between the Naval and Merchants' Services was immeasurable and complete.

If it was so to this date, it was perhaps more intense in the old days when common seafaring had not set as broad a distinction, as widely divergent a sea-practice, as our modern services shew. That such a contentious atmosphere existed we have ample witness. After experience as a merchants' man, Nelson wrote of his re-entry. "I returned a practical seaman

with a horror of the Royal Navy. . . . It was many weeks before I got the least reconciled to a man-o'-war, so deep was the prejudice rooted!" We have no such noted record of a merchant seaman re-entering from the Navy. Doubtless the laxity and indiscipline he might observe would produce a not dissimilar revulsion.

In the years that have elapsed since Nelson wrote, we have had few opportunities to compose our differences, to get on better terms with one another. The course of naval development took the great war fleets hull down on our commercial horizon, beyond casual intercommunication. On rare and widely separated occasions we fell into an expedition together, but the unchallenged power of the naval forces only served to heighten the barriers that stood between us. At the Crimea, in India, on the Chinese and Egyptian expeditions, during the Boer War, we were important links in the venture, but no more important than the cargoes we ferried. There was no call for any service other than our usual sea-work. The Navy saw to it that our comings and goings were unmolested. We were sea-civilians, purely and simply; there was nothing more to be said about it.

If little was said, it was with no good grace we took such a station. There were those who saw that seafaring could not thus arbitrarily be divided. Other nations were stirring and striving to a naval strength and power, drawing aid and personnel from their mercantile services. Sea-strength and paramountcy might not wholly come to be measured in terms of thickness of the armour-plating—in calibre of the great guns. Auxiliary services would be required. The Navy could no more work without us than the Army without a Service Corps.

The Royal Naval Reserve came as a link to our intercourse. Certain of our shipmates left us for a period of naval training. They came back changed in many particulars. They had acquired a social polish, were perhaps less 'sailor-like' in their habits. As a rule they were discontented with the way of things in their old ships; the quiet rounds bored them after the crowded life in a warship. We were frequently reminded of how well and differently things were done in *the* Service. Perhaps, in return, we took the wrong line. We made no effort to sift their experiences, to find out how we might improve our ways. Often our comrade's own particular shrewdness was cited as a reason for the better ways of naval practice. We

were rather irritated by the note of superiority assumed, perhaps somewhat jealous. Had commissions been granted on a competitive basis, we might have accepted such a tone, but we had our own way of assessing sea-values, and saw no reason why we should stand for these new airs. What was in it, what had wrought the change, we were never at pains to investigate. It was enough for us to note that, though his watch-keeping was certainly improved, our re-entered shipmate did not seem to be as efficient as a navigator or cargo supervisor as once we had thought him. All his talk of drills and guns and station-keeping considered, he seemed to have quite forgotten that groundnuts are thirteen hundredweights to the space ton and ought not to be stowed near fine goods!

On the other hand, he might reasonably be expected to see his old shipmates in a new light. Rude, perhaps. Of limited ideas. Tied to the old round of petty bickerings and small intrigues. He would note the want of trusty brotherhood. His sojourn among better-educated men may have roused his ideas to an appreciation of values that deep-sea life had obscured. The lack of the discipline to which he had become accustomed would appal and disquiet him. In time he would be worn to the rut again, but who can say the same rut? Unconsciously, we were influenced by his quieter manners. In self-study we saw faults that had been unnoticed before his return. Reviewing our hard sea-life, we recalled our exclusion from benefits of instruction that went a-begging on the beach. We stirred. There might yet be time to make up the leeway.

The influence of naval training was never very pronounced among the seamen and firemen of the Merchants' Service who were attached to the R.N.R. Their periods of training were too short for them to be permanently influenced by the discipline of the Navy (or our indiscipline on their return to us may have blighted a promising growth!) On short-term training they were rarely allotted to important work. The governing attitude was rather that they should be used as auxiliaries, mercantile handymen, in a ship. If there was a stowage of stores, cleaning up of bilges, chipping and scaling of iron rust—well, here was mercantile Jack, who was used to that kind of work; who better for the job? Generally, he returned to his old ways rather tired of Navy 'fashion' and discipline, and one saw but little influence of his temporary service on a cruiser. Usually, he was a good hand, to begin with:

he sought a post on good ships: with his papers in order we were very glad to have him back.

In few other ways did we come in touch with the Navy. At times the misfortune of the sea brought us into a naval port for assistance in our distress. Certainly, assistance was readily forthcoming, a full measure, but in a somewhat cold and formal way that left a rankling impression that we were not—well, we were not perhaps desirable acquaintances. The naval manner was not unlike that of a courteous prescribing chemist over his counter. "Have you had the pain—long?" "Is there any—coughing?" We had always the feeling that they were bored by our custom, were anxious to get back to the mixing of new pills, to their experiments. We were not very sorry when our repairs were completed and we could sail for warmer climates.

With the outbreak of war the R.N.R. was instantly mobilized. Their outgoing left a sensible gap in our ranks, a more considerable rift than we had looked for. Example drew others on their trodden path, our mercantile seamen were keen for fighting service; the unheralded torpedo had not yet struck home on their own ships. Commissions to a new entry of officers were still limited and capricious—the *Hochsee Flotte* had not definitely retired behind the booms at Kiel and Wilhelmshaven, to weave a web of murder and assassination. For a short term we sailed on our voyages, on a steady round, differing but little from our normal peace-time trade.

A short term. The enemy did not leave us long secure in our faith in civilized sea-usage. Our trust in International Law received a rude and shattering shock from deadly floating mine and racing torpedo. Paralysed and impotent to venture a fleet action, the German Navy was to be matched not only against the commercial fleets of Britain and her Allies, but against every merchant ship, belligerent or neutral. There was to be no gigantic clash of sea-arms; action was to be taken on the lines of Thuggery. The German chose his opponents as he chose his weapons. Assassins' weapons! The knife in the dark—no warning, no quarter, sink or swim! The 'sea-civilians' were to be driven from the sea by exercise of the most appalling frightfulness and savagery that the seas had ever known.

Under such a threat our sea-services were brought together on a rapid sheer, a close boarding, in which there was a measure of confusion. It could not have been otherwise. The only provision for co-operation, the R.N.R. organization, was directed to augment the forces of the Navy: there was no anticipation of a circumstance that would sound a recall. Our machinery was built and constructed to revolve in one direction; it could not instantly be reversed. Into an ordered service, ruled by the most minute shades of seniority, the finest influences of precedence and tradition, there came a need to fit the mixed alloy of the Merchants' Service. Ready, eager, and willing, as both Services were, to devote their energies to a joint endeavour, it took time and no small patience to resolve the maze and puzzle of the jigsaw. Naval officers detailed for our liaison were of varied moulds. Not many of the Active List could be spared; our new administrators were mostly recalled from fishing and farming to take up special duties for which they had few qualifications other than the gold lace on their sleeves. Some were tactful and clever in appreciation of other values than a mere readiness to salute, and those drew our affection and a ready measure of confidence. Others set up plumed Gessler bonnets, to which we were in no mood to bow. Only our devotion to the emergency exacted a jerk of our heads. To them we were doubtless difficult and trying. Our free ways did not fit into their schemes of proper routine. Accustomed to the lines of their own formal service, to issuing orders only to their juniors, they had no guide to a commercial practice whereby there can be a concerted service without the usages of the guard-room. They made things difficult for us without easing their own arduous task. They objected to our manners, our appearance, to the clothes we wore. Our diffidence was deemed truculence: our reluctance to accept a high doctrine of subservience was measured as insubordination.

The flames of war made short work of our moods and jealousies, prejudices, and dislikes. A new Service grew up, the *Temporary* R.N.R., in which we were admitted to a share in our own governance and no small part in combatant operations at sea. The sea-going section found outlet for their energy and free scope for a traditional privateering in their individual ventures against the enemy. Patrolling and hunting gave high promise for their capacity to work on lines of individual control. Minesweeping offered a fair field for the peculiar gifts of seamanship that mercantile practice engenders. Commissioned to lone and perilous service, they kept the seas in

fair weather or foul. Although stationed largely in the narrow seas, there were set no limits to the latitude and longitude of their employment. The ice of the Arctic knew them—riding out the bitter northern gales in their small seaworthy drifters, thrashing and pitching in the seaway, to hold a post in the chain of our sea-communications. In the Adriatic warmer tides lapped on their scarred hulls, but brought no relaxing variance to their keen look-out. For want of a match of their own size, they had the undying temerity to call three cheers and engage cruiser ordnance with their pipe-stems! A service indeed! If but *temporary* in title, there is permanence in their record!

Coincident with our actions on the sea—not alone those of our fighting cubs, but also those of our trading seamen—a better feeling came to cement our alliance. First in generous enthusiasm for our struggle against heavy odds, as they came to understand our difficulties, naval officers themselves set about to create a happier atmosphere. We were admitted to a voice in the league of our defence. Administration was adjusted to meet many of our grievances. Our capacity for controlling much of the machinery of our new movements was no longer denied. The shreds of old conservatism, the patches of contention and envy were scattered by a strong free breeze of reasoned service and joint effort.

We meet the naval man on every turn of the shore-end of our seafaring. We have grown to admire him, to like him, to look forward to his coming and association in almost the same way that we are pleased at the boarding of our favoured pilots. He fits into our new scheme of things as readily as the Port Authorities and the Ship's Husband. The plumed bonnets are no longer set up to attract our awed regard: by a better way than caprice and petulant discourtesy, the naval officer has won a high place in our esteem. We have borrowed from his stock to improve our store; better methods to control our manning, a more dispassionate bearing, a ready subordination to ensure service. His talk, too. We use his phrases. We 'carry on'; we ask the 'drill' for this or that; we speak of our sailing orders as 'pictures,' our port-holes are become 'scuttles.' The enemy is a 'Fritz,' a depth-charge a 'pill,' torpedoes are 'mouldies.' In speaking of our ships we now omit the definite article. We are getting on famously together.

AT SEA

ALTHOUGH our experience of their assured protection is clear and definite, our personal acquaintance with the larger vessels of the Navy is not intimate. Saving the colliers and the oilers and storeships that serve the Fleet, few of us have seen a 'first-rate' on open sea since the day the Grand Fleet steered north to battle stations. The strength and influence of the distant ships was plain to us in the first days of the war even if we had actually no sight of their grey hulls. While we were able to proceed on our lawful occasions with not even a warning of possible interference, the mercantile ships of the enemy—being abroad—had no course but to seek the protection of a neutral port, not again to put out to sea under their own colours.

The operation of a threat to shipping—at three thousand miles distance—was dramatic in intensity under the light of acute contrast. Entering New York a few days after war had been declared, we berthed alongside a crack German liner. Her voyage had been abandoned: she lay at the pier awaiting events. At the first, we stared at one another curiously. Her silent winches and closed hatchways, deserted decks and passages, were markedly in contrast to the stir and animation with which we set about unloading and preparing for the return voyage. The few sullen seamen about her forecastle leant over the bulwarks and noted the familiar routine that was no longer theirs. Officers on the bridge-deck eyed our movements with interest, despite their apparent unconcern. We were respectfully hostile: submarine atrocities had not yet begun. The same newsboy served special editions to both ships. The German officers grouped together, reading of the fall of Liège. Doubtless they confided to one another that they would soon be at sea again. Five days we lay. At eight o'clock 'flags,' our bugle-call accompanied the raising of the ensign: the red, white, and black was hoisted defiantly at the same time. We unloaded, re-loaded, and embarked passengers, and backed out into the North River on our way to sea again. The *Fürst* ranged to the wash of our sternway as we cleared the piers; her hawsers strained and creaked, then held her to the bollards of the quay.

Time and again we returned on our regular schedule, to find the German berthed across the dock, lying as we had left her, with derricks down and her hatchways closed. . . . We noted the signs of neglect growing on her; guessed at the indiscipline aboard that inaction would produce. For a while her men were set to chipping and painting in the way of a good sea-custom, but the days passed with no release and they relaxed handwork. Her topsides grew rusty, her once trim and clean paintwork took on a grimy tint. Our doings were plain to her officers and crew: we were so near that they could read the tallies on the mailbags we handled: there were no mails from Germany. Loading operations, that included the embarkation of war material, went on by night and day: we were busied as never before. The narrow water space between her hull and ours was crowded by barges taking and delivering our cargo; the shriek of steam-tugs and clangour of their engine-bells advertised our stir and activity. On occasion, the regulations of the port obliged the *Fürst* to haul astern, to allow working space for the Merritt-Chapman crane to swing a huge piece of ordnance to our decks. There were rumours of a concealed activity on the German. "She was coaling silently at night, in preparation for a dash to sea." . . . "German spies had their headquarters in her." The evening papers had a new story of her secret doings whenever copy ran short. All the while she lay quietly at the pier; we rated her by her draught marks that varied only with the galley coal she burnt.

At regular periods her hopeless outlook was emphasized by our sailings. Officers and crew could not ignore the stir that attended our departure. They saw the 'blue peter' come fluttering from the masthead, and heard our syren roar a warning to the river craft as we backed out. We were laden to our marks and the decks were thronged with young Britons returning to serve their country. The Fatherland could have no such help: the *Fürst* could handle no such cargo. For her there could be no movement, no canting on the tide and heading under steam for the open sea: the distant ships of the Grand Fleet held her in fetters at the pier.

While the Battle Fleet opened the oceans to us, we were not wholly safe from enemy interference on the high seas in the early stages of the war. German commerce raiders were abroad; there was need for a more tangible protection to the merchants' ships on the oversea trade routes. The older cruisers were sent out on distant patrols. They were our first associates of

the huge fleet subsequently detailed for our defence and assistance. We were somewhat in awe of the naval men at sea on our early introduction. The White Ensign was unfamiliar. Armed to the teeth, an officer from the cruiser would board us: the bluejackets of his boat's crew had each a rifle at hand. "Where were we from . . . where to . . . our cargo . . . our passengers?" The lieutenant was sternly courteous; he was engaged on important duties: there was no mood of relaxation. He returned to his boat and shoved off with not one reassuring grin for the passengers lining the rails interested in every row-stroke of his whaler. In time we both grew more cordial: we improved upon acquaintance. The drudgery and monotony of a lone patrol off a neutral coast soon brought about a less punctilious boarding. Our *procès-verbal* had unofficial intervals. "How were things at home? . . . Are we getting the men trained quickly? . . . What about the Russians?" The boarding lieutenants discovered the key to our affections—the secret sign that overloaded their sea-boat with newspapers and fresh mess. "A fine ship you've got here, Captain!" We parted company at ease and with goodwill. The boat would cast off to the cheers of our passengers. The great cruiser, cleared for action with her guns trained outboard, would cant in to close her whaler. Often her band assembled on the upper deck: the favourite selections were 'Auld Lang Syne' and 'Will ye no' come back again'—as she swung off on her weary patrol.

Submarine activities put an end to these meetings on the sea. Except while under ocean escort of a cruiser—when our relations by flag signal are studied and impersonal—we have now little acquaintance with vessels of that class. Counter-measures of the new warfare demand the service of smaller vessels. Destroyers and sloops are now our protectors and co-workers. With them, we are drawn to a familiar intimacy; we are, perhaps, more at ease in their company, dreading no formal routine. Admirals are, to us, awesome beings who seclude themselves behind gold-corded secretaries: commodores (except those who control our convoys) are rarely sea-going, and we come to regard them as schoolmasters, tutors who may not be argued with; post-captains in command of the larger escorts have the brusque assumption of a super-seamanship that takes no note of a limit in manning. The commanders and lieutenants of the destroyers and sloops that work with us are different; they are more to our mind—we look upon them as brother seamen. Like ourselves, they are 'single-ship' men. They are

neither concerned with serious plans of naval strategy nor overbalanced by the forms and usages of great ship routine. While 'the bridge' of a cruiser may be mildly scornful upon receipt of an objection to her signalled noon position, the destroyer captain is less assured: he is more likely to request our estimate of the course and speed. His seamanship is comparable to our own. The relatively small crew he musters has taught him to be tolerant of an apparent delay in carrying out certain operations. In harbour he is frequently berthed among the merchantmen, and has opportunity to visit the ships and acquire more than a casual knowledge of our gear and appliances. He is ever a welcome visitor, frank and manly and candid. Even if there is a dispute as to why we turned north instead of south-east 'when that Fritz came up,' and we blanked the destroyer's range, there is not the air of superior reproof that rankles.

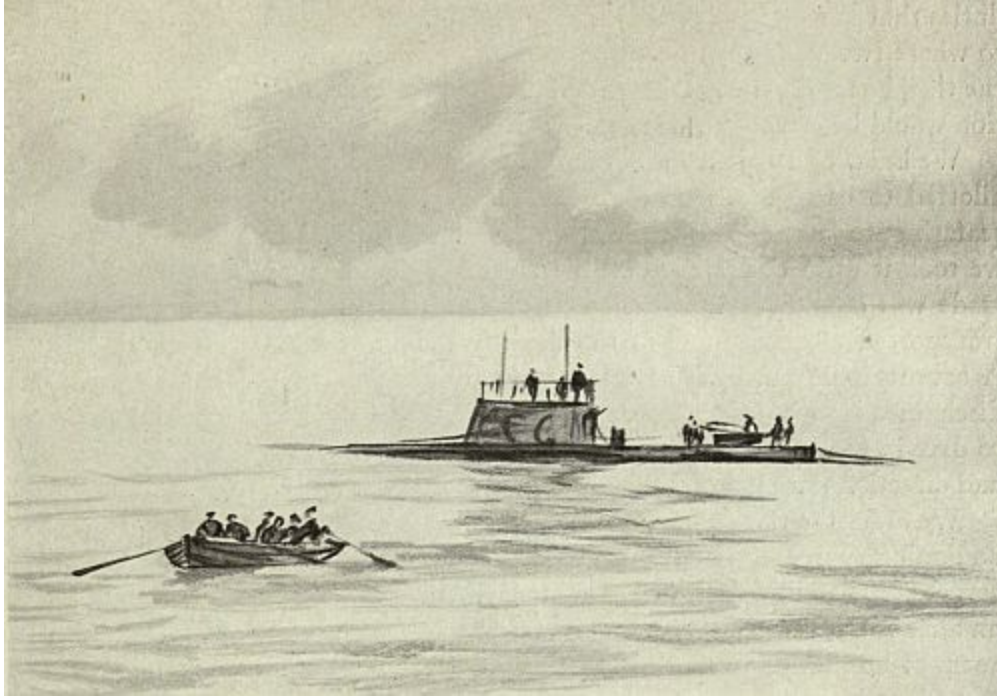
In all our relations with the Navy at sea there was ever little, if any, friction. We saw no empty plumed bonnet in the White Ensign. We were proud of the companionship and protection of the King's ships. Our ready service was never grudged or stinted to the men behind the grey guns; succour in our distress was their return. Incidents of our co-operation varied, but an unchanging sea-brotherhood was the constant light that shone out in small occurrences and deathly events.

Dawn in the Channel, a high south gale and a bitter confused sea. Even with us, in a powerful deep-sea transport, the measure of the weather was menacing; green seas shattered on board and wrecked our fittings, half of the weather boats were gone, others were stove and useless. A bitter gale! Under our lee the destroyer of our escort staggered through the hurtling masses that burst and curled and swept her fore and aft. Her mast and one funnel were gone, the bridge wrecked; a few dangling planks at her davits were all that was left of her service boats. She lurched and faltered pitifully, as though she had loose water below, making through the baulks and canvas that formed a makeshift shield over her smashed skylights. In the grey of the murky dawn there was yet darkness to flash a message: "*In view of weather probably worse as wind has backed, suggest you run for Waterford while chance, leaving us to carry on at full speed.*" An answer was ready and immediate: "*Reply. Thanks. I am instructed escort you to port.*"

The Mediterranean. A bright sea and sky disfigured by a ring of curling black smoke—a death-screen for the last agonies of a torpedoed troopship. Amid her littering entrails she settles swiftly, the stern high upreared, the bows deepening in a wash of wreckage. Boats, charged to inches of freeboard, lie off, the rowers and their freight still and open-mouthed awaiting her final plunge. On rafts and spars, the upturned strakes of a lifeboat, remnants of her manning and company grip safeguard, but turn eyes on the wreck of their parent hull. Into the ring, recking nothing of entangling gear or risk of suction, taking the chances of a standing shot from the lurking submarine, a destroyer thunders up alongside, brings up, and backs at speed on the sinking transport. Already her decks are jammed to a limit, by press of a khaki-clad cargo she was never built to carry. This is final, the last turn of her engagement. The foundering vessel slips quickly and deeper. "Come along, Skipper! You've got 'em all off! You can do no more! *Jump!*"

OUR WAR STAFF

SOME years before the war we were lying at an East Indian port, employed in our regular trade. The military students of the Quetta Staff College were in the district, engaged in practical exercise of their staff lessons. On a Sunday (our loading being suspended) they boarded us to work out in detail a question of troop transport. It was assumed that our ship was requisitioned in an emergency, and their problem was to estimate the number of men we could carry and to plan arrangement of the troop decks. Their inspection was to be minute; down to the sufficiency of our pots and pans they were required to investigate and figure out the resources of our vessel. The officer students were thirty-four in number; at least we counted thirty-four who came to us for clue to the mysteries of gross and register and dead-weight tonnage. In parties they explored our holds and accommodation, measured in paces for a rough survey, and prepared their plans. Their Commandant (a very famous soldier to-day) permitted us to be present when the officers were assembled and their papers read out and discussed. In general it was estimated that the work of alteration and fitting the ship for troops would occupy from eight to ten working days. Our quota—of all ranks—averaged about eleven hundred men.



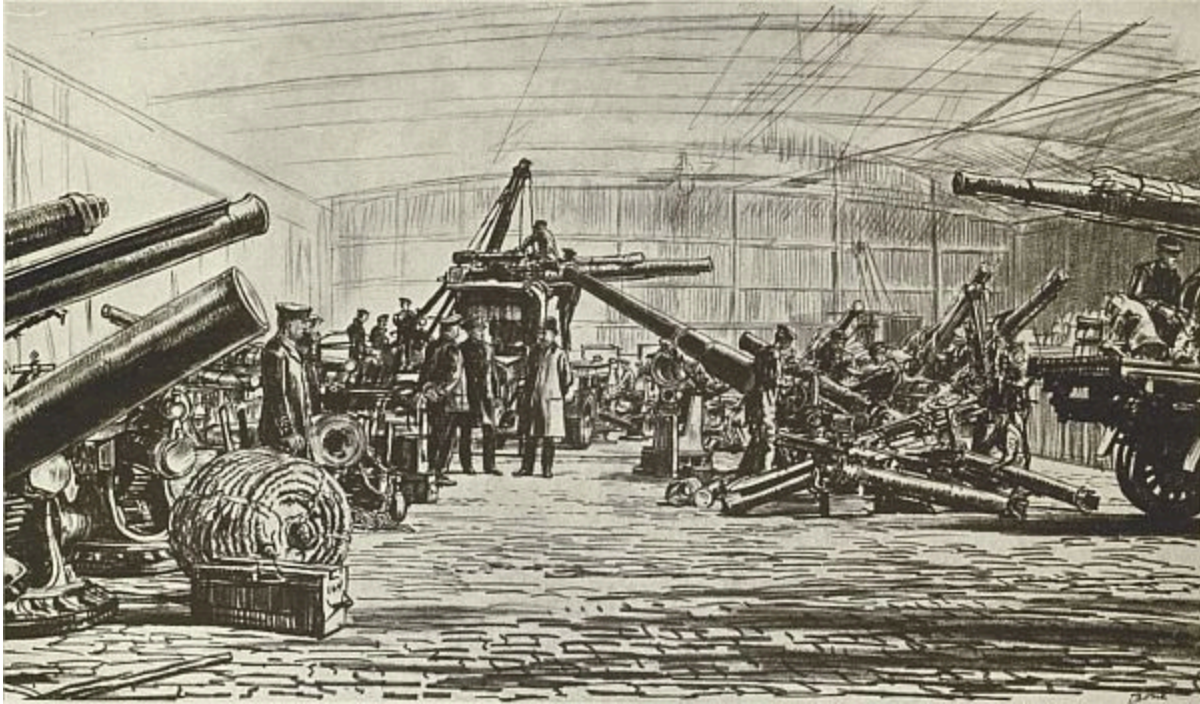
**A BRITISH SUBMARINE DETAILED FOR INSTRUCTION OF
MERCHANT OFFICERS**

The work was sound and no small ingenuity was advanced in planning adaptations, but the spirit of emergency did not show an evidence in their careful papers. The proposed voyage was distinctly stated to be from Newhaven to Dieppe, and it seemed to us that the elaborate accommodation for a prison, a guard-room, a hospital, were somewhat ambitious for a six-hour sea-passage. In conversation with the Commandant, we were of opinion that, to a degree, their work and pains were rather needless. Carrying passengers (troops and others) was our business; a trade in which we had been occupied for some few years. He agreed. He regarded their particular exercise in the same light as the 'herring-and-a-half' problem of the schoolroom: it was good for the young braves to learn something of their only gangway to a foreign field. "Of course," he said, "if war comes it will be duty for the Navy to supervise our sea-transport." We understood that their duty would be to safeguard our passage, but we had not thought of supervision in outfit. The Commandant was incredulous when we remarked that we had never met a naval transport officer, that we knew of no plans to meet such an emergency as that submitted to his officers. It was evident that his trained soldierly intendance could not contemplate a situation in which the seamen of the country had no foreknowledge of a war service; it was amazing to him that we were not already drilled for duties that might, at any

moment, be thrust upon us. Pointing across the dock to where two vessels of the Bremen Hansa Line were working in haste to catch the tide, he affirmed that they would be better prepared: *their* place in mobilization would be detailed, their duties and services made clear.

We knew of no plans for our employment in war service; we had no position allotted to us in measures for emergency. We were sufficiently proud of our seafaring to understand a certain merit in this apparent lack of prevision: we took it as in compliment to the efficiency and resource with which our sea-trade was credited. Was it not on our records that the Isle of Man steamers transported 58,000 people in the daylight hours of an August Bank Holiday. A seventy-mile passage. Trippers. Less amenable to ordered direction than disciplined troops. A day's work, indeed. Unequalled, unbeaten by any record to date in the amazing statistics of the war. There was no need for supervision and direction: we knew our business, we could pick up the tune as we marched.

We did. On the outbreak of war we fell into our places in transport of troops and military material with little more ado than in handling our peacetime cargoes. The ship on which the Staff students worked their problems set out on almost the very route they had planned for her, but with no prison or guard-room or hospital, and sixteen hundred troops instead of eleven: the time taken to fit her (including discharge of a cargo) occupied exactly four days. We saw but little of the naval authority.



THE D.A.M.S. GUNWHARF AT GLASGOW

Later, in our war work, we made the acquaintance of the naval transport officer. Generally, he was not intimate with the working of merchant ships. His duties were largely those of interpretation. Through him Admiralty passed their orders: it devolved on the mercantile shore staff of the shipping companies to carry these orders into execution. If, in transport services, our marine superintendents and ships' husbands did not share in the honours, it was not for want of merit. They could not complain of lack of work in the early days of the war when the transport officer was serving his apprenticeship to the trade. The absence of a keen knowledge and interest in commercial ship-practice at the transport office made for complex situations; hesitations and conflicting orders added to the arduous business. Under feverish pressure a ship would be unloaded on to quay space already congested, ballast be contracted for—and delivered; a swarm of carpenters, working day and night, would fit her for carriage of troops. At the eleventh hour some one idly fingering a tide-table would discover that the vessel drew too much water to cross the bar of her intended port of discharge. (The marine superintendent was frequently kept in ignorance of the vessel's intended destination.) Telegraph and telephone are handy—"Requisition cancelled" is easily passed over the wires! *As you were* is a simple order in official control, but it creates an atmosphere of misdirection almost as

deadly as German gas. Only our tremendous resources, the sound ability of our mercantile superintendents, the industry of the contractors and quay staffs, brought order out of chaos and placed the vessels in condition for service at disposal of the Admiralty.

Despite all blunders and vacillations our expedition was not unworthy of the emergency. How much better we could have done had there been a considered scheme of competent control must ever remain a conjecture. Four years of war practice have improved on the hasty measures with which we met the first immediate call. Sea-transport of troops and munitions of war has become a highly specialized business for naval directorate and mercantile executant alike. Ripe experience in the thundering years has sweetened our relations. The naval transport officer has learnt his trade. He is better served. He has now an adequate executant staff, recruited largely from the Merchants' Service. With liberal assistance he relies less on telegraph and telephone to advance his work: our atmosphere is no longer polluted by the miasma of indecision, and by the chill airs of the barracks.

Of our Naval War Staff, the transport officer was the first on the field, but his duties were only concerned with ships requisitioned for semi-naval service. For long we had no national assistance in our purely commercial seafaring. Our sea-rulers (if they existed) were unconcerned with the judicious employment of mercantile tonnage: some of our finest liners were swinging the tides in harbour, rusting at their cables—serving as prison hulks for interned enemies. Our service on the sea was as lightly held. We made our voyages as in peace-time. We had no means of communication with the naval ships at sea other than the universally understood International Code of Signals. Any measures we took to keep out of the way of enemy war vessels, then abroad, were our own. We had no Intelligence Service to advise us in our choice of sea-routes, and act as distributors of confidential information. We were far too 'jack-easy' in our seafaring: we estimated the enemy's sea-power over-lightly.

In time we learned our lesson. Tentative measures were advanced. Admiralty, through the Trade Division, took an interest in our employment. Orders and advices took long to reach us. These were first communicated to the War Risks Associations, who sent them to our owners. We received them as part of our sailing orders, rather late to allow of considered efforts

on our part to conform with their tenor. There was no channel of direct communication. When on point of sailing, we projected our own routes, recorded them in a sealed memorandum which we left with our owners. If we fell overdue Admiralty could only learn of our route by application to the holders of the memorandum. A short trial proved the need for a better system. Shipping Intelligence Officers were appointed at the principal seaports. At this date some small echo of our demand for a part in our governance had reached the Admiralty. In selecting officers for these posts an effort was made to give us men with some understanding of mercantile practice; a number of those appointed to our new staff were senior officers of the R.N.R. who were conversant with our way of business. (If they did, on occasion, project a route for us clean through the Atlantic ice-field in May, they were open to accept a criticism and reconsider the voyage.) With them were officers of the Royal Navy who had specialized in navigation, a branch of our trade that does not differ greatly from naval practice. They joined with us in discussion of the common link that held few opportunities for strained association. Certainly we took kindly to our new directors from the first; we worked in an atmosphere of confidence. The earliest officer appointed to the West Coast would blush to know the high esteem in which he is held, a regard that (perhaps by virtue of his tact and courtesy) was in course extended to his colleagues of a later date.

The work of the S.I.O. is varied and extensive. His principal duty is to plan and set out our oversea route, having regard to his accurate information of enemy activities. All Admiralty instructions as to our sea-conduct pass through his hands. He issues our confidential papers and is, in general, the channel of our communication with the Naval Service. He may be likened to our signal and interlocking expert. On receipt of certain advices he orders the arm of the semaphore to be thrown up against us. The port is closed to the outward-bound. His offices are quickly crowded by masters seeking information for their sailings: with post and telephone barred to us in this connection, we must make an appearance in person to receive our orders. A tide or two may come and go while we wait for passage. We have opportunity, in the waiting-room, to meet and become intimate with our fellow-seafarers. It is good for the captain of a liner to learn how the captain of a North Wales schooner makes his bread, the difficulties of getting

decent yeast at the salt-ports; how the schooner's boy won't learn ("indeed to goodness") the proper way his captain shows him to mix the dough!

On telegraphic advice the arm of the semaphore rattles down. The port is open to traffic again. The waiting-room is emptied and we are off to the sea, perhaps fortified by the S.I.O.'s confidence that the cause of the stoppage has been violently removed from the sea-lines.

Under the pressure of ruthless submarine warfare we were armed for defence. Gunnery experts were added to our war complement. A division for organization of our ordnance was formed, the Defensively Armed Merchant Ships Department of the Admiralty. We do not care for long titles; we know this division as the "Dam Ships." Most of the officers appointed to this Service are R.N.R. They are perhaps the most familiar of the war staff detailed to assist us. Their duties bring them frequently on board our ships, where (on our own ground) relations grow quickly most intimate and cordial. The many and varied patterns of guns supplied for our defence made a considerable shore establishment necessary, not alone for the guns and mountings, but for ammunition of as many marks as a Geelong wool-bale. In the first stages of our war-harnessing, the supply of guns was limited to what could be spared from battlefield and naval armament. The range of patterns varied from pipe-stems to what was at one time major armament for cruisers; we had odd weapons—*soixante-quinze* and Japanese pieces; even captured German field-guns were adapted to our needs in the efforts of the D.A.M.S. to arm us. Standardization in mounting and equipment was for long impossible. Our outcry for guns was cleverly met by the department. We could not wait for weapons to be forged: by working 'double tides' they ensured a twenty-four-hour day of service for the guns in issue, by a system that our ordnance should not remain idle during our stay in port. Incoming ships were boarded in the river, their guns and ammunition dismounted and removed to serve the needs of a vessel bound out on the same tide. The problem of fitting a 12-pounder on a 4.7 emplacement taxed the department's ingenuity and resource, but few ships were held in port for failure of their prompt action.

With the near approach to standardization in equipment (a state that came with increased production of merchant-ship arms) the division was able to reorganize on more settled lines. New types of armament were

issued to them and there was less adaptation for emplacements to be considered. With every ship fitted, the pressure on their resource was eased, the new ships being constructed to carry guns as a regular part of their equipment. While their activities are now less confused by the new methods, there is no reduction in their employment. Other defensive apparatus has been placed in their hands for issue and control, and their principal port establishments have grown from small temporary offices to large well-manned depots. To the surface guns have been added howitzers, bomb-throwers, and depth-charges for under-water action: smoke-screen fittings and chemicals form a part of their stock in trade: they issue mine-sinking rifles, and even control the supply of our zigzag clocks. The range of their work is constantly being extended. Their duties include inspection to ensure that darkening ship regulations may not fail for want of preparation in port. Makeshift screening at sea is dangerous.

Their establishments are at the principal seaports, with branch connections and transport facilities for reaching the smaller harbours. The gun-wharves may not present as splendid a spectacle as the huge store-sheds of our naval bases, but they have at least the busy air of being well occupied, a brisk appearance of having few 'slow-dealing lines' on the shelves. Their permanent staff of armourers and constructional experts are able to undertake all but very major repairs to the ordnance that comes under their charge. By express delivery—heavy motor haulage—they can equip a ship on instant requisition with all that is scheduled for her armament: down to the waste-box and the gun-layer's sea-boots, they can put a complete defensive outfit on the road almost before the clamour of a requesting telephone is stilled.

Another of our staff is the officer in charge of our 'Otter' installation, an ingenious contrivance to protect us against the menace of moored mines. For deadly spheres floating on the surface we have a certain measure of defence in exercise of a keen look-out, but our eyes avail us not at all in detecting mines under water moored at the level of our draught. Our 'Otters' may be likened to blind sea-dolphins, trained to protect our flanks, to run silently aside, fend the explosive charges from our course, bite the moorings asunder, and throw the bobbing spheres to the surface.

The 'Otter' expert is invariably an enthusiast. He claims for his pets every virtue. They run true, they bite surely: they can speak, indeed, in the complaint of their guide-wires when they are not sympathetically governed. While it is true that we curse the awkward 'gadgets' in their multitude of tricks, denounce the insistence with which they dive for a snug and immovable berth under our bilge keels—those of us who have come through a hidden minefield share the expert's affection for the shiny fish-like monsters. We cannot see their operation: we have no knowledge of our danger till it is past and over, a dark shape with ugly outpointing horns, turning and spinning in the seawash of our wake.



**INSTRUCTIONAL ANTI-SUBMARINE COURSE FOR MERCHANT OFFICERS AT
GLASGOW**

Adoption of the convoy system has brought a host to our gangways. Our war staff was more than doubled in the few weeks that followed the sinister April of 1917. If, at an earlier date, we had reasonable ground for complaint that our expert knowledge of our business was studiously ignored by the Admiralty, apparently they did not rate our ability so lightly when this old form of ship protection was revived. The additions to our staff included a large proportion of our own officers, withdrawn from posts where their knowledge of merchant-ship practice was not of great value. In convoy, measures were called for that our ordinary routine had not contemplated. The shore division of our new staff aid us in adapting our commercial sea-gear to the more instant demands of war service. They 'clear our hawse' from turns and twists in the chain of our landward connections. Repairs and adjustments, crew troubles, stores—that on a strict ruling may be deemed

private matters—became public and important when considered as vital to the sailing of a convoy. In overseeing the ships at the starting-line, indexing and listing the varying classes and powers of the vessels, the convoy section have no light task. To the longshore division, who compose and arrange the integrals of our convoys, we have added a sea-staff of commodores, R.N. and R.N.R., who go to sea with us and control the manœuvres and operations of our ships in station. For this, not only a knowledge of squadron movements is required: the ruling of a convoy of merchantmen is complicated as much by the range of character of individual masters as by the diverse capabilities of the ships.

It was not until the spring of 1917 that Admiralty instituted a scheme of instruction in anti-submarine measures for officers of the Merchants' Service. We were finding the defensive tune difficult to pick up as we marched. The German submarine had grown to be a more complete and deadly warship. Sinkings had reached an alarming height: a spirit almost of fatalism was permeating the sea-actions of some of our Service. Our guns were of little avail against under-water attack. Notwithstanding the tricks of our zigzag, the torpedoes struck home on our hulls. If our luck was 'in,' we came through: if we had bad fortune, well, our luck was 'out!' A considerable school—the bold 'make-a-dash-for-it-and-chance-the-ducks' section of our fellows—did not wholly conform to naval instructions. In many cases zigzag was but cursorily maintained; in darkening ship, measures were makeshift and inadequate.

Schools for our instruction were set up at various centres, in convenient seaport districts. At the first, attendance was voluntary, but it was quickly evident to the Admiralty that certain classes of owners would give few facilities to their officers to attend, when they might be more profitably employed in keeping gangway or in supervising cargo stowage. (The fatalistic spirit was not confined to the seagoers among us.) Attendance at the classes of instruction was made compulsory; it became part of our qualification for office that we should have completed the course.

Although our new schooling occupies but five days, it is intensive in its scope and application. The cold print of our official instructions has its limitations, and Admiralty circulars are not perhaps famous for lucidity. More can be done by a skilled interpreter with a blackboard in a few

minutes than could be gathered in half an hour's reading. At first assembly there is perhaps an atmosphere of boredom. Routine details and a programme of operations are hardly welcome to masters accustomed to command. In a way, we have condescended to come among our juniors, to listen with the mates and second mates to what may be said: we assume, perhaps, a detached air of constraint.

It is no small tribute to the lecturer that this feeling rarely persists beyond the opening periods. Only the most perversely immovable can resist the interest of a practical demonstration. The classes are under charge of an officer, R.N., who has had deep-sea experience of enemy submarine activities. Often he is of the 'Q-ship' branch, and can enliven his lectures with incidents that show us a side of the sea-contest with which not many are familiar. If we are informed of the deadly advantage of the submarine, we are equally enlightened as to its limitations. In a few minutes, by virtue of a plot on the blackboard, the vantage of a proper zigzag is made clear and convincing. Points of view—in a literal sense—are expounded, and not a few of us recall our placing of look-outs and register a better plan. Following the officer in charge, a lieutenant of the Submarine Service dissects his vessel on the blackboard, carefully detailing the action in states of weather and circumstance. The under-water manœuvres of an attack are plotted out and explained in a practical way that no handbook could rival. The personal magnetism of the expert rivets our attention; the routine of under-seafaring gives us a good inkling of the manner of man we have to meet and fight at sea; we are given an insight to the mind-working of our unseen opponent—the brain below the periscope is probed and examined for our education.

Nothing could be better illustrative of the wide character of our seafaring than the range of our muster in the lecture-hall. Every type of our trade appears in the class that assembles weekly to attend the instructional course. We have no grades of seniority or precedence. We are sea-republicans when we come to sit together in class. Hardy coasting masters, commanders of Royal Mail Packets, collier mates, freighter captains, cross-Channel skippers, we are at ease together in a common cause; on one bench in the classroom may be seafarers returned from foreign ports as widely distant as Shanghai and Valparaiso.

For instruction in gunnery and the use of special apparatus we come under tuition of a type of seaman whom we had not met before. If the backbone of the Army is the non-commissioned man, the petty officer of the Royal Navy is no less the marrow of his Service. Unfortunately, we have no one like him in the Merchants' Service. As Scots is the language of marine engines, the South of England accent may be that of the guns. That liquid ü! "Metal adapters, genelmen, lük. Metal adapters is made o' alüminium bronze. They are bored hoüt t' take a tübe, an' threaded on th' hoütside t' screw into th' base o' th' cartridge case—like this 'ere. Genelmen, lük. . . ." His intelligent demonstration of the gear and working of the types of our armament possesses a peculiar quality, as though he is trying hard to reduce his exposition to our level. (As a matter of plain fact, he is.)

The instructional course closes on a note of confidence. We learn that even 'inexorable circumstance' has an opening to skilled evasion. We go afloat for a day and put into practice some measure of our schooling. At fire-control, with the guns, we exercise in an atmosphere of din and burnt cardboard, aiming at a hit with the fifth shot in sequence of our bracket. (An earlier bull's-eye would be bad application of our lectures.) A smoke-screen is set up for our benefit, and we turn and twist in the artificially produced fumes and vapours in a practical demonstration of defence. A sea-going submarine is in attendance and is open to our inspection. Her officers augment the class instruction by actual showing. Every point in the maze of an under-water attack is emphasized by them in an effort to impress us with the virtue of the counter-measures advised. It must be hard indeed for the submarine enthusiast (and they are all enthusiasts) to lay bare the 'weaknesses' of his loved machine. We feel for them almost as if we heard a man, under pressure, admit that his last ship was unseaworthy.



THE LOSS OF A LINER

III

THE LONGSHORE VIEW

EARLY in November 1914, on return from the sea, I was invited to join His Majesty's Forces.

" . . . An' I can tell you this, mister," said the sergeant . . . "it ain't everybody as I asks t' join our corps. . . . Adjutant, 'e ses t' me this mornin', 'Looka here, Bates,' 'e ses, 'don't you go for to bring none o' them scallywags 'ere! We don't want 'em! We won't 'ave 'em at any price,' 'e ses! . . . 'Wot we wants is proper men—men with chests,' 'e ses!"

I felt somewhat commended; I trimmed more upright in carriage; he was certainly a clever recruiter. I told him I had rather important work to do. He said, with emphasis, that it must be more than important to keep a MAN out of the Army—these days! In sound of shrieking newsboys—"Ant—werp fallen! British falling back!"—I agreed.

I asked him what he did with the men recruited. He was somewhat surprised at my question, but told me that, when trained, they were sent across to the Front—he was hoping to *return* himself in the next draft. He thought all this talk was needless, and grew impatient. I mentioned that the men couldn't very well swim over there. He glared scornfully. "Swim? . . . Swim!. . . 'Ere! Wot th' hell ye gettin' at? You gotta hellova lot t' say about it, anyway!"

I explained that my business was that of putting the troops and the guns and the gear o' war across; that the drafts couldn't get very far on the way without our assistance. He glanced at my soft felt hat, at my rainproof coat, my umbrella, my handbag—said, "*Huh*" and went off in search of a more promising recruit. His broad back, as he strode off swinging his cane, expressed an entire disapproval of my appearance and my alleged business.

Good honest sergeant! His course was a clear and straight one. He would hold no more truck with one who wouldn't take up a man's job. His "*Huh*"

and the swing of his arm said plainly to me, "Takin' th' boys across, eh? A —— fine excuse, . . . a rare —— trick! Where's yer uniform? Why ain't ye in uniform, eh? You can't do me with that story, mister! I'm an old Service man, I am. I been out t' India. I been on a troopship. I seen all them gold-lace blokes a-pokin' their noses about an' growsin' at th' way th' decks was kep! *Huh!* A damn slacker, mister! That's wot I think o' you!"

The sergeant's attitude was not unreasonable. Where was our uniform? Where was any evidence of our calling by which one could recognize a seaman on shore? A sea-gait, perhaps! But the deep-sea roll has gone out since bilge-keels came to steady our vessels! Tattoo marks? These cunning personal adornments are now reserved to the Royal Artillery and officers of the Indian Army! Tarry hands? Tar is as scarce on a modern steamer as strawberries in December! Sea-togs? If there be a preference, we have a fondness for blue serge, but blue serges have quite a vogue among bankers and merchants and other men of substance! Away from our ships and the dockside waterfront, we are not readily recognizable; we join the masses of other workers, we become members of the general public. As such, we may lay claim to a common liberty, and look at our seafaring selves from an average point of longshore view.

. . . The sea? Oh, we know a lot about it! It is in us. We pride ourselves, an island race, we have the sea in our blood, we are born to it. Circumstances may have brought us to counting-house and ledger, but our heart is with the sea. We use, unwittingly, many nautical terms in our everyday life. We had been to sea at times, on a business voyage or for health or pleasure. We knew the captain and the mates and the engineers. The chief steward was a friend, the bos'n or quartermaster had shown us the trick of a sheepshank or a reef-knot or a short splice. Their ways of it! Port and starboard for left and right, knots for miles, eight bells, the watches, and all that! We returned from our sea-trip, parted with our good friends, feeling hearty and refreshed. We hummed, perhaps, a scrap of a sea-song at the ledgers. We regretted that our sea-day had come so quickly to an end. Anyway, we felt that we had got to know the sea-people intimately.

But that was on their ground, on the sea and the ship, where they fitted to the scheme of things and were as readily understood and appreciated as the little round port-holes, the narrow bunks, the cunning tip-up washstands, the rails for hand-grip in a storm. Their atmosphere, their stories, their habits, were all part of our sea-piece. Taken from their heaving decks and the round of a blue horizon, they seemed to go out of our reckoning. On shore? Of course they must at times come on shore, but somehow one doesn't know much about them there. There are our neighbours. . . . Yes! Gudgeon's eldest boy, he is at sea—a mate or a purser. He has given over wearing his brass buttons and a badge cap now: we see him at long intervals, when he comes home to prepare for examinations. A hefty sort of lad—shouldn't think he would do much in the way of study; a bit wild perhaps. Then Mrs. Smith's husband. Isn't he at sea, a captain or a chief engineer, or something? He comes among us occasionally; travels to town, now and then, in our carriage. A hearty man—uses rather strong language, though! Has not a great deal to say of things—no interest in politics, in the market, in the games. Never made very much of him. Don't see him at the clubs. Seems to spend all his time at home. At home! Oh yes; wasn't it only the other day his small daughter told ours her daddy was *going* home again on Saturday!

In war, we are learning. There are no more games; contentious politics are not for these days; the markets and business are difficult and wayward. We are come to see our dependence on the successful voyages of Mrs. Smith's husband. His coming among us, from time to time, is proof that our links with the world overseas are yet unbroken, that there may still be business to transact when we turn up at the office. Strangely, in the new clarity of a war vision, we see his broad back in our harvest-fields, as we had never noticed it before. He is almost one of our staff. He handles our goods, our letters, our gold, our securities, our daily bread. His business is now so near to us that——

But no! It cannot properly be done. We recall that there *is* one way for our ready recognition when we come on shore these days. We cannot appropriate a longshore point of view, we cannot conceal our seafaring and merge into the crowd. There *is* a mark—our tired eyes, as we come off the sea! True, there are now, sadly, many tired eyes on the beach, but few carry the distant focus, the peculiar intentness brought about by absence of

perspective at sea. We cannot adopt a public outlook owing to this obliquity in our vision, we are barred by the persistence of that vexed perspective in our views on shore.

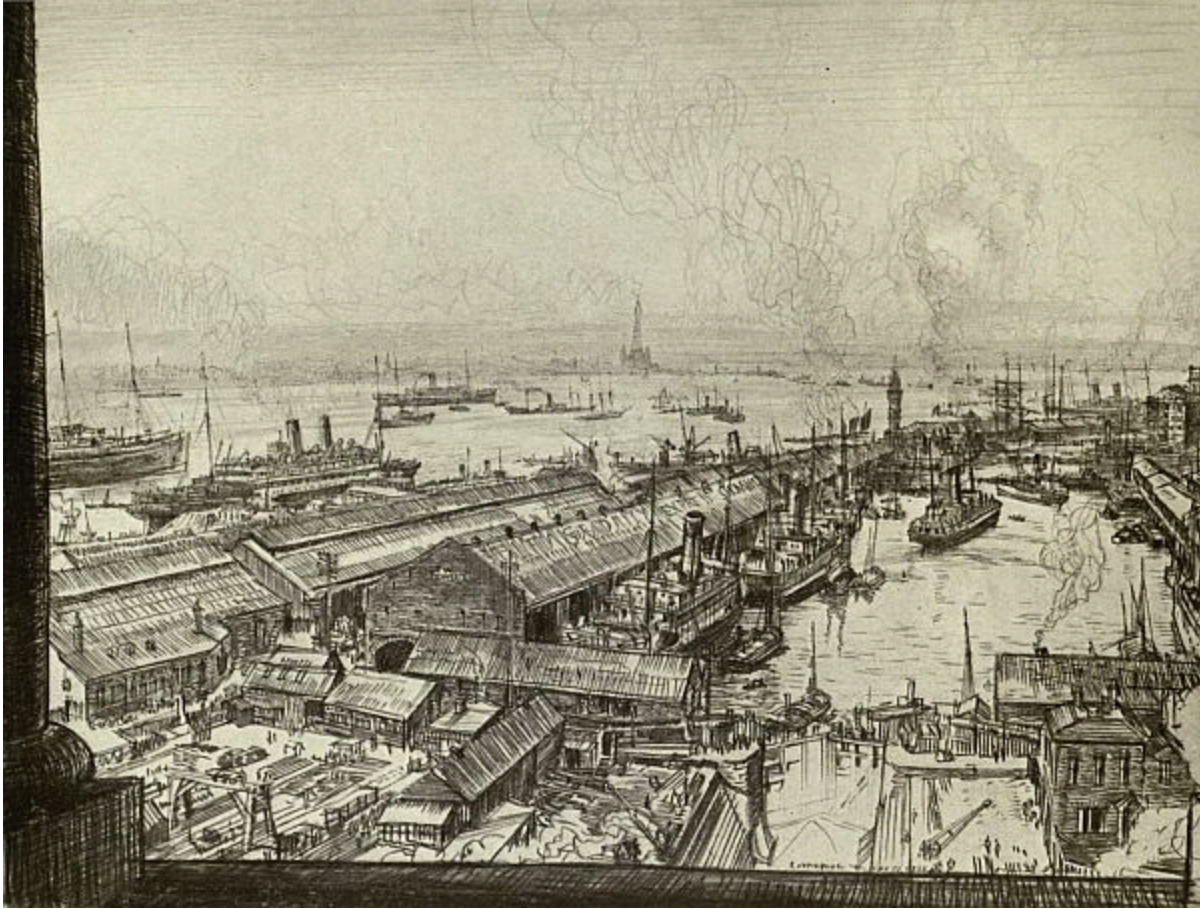
Still, the point may be raised that only in our actual seafaring are we recognized. We are poor citizens, nomads, who have little part with settled grooves and communal life on shore. The naval seaman is a known figure on the streets. His trim uniform, the cut of his hair, the swing of a muscular figure, his high spirits, are all in part with a stereotyped conception. He is the sailor; Mercantile Jack has lost his tradition in attire and individuality, he has vanished from the herd with his high-heeled shoes, coloured silk neckerchief, and sweet-tobacco hat.

In the round of shore communications there is exercise for assessing a measure of the other man's work: a large proportion of success hinges on easy fellowship, on an understanding and acquaintance not only with the technics of another's trade, but with his habits and his pursuits. All trades, all businesses, all professions have relations, near or distant, with the sea, but to them our grades and descriptions are dubious and uncertain. For this we are to blame. We are bad advertisers. We are content to leave our fraternization with the beach to the far distant day when we shall retire from the sea-service, 'swallow the anchor,' and settle down to longshore life. We cannot join and rejoin the guilderies on shore in the intervals of our voyaging. We preserve a grudging silence on our seafaring, perhaps tint what pictures we do present in other lights than verity. The necessary aloofness of our calling makes for a seclusion in our affairs: we make few efforts to remedy an estrangement; in a way, we adopt the disciplinary scourge of the flagellants, we glory in our isolation. If we share few of the institutions that exist for fellowship ashore, we have made no bid for admittance: if the tide of intercourse leaves us stranded, we have put out no steering oar on the drift of the flood. We are somewhat diffident. Perhaps we are influenced by a certain reputation that is still attached to us. Are we the prodigals not yet in the mood to turn unto our fathers?

Stout old Doctor Johnson enlarged on the sea-life—of his day—with a determination and no small measure of accuracy. "Sir," he said, "a ship is worse than a gaol. There is in a gaol better air, better company, better conveniency of every kind; and a ship has the additional disadvantage of

being in danger. When men come to like a sea-life they are not fit to live on land. . . . Men go to the sea before they know the unhappiness of that way of life; and when they have come to know it, they cannot escape from it, because it is then too late to choose another profession." At least he admitted the possibility of some of us coming to *like* a sea-life, though his postulate conveyed no high opinion of our intelligence in such a preference.

We have travelled far since the worthy Doctor's day. Not all his dicta may stand. There is still, perhaps, greater danger in a ship than in gaol, but Johnson himself admitted that "the profession of sailors has the dignity of danger"! For the rest, our air has become so good that invalids are ordered to sea; our conveniences are notably improved, our ships the last word in strength and comfort. Our company? Our company fits to the heave of our sea. If we have middling men for the trough, we have bold gallants for the crest. We draw a wide range to our service. The sea can offer a good career to a prizeman: we can still do moderately well with the wayward boy, the parents' 'heart-break,' the lad with whom nothing can be done on shore. Steam has certainly given a new gentility to our seafaring, but it cannot wholly smooth out the uneven sea-road. If we lose an amount of polish, of distinguished association, of education in our recruitment, we may gain just that essence that fits a man for our calling. Our company is, at any rate, stout and resolute, and, without that, we had long since been under German bondage.



THE MERSEY FROM THE LIVER BUILDINGS, LIVERPOOL

The war has brought a new prominence to our sea-trade. The public has become interested not alone in our sea-ventures, but in our landward doings. The astonishing fact of our civilian combatance has drawn a recognition that no years of peace could have uncovered. Not least of the revelations that the world conflict has imposed is the vital importance of the ships. Our naval fleets were ever talked of, read of, gloried in, as the spring of our national power, but not many saw the core of our sea-strength in the stained hulls of the merchants' ships. They were accepted without enthusiasm as an existing trade channel; they were there on a round of business and trade, not dissimilar to other transport services—the railways, road-carriage, the inland canals, the moving-van, the messengers. They were ready to hand for service; so near that their vital proportions were not readily apparent. Perhaps the greatest compliment the public has paid to the Merchants' Service lay in this abstract view. One saw an appreciation, perhaps unspoken, in the consternation that greeted the first irregularity in delivery of the oversea mails. Then, indeed, the importance of the ships was

brought sharply home. It was incredible: it was unheard of. Mercantile practice and correspondence had outgrown all duplications and weatherly precautions; the service was so sure and uninterrupted that no need existed for a second string to the bow. Bills of exchange, indents, invoices, the mail-letter, had long been confided to sea-carriage on one bottom. Pages could be written of the tangled skeins, the complex situations, the confusion and congestion that were all brought about by extra mileage of an ocean voyage. Fortunes, not alone in hulls and cargo, lie with our wreckage on the floor of the channels.

The sea-front suddenly assumed an importance in the general view, as the drain on our tonnage left vacant shelves in the bakehouse. Commodities that, so common and plentiful, had been lightly valued, were out of stock—the ships had not come in! Long queues formed at the shop doors, seeking and questioning—their topic, the fortunes of the ships! The table was rearranged in keeping with a depleted larder. Anxious eyes turned first in the morning to the list of our sea-casualties; the ships, what of the ships? The valiant deeds of our armies, the tide and toll of battles, could wait a second glance. Not all the gallantry of our arms could bring victory if our sea-communications were imperilled or restrained; on the due arrival of the ships centred the pivot of our operations.

Joined to the fortune of the ships, interest was drawn to the seamen. A new concern arose. Who were the mariners who had to face these deadly perils to keep our sea-lines unbroken? Were they trained to arms? How could they stand to the menace that had so shocked our naval forces? Daily the toll rose. Savagery, undreamt of, succeeded mere shipwreck: murder, assassination, mutilation became commonplace on the sea. Who were the mercantile seamen; of what stock, what generation?

To a degree we were embarrassed at such new attention. The mystery of sea-life, we felt, had unbalanced the public view. Our stock, our generation, was the same as that of the tailors and the candlestick-makers who were standing the enemy on his head on the Flanders fields; we differed not greatly from the haberdasher and the baby-linen man who drove the Prussian Guard, the proudest soldier in Europe, from the reeking shambles of Contalmaison. Indeed, we had advantage in our education for a fight. Our training, if not military, was at least directed to mass operations in

contest with power of the elements: torpedo and mine were but additions to the perils of our regular trade. If the clerk and the grocer could rise from ordered peaceful ways and set the world ringing with his gallantry and heroism, we were poltroons indeed to flinch and falter at the familiar conduct of our seafaring. We felt that our share in warfare was as nothing to the blaze of fury on the battle-fronts, our sea-life was comparative comfort in contrast to the grisly horrors of the trenches.

With universal service, opportunity for acquaintance with our life and our work was extended beyond the numbers of chance passengers. The exodus oversea of the nation's manhood brought the landsman and the seaman together as no casual meeting on the streets could have done. Millions of our country-men, who had never dreamed of outlook on blue water bounded by line of an unbroken horizon, have found themselves brought into close contact with us, living our life, assisting in many of our duties, facing the same dangers. In such a firm fellowship and communion of interest there cannot but be a bond between us that shall survive the passage of high-water mark.



THE MASTER OF THE GULL LIGHTSHIP WRITING THE LOG

IV

CONNECTION WITH THE STATE

TRINITY HOUSE, OUR ALMA MATER

OF all trades, seafaring ever required a special governance, a unique Code of Laws, suited to the seaman's isolation from tribunal and land court, to the circumstance of his constant voyaging. On sea, the severance from ordered government, from reward as from penalty, was irremediable and complete. No common law or enactment could be enforced on the wandering sea-tribesmen who owned no settled domicile, who responded only to the weight of a stronger arm than their own, who had an impenetrable cloak to their doings in the mystery of distant seas. The spirit and high heart that had called them to the dangers and vicissitudes of a sea-life would not brook tamely the dominance and injunction of a power whose authority was, at sea, invisible—and even under the land, could carry but little distance beyond high-water mark. To the bold self-enterprise of the early sea-venturers, the unconfined ocean offered a free field for a standard of strength, for a law of might alone. Kings and Princes might rule the boundaries of the land, but the sea was for those who could maintain a holding on the troubled waters. Were the 'Rectores' not Kings on their own heaving decks, their province the round of the horizon, their subjects the vulgar 'shippe-men,' their slaves the unfortunate weaker seafarers, whom chance or the fickle winds had brought within reach of their sea-arms? The sea-rovers were difficult to bridle or restrain. *Spurlos versenkt* might well have been their motto—as that of later pirates. No trace! The sea would tell no tales. They were alone on the breadth of the ocean, no ordered protection was within hail, the land lay distant under rim of the sea-line. Blue water would wash over the face of robbery and crime: the hazards of the sea could well account for a missing ship!

Reverse the setting and the same uncharity could similarly be masked. In turn, the humanity the seamen contemned was denied to them. Driven on shore, wrecked or foundered on coast or shoal, the laws they scorned were powerless to shield or salve the wreckage of their vessels, to save their weary sea-scarred bodies. 'No trace' was equally a motto for the dwellers on the coast: blue water would wash as freely over their bloody evidence, the miserable castaways could be as readily returned to the pitiless sea: an equal hazard of the deep could as surely account for missing men!

Only special measures could control a situation of such a desperate nature, no ordinary governance could effect a settlement; no one but a powerful and kingly seafarer could frame an adjustment and post wardens to enforce a law for the sea. When Richard Cœur de Lion established our first Maritime Code, he had his own rude sea-experience to guide him. On perilous voyaging to the Holy Land, he must have given more than passing thought to the trials and dangers of his rough mariners. Sharing their sea-life and its hardships, he noted the ship-measures and rude sea-justice with a discerning and humane appreciation. In all the records of our law-making there are few such intimate revelations of a minute understanding as his *Rôles d'Oléron*. The practice of to-day reflects no small measure of his wisdom; in their basic principles, his charges still tincture the complex fabric of our modern Sea Codes. Bottomry—the pledging of ship and tackle to procure funds for provision or repair; salvage—a just and reasonable apportionment; jettison—the sharing of another's loss for a common good; damage to ship or cargo—the account of liability: many of his ordinances stand unaltered in substance, if varied and amplified in detail.

The spirit of these mediæval Shipping Acts was devoted as well to restrain the lawless doings of the seamen as to check the inhuman plunderings of the coast dwellers. The rights and duties of master and man were clearly defined: in the schedule of penalties, the master's forfeit was enhanced, as his was assumed to be the better intelligence. For barratry and major sea-crimes, the penalty was death and dismemberment. All pilots who wrecked their charges for benefit of the lords of the sea-coast were to be hung on a gibbet, and so exhibited to all men, near the spot where the vessels they had misdirected were come on shore. The lord of the foreshore who connived at their acts was to suffer a dire fate. He was to be burned on a stake at his own hearthstone, the walls of his mansion to be razed, and the

standing turned to a market-place for barter of swine! Drastic punishment! Doubtless kingly Richard drew abhorrence for the wrecker from his own bitter experience on the inhospitable rocky coast of Istria!

Little detail has come down to us of the means adopted to enforce these just acts. Of the difficulties of their enforcement we may judge a little from the character of the seamen as presented by contemporary chronicles. . . .

*"Full many a draught of wyn had he drawe
From Burdeux-ward, whil that the chapman
sleep.
Of nyce conscience took he no keep.
If that he foughte, and hadde the heigher
hand,
By water he sent hem hoom to every land."*

. . . Thus Chaucer; but Chaucer was a Collector of Customs, and would possibly assess the stolen draught of Bordeaux as a greater crime than throwing prisoners overboard! From evidence of the date, Richard's shipping laws seem to have been but lightly regarded by the lords of the foreshore. In the reign of King John, wrecking had become a practice so common that prescriptive rights to the litter of the beaches was included in manorial charters, despite the Rôle that . . . "the pieces of the ship still to belong to the original owners, notwithstanding any custom to the contrary . . . and any participators of the said wrecks, whether they be bishops, prelates, or clerks, shall be deposed and deprived of their benefices, and if lay people they are to incur the penalties previously recited."

It was surely by more than mere chance the churchmen were thus specially indicted! Perhaps it was by a temporal as well as a spiritual measure that Stephen Langton, Archbishop of Canterbury, strove to remove a reproach to the Church. He founded a Guild of sea-samaritans, a Corporation

"of godly disposed men, who, for the actual suppression of evil disposed persons bringing ships to destruction by the shewing forth of false beacons, do bind themselves together in the Love of our Lord Christ, in the name of the Masters and Fellows of Trinity Guild to succour from the dangers of the sea all who are beset upon the coasts of England, to feed them when

ahungered and athirst, to bind up their wounds, and to build and light proper beacons for the guidance of mariners."

An earnest and compassionate Charter: a merciful and honourable Commission.

In this wise was formed our Alma Mater, the ancient guild of shipmen and mariners of England. Subsequent charters advanced their titles as they enlarged their duties and charges. In 1514, Henry VIII confirmed their foundation under style of . . . "Master, Wardens, and Accistants of the Guild or Fraternity of the Most Glorious and Undivided Trinity, and of St. Clement, in the Parish of Deptford Strond, in the County of Kent." Some years later, the 'accistants' were subdivided as Elder and Younger Brethren, the Foundation being familiarly referred to as the Corporation of Trinity House.

In early days, their efforts were directed in charity to stricken seafarers, in humane dispensation, in erection and maintenance of sea-marks, in training and provision of competent sea and coast pilots—a line of endeavour directed by the Godly Primate, in his Commission. Beacons were built on dangerous points of the coast, keepers appointed to serve them, watchers detailed to observe the vessels as they passed and restrain the activities of the wrecker. The magnitude of the task, the difficulties of their office, the powerful counter-influences arrayed against their beneficent rôle, may be judged by an incident that occurred as late as little over a hundred and twenty years ago. . . . "When Ramsgate Harbour, as a port of refuge from storm and stress, was intended, and the business was before Parliament, a petition from the Lord of the Manor tended to accelerate matters. He represented to the House, while the Bill was depending, that, *as the wrecks on the coast belonged to him and formed a considerable part of his property, he prayed that the Bill would not pass!* "

Established in charity for the guardianship of the coasts, the Brethren of Trinity passed to a supervision of the ships and the seamen. Although a closely guarded Corporation, qualifications for entry were simply those of sea-knowledge. The business of shipping, if more hazardous and difficult on the sea, was less complicated in its landward connections than is its modern conduct. The merchants were well content to be guided in their

affairs by their sea-partners, the men who actually commanded and sailed the ships. The voyages, ship construction, refitment and victualling were matters that could only be advised by the skilled seamen. Jealous for professional advancement, the Brethren of Trinity held their ranks open only to skilled master seamen and to kindred sea-tradesmen—the shipwrights and rope-makers. While attracting leaders and statesmen to the higher and more ornamental offices, control was largely vested in the Elder and Younger Brethren—technical advisers, competent to understand sea-matters.

In no small measure, the rise and supremacy of our shipping is due to their wise direction and control. They were the sole machinery of the State for control of the ships and the seaman. Survey and inspection of sea-stores, planning and supervision of ship construction, registry and measurement of vessels, had their beginning in the orderly efforts of the Brethren. Examination of the competence of masters was part of their duties—as was their arbitration in crew disputes. They licensed and supplied seafarers of all classes to the 'King's Ships,' tested their ordnance and examined the ammunition. Their reading of the ancient charter of their foundation was wide and liberal in its scope—"*to build, and light proper beacons for the guidance of mariners*" was their understanding. In construction and equipment and maintenance of sea-marks, in licence and efficient service of their coastal pilots, they carried out to the letter the text of their covenant; in spirit, they understood a guidance that was less material if equally important. Their beacons were not alone standing structures of stone and lime, but world-marks in precept and ordinance, in study and research. They held bright cressets aloft to illuminate the difficult seaways in the paths of navigation and science of the seafarer. They placed facilities for the study of seamanship before the mariners and sought to advance the science of navigation in line with the efforts of our sea-competitors. The charts and maps of the day—most of them being rude Dutch draft sheets—were improved and corrected, and new surveys of the coastal waters were undertaken at charge and patronage of the Brethren. Captain Greenville Collins, Hydrographer to Charles II, bears witness to their high ideals in presenting to the Corporation the fruits of his seven years' labour in survey and charting of the coast. The preface to his work is made noteworthy by

his reference to the practice of the day—the haphazard alterations on the charts that brought many a fine ship to grief.

". . . I then, as in Duty bound (being a Younger Brother) did acquaint you with it, and most humbly laid the Proposals before you; whereupon you were pleased not only to approve of them, but did most bountifully advance towards the charge of the work. . . . I could heartily wish that it might be so ordered by your Corporation, that all Masters of Ships, both using Foreign and Home Voyages, might be encouraged to bring you in their Journals, and a Person appointed to inspect them; which would be a great Improvement of Navigation, by imparting their Observations and Discoveries of the true Form and Prospect of the Sea Coast . . . and other dangerous Places. . . . And that those Persons who make and sell Sea Charts and Maps, were not allowed to alter them upon the single Report of Mariners, but with your approbation; by which means our Sea Charts would be more correct and the common Scandal of their Badness removed."

In all her enactments and activities, our Alma Mater ever preserved a worthy pride in her sons. Enthusiasm for a gallant profession, patronage for advancement in sea-skill and learning, a keen and studied interest in whatever tended to elevate and ennoble the calling of the sea, were her inspiring sentiment. Even in wise reproof and cautionary advice, her words were tempered by a brave note of pride—as though, under so many difficulties and serious dangers, she gloried in our work being worthily undertaken. In charge to the seaman, Captain Collins continues his kindly preface:

"It sometimes happens, and that too frequently, that when Ships which have made long and dangerous Voyages, and are come Home richly laden, have been shipwrecked on their native Coast, whereby both Merchants, Owners, and Mariners have been impoverished. All our neighbours will acknowledge, that no Nation abounds more with skilful and experienced Seamen than our own; none meeting a Danger with more Courage and Bravery . . . so a Master of a ship has a very great Charge, and ought to be a sober Man, as well as a skilful Mariner: All Helps of Art, Care, and Circumspection are to be used by him, that the Lives of Mariners (the most useful of their Majesties' Subjects at this juncture) and the Fortunes of honest Merchants under his Care may be preserved."



AT GRAVESEND: PILOTS AWAITING AN INWARD-BOUND CONVOY

For over three hundred years, our Alma Mater flourished as the spring of our seafaring—a noble and venerable Corporation, concerned solely and alone with the sea and the ships and the seamen. The Brethren saw only one aim for their endeavours—the supremacy of the sea-trade, the business by which the nation stood or fell. Nor was theirs an inactive part in all the long sea-wars and crises that reacted on our commerce. Before a navy existed, the stout old master-seamen of Deptford Strond were charged with the sea-defences of the capital. The new naval forces came under their control at a later date, and we have the record of an efficiency in administration that showed prevision and thought well in advance of that of their landward contemporaries. Piracy, privateering, the restraints of rulers and princes, were dealt with in their day. At critical turns in the courses of our naval conduct, it was to the steersmen of Trinity that the Ministers of the State relied for prompt and seamanlike action. The 'sea to the seamen' was the rule. Adapting their resources to the needs of the day, the Brethren were

held fast by no conventional restraint. They assisted peaceful developments in trade in the quieter years, but could as readily mobilize for war service under threat of invasion, or turn their skilled activities to removal of the sea-marks to prevent the sailing of a mutinous fleet. In the long and stormy history of Trinity House there were many precedents to guide the action of the Brethren on the outbreak of war. As guardians of the sea-channels and the approaches to our coasts, they manned these misty sea-trenches on the outbreak of war in 1914. Weaponless, by exercise of a skill in pilotage and a resolution worthy of great traditions, the Trinity men have held that menaced line intact. That little has been said about their great work is perhaps a tradition of their service.

We are parted now. The Merchants' Service is no longer a studied and valued interest of the ancient corporation. In an assured position as arbiters between the State and the shipping industry, the Trinity Brethren could combine a just regard for the merchants' interest with a generous and understanding appreciation of the seamen's trials and difficulties. If for no other reason than the record of past endeavours, they should still control the personnel of the Merchants' Service, in regulating the scheme of our education, the scope of our qualification for office, the grades of our service, the essence of our sea-conduct. But in the fickle doldrums of the period when steam superseded sail as our motive power, we drifted apart. Shipping interests have become complicated with land ventures, as widely different from them as the marine engine is from our former sail plan. In 1850 the Merchants' Service was placed under control of the Board of Trade; we were handed over to a Board that is no Board—a department of the State with little, if any, sea-sentiment, and that is sternly resolved to repress all our efforts to regain a voice in the control of our own affairs.

THE BOARD OF TRADE

IF we may claim the ancient Corporation of Trinity House as the Alma Mater of the Merchants' Service, we may liken our comparatively new directorate, the Board of Trade, to our Alma step-Mater—an austere, bureaucratic dame, hard-working and earnest, perhaps, but lacking the kindly spirit of a sea-tradition. She is utterly out of touch and sympathy

with a sea-sense—her arms, overstrained perhaps by the tremendous burden of charge upon charge that comes to her for settlement, are never open to the seamen. Sullenly, we resent her dictation as that of a usurper—a lay impropiator of our professional heritage. Under her coldly formal direction, we may attend our affairs in diligence and prudence, but for us there is no motherly licence; she has no pride in our doings (if one counts not the vicious insistence of her statistics)—we are only the stepchildren of her adoption, odd men of the huge and hybrid family over whom she has been set to cast a suspicious, if guardian, eye. While Trinity House was concerned alone with the conduct of shipping and sea-affairs, our new controllers of the Board of Trade have interests in charge as widely apart as the feeding of draught-horses and the examination of a bankrupt cheesemonger. We are but a Department. The sea-service of the nation, the key industry of our island commerce, is governed by a subdivision in a Ministry that has long outgrown the limits of a central and answerable control. Instead of settlement by a contained and competent Ministry of Marine, our highly technical sea-conduct is ruled for us in queue with longshore affairs, sandwiched, perhaps, between horse-racing and the period of the dinner table.

"The President of the Board of Trade has intimated to the Stewards of the National Hunt Committee that . . . it is not possible to sanction a list of fixtures for the season."

"Mr. Peto asked the President of the Board of Trade whether his attention has been called to the decision of Mr. Justice Rowlatt . . . in which judgment was given for the plaintiff company, owners of the steamship X —, sunk in collision, due to steaming without lights."

"The President of the Board of Trade announces modifications of the Lighting Order during the present week, one effect being that the prohibition of the serving of meals in hotels after 9.30 p.m. is temporarily suspended."

Perhaps we were rather spoilt by the pride that was in us when our seafaring was ruled by the appreciative Brethren of Trinity, and it may be as a repressive measure of discipline the Board of Trade extends no particular favour to our sea-trade, and has indeed gone further in being at pains to

belittle our sea-deeds, and disparage a recognition of our status. Our controllers are anxious that their ruling of award and reward should suffer no comparison. For gallantry at sea, the grades of their recognition may vary from the Silver Medal (delivered, perhaps, as in a recent case, with the morning's milk) to a sextant or a pair of binoculars.

In 1905 a very gallant rescue was effected by the men of the Liverpool steamer *Augustine*. The crew of a Greek vessel were taken from their foundering ship in mid-Atlantic under circumstances of great peril. Not only was boat service performed in tempestuous weather, but the officers of *Augustine* themselves jumped overboard to try to save the Greek seamen, who were too far exhausted to hold on to the life-lines and buoys thrown to them. The King of Greece, in recognition of the gallantry and humanity displayed, signed a decree conferring on the British master and his officers the Gold Decoration of the Redeemer.

A general view would be that this was an award quite appropriate to the services rendered, an expression by the Greek Government that they wished to place the names of the gallant savers of their seamen on the Roll of their Honour. Our Board of Trade objected. Through the Foreign Office, they appear to have informed the Greek Government that such distinguished awards were unusual and might prove a source of dissatisfaction in future cases. Possibly they viewed the appearance of a ribbon on the breast of a merchant seaman as an encroachment on the rights of their own permanent officials. The awards were not made; silver medals were substituted, which Captain Forbes and his officers, learning of the Board's action, did not accept. On a later occasion the same unsympathetic influence was exercised; the Russian Order of St. Stanislaus was withdrawn and replaced by a gold watch and chain!

In supervision of our qualifications as masters and mates, the Board of Trade has followed the lines of least resistance. It is true that they have established certain standards in navigation and seamanship that we must attain in order to hold certificates, but the training to these standards has never been an interest of their Department. While our shipmate, the marine engineer, has opportunity in his apprenticeship on shore to complete his education, we are debarred from the same facility. Apprenticed to the sea at from fourteen to sixteen years of age, our youth bid good-bye to their

school books and enter on a life of freedom from scholarly restraint—a 'kindergarten' in which their toys are hand-implements of the sea. There is no need to worry; there is no study required for four years; a week or two at the crammer's will suffice to satisfy the Board of Trade when apprenticeship days were over. And the fault does not lie with the 'crammer.' Scholarly and able and competent, as most of them are, to impart a better and more thorough instruction, the system of leaving all to the voyage's end offers to them no alternative but to present the candidate for examination as rapidly as possible. Sea-apprentices of late years did not often share in a scheme of instruction afloat. Rarely were they carried as complements to a full crew; for the most part they were workmen in a scant manning—'greenhorns'—drudges to the whim of any grown man. In a rough measure, the standard of such seamanship as they *gathered* was good—else we had been in ill case to-day—but it was without method or apprehension—a smattering—the only saving grace of which lay in the ready resource that only seafaring engenders. The exactions of a busy working sea-life left little leisure for self-advancement in study; the short, and ever shortening, intervals of a stay in port provided small opportunity for exercise of a helping hand from the shore. By deceptive short cuts that gave small enlightenment, by rules—largely mnemonic—we passed our tests and obtained our certificates. On shore, the landward youth fared better. The spirit of the times provided a free and growing opportunity for the study of technics and advance of scientific craftsmanship. The Navy took full advantage of this tide. The Board of Admiralty saw the futility of the old system of sea-training, having regard to the complete alteration of the methods in seamanship and navigation. Naval education could no longer be compensated by a schedule of bugle-calls and the exactitude of a hammock-lashing. Concurrent with a sound sea-training, general education was insisted upon. Zealously Admiralty guided their youth on a path that led to a culture and appreciation of values, wide in scope, to serve their profession. If it was essential, in the national interest, that the general education and sea-training of naval officers should be so closely supervised, it was surely little less important that that of the merchants' officers should receive some measure of attention. But for the private efforts of some few shipowners, nothing on the lines of a considered scheme was done. No assistance or advice or grant in aid was made by the Board of Trade. While drawing to their coffers huge sums, accumulations of fines and forfeitures, deserters' wages, fees, the

unclaimed earnings of deceased seamen, they could afford no assistance to guide the youthful seaman through a course of right instruction to a better sea-knowledge; they made no advances to place our education on a less haphazard basis. It may be cited as an evidence of *their* indifference that a large proportion of unsuccessful candidates for the junior certificates fail in a test of *dictation*.

With our entry to the war at sea in 1914, the same indifference was manifest. There was no mobilization or registration of merchant seamen to aid a scheme of manning and to control the chaos that was very soon evident. Despite their intimate knowledge of the gap in our ranks made by the calling-up of the Naval Reserve—accentuated by the enlistment of merchant seamen in the Navy—the Board of Trade could see no menace to the sea-transport service in the military recruitment of our men. It was apparently no concern of theirs that we sailed on our difficult voyages short-handed, or with weak crews of inefficient landsmen, while so many of our skilled seamen and numbers of our sea-officers were marking time in the ranks of the infantry. Under pressure of events, it was not until November 1915 they took a somewhat hesitating step. This was their proclamation; it may be contrasted with Captain Greenville Collins's preface.

"MAINTENANCE OF BRITISH SHIPPING

"At the present time the efficient maintenance of our Mercantile Marine is of vital national interest, and captains, officers, engineers, and their crews will be doing as good service for their country by continuing to man British ships as by joining the army.

"THE PRESIDENT OF THE BOARD OF TRADE."

"At the present time"! Possibly our Board was writing in anticipation of the completion of the Channel tunnel, or of a date when our men-at-arms and their colossal equipment, the food and furnishings of the nation, the material aid to our Allies, could be transported by air. "As good service"! An equality! An option! Was it a matter of simple balance that a seaman on military service was using his hardily acquired sea-experience as wisely as

in the conduct of his own skilled trade, as efficiently as in maintaining the lines of our oversea communications? Events at this date were proving that we had no need to go ashore for fighting service.

In the first violence of unrestricted submarine warfare, the Board advanced little, if any, assistance to the victims of German savagery. Their machinery existed only to repatriate torpedoed crews under warrant as "distressed British seamen"; they were content to leave destitution, hunger—the rags and tatters of a body covering—to be relieved and refitted by the charitable efforts of philanthropic Seamen's Societies. To them—to the kindly souls who met us at the tide-mark—we give all honour and gratitude, but it was surely a shirking of responsibility on part of our Board that placed the burden of our maintenance on the committee of a Seaman's Bethel. As a tentative measure, our controllers advanced a scheme of insurance of effects—a business proposition, of which many took advantage. Later, this was altered to a gratuitous compensation. Cases occurred in which distressed seamen had a claim under both schemes: their foresight was not accounted to them. Although proof might be forthcoming of the loss of an outfit that the small compensation could not cover, they could claim only on one or the other, the insurance or the gratuitous compensation. It was evident that the Board derived some measure of assistance from the examiners in bankruptcy on their staff.

In certain seaports—notably at Southampton—Sailors' Homes (built and endowed for the comfort and accommodation of the merchant seamen) were permitted, without protest, to be requisitioned by Admiralty for the sole use of their naval ratings. The merchantmen, on service of equal importance and equal danger, were turned out to the streets, and our Board took no action, registered no complaint.

To await popular clamour was evidently a guiding principle with our controllers. Their view was probably that we were private employees in trading ventures, that their concern was only to see the sea-law carried out. Sea-law, however, was not in question in the case of the master and officers of *Augustine*, and, if they could assume the right to interfere in that personal matter, they accepted a position as curators of the personnel of the Merchants' Service. They cannot complain if our understanding of their

duties does not agree with theirs. Deliberately, they have asserted that our sea-conduct is within their province.

An extraordinary matter is the character and calibre of the Board's marine officials. Unquestionably able and personally sympathetic as they are, it remains the more incomprehensible that our governance is so stupidly controlled. Perhaps their submissions fail of acceptance in the councils of a higher control—that has also to decide on horse-racing and bankruptcy. Under a less heavily encumbered Ministry, our affairs should receive the consideration that is their due. It required but little experience of the new sea-warfare to establish our claim to be considered a national service with a mission and employment no less vital and combatant than that of the enlisted arms. Master and man, we have earned the right to no small voice in the control of our own affairs. Our sea-interests are large enough to require a separate Department of the State, a Ministry of Marine, in which we should have a part.

The Board of Trade has failed us, they have proved unworthy of our confidence. Quite lately they began to mobilize and register the mercantile seamen of the country. *Three years and nine months after the outbreak of war, they sounded the 'assembly' of the Merchants' Service.* Let that be their epitaph!



**TRANSPORTS LEAVING SOUTHAMPTON ON THE NIGHT PASSAGE
TO FRANCE**

V

MANNING

SEA-LABOUR cannot be likened to employment on shore. Once signed and boarded and to sea, there can be no dismissal and replacement of the men such as may be seen any morning at the street gates of a workshop or shipyard. Good or bad, we are bound as shipmates for a voyage. Ordinary laws and regulations cannot reach us in our sailing; we are given the Merchant Shipping Act for our guidance, the longest and wordiest Act on the Statute Book, a measure that presupposes a discipline that no longer exists. Our ships, in size and power—our complement, in number and character—have altered greatly beyond the views of the Act. That statute, that in its day may have sufficed to set a standard of law and order to the moderate crews of our sailing ships, is utterly inadequate to control effectively the large ship's company of our modern steam vessels. The men, too, are changed—the sailormen, perhaps, not greatly—but, with the thundering evolution of steam-power, we have drawn grown men to the fires, ready-made men, uninfluenced by traditions of sea-service. We had no hand in their making—in the early years when discipline may be inculcated and character be formed. The drudgery and uninterest of their heavy work makes for a certain reaction that frequently finds its expression in violence and criminal disorder. The short voyage system and the grossly inadequate provisions of the Act afford no opportunity to guide the reaction in a less vicious direction. We hailed as a benefactor to the sea the inventor of single topsails; the statistics of our sea-fatalities give a definite date to their introduction. Daily we pray for an inventor to emancipate our stokehold gangs.

It would be idle to pretend that, as master-seamen, we were not disquieted by our manning problem, following upon the outbreak of war. While mobilization of the Army Reserve drew men from all industries in a proportion that did not affect seriously any one employment, the calling-up of the Royal Naval Reserve strained our resources in men to the utmost. Seamen, naval or mercantile, are of one great trade: the balance of our

activities being thrown suddenly and violently to one side of our engagement could not fail in disorganizing the other. Added to the outgoing of the retained Reserve seamen, recruitment of a new Reserve to man Auxiliaries and Special Service vessels was almost instantly begun. There were many applicants; the choice naturally fell upon our best men remaining. In and after August 1914, we were short-handed in the Merchants' Service. We were, indeed more than short-handed, for the loss of our steadiest men had effect in removing a certain check upon indiscipline. We missed just that influence upon which, for want of adequate authoritative powers, we counted to preserve some measure of subordination in our ranks.

Large vessels were most seriously affected. The service of troop transport suffered and was delayed. On occasion, there was the amazing instance of some 1500 trained and disciplined troops standing by to await the sobering-up and return to duty of a body of seamen and firemen. Drunkenness is not yet accounted a crime, but the holding up of vital reinforcements was no petty fault. Under the Act we were empowered to inflict a fine of exactly five shillings on each offender. The offence that held 1500 soldiers in check was met by a mulct of two half-crowns.



LIVERPOOL: MERCHANTMEN SIGNING ON FOR OVERSEA VOYAGES

The Army and the Naval Authorities were startled, as at a situation they had not contemplated. Masters and officers, if not actually challenged, were deemed to be responsible for such a state of insubordination among their crews. While such an assumption was, to a degree, unjust, it is true that we were not wholly blameless. For the sake of a quiet commercial life, we had accepted the difficulties of our manning without protest. In this we erred. Had we been an independent and economically fearless body, we would, in the days before the war, have refused to proceed to sea with any less than the summary powers held by a magistrate on shore to enforce law and order in his district. It is true that no magisterial powers will prevent drunkenness, but that condition on the ships was due directly to the general indiscipline that we were unable wholly to control.

The state of affairs called for more than a merely temporary measure, but our controllers advanced no settlement—only they devised an expedient. The situation was met, not by a firm action that would affect all merchant ships and seamen alike, but by a Defence of the Realm regulation that operated only when ships were chartered directly by Government. The

opportunity to make the merchantmen's forecastle a place for decent men to earn a living was passed by. While admitting, by their concern, that the matter called for redress, Government could only take action in cases where their bureaucratic interests were threatened. Vessels on purely commercial voyages, including carriage of the mails and millions in the nation's securities, were left without the regulation: we had to carry on as best we could. It entailed hardship on the better-disposed members of our ships' companies: in whatever fashion, the work had to be carried on: we taxed our steady men to the limit. The effect upon them may be judged when they realized that the delinquency of their shipmates, whose duty they had undertaken, was assessed at the price of a pound of 'Fair Maid' tobacco.

While the quality of our men was thus affected, we suffered in their diminished numbers. Without a protest from our governing body, the Board of Trade, the army took a toll of our seamen. Thus early, it was not realized that we merchantmen would have to fight for our ships and our lives at sea. The drums of field-war set up a note that was heard outside of six fathoms of blue water; large numbers of our seamen and many ships' officers joined up for military service. There was a certain measure of compensation afforded by the industrial situation ashore. As the magnitude of the world conflict was realized, nervous employers of labour reduced their staffs. All workmen suffered, the building trades being perhaps most affected. As needs must, we were open to recruit able-bodied men: we had to make seamen, and that quickly. Masons, brick-layers, tilers, slaters—they reached tide-mark in their quest for employment. We were glad enough to sign them on to make up our complements. At the first they were not of great value. Unused to the sea and ship-life, they had to be nursed through stormy weather: a source of anxiety to the watch-keeper when the seas were up. In time they became moderately efficient. As good tradesmen, they had a self-respect that could be encouraged: they were not difficult to control.

Of these, perhaps 50 per cent. made a second voyage, but not more than 10 per cent. remained at sea permanently. Their reasons for returning to the beach were always the same. Not the hard work or the seas appalled them, but the class of men with whom they had to live and work. Some of our recruits had other objects in view than a desire for a sea-life. At ports abroad, notably in the United States, they deserted. Strict as the Federal machinery is for regulating immigration into the United States, there

appeared to be no keen desire on the part of the authorities to embarrass the improper entry of our men. It was not difficult to assign a cause for their laxity. Technically, the men were seamen. Our Uncle Sam was stirring towards true sea-power—the acquisition of large mercantile fleets. The native American could see no prosperous commercial career in the forecastle: only from abroad might labour be obtained for operation of the ships. We had done the same in our time. Desertions were not confined to the landsmen of our crews. A situation arose quickly, in which it became profitable for our men to desert abroad and re-sign on another ship at an enhanced pay. As though to facilitate their breach of agreement, it was not long before the United States Seamen's Act came into force. By some international process that we seamen are not yet able to understand, this Act became operative on every vessel entering an American port. It establishes, for all seamen, the 'right to quit.' Strangely, our men did not all abandon ship. Some stirring of the patriotism that, later, became pronounced among them must have had effect in restraining wholesale disembarkation. Short-handed by perhaps an eighth of a full crew, we made our return voyages. By shift and expedient, we kept a modest head of steam. The loss was almost wholly at the fires. Stewards were set to deck duties and the look-out, the released sailormen went below to the stokehold—on occasion, passengers were recruited on board to bear a hand. Perhaps the public grumbled at receiving their letters an hour or two behind time.

It is not easy to advance reasons for the new and better spirit that came to us coincident with the appearance of German savagery at sea. Restrictions of the supply of drink had effect in enabling us to commence a voyage under good conditions, without brawling and bloodshed in the forecastle. An atmosphere of determination was, perhaps, introduced by the tales of undying heroism in the trenches that reached us. The losses in ships served partially to supplement the numbers of men available: a choice could be made in engagement of a crew. Over all, there was the menace to our seafaring—the threat and challenge to our sea-pride, as compelling and remedial as the draught of a free breeze. In his action, the enemy made many miscalculations; not the least was when he roused a spirit of readiness to service in our merchantmen; he blew more than the acrid fumes into us with the shattering explosion of his torpedoes.

If we may claim a patriotic influence acting upon our white seamen as reason for good service in the war, how shall we assess the lascar's quiet employment in a conflict that, perhaps, only dimly he understood? Of its operation he could have no ignorance. *Schrecklichkeit* was particularly to be employed against the native seaman. Shell and torpedo took toll of his numbers, but there was little hesitancy when he was invited to sign for further voyages. It was ever a point of prophecy with his detractors in the days of peace that he would be found wanting under stress. Not boldly or magnificently or in a spirit of vainglory, but in a manner that is not the less impressive because few have spoken of it, he has given them the lie.

The attitude of the naval authorities in regard to our manning is peculiar. They seem to be unable to think of ships' crews in any other terms than that of their own large complements. There is one part in the lectures of our instructional course that never fails to arouse rude merriment among the master-seamen attending—as it produces a shamefaced attitude on part of the naval lecturer (now intimate with our difficulties). In instructions for detailing our men to 'action stations' the phrases occur: "a party to be detached for attention to wounded," "a party to serve hoses at fire stations," "an ammunition supply party," "party to put the provisions and blankets in the boats." In practice, we are also working the guns, attending the navigation, spotting the fall of shot, keeping post at wheel and look-out. The average cargo vessel rarely carries more than eight men on deck: we cannot afford to have many wounded!

PART II



THE RULER OF PILOTS AT DEAL

VI

THE COASTAL SERVICES

THE HOME TRADE

"We're a North-country ship, an' a deep-water crew.

A—way, i-oh!

*Ye can stick t' th' coast, but we're damned if we do.
An' we're bound t' Rio Grande!"*

SO we sang—sounding a bravery at the capstan as we hove around and raised anchor to begin a voyage. We had our ideas. We were foreign-going sailors, putting out on a far venture. In pride of our seafaring—of rounding the Horn, of crossing Equator, perhaps of a circumnavigation—we looked down upon the coaster. He was a hoveller, a tidesman, a mud-raker—his anchors could shew no coral on the flukes as they came awash. We carried these ideas to the beach. Deliberately, we produced an atmosphere that is unjust to the cross-channel man.

The oversea voyage possesses a greater appeal to the imagination. Long distances, variation of the climes, storm and high ocean seas—a burthen of goods brought from a far country, all contribute to make an impression that the tale of a coasting voyage could not produce. Familiarity, perhaps, has robbed the short-carriers' sea-trip of what shreds of romance existed. In tide and out, the smaller vessels have grown to the sight as almost part of the familiar quays and wharves they frequent. A voyage from Tyne to the Thames or from Glasgow to Liverpool is so common and everyday that little remark is excited. We are unconcerned at its incident; the gale that wrecked a collier on the Black Middens may have blown a tile or two from our roof; the fog that bound the Antwerp boat for a tide is, perhaps, the same that held us in the City for an hour over time. We may entertain our

friends with recital of a sea-voyage, but we have not a great deal to say of a Channel passage.

At war, this focus of the public outlook has persisted. The threat to our sea-communications, to the source by which the nation gains its daily bread, has drawn an intense interest to the fortunes of the ships, but that interest has rarely been extended to the coasting vessels and the seamen who man them; there is little said of the work of the coastal pilots, on whose skill and local knowledge so much depends. We are concerned for our *Britannics* and *Justitias*, but the fate of the *Sarah Pritchard* of Beaumaris, or the escape of *Boy Jacob* are small events in relation to the toll of our tonnage. Their utility has not been brought before us in the same way as the direct service of the great ocean carriers. It is not difficult to understand that a breakdown of that source of supply would mean starvation and disaster. Our dependence on the coasting vessels is not so apparent. The vital needs served by them are, in part, obscured. We are, perhaps, satisfied that alternative channels exist for passage of the tonnage they transport: road and rail are open for inland carriage.

The situation is not quite so clear. Pressure at the rail-heads, at the collieries, at the steelworks and the manufactories, has thrown a burden on our island railways that they are unable to bear. But for the service of the coasters and the resolution of the home-trade seamen, the block to our traffic could not have been other than fatal. By relieving the congestion on the lines, they made possible the expansion of our output of munitions. Millions of tons that would otherwise have been put upon land transport (and have lain to swell the accumulations), are brought to tide-mark to be handled and cleared and ferried between home ports and across the channels by the coasting vessels. The Fleet is coaled and stored almost entirely by sea. Our men in France and Flanders are carried and fed and refitted by light-draught steamers. Power is transmitted to our Allies from British coalfields by our grimy colliers. Constant voyaging, dispatch at the ports of lading and discharge, seagoing through all weathers, make huge the total of their tonnage, but their individual cargoes rank small against the mammoth burdens of the oversea merchantmen. The sea-ants (however busily they throng the ports) are seldom remarked; their work is carried on in the shadow of more spectacular and lengthy voyaging. On occasion, a stray beam of popular recognition is turned on the smaller craft—as when

Wandle steams up Thames after her gallant fight, or when *Thordis* (Bell, master) rams and sinks a U-boat—but the light is quickly slewed again to illuminate the seafaring of the oversea vessels. Similarly—with the men—interest has centred on the deep-water mariner; the coasting masters and their crews, together with the pilots, are little heard of. Their navigations, steering by the land on a short passage of a tide or two, have not the compelling emphasis of long voyaging on distant seas. Chroniclers of our deeds and fates have set out the drawn agony of the raft and the open boat in mid-Atlantic; they are less insistent on the tragedies (as bitter and prolonged) of inshore waters. Perhaps they are influenced by a common misconception that succour is ever ready at hand in the narrow sea. There are the lifeboats on the coast, patrols on keen look-out in the channels, vessels are ever passing up and down the fairways; the land, in any case, is not far distant. Such assurance has but slender warrant. Gallant, unselfish, and thorough as are the services of the lifeboatmen, their operations in the main are intended to serve known wrecks and strandings. A flare in the darkness or a flash of gunfire in the channels is now no special signal; the new sea-casualty gives little time or warning for a muster of resources. The ready succour of the patrols is, perhaps, more instant and alert, but the channel seaways cover an area that no system could place under a quartered post or guard. No vigilance could prevent the capture of *Brussels* and the martyrdom of Captain Fryatt; the crew of the *Nelson* smack were for over thirty hours adrift in the narrow seas ere they were sighted and rescued. In the busy waters of the Irish Sea, three men of the ketch *Lady of the Lake* made ten miles in eight hours under oars, after their vessel had been sunk by gunfire. A weary progress, with ships passing near and far, but none daring too close the boat that might, for all they know, be trap for an enemy mine or torpedo.

It is time we ceased to sing that Rio Grande chanty: an *amende* is overdue.

While we, the foreign-going men, have our 'ins and outs' of the most dangerous seas—serving our turn in the front-line sea-trenches, then retiring to a rest in safer and more distant waters—the coastal seaman has no such relief. His daily duty lies in the storm-centre, in the very midst of the sea-war. From harbour mouth to the booms of his port of entry, no

course can be steered that does not drive his keel through minable areas and across the ranges of lurking submarines.

The new sea-warfare has developed a scheme of offence that renders our inshore waters peculiarly fraught with peril to navigators. The coast-line is no longer a defence and protection; rather, by limiting sea-room in manœuvre, the shoals and rock-bound beach have turned ally to the enemy. Sea-mark and headland provide a guide in estimating the run of a torpedo; note of a point definite, on which sea-routes converge, is of value to a submarine commander. Even in the shallower waters—depths in which a torpedo attack would be difficult—an equally deadly offence may be maintained. The run of the sea-bottom in the channels offering a good hold to slipped mine-moorings, it was not long before the enemy had adapted submarines to continue the minelaying that our command of the surface had stopped. While new and larger U-boats are sent abroad on the trade routes, special submarines, less encumbered by the stores and equipment that longer passages would demand, make frequent visits to the fairways to sow a freight of mines. No section of the channels holds sanctuary for the coaster. Close inshore, as in the offing, is all a danger area, open to the stealthy visits of the submarine minelayers. Right on the Mersey Bar, the Liverpool pilot steamer went up with a loss of forty lives; remote West Highland bays have echoed to the crash of mines exploded; seaward of the Irish banks, the deeps are alike dangerous. Counter-measures there are (services as efficient and resourceful in life-saving as those of the enemy are cunning and viciously ingenious in murder), but even the gallantry and skill and untiring efforts of our minesweepers cannot wholly clear the immense water-spaces. Mechanical contrivances—the Otters—are valuable, and aid in fending the mines, but (the sea-bottom being foul with wreckage) they are often a danger to their carriers. There is ever the harassing uncertainty which no vigilance may allay. The sheer relief of passing over the hundred-fathom line to the comparative safety of the deeps of ocean is never experienced by the cross-channel captain.

Favoured by their light draught and smaller proportions, the coasters are perhaps less exposed to successful torpedo attack than their larger and deeper ocean sisters. In the early days of submarine activity, the enemy was loath to use his deadlier and more expensive weapon on the small craft. He relied on gunfire to produce effects. The channel seas were not then as well

patrolled as now by armed auxiliaries: he could have a leisurely exercise in frightfulness at little risk to himself—there was no return to his fire—it was an easy target practice. *Cottingham* was shelled at short ranges when off the Bristol Channel. Unarmed and outdistanced, the master stopped his engines, lowered the two boats, and abandoned ship. The shelling continued, but was directed on the sinking ship; the submarine commander evidently thought the bitter wintry weather would accomplish a more refined *Schrecklichkeit* than the summary execution of his shell-bursts. In the heavy battery of a sou'west gale, the boats drove apart. The master's boat was sighted by a patrol, and the crew of six rescued after some hours' exposure. The mate's boat came ashore at Portliskey in Wales, bottom up and shattered; of the seven men who had manned her there was no trace. Six of *Cottingham's* crew survived the bitter weather—six hardy seamen were spared to return to service afloat. The German became dissatisfied with a frightfulness that murdered only half a merchant ship's crew when it was possible to murder all. It was not enough to destroy the ships and leave the seamen to the wind and sea and bitter weather. If they were not to be driven from their calling by fear, there were other measures—sure, definite, final. There was to be no weakness among the apostles of the new creed, no shrinking, no humanity—British seamen were to follow their shattered ships to the litter of the channel bottom. The *Kölnische Zeitung* set forth that "in future, our German submarines and aircraft would wage war against British mercantile vessels without troubling themselves in any way about the fate of the crews." The *Kölnische Zeitung* could not have been well informed. Their submarine commanders troubled themselves greatly about the fate of our crews. They shelled the boats in many subsequent attacks. They expended ammunition in efforts to secure that no further seafaring would be possible to their victims. Sheer individual murder took the place of an illegal act of war. ". . . We were unarmed, a slow ship. The submarine hit us with a shot on the bow and then ran up the signal to take to the lifeboats. We did so, and several shots were fired at the *Palermo*. They did not take effect, however, and a torpedo was sent into her side. She sank within a few minutes. Whether the fact that he had to use a torpedo to send our vessel to the bottom angered the commander I do not know, but the submarine came directly alongside of our lifeboats. The commander was on the deck, and yelled, 'Where is the captain of that ship?' The captain stood up and made his way to the side where the German was standing. The German held his revolver close to our

captain's head. 'You will never bring *another ship across this ocean*,' he said, using several oaths, then he pulled the trigger. Our captain fell dead, and we were permitted to continue."

The new campaign was directed particularly against the coasters and fishermen. The procedure was simple. No great speed or gun-range was required. There was no risk, if a good look-out was kept for patrols and war craft. The helpless, unarmed vessel, outsped and hulled, was brought-to within easy range, and shelling could be continued to augment the confusion of boat-lowering in a seaway. If by resolution and fine seamanship the boats were got away, there was further target practice with shrapnel or machine-gun. The schooner *Jane Williamson* of Arklow was attacked without warning. The first shot smashed one of her boats, the second killed one of the crew. At shouting distance—a hundred yards range—point-blank under the submarine's gun—there could be no question of defence or escape. The remaining five hands put over the second boat, tumbled into her and shoved clear. To hit the boat the submarine's gun must have been slewed deliberately from the larger target: bad shooting could not have occurred. Afloat and helpless, a shell struck her, killing one man outright, mortally wounding the master and another, and damaging the frail row-boat. The Germans beckoned the boat to them, but it was only to laugh at the throes of the dying men. The U-boat submerged, leaving the three survivors to ship oars and face the long weary pull towards the distant land. The *William* was sunk by gunfire; the gun's crew of the U-boat then loaded shrapnel and turned the gun on the open boat, wounding a man of the crew. *Redcap* was hauling her trawl when without any warning shrapnel burst on board. There was no challenge, the fishermen had made no attempt to get under way and escape. Busied with the gear, all hands were grouped together, when the shell exploded among them. One hand was killed instantly, the mate's leg was blown off, two seamen were wounded. Under fire, the survivors put the boat over and removed the wounded; the Germans gave no thought to their distress, but centred rapid fire on the trawler, sunk her, and disappeared.



A HEAVILY ARMED COASTING BARGE

When guns were served to merchant ships, the coasters shared in their issue. Encounters with enemy submarines were no longer one-sided and hopeless. Effects could not be secured by the Germans at so small a cost. Frequently the effects were those that the submarine commander was most anxious to avoid. *Atalanta* picked up the crew of *Maréchal de Villars*, then fought off the U-boat that had sunk that vessel. Watchers on the coastal headlands saw many a running fight between handy little home-traders and the under-sea pirates. Nor were the fishermen slow in action. Once armed

for defence, they proved that they could use their weapons with skill and precision. Off Aberdeen in stormy weather, a German submarine hove up from his depths for practice on a fleet of trawlers. It was to be a *Redcap* diversion: rapid fire, shrapnel, boats thrown out hastily, common shell on the hulls of the trawlers—wholesale destruction. But there was a mistake. A 'watch-dog' was among the fleet—*Commissioner*, armed and alert. At an opportune moment she cut her gear adrift, canted under speed and helm, returned the U-boat's fire and sank her in five rounds. Submarine commanders soon realized that 'diversions' were risky, the target could now hit back. It was safer to submerge when within range of anything larger than a row-boat. Even the sailing barges acquired a sting. In proportion to her tonnage, *Drei Geschwister*—a captured German, refitted to our coastal service—is probably the heaviest armed vessel afloat.

In channel waters, look-outs must not be confined to the round of the sea. To the U-boat's gunfire and torpedo, to the menace of moored and drifting mines, is added a danger that rarely threatens the oversea trader—an attack from the air. Striking distance from enemy bases has given opportunity for exercise of aircraft. Zeppelin and seaplane have their turns of activity in the North Sea and the Straits. Steering a careful course in a sea 'foul with floating mines,' the Cork steamship *Avocet* was attacked by three aeroplanes. The action lasted for over half an hour. Bombs exploded alongside, the bridge and upper decks were scarred and pitted by a hail of machine-gun bullets. The master and mate kept the aircraft at a respectful height by using their rifles—the only arms carried. By skilful handling, Captain Brennell saved his ship. He is probably the only seaman who has steered a deliberate course between a 'fall' of bombs; swinging on starboard helm, 'three bombs missed the starboard bow and three the port quarter by at most seven feet.' The *Birchgrove* was attacked by two seaplanes carrying torpedoes—a novel adaptation. Again the use of ready helm proved a moving ship a difficult target. Both torpedoes missed. Less fortunate was the *Franz Fischer*, an ex-German collier. Anchored off the Kentish Knock, the night black dark, the thunder of a Zeppelin's engines was heard overhead. Before there was time to extinguish all lights, the huge airship was able to take up a position for attack. One heavy bomb sufficed. *Franz Fischer* reeled to a tremendous explosion, heeled over, and sank. Only three survived of her crew of sixteen.

Constant sea-perils are enhanced by war measures in the channels. On open sea there is less confusion; the issue is narrowed to contest between ship and submarine and the hazard of a derelict or floating mine—there is ample sea-room in which to 'back and fill.' The coaster has a harder task. His navigational problem is complicated by the eight hundred odd pages of 'Notices to Mariners'—the amends and addends and cancellations of Admiralty instructions relating to the seafaring of the coast. Inner channels are confused by 'friendly' minefields or by alteration of the buoyage; aids to navigation are suspended or rearranged on scant notice; coastwise lights are put out or have their powers reduced to small efficiency in the mists and grey weather. Unmarked wrecks, growing daily in numbers, litter the seabottom; areas are to be avoided to leave a fair field for the hunters; zigzag courses in close proximity to the land sustain a constant anxiety. Above all, navigation without lights increases the danger to all merchantmen and to the patrols and naval craft that crowd the seaways of the coast.

Through all that the enemy can set against them, the home-trade vessels proceed on their voyages. Their losses are heavy in numbers (if the sum of their tonnage be not great), but the press of short sea-carriers that passes up Channel or down shews no evidence that frightfulness achieves an effect in holding them, loath, at their moorings. There is freight enough for all. Every vessel that has a sound keel and a helm to steer her is actively employed. Old craft and odd are come on the sea to serve turn in our emergency. Barges and inland watermen, Hudson Bay sloops, whilom pleasure craft, mud-hoppers reshelled, hulks even, are used; if they can neither sail nor steam, the ropemakers can supply a hawser—there is trade and bargain for a tow. After peace-years of grinding competition with the freight-grabbing steam coasters, the sailing craft of the smaller ports have found a new prosperity, from which no risks can daunt them. Sailmakers and rigging-cutters, the block and spar makers, have taken up their old tools again, and the gallant little topsail schooners, brigantines, cutters, and ketches are out under canvas.

The German boast that he can achieve victory by submarine policy could be nowhere more plainly refuted than in the War Channel that extends from the Thames to the Tyne. The evidence is there for all to judge. The seaway is foul with wrecks, foundered on beach and sandbar—the tide vexed by under-water obstructions. Topmast spars with whitened cordage whipping

in the wind stand out above the swirl of the tides; a shattered bow-section or gaunt listed shell of a wrecked vessel sets the turn to a new shoal drift; crazy funnels, twisted and arake by the broken hulls below, stud the angles of the buoyage that marks the fairway. Disaster to our shipping is plainly shewn, grouped in a way that no figures or statistics could rival. But there is other evidence. Daybreak in the Channel gives light to a progress of seaworthy craft that seems in no way diminished by the worst that the enemy can do. He has failed, despite the sinister sea-marks that litter the fairway. Down the river estuaries and out from the sea-harbour and roadstead, the coasters still join in company through the channels. An unending procession; the grey seascape is never free of their whirling smoke-wreaths. Passing and turning in the deeps, they steam close to the red-rusted, shattered hulls of their sister ships. The gaunt masses of tortured steel stand out as monuments to an indomitable spirit—or to an influence that calls their sea-mates out to steer by the loom of their wreckage.

PILOTS

IF we may count antiquity and precedence a claim, the pilot is the real senior of our trade. Before the ship and her tackling—the rude coracle, setting across the river bars or steering on a short passage by sea-marks on the coast, before the oversea venturer with his guide in sun and star—the lodesman, who marked the deeps and the shallows.

The pilot's departure and boarding are definite and well-marked incidents in the course of a voyage, and have a significance and interest few other ship-happenings claim. He is our last and first connection with the shore. His leaving is attended by a sober emotion, a compound of regret and impatience; regret that his sure support is withdrawn—impatience to go ahead to open sea. He backs over the rail and lurches down the swaying side-ladder to his dinghy to an accompaniment of cordial good-byes. Passengers crowd the bulwarks to watch his small boat go a-bobbing in the stern-wash as we gather way. It hardly occurs to them that their farewell letters, now in his weather-stained bag, may be for days or weeks unposted; to them he is the last post—the link is snapped, the voyage now really begun.

There may be masters who affect a fine aloofness when the pilot boards them on incoming, others who preserve a detached air—but there are few who do not feel relief in answering the cheerful hail—'All well aboard, Captain?'—as the pilot puts a cautious testing foot on the side-ladder. Here is the voyage practically at an end with the coming of an expert in local navigation. The anxiety of a landfall is over. The channel buoys, port hand and starboard, stretch out ahead to mark definite limits to shoal and sandbank; familiar landmarks loom up through the drift of distant city haze; the outer lightship curtsies in the swell, beckoning us into port to resume the brief round of longshore life. After a lengthy period of silence and detachment, we are again in touch with the affairs of the beach; the news of the day and of weeks past is told to us in intervals of steering orders—sailor news, edited by a competent understanding of our professional interests. The tension of the voyage is unconsciously relaxed. We are in good hands. The engines turn steadily and we come in from sea.

If the pilot was ever a welcome attendant in the peaceful days, his services in the war earn for him an even warmer appreciation. War measures in their operation have rendered our seaports difficult of entry. The buoyage has, perhaps, been reset in the interval of a voyage's absence. Boom defences and examination areas exist, channels are closed or obstructed; certain of the lightships or floating marks may be withdrawn on short warning. Amid all our doubts and uncertainties, we look for the one assured sea-mark on the unfamiliar bars—the red-and-white emblem of a pilot vessel on her boarding station. Undeterred by the risk of mine or torpedo while marking time on their cruising ground, the pilots are constantly on the alert to board the incoming vessels as they approach from seaward. No state of the weather drives the cutter from her station to seek shelter in safer waters. If the seas are too high for boatwork, she steams ahead and offers a lead to a quieter section of the fairway where boarding may be attempted.

Turn and turn of the pilots in service can no longer be effected. The even balances of their roster (that worked so well in peace-time) have been rudely disturbed by war. The steady round of duty, in which every man knew the date of his relief, has given place to a state of 'feast and famine'; all hands are frequently mustered to meet the sudden and unheralded demands of an inward-bound convoy, or the limited accommodation of the

cutter is taxed and overloaded by the release of pilots from an outward mass sailing.

There are grades of pilotage—from that of the rivers and protected waters to the more hazardous voyages between coastal ports. It is, perhaps, to the sea-pilots of Trinity we are most intimately drawn. While the river pilot is with us for the short term of the tide, the Trinity man is of our ship's company for a day or days. His valued local knowledge is at our service to set and steer fair courses in the perplexing tangents of unfamiliar tideways; operations of the minesweepers and patrols—that alter and multiply beyond counting in the course of a voyage abroad—are a plain book to him. If we meet disaster in the channels, we have a prompter at our elbow to advise a favourable beaching. We have a peer to confide in throughout our difficulties. After days of anxious watchkeeping on the bridge we are well served by a competent relief.

Ship movements in the western waters are controlled by the naval authorities in a manner that allows of independent sailings, but the Trinity pilots' duties lie in the Channel and the North Sea, where a more exacting regime is in force. From the Downs to the north, measures adopted for protection of the ships call for a time-table of sailings and arrivals that can only be adhered to by the pilot's aid. A 'War Channel' is established, a sea-lane of some two hundred and eighty miles that has constantly to be swept and cleared in advance of the traffic. Navigation in the channel obstructs an efficient search for mines; sweeping operations interfere with the passage of the ships. No small amount of control and management is necessary to reconcile conflicting actions and expedite the safe conduct of the shipping. Latterly, sailings were restricted to the hours of daylight; a system of sectional passages is enforced, by which all vessels are scheduled to make a protected anchorage before nightfall. An effect of this is to group the vessels in large scattered convoys, forming a pageant of shipping that even the busiest days of peace-time could not rival.

In all the story of the Downs, the great roadstead can rarely have presented such a scene as when, on a chill winter morning, we lay at anchor awaiting passage. Overnight, we had come in under convoy from the westward, eighteen large ships, to swell the tonnage that had gathered from the Channel ports. From Kingsdown to the Gull, there was hardly water-

space to turn a wherry. Even in the doubtful holding ground of Trinity Bay some large ships were anchored, and the fairway through the Roads was encroached upon by more than one of us—despite the summary signals from the Guardship. All types were represented in our assembly; we boasted a combination in dazzle paint to set us out, and our signal flags carried colour to the mastheads to complete the variegations of our camouflage. Troop transports from the States, standard cargo ships, munition carriers come over in the night from the French ports, high-sided empty colliers returning to the north for further loads, deep-laden freighters for London, ammunition and store ships for the Fleet, coasters and barges, made up the mercantile shipping riding at anchor, while naval patrols and harbour craft under way gave movement to the spectacle. Snow had fallen, and the uplands above Deal and Walmer had white drifts in the quartered fields. To seaward, we could see twin wreaths of smoke blowing low on the water, marking the progress of a flotilla of minesweepers, on whose operations we waited. A brisk north wind held out our signal flags, shewing our ports of destination, and the pilot cutter, busily serving men on the inward bound, took note of our demands. In time, the punt delivered our pilot, and we hove short, awaiting a signal from the Guardship that would release the traffic.

The teeth of the Goodwins had bared to a snarl of broken water that shewed the young flood making when movement began among the ships. Long experience had accustomed the pilots to the ways of the minesweepers, and when the clearing signal 'Vessels may proceed' was hoisted at the yard-arm of the Guardship, there were few anchors still to be raised. Crowding out towards the northern gateway, we found ourselves in close formation. Variations of speeds rendered the apparent confusion difficult to steer through, but the action of a kindred masonry among the pilots seemed to clear the narrow sea-lane. There was little easing of speed; with only a few hours of winter daylight to work in, shipping was being driven at its utmost power to make the most of the precious time. 'All out,' stoking up and setting a stiff smoke-screen over the seascape, we thinned out to a more comfortable formation, while the smaller craft, taking advantage of the rising tide, cut the inner angles of the channel to keep apace.

With flood tide to help us, we made good progress. The press of shipping gradually dropped astern till only the troop transport, our sea-neighbours of the convoy, kept company with us. Satisfied with the speed made, the pilot reckoned up the mileage and the tide. We were for Hull and, with luck, he expected to make Yarmouth Roads before darkness and the Admiralty regulations obliged us to bring up. Like all who serve the tide, he was prepared for an upset to his plans. "Not much use figuring things out in these days, Capt'n," he said. "A lot o' happenings come our way. In spite o' these fellows out there"—he pointed to a group of destroyers lining out on our seaward beam—"the U-boat minelayers get in on the channels to lay 'eggs'; as fast as we can sweep them up, sometimes. But"—cheerfully—"they don't always get back for another load: saw the bits o' one being towed into Harwich last week."

Happenings came our way. At the Edinburgh Channel, where the troop transports parted company and turned away for London, we were halted by an urgent signal from a spurring torpedo-boat. 'Ships bound north to anchor instantly,' was the reading of her flags; we rounded to and obeyed. In groups and straggling units, we were joined by the larger number of the fleet that had left the Downs with us. Some few were for the Thames and steamed ahead in wake of the troop-ships, but the most were bound for east-coast ports and anchored near the Channel Lightship. Two hours of precious daylight were lost to us as we rode out the last of the flood. High water came and we swung around on the cant of the wind. The pilot grew visibly impatient. The traverse of his reckoning lessened in mileage with every hasty step or two up and down the bridge. Yarmouth Roads receded into the morrow; Lowestoft (if the chief could crack her up to thirteen) was possible, but unlikely. Time passed, with no clearing signal—we were to be 'nipped' on the long stretch with no prospect but to dodge into Hollesay Bay before black night came.

By some mysterious agency, the coasters developed a foreknowledge of permission to proceed. Feathers of white steam curled from their windlasses, and their anchors were awash before the block was signalled clear. They had start of us. Less handily, we got under way and stood on into the Black Deep, where the smaller craft were throwing green smoke in their efforts to get ahead. The tide had now turned ebb to set us on our way. As we surged past the channel buoys the pilot was reassured. The prospect

of windy Lowestoft Roads beckoned him on with every coaster we overhauled and passed; the outlook improved as we timed our passage between the sea-marks. Off the Sunk, we came on the cause of our stoppage. The pilot noted a new wreck on the sands, one that had not been there when last he steered over this route. Beached at high water, he said. She had not been long on. The wreck lay listed on a spit of the sandbank. Her bows were blown open, exposing the interior of forecastle and forehold. Neutral colours were painted on her topside; the boats were gone and dangling boat-falls streamed alongside in the tideway. There was no sign of life on her, but a patrol drifter was standing by with a crowd of men on her decks. Out to seaward a flotilla of minesweepers was busily at work. Turning no more than a curious eye on the mined neutral, the pilot paid attention to the steering. That we were over a mined area had no grave concern for him. Relying on the minesweepers, he kept course and speed—the channel was reported clear.

LIGHTSHIPS

DEVOTED to the service of humanity, in a bond that linked all seafarers, lightships and isolated sea-beacons were regarded as exempted from the operation of warlike acts. The claim of the 'beacons established for the guidance of mariners' rested upon a high conception of world-wide service to mankind. Their duties were not directed to military uses or to favouring alone the nation who manned them. Their upkeep was met by a universal levy. Their warning beams were not withdrawn from foreign vessels; no effort was made to establish the nationality of a ship in distress ere setting portfire to the signal-gun to call out the lifeboat. On rare occasions sea-rovers interfered with the operation of the guide-marks. Retribution overtook them; they were outlawed by even the loose opinion of the period. There is surely more than legend in the ballad of Sir Ralph the Rover; if death by shipwreck was not actually his fate, it is at least the penalty adjudged to him by popular acclaim. Smeaton, in his Folio, records an instance of reparation for a similar 'diversion.'

"Lewis the Fourteenth being at war with England during the proceeding with this building, a French privateer took the men at work upon the

Eddystone Rock, together with their tools, and carried them to France, and the Captain was in expectation of a reward for the achievement. While the captives lay in prison, the transaction reached the ears of that monarch. He immediately ordered them to be released and the captors to be put in their place: declaring that though he was at war with England, he was not at war with mankind. He therefore directed the men to be sent back to their work with presents, observing that the Eddystone Lighthouse was so situated as to be of equal service to all nations having occasion to navigate the Channel."

A lightship is as peaceful and immobile as the granite blockstones of a lighthouse. She requires an even greater protection, exposed as she is to dangers on the sea that do not threaten the landward structure. She is incapable of offence or defence. Unarmed, save for the signal-gun that is only used to warn a vessel from the sands or to summon assistance to a ship in distress, she can offer no resistance to a show of force. She is moored to withstand the strongest gales, and cannot readily disengage her heavy ground-tackle. She has no efficient means of propulsion; parted from her stout anchors, she would drive helplessly on to the very shoals she had been set to guard. To all seafarers, in war as in peace, she should appeal as a sea-mark to be spared and protected; in the service of humanity, she is exposed to danger enough—to the furious gales from which she may not run.

Unlike the Grand Monarch, the Germans are bitterly at war with mankind. As one of their first war acts at sea, they shelled the Ostend Lightship. Like the Lamb, she was using the water; the Wolf would suffer no protestation of her innocence. Was she not floating placidly on the same tides that served the German coast?

In view of his subsequent atrocities in torpedoing hospital ships and shelling rafts and open boats, it is probable that our light-vessels would have been similarly destroyed by the enemy, but that his submarine commanders found under-water navigation required as accurate a check as in coasting on the surface. The fury of the Wolf was, in his own interest, tardily suppressed. He recognized that the value of the lightships in establishing a definite position was an asset to him. Withal—his 'fix' decided—he had no qualms in sowing mines in the area of these signposts; nor did he stay his hand in the case of a sea-mark that was not vital to his

plans. Two lightships on the east coast were blown up by mines; one, off the coast of Ireland, was deliberately torpedoed.

The menace of the German sea-mine remains the greatest war danger to which the lightships are exposed. Zeppelin and seaplane pay visits to the coastal waters, but the sea is wide for a chance missile from the air, and no great success has attended their bombing efforts. But the enemy mine has no instant aim. Full-charged and deadly, its activity is not confined—as the British mine is—to the area of the mooring. Their minelayers,

creeping in to the fairways in cloak of the darkness, are anxious to settle their cargo of high explosive as quickly as possible. Not all of the mines they sow hold to the hastily slipped 'sinkers' till disaster to our shipping or the untiring search of the minesweepers reveals their presence. Many break adrift and surge in the tideways, moving as the set of the current takes them.



THE LAMPMAN OF THE GULL LIGHTSHIP

Vessels under way, by keen look-out and ready helm, can sight and avoid the drifting spheres, but the lightships have no power to steer clear. Moored on the offset of a shoal or sandbank (their position, indeed, a guide to the minelayer), their broad bows offer contact to all flotsam that comes down on swirl of the tide. The authorities were unwilling to expose their men to a danger that could not be evaded, however gallant the shipmen or skilled their seamanship. It was not a seagoing risk that could be met; no adequate protection consistent with the lightship's mission could be devised. As the submarine war became intensified, the more distant vessels were withdrawn; new routes were set to divert shipping from the outer passages; only those floating sea-marks are now maintained whose removal would entail disaster to the traffic that passes by night and day.

Holding station in waters that are patrolled and, in part, protected, the Trinity men who form the crews of the lightships have readjusted their manning. A large proportion of the able-bodied men have joined the naval forces, leaving the older hands (and some few who have a physical disability) to tend the lights. War risks still remain, for the German minelayers have followed the shipping to the inner channels, but the greybeards have grown stolid and immovable in a service that was never at any time a safe and equable calling. They have become sadly familiar with the new sea-warfare—with disaster to the shipping in the channels. While they have incident enough, in the movement and activity of patrols and war craft, in the ceaseless sweeping of the channels, to judge our sea-power and take pride in its strength, they have all too frequent experience of the murderous under-water mechanics of the enemy. Living in the midst of sea-alarms, the old placid tedium of their 'sixty days' has given place to an excitement that even the monotonous rounds of their small ship-life cannot suppress. The men on the 'Royal Sovereign' were observers of the terrific power of the sea-mine; three ships in sight being blown to small wreckage within an hour. 'Shambles' jarred to distant torpedoings off the Bill. The 'South Goodwin' saw *Maloja* brought up in her stately progress by a thundering explosion, then watched her list and settle in the stormy seaway; a second crash and upheaval drew the eyes of the watch on deck to the fate of the *Empress of Fort William* as she was hastening to succour the people of the doomed liner. Up Channel and down, the lightshipmen were

observers of the toll exacted by the enemy—the price we paid for the freedom of the seas.

But not all their observations of sea-casualties brought gloom to the dog-watch reckoning. If there remained no doubt of the intensity and power of German submarine activity, they were equally assured of the efficiency of our surface offence, and the deadly precision of our own under-water counter-measures. On occasion, there were other sea-dramas enacted under the eyes of the lightshipmen—short, swift engagements that set an oily scum welling over the clean sea-space of the channel, or an affair of rapid gunfire that cleared a pest from the narrow waters. There is at least one instance of a lightship having a commanding, if uncomfortable, station in an action between our drifters and a large enemy submarine. The lampman of the 'Gull' had a front view. . . . "Misty weather, it was. Day was just breakin', about seven o' th' mornin' when I see him. I see him just over there—a little t' th' nor'ard o' that wreckage on th' Sands. A big fella, about th' size o' them oil-barges as passes hereabouts. I didn't make him out at first—account o' th' mornin' haze, but there was somethin' over there where no ship didn't oughta be. I calls down th' companion—'Master,' I says, 'there's somethin' on th' north end o' th' Sands.' He comes up an' has a look. Then we made 'im out what he was, a big German sub.—but he hadn't no flag flyin'. Jest then we hears firin', an' th' shells goes over us an' lands nigh him. They was three drifters jes' come out o' th' Downs t' start sweepin' an', all three, they goes for him like billy-o—firin' as they comes. We was right atween them an' th' shots passes over th' lightship. One as was short just pitches clear an' 'undred yards ahead o' us. Two guns he had—th' sub.—an' they didn't half make a din as they goes at it—*bang-bang-bang!* Th' drifters passes us, goin' a full clip. The first one, she got hit a-top th' wheelhouse, but they didn't stop for nothin'. The' keeps bangin' away with th' gun. . . . Yes. Some shots landed hereabouts, but we was busy watchin' th' drifters. . . . I see their shots hittin', too. I see one blaze up on th' submarine's deck, an' one o' his guns didn't talk back no more. Th' drifters was steerin' straight for him. I dunno how one o' them didn't go ashore herself—near it, she was. The sub. was hard on by this time, an' he stands high—with a list, too, but fightin' away like he was afloat.

"Two more drifters come up an' they joins in, an' th' shells goes *who-o-o-o!* overhead again. Then a destroyer, he comes tearin' along at full speed, an'

he puts th' finishin' touch to him. There was an explosion on th' submarine, an' th' nex' we see—we see his men tumblin' out o' him overside t' th' Sands. . . . Them up t' their middles in th' water an' holdin' their hands up."

The lampman was, of his service, a trained observer. He said nothing of the scene on the deck of the lightship—the watch tumbling up from below, their clothing hastily thrown on—the questioning, the alarmed cries. His concern was directed to the happenings on spit of the Sands. "Some shots landed hereabouts," he said; but his interest was on the Goodwins.



MINESWEEPERS GOING OUT

VII

'THE PRICE O' FISH'

THE inshore patrol hailed us and reported the channel clear as far as the Nore, and we stood on at full speed, making the most of the short winter daylight. Past the Elbow buoy, we met the minesweepers returning from a sweep of their section. They were steaming in two columns, line ahead, and we sheered a little to give them room; within the reading of our Admiralty instructions, they were a 'squadron in formation,' to whose movements we were advised to give way. They passed close. The leader of the port column was *Present Help*; we read the name on a gilt scroll that ornamented her wheelhouse. For the rest, she was trim in a coat of iron-grey, with her port and number painted over. A small gun—a six-pounder, perhaps—was mounted on her bows, and she carried a weather-stained White Ensign aloft. She scurried past us, pitching to our bow wash in an easy sidling motion that set her wheelhouse glasses flashing a cheery message. The skipper leaned from an open doorway, in an attitude of ease that, somehow, assured us of his day's work being well done—with no untoward happenings. He waved his cap to our greeting. *Present Help* and her sisters went by, and we returned to our course in the fairway.

"These lads," said the pilot, waving his arm towards the fast-receding flotilla. "If it wasn't for these lads, Capt'n, you and I wouldn't feel exactly comfortable on the bridge in channel waters. Two went up this week, and one a little while ago." He turned his palms upward and raised both arms in an expressive gesture. . . . "Three gone, one with all hands, but only one merchant ship done in by mines hereabouts in the last month. (*Starboard, a little, quartermaster!*). . . I dunno how we could carry on without them. Out there in all weathers, clearing the fairways and—Gad!—it takes some doing. . . . I was talking to one of the skippers in Ramsgate the other day. Saying what I'm saying—(*Steady, now, steady's you go!*)—what I'm saying now, and all he said was—'Right, pilot,' he says. 'If you feels that way, remember it when we gets back to th' fishin' in peace-time, an'—for th' Lord's sake—keep clear o' our gear when th' nets is down! I lost a tidy lot o'

gear,' he says, 'with tramps an' that bargain' about on th' fishin' grounds.' . . . He didn't think nothing of this minesweeping. His mind was bent on his nets and the fish again." A pause, while he coned the ship on a steady course, then, reflectively, "An' there's some folks—there's folks ashore growling about the price o' fish!"

Of courage in the war, on land as on sea, there are few records comparable to the silent devotion of the fishermen. The heat of attack and fury of battle may call out a reckless heroism that has no bounds to individual gallantry, but the sustained courage required for a lone action under heavy odds—every turn of the engagement being assessed and understood—is of a rarer quality; mere physical health and high spirit cannot generate it; tradition of a sea-inheritance and long self-training alone can bring it forth. That the fishermen (inured to a life of bold hazard and hardship) would offer valuable service in emergency was never doubted, but that the level of their gallantry should reach such heights, even those who knew them were hardly prepared to assume. And we were weak in our judgment, for their records held ample evidence by which we should have been able to predict a bravery in war action no less notable than their courage in the equally perilous ways of their trade. For a lifetime at war with the sea, wresting a precarious living from the grudging depths, their skill and resolution required no stimulus under the added stress of sea-warfare. In the fury of the channel gales, shipwreck and disaster called forth the same spirit of dogged endurance and elevating humanity that marks their new seafaring under arms. The countless instances of their service to vessels in distress, to torpedoed merchantmen and warships, in the records of strife, are but repetitions of their sea-conduct throughout the years of their trading. When Rozhdestvensky's panic-stricken gunlayers opened fire on the 'Gamecock' fleet on the Dogger, the story of that outrage was distinguished by the same heroism of the trawlermen that ennobles their diary to-day. When the *Crane* was sinking, the crew of *Gull*, themselves suffering under fire, boarded her to rescue the survivors. . . . "When they got on board the *Crane* they found the living members of the crew lying about injured. The vessel was in total darkness, and it was known that at any moment she might founder; yet Costello (the *Gull's* boatswain) went below to the horrible little forecabin to bring up Leggatt's dead body. Smith (the second hand), who took charge of the *Crane* when the skipper was

killed, refused to leave her till every man had been taken off. Rea (the engineer) showed unyielding courage when, in spite of the fact that the little ship was actually foundering, he groped back to the engine-room, which was in total darkness, to reach the valves. The stokehold was flooded with water, and Rea could do nothing. He went on deck, where the skipper was lying dead, and all the survivors, except the boy, were wounded."

In all its bearings, the comradely action of the *Gull* was but a foreshadowing of *Gowan Lea's* assistance to *Floandi* in the raid by Austrian cruisers on the drifter line in the Adriatic. The circumstances were curiously alike—the actual occurrence, the individual deeds. We have Skipper Nichols refusing to leave until his wounded were embarked, and Engineman Mobbs groping (as Rea did) through the scalding steam of *Floandi's* wrecked engine-room to reach the stokehold and draw the fires. Then, as in the Russians' sea-panic of October 1904, the fishermen (fighting seamen now) came under a sudden and murderous gunfire at close range. Overpowered by heavy armament, there was no flinching, no surrender. *Gowan Lea* headed for the enemy with her one six-pounder spitting viciously. The issue was not considered—though Skipper Joseph Watt must have had no doubt that he was steering his drifter towards certain destruction. Her gun was quickly put out of action. Her funnel and wheelhouse were riddled and shot to pieces. Water made on her through shot-holes in the hull. On the gun-platform, her gunlayer struggled to repair the mechanism of the breech—his leg dangling and shattered. Shell-torn and incapable of further attack, she drifted out of the line of fire. Bad as was her own condition, there were others in worse plight. *Floandi* had come under direct point-blank fire, and her decks were a shambles. Out of control—her main steam-pipe being shot through—seven dead or badly wounded, and only three remaining to work her, she was in dire need of assistance. Skipper Watt observed the distress of his sea-mate and steered *Gowan Lea* down to her to offer the same brotherhood as of the *Gull* to *Crane*. The analogy is peculiarly complete: the boarding, the succour to the wounded, the reverent handling of the dead. Not as a new spirit born of the stress of war, but as the outcome of an age-old tradition, *Gowan Lea* stood by.

After four years of warfare at sea, serving under naval direction and discipline, one would have expected the fisherman sailing under the White Ensign to lose at least a certain measure of his former character—to have

become a naval seaman in his habits of thought, in his actions, his outlook. Four years of constant service! A long term! He has come under a control that differs as poles apart from the free days of 'fleeting' and 'single boating.' He is set to service in unfamiliar waters and abnormal climates, but the habits of the old trade still cling to him. New gear comes to his hands—sweeps, depth-keepers, explosive nets, hydrophones, and paravanes—but he regards them all as adaptations to his fishing service. He is unchanged. He is still fishing; that his 'catch' may be a huge explosive monster capable of destroying a Dreadnought does not seem to have imposed a new turn to his thoughts. He is apart from the regular naval service. The influence of his familiar little ship, the association of his kindred shipmates, the technics of a common and unforgettable trade, have proved stronger than the prestige of a naval uniform. In his terms and way of speech, he draws no new farrago from his brassbound shipmate. Did not the skipper of the duty patrol hail *Aquitania* on her approach to the Clyde booms and advise the captain? . . . 'Tak' yeer *bit boatie* up atween thae twa trawlers!'

The devotion and gallantry and humanity of the fishermen is not confined to the enlisted section who man the patrol craft and minesweepers. The regular trade, the old trade, works under the same difficulties and dangers that ever menaced the ingathering of the sea-fishery. Serving on the sea in certain areas, the older men and the very young still contrive to shoot the nets and down the trawls. Their contribution to the diminished food-supply of the country is not gained without loss; 'the price o' fish' is too often death or mutilation or suffering under bitter exposure in an open boat. The efforts of the enemy to stop our food-supply are directed with savage insistence towards reducing the rations drawn from the depths of the sea; brutality and vengeful fury increase in intensity as the days pass and the indomitable fishermen return and return to their grounds. In August 1914, fast German cruisers and torpedo-boats raided our fleets on the Dogger Bank. Twenty fishing vessels were sunk, their crews captured. There was no killing. ". . . The sailors [of the torpedo-boat] gave us something to eat and drink, and we could talk and were pretty free," said the skipper of *Lobelia*. Later, on being taken ashore ". . . with German soldiers on each side of us, and the women and boys and girls shouting at us and running after us and pelting us, we were marched through the streets of Wilhelmshaven to a prison." Hardship, abuse! Now ridicule! ". . . The Germans stripped us of

everything we had. . . . But they were not content with that—they disfigured us by cutting one half of the hair of our heads off and one half of the moustache, cropping close and leaving the other half on, making you as ugly as they could. . . . It was a nasty thing to do; but we made the best of it, and laughed at one another."

Hardship, abuse, ridicule! The fishermen still served their trade at sea. Now, brutality! The third hand of *Boy Ernie* details the callous precision of German methods in September 1915. The smack was unarmed. ". . . It was very heavy and deliberate fire. [There were two enemy submarines.] The shots . . . were coming on deck and going through the sails. We threw the boat overboard and tumbled into her. . . . I started sculling the boat away from the smack, all the time under fire; but the Germans were not content with firing shells at a helpless craft—they now turned a machine-gun on to defenceless fishermen in a boat on the open sea. . . . The boat was getting actually riddled by the machine-gun fire, and before I knew what was happening, I was struck by a bullet on the right thigh, and began to bleed dreadfully. . . . The smack was blown to pieces and went down. This was the work of one of the submarines—while she was sinking the smack the other was firing on us."

Throughout all the malevolent and calculated campaign of destruction, the fishermen remain steadfast to their old traditions of humanity. When *Vanilla* is torpedoed without warning and vanishes in a welter of broken gear, her sea-mate, *Fermo*, dodging a second torpedo, steams to the wreckage to rescue the survivors—but finds none. In a heavy gale, *Provident* of Brixham risks her mast and gear, gybing to close the sinking pinnacle of the torpedoed *Formidable*, and rescue the exhausted seventy-one men who crowded her. The instances of fisher help to merchantmen in peril are uncounted and uncountable.

In the distant days when the Sea Services were classed apart, each in its own trade and section—working by a rule that admitted no co-partnery—we foreign traders had little to do with those whom (in our arrogance) we deemed the 'humble' fishermen. In the mists of the channel waters, we came upon them at their trawls or nets. Their floats and buoys obstructed our course; the small craft, heading up on all angles, confused the operation of a 'Rule of the Road.' Impatient of an alteration that took us miles from a

direct course, we felt somewhat resentful of their presence on the sea-route. That they were gathering and loading a cargo under stress and difficulty that contrasted with *our* easy stowage in the shelter of a dock or harbour, did not occur to us; they were obstructionists, blocking our speedy passage with their warps and nets and gear. Although most masters grudgingly steered clear, there were those in our ranks who elected to hold on through the fleets, unconcerned by the confusion and risk to the fishermen's gear that their passage would occasion. There were angry shouts and protests; the gear and nets were often the sole property of the fishermen; serious losses were sustained.

At war, we have incurred debts. When peace comes and the seas are free again, we shall have memories of what we owe to the fishermen in all the varied services they have paid to us. The minesweepers toiling in the channels, that we may not meet sudden death; patrols riding out bitter weather in the open to warn us from danger, to succour and assist the remnants of our manning when a blow goes home. War has purged us of many old arrogant ways. When next we meet the fishing fleet at peaceful work in the channels, we shall recall the emotion and relief with which we sighted their friendly little hulls bearing down to protect us in a menaced seaway. We shall 'keep clear o' th' gear when th' nets is down.'



SOUTHAMPTON WATER

VIII

THE RATE OF EXCHANGE

THE Bank of England official, who had been a close attendant on the bridge during the early part of the voyage, seems now to be reassured. We are nearing land again. Another day should see us safely berthed at New York, where—his trust discharged—a pleasant interval should open to him ere returning to England. The gold and securities on board are reason for his passage; he is with us as our official witness, should the activity of an enemy raider compel us to throw the millions overboard. Nothing has happened. The 'danger zone' has been passed without event. Stormy weather on the Grand Banks has given way to light airs and a smooth sea as we steer in to make our landfall.

Together on the navigation bridge, we are discussing the shipment. ". . . It is the exchange, Captain," he says. "The exchange is against us. These huge war purchases in the States cannot be balanced by the moderate exports we are able to send over. When we left Liverpool the sovereign was worth four dollars, seventy-one cents in America. I don't know where it is going to end. We can't make securities. There must be a lim——" Drumming of the wireless telephone cuts in on his words. "Operator wishes to know if he can leave the 'phones, sir? Says he has to see you."

The bridge messenger turns aside inquiringly, holding out the receiver of the telephone as a context to his words. The request, that would have aroused an instant disquiet six days ago, now appears trivial and normal. There may be receipts to be signed. Approaching port the operator will be completing his accounts. We are unconcerned and resume our conversation until he arrives.

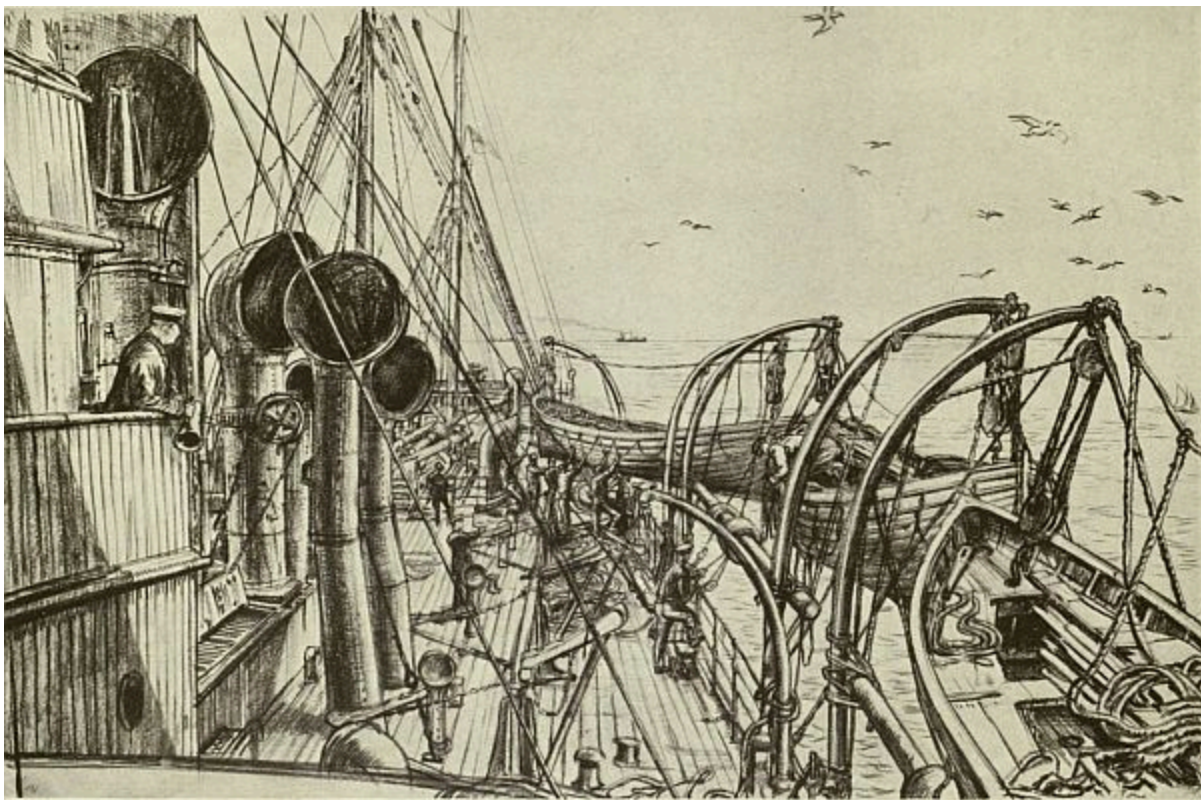
He is insistent that it cannot be due to atmospherics. "A queer business, sir. Thought it best to report instead of telephoning. Some station addressing a message to ABMV [all British merchant vessels], and another trying to jam it out. Can't get more than the prefix, when jamming begins. No, not

atmospherics. I've taken ABMV, though distant, twice in this watch, and, looking up the junior's jottings for the last watch, I see he had traces. Whatever is jamming the message out is closer to us than the sender. I dunno what to make of it!"

"You mean that a message from a land station to us is being interfered with, deliberately, from somewhere near at hand?"

He produces the slip of his junior's scribbles. Among the jumble of noughts and crosses, there is certainly a hastily scrawled ABMV, then x's and x's. "What else, sir? At first I thought it was atmospherics—x's were fierce last watch—but x's can't happen that way twice running!"

"All right! Carry on again. Let me know at once if anything further. Gear to be manned continuously from now on. Keep your junior at hand."



'OUT-BOATS' IN A MERCHANTMAN

A queer business! We trim the possibilities in our mind. It is now nearly dark. As we go, we should make Nantucket Lightship at daybreak; our usual landfall on the voyage. There is not much to work on. 'A message being sent, and some one making unusual efforts to prevent receipt.' A

raider? It is now some months since *Kronprinz Wilhelm* was driven into Norfolk; she cannot surely, have escaped internment. *Karlsruhe*? Nothing has been heard of her for a long term. A submarine? Perhaps *Deutschland*, with his torpedo-tubes refitted and a gun mounted? He knows the way; he could carry oil enough to reach the coast, do a strafe, and sneak into a port for internment. . . . Figuring on the chart, measuring distance and course and speed, it comes to us that enemy action would best succeed off Nantucket or the Virginia Capes. We resolve to cut in between the two, to make the land below Atlantic City, and take advantage of territorial waters. If there is no serious intention behind the jamming of the wireless, there will be no great harm done—we shall only lose ten hours on the passage; if a raider is out, we shall, at least, be well off the expected route. We pass the orders.

A quiet night. We are steering into the afterglow of a brilliant sunset. The mast and rigging stand out in clear black outline against lingering daylight as we swing south four points. The look-out aloft turns from his post and scans the wake curving to our sheer; anon, he wonders at the coming of a mate to share his watch. Passengers, on a stroll, note unusual movement about the boat-deck, where the hands are swinging out lifeboats and clearing the gear. As the carpenter and his mates go the rounds, screwing blinds to the ports and darkening ship, other passengers hurry up from below and join the groups on deck; an excitement is quickly evident. They had thought all danger over when, in thirty degrees west, we allowed them to discard the cumbersome life-jackets that they had worn since leaving the Mersey. And now—almost on the threshold of security and firm land—again the enervating restrictions and routine, the sinister preparations, the atmosphere of sudden danger. Rumours and alarms fly from lip to lip; we deem it best to publish that the wireless has heard the twitter of a strange bird.

Before midnight, the bird is identified. Our theories and conjectures are set at rest. The operator, changing his wave-length suddenly from 600 to 300 metres, succeeds in taking a message. '*From Bermuda*'—of all places—'*to ABMV German armed submarine left Newport eighth stop take all precautions ends.*' A submarine! And we had thought the limits of their activity stopped at thirty degrees west. Even the Atlantic is not now broad enough! The definite message serves to clear our doubts. A submarine from

Newport will certainly go down off Nantucket. Our course should now take us ninety miles south of that. There remains the measure of his activity. A fighting submarine that can navigate such a distance is new to us. His speed and armament are unknown. We can hardly gauge his movements by standards of the types we know. We are unarmed; our seventeen knots top speed may not be fast enough for an unknown super-submarine. Crowded as we are by civilian passengers, we cannot stand to gunfire. A hit will be sheer murder. It is a problem! We return to the deck and make three figures of that ninety miles.

The pulse of the ship beats high in the thrust and tremor of the engines, now opened out to their utmost speed; the clean-cut bow wave breaks well aft, shewing level and unhindered progress. In the calm weather, the whirl of our black smoke hangs low astern, joining the sea and sky in a dense curtain; we are prompted by it to a wish for misty weather when day breaks—to make a good screen to our progress. Though dark, the night is clear. A weak moon stands in the east, shedding sufficient light to brighten the lift. We overhaul some west-bound vessels in our passage and warn them by signal. Two have already taken Bermuda's message and are alert, but one has no wireless, and is heading up across our course. We speak her; her lights go out quickly, and she turns south after us.

Daybreak comes with the thin vapours of settled weather that may turn to a helpful haze under the warm sun. We zigzag in a wide S from the first grey half-light, for we are now due south of the Lightship. In the smooth glassy surface of the sea we have an aid to our best defence—the measure of our eyes. We note a novel vigilance in the watchkeepers, a suppressed anxiety that was not ours in the infinitely more dangerous waters of the channels. The unusual circumstance of zigzagging and straining look-out for a periscope almost in American waters has gripped us. Every speck of flotsam is scanned in apprehension. The far-thrown curl of our displacement spitting on the eddy of the zigzag, throws up a feather that calls for frequent scrutiny. We have no lack of unofficial assistance in our look-out. From early morning, the passengers are astir—each one entrammelled in a life-jacket that reminds them continually of danger. For the children, it is a new game—a source of merriment—but their elders are gravely concerned. Gazing constantly outboard and around, they add eyes to our muster. Every hour that passes without event seems to increase the

tension; the size and numbers of enemy vessels grow with the day. A telegraph-cable ship at work is hailed as 'a raider in sight'—a Boston sea-tug, towing barges south, is taken for a supply-ship with submarines in tow.

The wireless operator reports from time to time. The 'humming bird' (whoever he is) has ceased jamming. The air is full of call and counter-call. Halifax is working with an unknown sea-station—long messages in code. Coastal stations are joined in the 'mix-up.' Cape Cod is offering normal 'traffic' to the American steamer *St. Paul*, as though there was no word of anything happening within reach of the radio. It is all very perplexing. Perhaps the Bermuda message was a hoax; some 'neutral' youth on the coast may have been working an unofficial outfit, as had been done before. Anon, an intercepted message comes through. A Hollands steamer sends out 'S.O.S.. . . S.O.S.. . .' but gives no name or position. Then there is silence; nothing working, but distant mutterings from Arlington.

Throughout the day we swing through calm seas, shying at each crazy angle of the zigzag in a turn that slows the measured beat of the engines. Night coming and the haze growing in intensity, we use the lead—sounding at frequent intervals—and note the lessening depth that leads us in to the land. At eight, we reach six fathoms—the limit of American territorial waters. It is with no disguised relief we turn north and steer a straight course.

Although now less concerned with the possibility of enemy interference, we have anxiety enough in the navigation of a coastal area in hazy weather. We reduce speed. The mist has deepened to a vapour that hangs low in the direction of the shore. House lights glimmer here and there, but only by the lead are we able to keep our distance. A glow of light over Atlantic City shews itself mistily through a rift in the haze and gives an approximation of our latitude, but it is Barnegat's quick-flashing lighthouse beam that establishes our confidence and enables us to proceed at better speed. We shew no lights. For all we are in American waters, we have not forgotten *Gulflight* and *Nebraskan* and other international 'situations'; we look for no consideration from the enemy and preserve a keen look-out. Vessels pass us in the night bound south with their deck lights ablaze, but we stand on up the coast with not a glimmer to show our presence. Turning wide out to the

shoal-water off Navesink, we sight the pilot steamer lying to. We switch on all lights and steer towards her.

It is not often one finds the New York pilots unready, but our sudden arrival has taken them aback. We have to wait. Daybreak is creeping in when the yawl comes alongside with our man. He is an old Swedish-American whom we had long suspected of pro-German leanings, but the relief and enthusiasm on his honest old face is undisguised. "Gott! I am glat to see yo, Cabtin," he calls. "Dere was a rumour dat yo was down too! Yoost now, ven yo signal de name of de ship, I was glat—glat!" He is full of his news; there are rumours and rumours. 'The White Star mailboat is down,' 'a Prince liner is overdue,' 'there are fears for a Lamport and Holt boat.' In view of our safe arrival, he is prepared to discount the rumours. What is certain is that U 53 has arrived in these waters, and has already sunk six large ships off Nantucket.

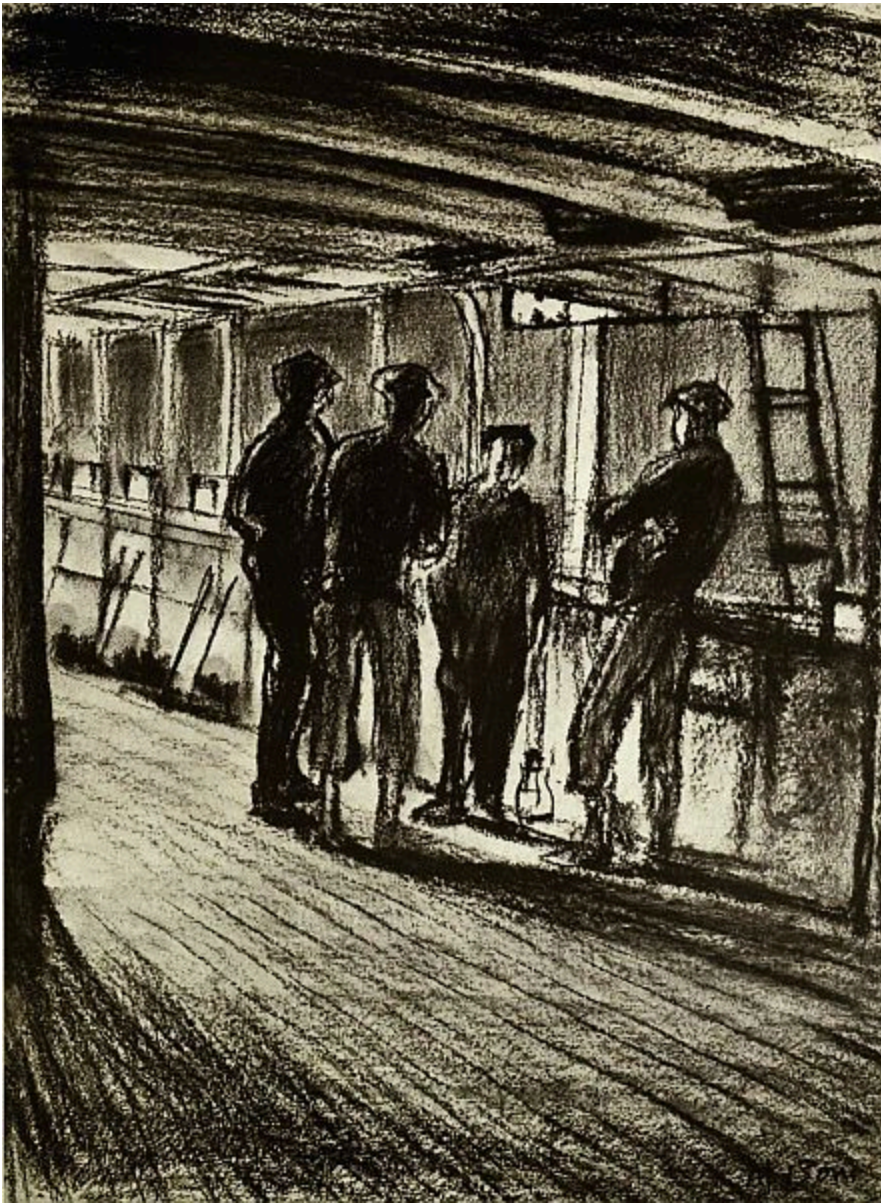
A day later we turn to the commercial pages of the *New York Herald*. Our arrival is reported, and it seems that the sovereign is now worth \$4.72-1/16!

IX

INDEPENDENT SAILINGS

UNTIL nearly three years of war had gone on, we sailed independently as 'single' ships, setting our speeds and courses and conforming only to the general route instructions of the Admiralty. The submarine menace did not come upon us in a sudden intensity. Its operation was gradually unfolded and counter-measures were as methodically advanced to meet it. The earliest precaution took the form of a wide separation of the ships, branching the sea-routes apart on the sound theory that submarines would have voyaging to do to reach their victims. While this was a plan of value on the high seas, it could not be pursued in the narrower waters of the channels. Destroyers in sufficient numbers not being available to patrol these waters, fishing craft—trawlers and drifters—were commissioned to that service. Being of moderate speed, their activities were not devoted to a mass operation, by which they could group the merchantmen together for protection. The custom was still to separate them as widely as possible, each zigzagging on her own plan. Until the convoy system was established, measures for our protection did not take the form of naval escorts sailing in our company: such vessels were only provided for transports or for ships on military service: vessels on commercial voyages were largely left to their own resources when clear of harbour limits.

That all sea-going vessels should carry a wireless installation was one of the first measures enforced by Admiralty. The magnificent resources of the Marconi Company, though strained, were equal to the task. There was a life-labour alone in the technical education of their operators, but they drilled the essentials of their practice into landward youths in a few months—blessed them with a probationer's licence—and sent them to sea. It is idle to speculate on what we could have done without this communication with the beach: it is inconceivable that we could have served the sea as we have done. Throughout the length of channel waters, we were constantly in visual touch with the patrols, but in the more open seas we relied on the wireless to keep us informed of enemy activities. At first, we were lavish in



FIREMEN STANDING BY TO RELIEVE THE WATCH

its use. The air was scored by messages—'back chat' was indulged in by the operators. An S.O.S. (and they were frequent) was instant signal for a confusion of inquiries—a battery of call and counter call—that often prevented the ready succour of a vessel in distress. We grew wiser. We put a seal on the switch. Regulations came into force to restrain unnecessary 'sparking'; we sat in to listen and record, and only to speak when we were spoken to.

Codes were issued by the Admiralty for use at sea. Their early cryptogram was easily decoded by friend and enemy alike. Knowing that certain words would assuredly be embodied in the text of a message (words such as, *from—latitude—report—submarine—master*), it was not difficult to decipher a code of alphabetical sequence. There were famous stories of traitors and spies, but our authoritative simplicity was responsible for the occasional leakage of information. At this date, 1915-16, wireless position-

detectors came into use by the enemy. A spark-group, repeated after an interval, could give a fair approximation of distance and course and speed. More than ever it was necessary to maintain silence when at sea. Withal, the air was still in strong voice. At regular periods the great longshore radios threw out war warnings to guide us in a choice of routes and warn us away from mined areas. Patrols and war craft kept up an incessant, linking report. Distress signals hissed into the atmosphere in urgent sibilance, then faltered and died away. On occasion, the high note of a *Telefunken* set invited a revealing confidence that would lead us, 'chicky-chicky,' to the block. We were well served by Marconi.

Extension of the power of enemy submarines brought new practice to our seafaring. We had made the most of a passage by the land, steering so close that the workers in the fields paused in their toil and waved us on; but the new under-water craft crept in as close, and mined the fairways. We were ordered to open sea again, to steer the shortest course by which we could reach a depth of water that could not be mined. Zigzag progress now assumed the importance that was ever its right. It had been but cursorily maintained. The 'shortest distance between two points' had, for so long, been our rule that many masters were unwilling to steer in tangents. On passage in the more open sea, they were soon converted to a belief in the efficacy of a crazy course. Statistics of our losses proved the virtue of the tangent: of a group of six vessels sunk in a certain area only one—a very slow vessel—was torpedoed while maintaining a zigzag. Extracts from the diary of a captured submarine commander were circulated among us, giving ground for our confidence, in the frequent admissions of failure—"owing to a sudden and unexpected alteration of course."

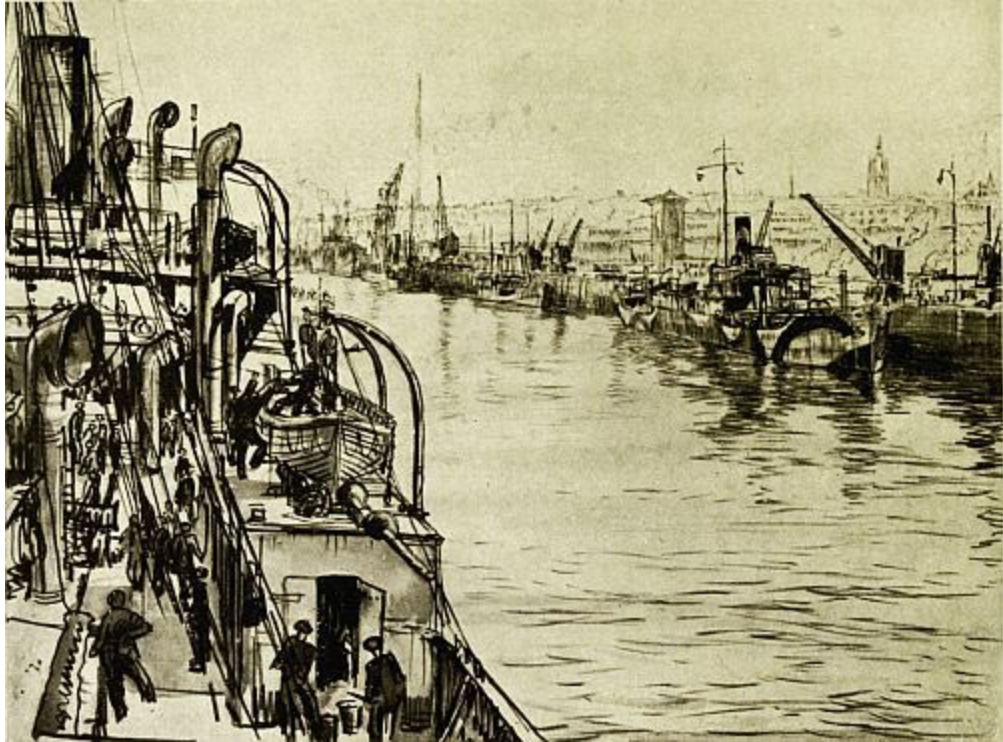
Still, we were unarmed. If, by zigzag and a keen look-out, we were fortunate in evading torpedo attack, the submarine had by now mounted a surface armament, and we were exposed to another equally deadly offence. For our protection, Admiralty placed a new type of warship on the routes approaching the channels. Built originally for duty as minesweepers, the sloops were faster and more heavily armed than the drifters. They patrolled in a chain of five or six over the routes that we were instructed to use. During the daylight hours we were rarely out of sight of one or other of the vessels forming the chain. Our route orders were framed towards a definite point of departure into the high seas when darkness came. There, the patrol

of the sloops ended: we had the hours of the night to make our offing and, by daybreak again, were assumed to be clear of the 'danger zone.' But the 'danger zone' was being extended swiftly; it was not always possible to traverse the area in the dark hours of a night: only the fast liners could stretch out a speed that would serve. Profiting by experience that was constantly growing, the *Reichsmarineamt* constructed larger submarines capable of remaining long at sea, and of operating in ocean areas that could not adequately be patrolled. Twelve, fifteen—then twenty degrees of longitude marked their activity advancing to the westward: they went south to thirty-five: in time the Mediterranean became a field for their efforts. Gunfire being the least expensive, they relied on their deck armament to destroy unarmed shipping. The patrols were but rarely in sight; the submarine became a surface destroyer. There was no necessity for submergence on the ocean routes: under-water tactics were held in reserve for use against fast ships—the slower merchantmen were brought-to in a contest that was wholly in favour of the U-boat. In a heavy Atlantic gale, *Cabotia* was sunk by gunfire, 120 miles from land. She had not the speed to escape. Despite the heavy seas that swept over the submarine and all but washed the gunner from the deck, the enemy was able to keep up a galling fire that ultimately forced the master to abandon his ship. *Virginia* was fired upon at midnight when steering for the Cerigo Channel. Notwithstanding the courage of Captain Coverley, who remained on board to the last, there could be but one end to the contest. *Virginia* was sunk. A strong ship; the enemy had to expend two of his torpedoes to destroy her.

Against such attacks only one measure could be advocated—the measure we had for so long been demanding. It was impossible to patrol adequately all the areas of our voyaging. Guns were served to us and we derived a confidence that the enemy quickly appreciated. We did not expect wholly to reduce his surface action, but we could and did expose him to the risk he had come so far out to sea to avoid. On countless occasions our new armament had effect in keeping him to his depths, with the consequent waste of his mobile battery power. Even in gun action he could no longer impose his own speed power on a slow ship. Under conditions that he judged favourable to his gunnery, the submarine commander still exercised his ordnance—usually after a torpedo had failed to reach its mark. Many of the hazards were against us, but our weapons brought the contest to a less

unequal balance. If we did present the larger target, we had—in our steady emplacement—a better platform from which to direct our fire. From the first it was a competition of range and calibre. Six-pounders led to twelves; these in turn gave way to 4.7's. Anon, the enemy mounted a heavier weapon, to which we replied by a new type of 4-inch, sighted to 13,000 yards.

Thus armed and equipped, we were in better condition to meet the enemy in our independent sailings. He was again obliged largely to return to the use of his torpedoes, with all the maze of under-water approach that that form of attack involved. If outranged in a surface action, we had our smoke-producing apparatus to set up a screen to his shell-fire, and that form of defence had the added value of forcing him to proceed at a high and uneconomical speed to press an attack. Some of our gun actions resulted in destruction of a sea-pest, but all—however unsuccessful—contributed to lessen his power of offence. Every torpedo fired, every hour of submergence, every knot of speed expended in a chase, was so far a victory for us as to hasten the date when he would be obliged to head back to his base. His chances of survival in that passage through the patrols and the nets and mines could not be considered as good.



QUEEN'S DOCK, GLASGOW

X

BATTLEDORE AND SHUTTLECOCK

*"All vessels are prohibited from approaching within four miles of Rathlin Island
between sunset and sunrise"*

IN view of Admiralty instructions, we are 'proceeding as requisite'—turning circles, dodging between Tor Point and Garron Head—and awaiting daybreak to make a passage through Rathlin Sound. Steering south from the Clyde, we had reached Skullmartin when the wireless halted us. Enemy activity off the south coast of Ireland had become intensified, and all traffic from west-coast ports was ordered to proceed through the North Channel. In groups and singles, the ships from Liverpool and the Bristol Channel join us, and we make a busy channel-way of the usually deserted coastal waters. We show no lights, but the moon-ray reveals us, sharply defined, as we pass and repass on the lines of our courses. We keep well within the curve of the coast until the light grows in the east, then turn finally to the north. The sun comes up as we reach Fair Head, and we stand on towards the entrance of the Sound.

In the first hour of official clearance, the North Channel is busy with the traffic. Outside as well as within, ships have been gathering in anticipation of Admiralty sunrise. The seaway over by the mainland shore is scored and lined by passage of the inward-bound vessels, all pressing on at their best speed to make their ports before nightfall. A strong ebb tide runs through, favouring our company of outward-bounders. We swing past Rue Point in a rip and whirl that gives the helmsman cause for concern, cross the bight of the Bay at a speed our builders never contemplated, and round the west end of the Island before the sun has risen high.

It is fine weather in the Atlantic. Only the slight heave of an under-running swell, and the rips and overfalls of the tide, mark the smooth surface of the sea: the light north airs that come and go have no strength to

ruffle the glassy patches. Everything promises well for speedy progress. The engines are opened out to their utmost capacity. Already we have drawn ahead of the press of shipping that marked time with us on the other side of the channel. Our only peer, a large Leyland liner, has opened out abeam of us and the whirl of black smoke at his funnel-tip shows that he is prepared to make and keep the pace. 'To proceed at such a time as to reach 56° 40' North, 11° West, by nightfall'—is the reading of our new route orders. We shall have need of the favour of the elements if we are to reel off 200 miles between now and 10 p.m. Anon, we pass Oversay and the Rhynns of Islay and head for a horizon that has no blue mountain-line to break the level thread of it. Our sea-mates of the morning are hull down behind us—the slower vessels already turning west on the inner arms of the fan formation that is devised to keep us widely separated in the 'danger area.' Only the Leyland boat remains with us. We steer on a similar mean course, but the angles of our independent zigzags make our progress irregular in company. At times we sheer a mile or more apart, then close perceptibly to crossing courses. She has perhaps the better speed, but her stoking is irregular. Drawing ahead for a term, she shows us her broad sternwash in a flurry of disturbed water; then comes the cleaning of the fires—we pull up and regain a station on her beam.

So, till afternoon, we keep in company—pressing through the calm seas at a speed that augurs well for our timely arrival in 11° West. We sight few vessels. A lone drifter on patrol speaks us and reports no enemy sighted in the area: an auxiliary cruiser with a destroyer escorting her passes south on the rim of the landward horizon. A drift of smoke astern of us hangs in the clear air, then resolves to a fast Cunarder that speedily overhauls and passes us. As though impressed by the mail-boat's progress, our sea-mate puts a spurt on and maintains a better speed than any she has shown since morning. She draws ahead and we are left with clear water to exercise the cantrips of our zigzag.

An *allo* is intercepted by the wireless in the dog-watch. (We have coined a new word to report an enemy submarine in sight, a word that cannot offer a key to our codes.) It comes from the Cunarder, now out of sight ahead. We figure the radius on the chart, and bear off six points on a new course to keep well clear of the area. The Leyland liner is by now well ahead and we note she has turned to steer west. There is a slight difference in our courses

and we draw together again as we steam on. The wireless operator now reports that a vessel near at hand has acknowledged the Cunarder's *allo*. Shortly a man-o'-war sloop appears in sight and passes north at high speed, steering towards the position we are avoiding.

The second officer keeps a keen look-out. He has had bitter experience of the power of an enemy submarine and is anxiously desirous that it should not be repeated. A 'check' on the distant sea-line (that we had taken for the peak of a drifter's mizen) draws his eye. He reports a submarine in sight—broad on the port bow. The circle of our telescope shows the clean-cut horizon ruling a thread on the monotint of sea and sky. Sweeping the round, a grey pinnacle leaps into the field of view. It is over-distant for ready recognition. Only by close scrutiny, observing a hair-line that rises and falls on either side of the grey upstanding point, are we able to recognize our enemy. He is pressing on at full speed, trusting to our casual look-out, that he may secure a favourable position to submerge and attack. Our fine confidence with which we have anticipated such a meeting gives place to a more sober mood. Though not yet in actual danger, there is the former *allo* to be thought of—the possibilities of a combination. Quick on recognition, we alter course, steering to the north again. The gun, already manned, is brought to the 'ready,' and the intermittent crackle of the wireless sends out an urgent warning. The Leyland steamer starts away at first sight of our signals: ahead, grey smoke on the horizon marks where the patrol sloop has gone hull-down.

A spurt of flame throws out from the distant submarine. He has noted our sudden alteration of course and knows that he has now no prospect of reaching torpedo range unobserved. His shell falls short by about a thousand yards. We reply immediately at our extreme elevation, but cannot reach him. The next exchange is closer—he is evidently overhauling us at speed. Mindful of our limited fifty rounds, we telephone to the gun-layer to reserve his fire until he has better prospect of a hit. Two shots to our one; the enemy persists though he does not now seem to be closing the range. Our seventh shot pitches close to him, and ricochets. There is a burst of flame on his deck—whether from his gun or the impact of our shell we shall never know; when the spume and spray fall away he has dived.

Suddenly, it is recalled to us that we have been, for over half an hour, steering into the radius of the Cunarder's *allo*. The patrol sloop has turned to close us and is rapidly approaching. A decision has quickly to be made. If we stand on to keep outside torpedo range of our late antagonist, we may blunder into the sights of number two. North and east and west are equally dangerous: we may turn south-east, but our course is for the open sea. The sloop sheers round our stern and thunders up alongside. Receiving our information, her helm goes over and she swings out to investigate the area we have come from. We decide to steer to the north-west as the shortest way to the open sea.

We have the luck of the cast. As we ease helm to our new course, the ship jars and vibrates—a thundering explosive report comes to our ears. The Leyland liner close on our starboard quarter has taken a torpedo and lies over under a cloud of spume and debris.

XI

ON SIGNALS AND WIRELESS

FOR war conditions our methods and practice of signalling were woefully deficient. In sailing-ship days the code was good enough; we had no need for Morse and semaphore. We had time to pick and choose our signals and send them to the masthead in a gaudy show of reds and blues and yellows. Our communications, in the main, were brief and stereotyped. "What ship? Where from? How many days out? Where bound? Good-bye—a pleasant passage!" Occasionally there was a reference to a coil of rope or a tierce of beef, but these were garrulous fellows. The ensign was dipped. We had 'spoken'; we would be reported 'all well!'

Good enough! There were winches to clean and paint, bulwarks to be chipped and scaled, that new poop 'dodger' to be cut and sewn. "Hurry up, there, you sodgerin' young idlers! Put the damned flags in the locker, and get on with the *work!* "

With steam and speed and dispatch increasing, we found need for a quicker and more instant form of signal correspondence. New queries and subjects for report grew on us, and we had to clip and abbreviate and shorthand our methods to meet the lessening flag-sight of a passing ship. We altered the Code of Signals, adding vowels to our flag alphabet. We cut out phrases like 'topgallant studding sail boom' and 'main spencer sheet blocks,' and introduced 'fiddley gratings' and 'foo-foo valve.' Even with all our trimming, the book was tiresome and inadequate. We began to fumble with Morse and semaphore, with flashlights and wig-wags and hand-flags.

We did it without a proper system. As a titbit to our other 'snippings,' medicine, the Prayer Book, the law, ship's business, the breeches buoy, ship-cookery! Fooling about with flags and tappers and that, was all very well for the watch below, but there was *work* to be done—the binnacles to be polished, the sacred *suji-mudji* to be slapped on and washed off!

Hesitating and slipshod and inexact as we were, at least we made, of our own volition, a start; a start that might, under proper and specialized direction, have made an efficient and accurate addition to the sum of our sea-lore. But we were wedded to titbits. Late on the tide, as usual, the Board of Trade woke up to what was going on. They added a 'piece' to our lessons, without thought or worry as to the provision of facilities for right instruction. We crammed hard for a few days, fired our shot at the right moment, and forgot all about it.

Withal, in our own amending way, we were enthusiastic. We learned the trick of *Ak* and *Beer* and *Tok* and *Pip*. We slapped messages at one another (in the dog-watches), in many of which a guess was as good a translation as any. Our efforts received tolerating and amused recognition from naval officers (secure in possession of scores of highly trained signal ratings). If we came, by chance, across an affable British warship, she would perhaps masthead an E (exercise), to show that there was no ill-feeling. Then was the time to turn out our star man, usually the junior-most officer, and set him up to show that we were not such duffers, after all! Alas! The handicaps that came against us! The muddled backgrounds (camouflage, as ever was!), the fatal backthought to a guess at the last word! The call and interfering counter-call from reader to writer, and writer to reader, and, finally, the sad admission—an inevitable *Eye, emmer, eye* (I.M.I.—please repeat), when our scrawl and jumble of conjectural letters would not make sense! We have yet a mortifying memory of such an incident, in which a distant signalman spelt out to us, clearly and distinctly, "*Do you speak English?*"



**THE BRIDGE-BOY REPAIRING
FLAGS**

Under the stress of war we have improved. Fear for the loss of important information has spurred us to keener appreciation. If you promise not to flirt

the flags backhanded (a most damnably annoying habit of superior, *flic-flac* Navy men) we can read you in at ten or twelve words a minute. For single-ship work, that was good enough; if we had a press of signalling to attend, we could make up for our busy time in leisurely intervals. But convoy altered that. In the Naval Service a signalman has nothing whatever to do in the wide world but attend to signals. It is his only job: a highly trained speciality. With us the demands of ship work on our bare minimum crews do not allow of a duty signaller; he must bear a hand with the rest to straighten out the day's work. In convoy, with signals flying around like crows at the harvest, we found our way of it unworkable. It resolved itself to what used to be called a 'grand rally' in pantomime—all hands on the job, and the officer of the watch neglecting a keen look-out to see that note of the message was kept properly.

The naval authorities took counsel. The experiment had been a 'try on,' in which they (with their large staff of special signalmen) had assessed our ability as greater than their own! It was decided to train signalmen—R.N.V.R.—for our service. Pending their formation and development, we were given skilled assistance from the crews of our ocean escorts. But for our gun ratings, and they mostly R.N.R., we had no experience of the regular Navy man in our muster. He spun a bit, trimming the grass, before he found rest and a level. With us only for a voyage, we did not get to know him very well, but in all he was competent enough.

One we had, from H.M.S. *Ber*—*Sharpset*, Private Henry Artful, R.M.L.I. Drouthy, perhaps, but a good hand. At the end of sailing day, when the flags were made up and stowed, he came on the bridge.

"Fine night, sir!" We assented, curiously; democratic and all as we are, it is rather unusual for our men to be so—so sociable. "Larst capt'in I wos with, sir, 'e allus gimme a drink after th' flag wos stowed."

We stared, incredulous. "What! Do you say the captain of *Sharpset* gave you a drink when your work was done?" He started in affright. "Not the capt'in o' *Sharpset*, sir! Oh no, sir!—Gawd!—No! Th' capt'in o' th' larst merchant ship wot I wos signallin' in!"

His horror, genuine and unconcealed, at our suggestion of such an unheard-of transaction, gave illustration alike of the discipline in His

Majesty's ships and, sadly, the lack of it in ours.

In time our quickly trained R.N.V.R.'s joined. They came from Crystal Palace, these new shipmates. Clean fellows—smart. Bacon-curers, Cambridge men, lawyers, shopmen, clerks, haberdashers—trimmed and able and willing to carry on, and lacking only a little ship practice, and a turn of sea-legs, to fit them for a gallant part in delivering the goods. With their coming we are introduced to a line of longshore life that had escaped us. There is talk and ado of metropolitan habits and styles, of 'Maudlen' and high life, of music scores, the latest revue, the quips of the music-halls. ("When Pa—says—*turn*" is now the correct aside, when Commodore gives executive for a new angle on the zigzag!)

At the first we were somewhat concerned at the apparent 'idleness' of our signalman. He was on our books for but one employment—the business of flags and signals. In intervals of his special duties he made an odd picture on the bridge of a merchant ship—a man without a 'job.' The firemen, on deck to trim ventilators, would take a peep at him as at some strange alien; seamen, passing fore and aft on their reliefs, would nod confidently. "Still diggin' wet sand, mate? . . . Wish I 'ad your job!" There were days when he was busy enough—'windmilling' with the hand-flags, or passing hours in hoist and rehoist when Commodore was sharpening the convoy to a precision in manœuvre, but on open sea his day was not unduly crowded. There were odd hours of 'stand-by' under screen of the weather-cloth, intervals of leisure which he might use as he liked, provided he kept a ready ear for the watch officer's call. Reading was usual. In this his taste was catholic. *Tit-Bits* and *My Dream Novelettes* found favour; one had back numbers of the *Surveyor and Municipal and County Engineer*, old volumes of *Good Words* from the Bethel box found a way to the bridge; we saw a pocket volume of Greek verse that belonged to the bold lad who altered our signalled 'will' to 'shall'!

For all his leisured occupation he was quick enough when the call of "*Signals*" brought him to business. His concentration on the speciality of the flags brought an accuracy to our somewhat haphazard system of signalling. We benefited in more than his immediate work by promoting his instruction of our young seamen. Spurred, perhaps, by the knowledge of our quondam haberdasher's efficiency, the boys improved rapidly under his tuition. We

paid a modest bonus on results. We are looking forward. We shall not have our duty signalman with us when there is 'peace bacon' to be cured.

Another new shipmate who has signed with us is the wireless operator, the lieutenant of Signor Marconi, our gallant *salvator* in the war at sea. If we may claim for our sea-service a foremost place in national defence, it is only by grace of our wireless we register a demand. Without it, we were undone. No other system of communication would have served us in combat with the submarine; *spurlos versenkt*, without possibility of discovery, would have been the triumph of the enemy. If to one man we seamen owe a debt unpayable, Marconi holds the bond.

Unthinking, we did not accept our new shipmate with enthusiasm. Before the war he could be found on the lordly liners, tapping out all sorts of messages, from the picture-post-card-like greetings of extravagant passengers to the deathless story of *Titanic* and *Volturno*. We looked upon him as a luxury, only suited to the large passenger vessels. We could see no important work for him in the cargo-carriers; we could get on very well without a telegraph to the beach. A week of war was sufficient to alter our views; we were anxious to have him sign with us. Although he is now an important member of the crew, his reception at first was none too cordial. The apparent ease and comfort of his office rankled in contrast to the rigours of the bridge and the hardships of the engine-room. His duties—specialized to one operation—we deemed unfairly light in comparison with our jack-of-all-trades routine. In port, he was a lordling—no man his master—able to come and go as the mood took him. Frankly, we were jealous. Who was this to come among us with the airs of a full-blown officer, and yet not a dog-watch at sea? Messed in the cabin too, and strutted about the decks with his hands in his pockets, as bold and unconcerned as any first-class passenger! We were puzzled to place him. He talked airily of ohms and static leaks, ampere-hours and anchor-gaps, and yet, in an unguarded moment, had he not told us of his experiences in a Manchester broker's office, that could have been no more than six months ago? The airs of him! Absurd assumption of an official confidence between the Old Man and himself, as if *he* had the weight of the ship's safety on his narrow shoulders! As for his baby-brother assistant—that kid with the rosy cheeks—everybody knows that all he does is to screw up his 'jimmy fixin's' and sit

down good and comfortable to read "The Rosary," with his dam mufflers on his ears! *Huh!*

But we are wiser now! Here is a text for our conversion. It is a record of a wireless conversation between a merchantman attacked and a British destroyer steaming to her assistance from somewhere out of sight.

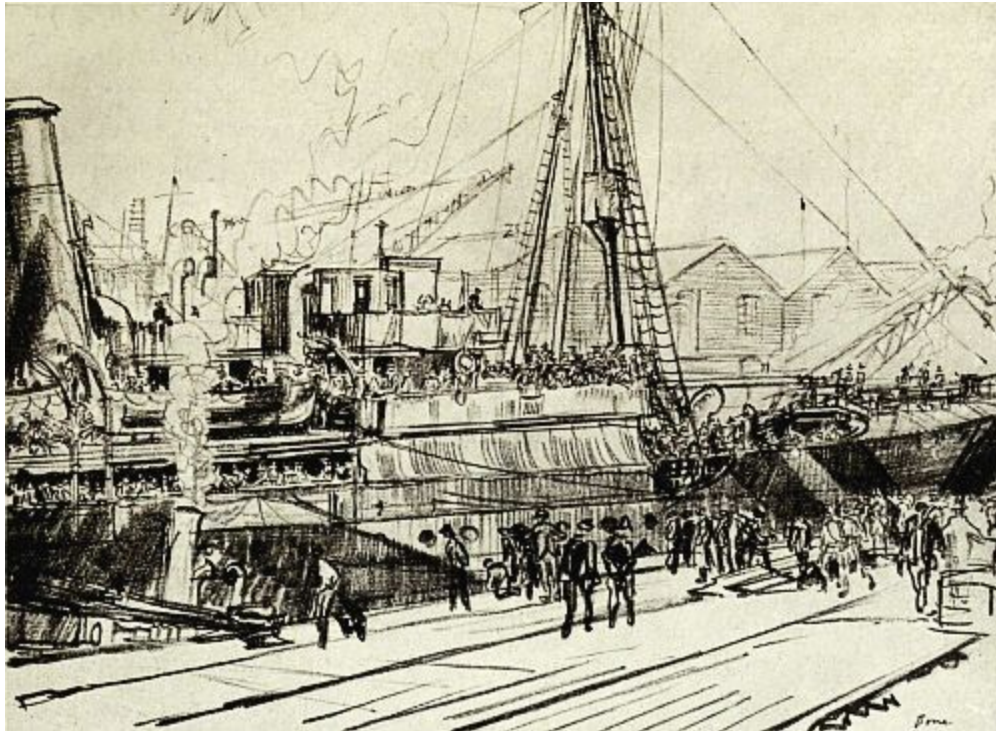
"Are you torpedoed?"

"Not yet. . . . Shots in plenty hitting. Several wounded. Shrapnel, I believe. Broken glass all round me."

"Keep men below. Stick it, old man!"

"Yes, you bet. Say, the place stinks of gunpowder. Am lying on the floor. . . . I have had to leave 'phones. My gear beginning to fly around with concussion. . . . Captain is dead. . . ."—an interval—"Submarine has dived! Submarine has dived!"

Yes, we are wiser now! We admit him to full fellowship at sea. And on land, too! We admit him the right to trip it in Kingsway or the Strand, with his kid gloves, and his notebook, and his neat uniform, for his record has shown that it does not require a four-years' apprenticeship to build up a stout heart; that on his 'jimmy fixin's' and their proper working depends a large measure of our safety; and if the crack does come and the air is thick with hurtling debris, broken water and acrid smoke, our first look will be aloft to see if his aerial still stands. We do him and baby brother the honour that we shall not concern ourselves to wonder whether they be ready at their posts!



A TRANSPORT EMBARKING TROOPS FOR FRANCE

XII

TRANSPORT SERVICES

THE first State control of the merchants' ships began with the transports employed to convey the Expeditionary Force to France in the early days of August 1914. Vessels of all sizes and classes were commandeered at the dockside to serve in the emergency. The comparatively short distance across the channels did not call for elaborate preparation and refitment: the times would admit of no delay. Ships on the point of sailing on their trading voyages were held in dock, their cargo discharged in quantity to make space for troops and their equipment. Lining-up on the quays and in the littered dock-sheds, troops awaited the stoppage of unloading operations. With the last sling of the 'tween-deck lading passed to the shore, they marched on board. As the tide served, the vessels steamed out of dock and turned, away from their normal routes, towards the coast of France.

To serve as ballast weight, the stowage of cargo in the lower holds was frequently left in place for the term of the vessel's troop service. Months, perhaps a year later, the merchandise arrived at its destination. Consignees would wonder at its tardy delivery—they could see no record of its itinerary as shewn by the bills of lading, unless they read into the fine prefix—'War: the King's Enemies: restraints of Rulers and Princes'—the romance of its voyaging with the heroes of Mons.

To transport the overseas troops from India and Canada and Australia, different measures were necessary. The ships requisitioned for this service had to be specially fitted for the longer voyage. The State was lavish and extravagant under the sudden pressure of events. The many-handed control at the ports made for an upheaval and dislocation of shipyard labour that did not hasten the urgent dispatch of the vessels. The hysteria of the times gave excuse for a squandering of valuable ship-tonnage that was without parallel. Large liners, already fitted for carriage of passengers, were employed as prison and internment ships. Curious situations arose in the disposal of

others. At the north end, a large vessel might suddenly be requisitioned and taken from her trade—with all the consequent confusion and relay; by day and night the work of fitting her would go on. South, a vessel of similar size and build might be found, having her troop-fittings removed, in preparation for an ordinary trading voyage. Still, if the end justifies the means, the ultimate results were not without credit. The garrison troops from Malta and Egypt and Gibraltar and South Africa were moved with a celerity that is unexampled; a huge contingent from India was placed on the field in record time. A convoy of thirty-one merchantmen brought Canadian arms to our assistance: Australians, in thirty-six ships, crossed the Indian Ocean to take up station in Egypt. The unsubsidized and singular enterprise of the merchants was proving its worth: as vital to the success of our cause as the great war fleet, the merchants' ships aided to stem the onrush in France and Flanders.

Considerations of economy followed upon the excited measures with which the first transport of available troops was effected. In the period of training and preparation for the long offensive, the Transport Department had opportunity to organize their work on less stressful lines. It was well that there was breathing-space at this juncture. Enemy interference, that had so far been almost wholly a surface threat to our communications, grew rapidly to a serious menace from under water. The engagement and organization of naval protection underwent an immediate revisal. Heavily armed cruisers and battleships could afford little protection against the activity of the German submarines, now at large in waters that we had thought were overdistant for their peculiar manœuvres. Destroyers and swift light craft were needed to sail with the transports.

The landing at Gallipoli, under the guns of the enemy, was a triumph for the Transport Service. In the organization and disposal of the ships, the control and undertaking that placed them in sufficient numbers in condition for their desperate venture, the Department redeemed any earlier miscalculations. The efficient service of the merchant masters and seamen was equally notable. Under heavy fire from the batteries on shore they carried out the instructions given to them in a manner that was "astonishingly accurate" and impressed even the firebrands of the naval service. Strange duties fell to the merchant seamen on that day. Compelled by the heavy draught of their ships to remain passive spectators of the deeds

of heroism on the beach, they saw ". . . whole groups swept down like corn before a reaper, and to realize that among these groups were men who only a short time before had bid us good-bye with a smile on their lips, was a bitter experience.

"Our vessel was used to re-embark the wounded, and we stood close inshore to make the work of boating them off less hazardous. We had three doctors on board, but no nurses or orderlies, and the wounded were being brought on board in hundreds, so it was a relief to us to doff our coats and lend a hand. We had to bury the dead in batches; officers and men were consigned to the deep together. On one occasion the number was exceptional, and the captain broke down while reading the service. . . ." It was surely a bond of real brotherhood that brought the shattered remnants of the complement she had landed earlier in the day to meet their last discharge at the hands of the troopship's seamen—their committal to the deep at the broken words of the vessel's master.

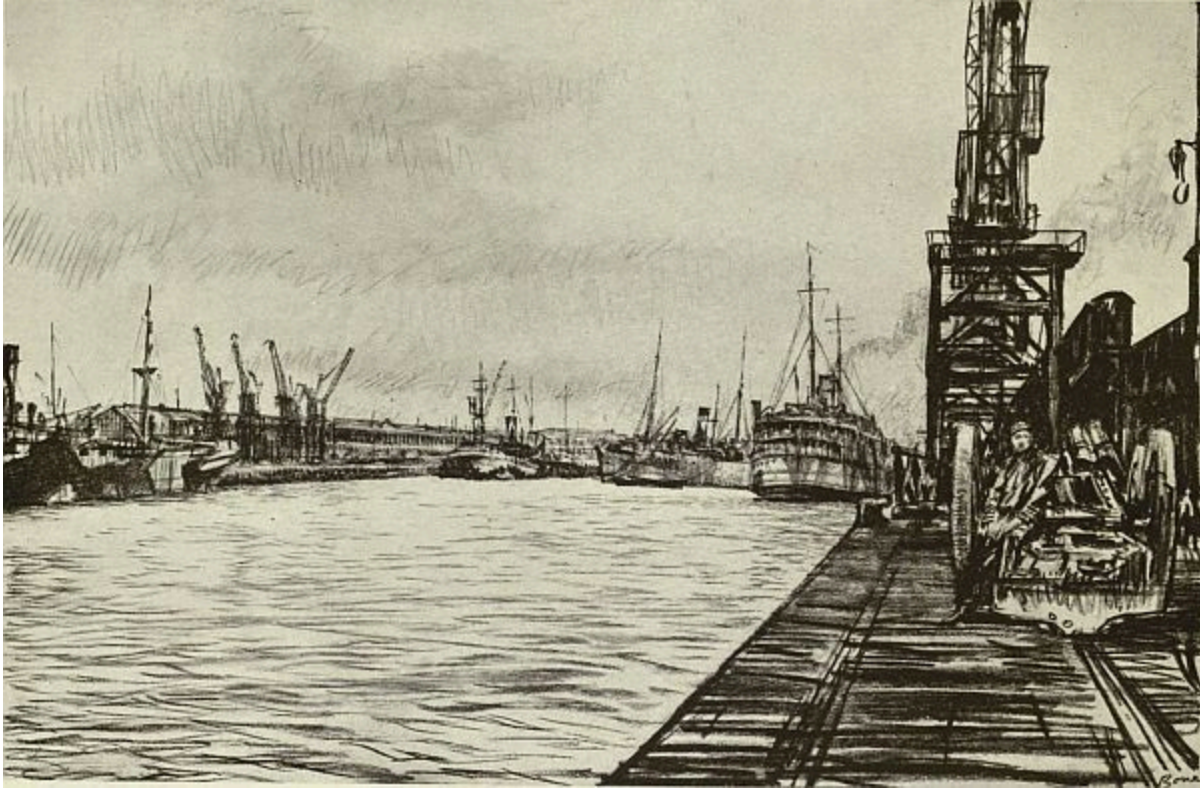
While the transport of troops in the Channel and the narrow seas was not, at any time, seriously interfered with, the movements of the larger ocean transports were not conducted without loss. *Royal Edward* was the first transport to be torpedoed. She went down with the sacrifice of over a thousand lives. The power of the submarine had been over-lightly estimated by the authorities: measures of protection were inadequate. Improved U-boats were, by now, operating in the Mediterranean, and their commanders had quickly acquired a confidence in their power. More destroyers were required to escort the troopships.

By a rearrangement of forces a more efficient measure of naval protection was assured. Although the provision of a swift escort did not always prevent the destruction of ships, the loss of life on the occasion of the sinking of a transport was sensibly reduced by the presence of accompanying destroyers. The skill and high gallantry of their commanders was largely instrumental in averting complete and terrible disaster. As the numbers of ships were reduced by enemy action there came the need to pack the remaining vessels to a point of overloading. Boat equipment on the ships could not be other than inadequate when the certified complement of passengers was exceeded by 100 per cent. In any case, the havoc of a torpedo left little time to put the huge numbers of men afloat. With no

thought of their own hazard—bringing up alongside a torpedoed vessel and abandoning the safeguard of their speed and manœuvring power—the destroyer men accepted all risks in an effort to bring at least the manning of their charge to port.

Every casualty added grim experience to the sum of our resources in avoiding a great death-roll. Life-belts that we had thought efficient were proved faulty of adjustment and were condemned: methods of boat-lowering were altered to meet the danger of a sudden list: the run of gangway and passage to the life-apparatus was cleared of impediment. When on a passage every precaution that could be taken towards a ready alert was insisted upon. Despite the manly grumbling of the very young military officers on board, certain irksome regulations were enforced. Life-belts had to be worn continuously; troops were only allowed below decks at stated hours; systems of drill, constantly carried through, left little leisure for the officers and men. Although no formal drill can wholly meet the abnormal circumstances of the new sea-casualty, we left nothing undone to prepare for eventualities. That our efforts were not useless was evident from the comparatively small loss of life that has resulted from late transport disasters.

The system of escort varies largely in the different seas. Homeward from Canada and, latterly, from the United States the troopships are formed in large convoys under the ocean escort of a cruiser. On arrival at a position in the Atlantic within working distance of the destroyers' range of steaming, the convoy is met by a flotilla of fast destroyers who escort the ships to port. For transport work in the Mediterranean no such arrangement could be operated. Every sea-mile of the great expanse is equally a danger zone. Usually, vessels of moderate speed are accompanied by sloops or armed drifters, but the fast troopships require destroyers for their protection. The long courses call for relays, as the destroyers cannot carry sufficient fuel. Marseilles to Malta, Malta to Suda Bay, Suda Bay to Salonika—a familiar voyage of three stages—required the services of no less than five destroyers. The numbers of our escorting craft were limited: it called for keen foresight on the part of the Naval Staff and unwearying sea-service on that of the war craft to fit their resources to our demands.



TRANSPORTS IN SOUTHAMPTON DOCKS

In the narrow seas, with the patrols more numerous and closely linked, the short-voyage transports proceed on a time-table of sailings that keeps them constantly in touch with armed assistance. The vessels are mostly of light draught and high speed. Whilom railway and pleasure craft, they make their voyages with the exactitude of the rail-connections they served in the peaceful days. Although many of them are built and maintained (and certificated by the Board of Trade) for smooth-water limits only, the emergency of the times has given opportunity of proof that their seaworthy qualities are underestimated by the authorities. The high gales and dangerous short seas of the Channel are no deterrent to their voyages; under the pressure of the continual call for reinforcements on the Western Front, and serving the line of route from England to the Continent, to Marseilles and beyond, they stand no hindrance. They are specially the objects of enemy attention. Their high speed and rapid turning power enables them to run moderately free of torpedo attack—though the attempts to sink them by this weapon are frequent enough—but in the German sea-mines they have a menace that cannot so readily be evaded. Many have fallen victims to this danger, but the ready succour of the patrols has prevented heavy loss of life.

Though armed for defence, they have not had many opportunities for gun action. Their keen stems are weapon enough, as Captain Keith considered when he drove *Queen Alexandra* at full speed into an enemy submarine, sinking him, and nipping a piece of his shorn hull for trophy.

Southampton is the principal base for the smaller transports. Large vessels—the *Olympic* and her sisters—come and go from the port, but it is by the quick turns of the smaller vessels that the huge traffic of the base is cleared. Tramping through the streets of the ancient town to turn in at the dock gates, company after company of troops file down the quayside to embark on the great adventure. The small craft are berthed at the seaward end of the docks, and the drifting white feathers at their funnel-tips marks steam up in readiness for departure. The drab-grey of their hulls and decks is quickly lined by ochre tint of khaki uniforms. There is no halt to the long lines of marching men, save on the turn of the stream to another gangway. By long practice, the Naval Transport Staff and the embarkation officers have brought their duties to a finished routine. There is not here the muster, the enumeration, the interminable long-drawn march and counter-march on the wharf-side, that is the case with the larger ocean transports. Crossing the gangway, carrying pack and equipment, the troops settle down on the decks in a closely packed mass.

Anon, with no undue advertisement, the transports unmoor from the quay and steam down Southampton Water. Off St. Helens, the night covers them and they steal out swiftly on the Channel crossing.

INTERLUDE

BUT for the flat-topped dwellings, the domes and minarets, of the town that stands in the alluvial valley, Suda Bay is not unlike a Highland loch in its loneliness and rugged grandeur. The high surrounding mountains, the lofty snow-capped summit of Psiloriti standing up in the east, the bare hill-side sloping to the water with no wooded country to break the expanse of rock and heath, the lone roadway by the fringe of the sea that leads to the wilds, are all in likeness to the prospect of a remote Sutherland landscape. The darkling shadows on the water, the play of sun and cloud on the distant

uplands, completes the picture; sheep on the hill-side set up plaintive calls that echo over the Bay.

The heavy westerly gale that was reason for our being signalled in from sea has blown itself out, and the water of the Bay stands still and placid. All that is left of the furious squalls of yesterday has not strength to keep us wind-rod in the anchorage, and we cast about to the vagaries of the drift.

We were bound down from Salonika to Marseilles when ordered in. We had expected to meet the relieving escort of destroyers at the Cerigo Channel, but the bad weather had prevented them from proceeding at any but a slow speed, and there was no prospect of their arrival at the rendezvous. So we turned south to seek protection behind the booms at Suda Bay. We are a packed ship. The shortage of transports has had effect in crowding the vessels in service to a point far beyond the limits of their accommodation. We have had to institute a watch-and-watch system among our huge complement. While a proportion are seeking rest below, others crowd the upper decks, passing the time as best they may until their turn of the hammocks comes round.

The fine weather after the late gale has brought every one on deck. The doings of the ships in the anchorage have interest for the landsmen. Naval cutters and whalers are out under oars for exercise, and thrash up and down the Bay with the long steady sweep of practised rowers. Our escort of two destroyers arrives—their funnels white-cruised from the heavy weather they have experienced on passage from Malta. They engage the flagship with signals, then steam alongside an oiler to take fuel for the return voyage. A message from the senior officer is signalled to us to have steam raised, to proceed to sea at midnight.

Standing in from the Gateway, a British submarine comes up the Bay. She moves slowly, as though looking for the least uncomfortable berth in the anchorage. The oil-ship, having already the two destroyers alongside, cannot offer her a place: she will have to lie off and await her turn. We put a signal on her, inviting her people to tie up alongside and come stretch their legs on our broad decks. Instant compliance. She turns on a long curve, rounds our stern, and her wires are passed on board.

The commander of the submarine gazes about curiously as he comes on board. He confesses that he has had no intimate acquaintance with merchants' ships. The huge number of our passengers impresses him, accustomed as he is to the small manning of his own vessel. Standing on the navigation bridge, we look out over the decks below at the khaki-clad assembly. The ship seems brimming over with life and animation. There is no corner but has its group of soldiers. They are everywhere; in the rigging, astride the derricks, over the top of boats and rafts they are stretched out to the sun. Mess-cooks with their gear push their way through the crowds; there is constant movement—the men from aft barging forward, the fore-end troops blocking the gangways as they saunter aft. Noisy! Snatches of song, hails, and shouts—the interminable games of 'ouse with '*Clikety-clik* and *blind-forty*' resounding in the many local dialects of the varied troops. High in spirit! We are the leave-ship, and they are bound home for a long-desired furlough after the deadly monotony of trench-keeping on the Doiran Front.

"Gad! What a crowd," he says. "I had no idea you carried so many. They look so big—and so awkward in a ship. Of course, on a battleship we muster a lot o' men, twelve hundred in the big 'uns, but—somehow—one never sees them about the decks unless at divisions or that. Perhaps it's khaki does it; one gets accustomed to blue in a ship."

A 'diversion' has been arranged for the afternoon. Dinner over, all troops are mustered to a boat drill that includes the lowering of the boats. Since leaving Salonika there has been no such opportunity as now offers. Despite foreknowledge of the time of assembly it is a long proceeding. Our complement is made up of small details—a handful of men from every battalion on the Front. Officers set to their control are drawn from as many varied branches of the service. The valued personal 'grip' of non-commissioned officers is not at our disposal. There is no such order and discipline as would be the case if we were manned by complete battalions. The routine of military movements seems dull and lifeless at sea, however efficient it may prove on land. We are long on the job.

By dint of check and repetition the grouping of the men at their boat stations is brought to a moderate proficiency. The seamen at the boats swing out and lower, and we set the boats afloat, each with a full complement of

troops. Embarked, and left to their own resources—with only one ship's rating to steer—the men make a better show. The division of the mass into smaller bodies induces a rivalry and spirit of competition: they swing the oars sturdily and make progress to and fro on the calm water of the Bay.

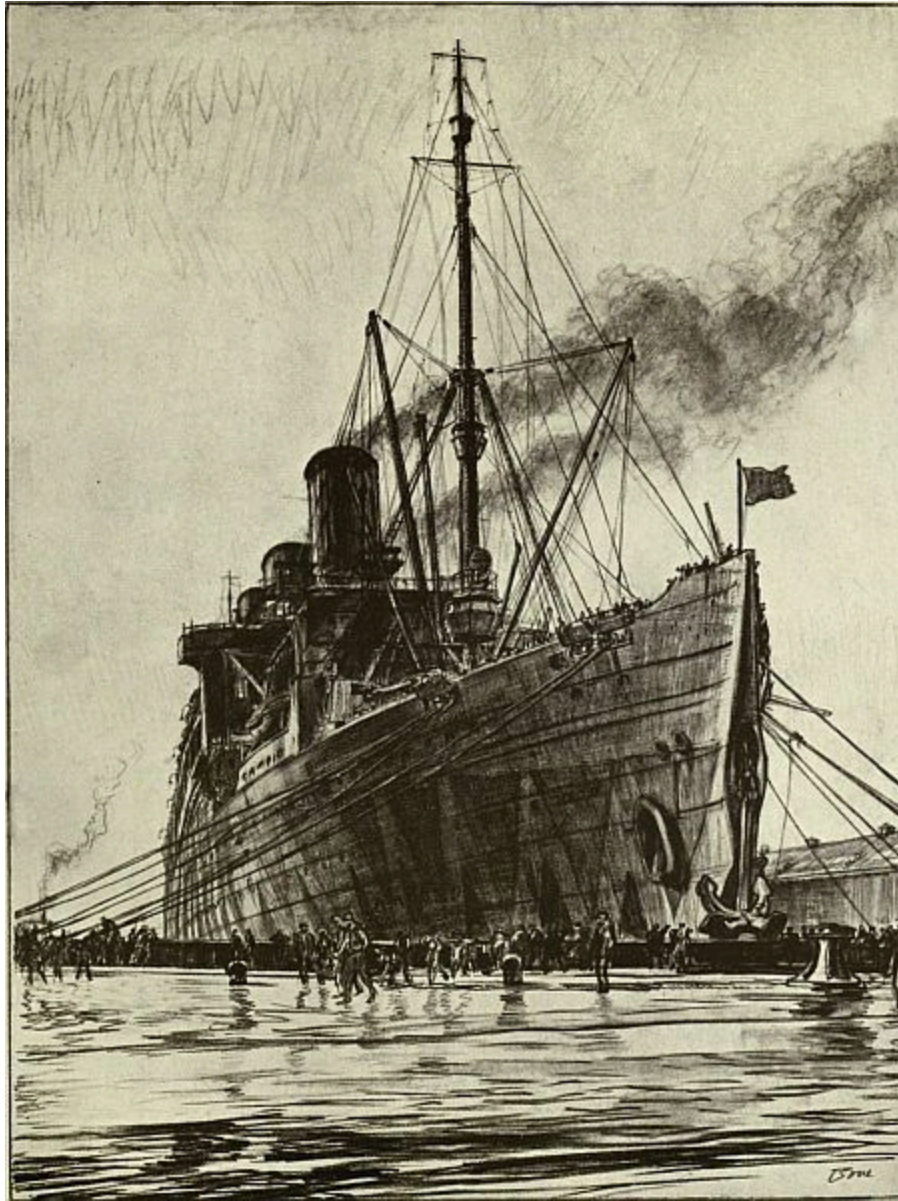
With the boats away full-loaded, we take stock of the numbers still mustered on the deck. Considerably reduced, they are still a host. The boat deck, the forecastle head, the poop—are all lined over by the waiting men: the empty boat-chocks and the dangling falls inspire a mood of disquiet. Standing at ease, they seem to be facing towards the bridge. Doubtless they are wondering what we think of it all. The submarine's commander has been with us at our station during the muster. We look at one another—thoughtfully.

'THE MAN-O'-WAR 'S 'ER 'USBAND'

A SENSE of security is difficult of definition. Largely, it is founded upon habit and association. It is induced and maintained by familiar surroundings. On board ship, in a small world of our own, we seem to be contained by the boundaries of the bulwarks, to be sailing beyond the influences of the land and of other ships. The sea is the same we have known for so long. Every item of our ship fitment—the trim arrangement of the decks, the set and rake of mast and funnel, even the furnishings of our cabins—has the power of impressing a stable feeling of custom, normal ship life, safety. It requires an effort of thought to recall that in their homely presence we are endangered. Relating his experiences after having been mined and his ship sunk, a master confided that the point that impressed him most deeply was when he went to his room for the confidential papers and saw the cabin exactly in everyday aspect—his longshore clothes suspended from the hooks, his umbrella standing in a corner as he had placed it on coming aboard.

Soldiers on service are denied this aid to assurance. Unlike us, they cannot carry their home with them to the battlefield. All their scenes and surroundings are novel; they may only draw a reliance and comfort from the familiar presence of their comrades. At sea in a ship there is a yet

greater incitement to their disquiet. The movement, the limitless sea, the distance from the land, cannot be ignored. The atmosphere that is so familiar and comforting to us, is to many of them an environment of dread possibilities.



THE LEVIATHAN DOCKING AT LIVERPOOL

It is with some small measure of this sense of security—tempered by our knowledge of enemy activity in these waters—we pace the bridge. Anxiety is not wholly absent. Some hours past, we saw small flotsam that may have come from the decks of a French mail steamer, torpedoed three days ago. The passing of the derelict fittings aroused some disquiet, but the steady

routine of our progress and the constant friendly presence of familiar surroundings has effect in allaying immediate fears. The rounds of the bridge go on—the writing of the log, the tapping of the glass, the small measures that mark the passing of our sea-hours. Two days out from Marseilles—and all well! In another two days we should be approaching the Canal, and then—to be clear of 'submarine waters' for a term. Fine weather! A light wind and sea accompany us for the present, but the filmy glare of the sun, now low, and a backward movement of the glass foretells a break ere long. We are steaming at high speed to make the most of the smooth sea. Ahead, on each bow, our two escorting destroyers conform to the angles of our zigzag—spurring out and swerving with the peculiar 'thrown-around' movement of their class. Look-out is alert and in numbers. Added to the watch of the ship's crew, military signallers are posted; the boats swung outboard have each a party of troops on guard.

An alarmed cry from aloft—a half-uttered order to the steersman—an explosion, low down in the bowels of the ship, that sets her reeling in her stride!

The upthrow comes swiftly on the moment of impact. Hatches, coal, shattered debris, a huge column of solid water go skyward in a hurtling mass to fall in torrent on the bridge. Part of a human body strikes the awning spars and hangs—watch-keepers are borne to the deck by the weight of water—the steersman falls limply over the wheel with blood pouring from a gash on his forehead. . . . Then silence for a stunned half-minute, with only the thrust of the engines marking the heart-beats of the stricken ship.

Uproar! Most of our men are young recruits: they have been but two days on the sea. The torpedo has gone hard home at the very weakest hour of our calculated drill. The troops are at their evening meal when the blow comes, the explosion killing many outright. We had counted on a proportion of the troops being on the deck, a steadying number to balance the sudden rush from below that we foresaw in emergency. Hurrying from the mess-decks as enjoined, the quick movement gathers way and intensity: the decks become jammed by the pressure, the gangways and passages are blocked in the struggle. There is the making of a panic—tuned by their outcry, "*God!*

O God! O Christ!" The swelling murmur is neither excited nor agonized—rather the dull, hopeless expression of despair.

The officer commanding troops has come on the bridge at the first alarm. His juniors have opportunity to take their stations before the struggling mass reaches to the boats. The impossibility of getting among the men on the lower decks makes the military officers' efforts to restore confidence difficult. They are aided from an unexpected quarter. The bridge-boy makes unofficial use of our megaphone. "Hey! Steady up you men doon therr," he shouts. "Ye'll no' dae ony guid fur yersels croodin' th' ledders!"

We could not have done it as well. The lad is plainly in sight to the crowd on the decks. A small boy, undersized. "Steady up doon therr!" The effect is instant. Noise there still is, but the movement is arrested.

The engines are stopped—we are now beyond range of a second torpedo—and steam thunders in exhaust, making our efforts to control movements by voice impossible. At the moment of the impact the destroyers have swung round and are casting here and there like hounds on the scent: the dull explosion of a depth-charge—then another, rouses a fierce hope that we are not unavenged. The force of the explosion has broken connections to the wireless room, but the aerial still holds and, when a measure of order on the boat-deck allows, we send a message of our peril broadcast. There is no doubt in our mind of the outcome. Our bows, drooping visibly, tell that we shall not float long. We have nearly three thousand on board. There are boats for sixteen hundred—then rafts. Boats—rafts—and the glass is falling at a rate that shows bad weather over the western horizon!

Our drill, that provided for lowering the boats with only half-complements in them, will not serve. We pass orders to lower away in any condition, however overcrowded. The way is off the ship, and it is with some apprehension we watch the packed boats that drop away from the davit heads. The shrill ring of the block-sheaves indicates a tension that is not far from breaking-point. Many of the life-boats reach the water safely with their heavy burdens, but the strain on the tackles—far beyond their working load—is too great for all to stand to it. Two boats go down by the run. The men in them are thrown violently to the water, where they float in the wash and shattered planking. A third dangles from the after fall, having

shot her manning out at parting of the forward tackle. Lowered by the stern, she rights, disengages, and drifts aft with the men clinging to the life-lines. We can make no attempt to reach the men in the water. Their life-belts are sufficient to keep them afloat: the ship is going down rapidly by the head, and there remains the second line of boats to be hoisted and swung over. The chief officer, pausing in his quick work, looks to the bridge inquiringly, as though to ask, "How long?" The fingers of two hands suffice to mark our estimate.

The decks are now angled to the deepening pitch of the bows. Pumps are utterly inadequate to make impression on the swift inflow. The chief engineer comes to the bridge with a hopeless report. It is only a question of time. How long? Already the water is lapping at a level of the foredeck. Troops massed there and on the forecastle-head are apprehensive: it is indeed a wonder that their officers have held them for so long. The commanding officer sets example by a cool nonchalance that we envy. Posted with us on the bridge, his quick eyes note the flood surging in the pent 'tween-decks below, from which his men have removed the few wounded. The dead are left to the sea.

Help comes as we had expected it would. Leaving *Nemesis* to steam fast circles round the sinking ship, *Rifleman* swings in and brings up alongside at the forward end. Even in our fear and anxiety and distress, we cannot but admire the precision of the destroyer captain's manœuvre—the skilful avoidance of our crowded life-boats and the men in the water—the sudden stoppage of her way and the cant that brings her to a standstill at the lip of our brimming decks. The troops who have stood so well to orders have their reward in an easy leap to safety. Quickly the foredeck is cleared. *Rifleman* spurts ahead in a rush that sets the surrounding life-boats to eddy in her wash. She takes up the circling high-speed patrol and allows her sister ship to swing in and embark a number of our men.

It is when the most of the life-boats are gone we realize fully the gallant service of the destroyers. There remain the rafts, but many of these have been launched over to aid the struggling men in the water. Half an hour has passed since we were struck—thirty minutes of frantic endeavour to debark our men—yet still the decks are thronged by a packed mass that seems but little reduced. The coming of the destroyers alters the outlook. *Rifleman's*

action has taken over six hundred. A sensible clearance! *Nemesis* swings in with the precision of an express, and the thud and clatter of the troops jumping to her deck sets up a continuous drumming note of deliverance. Alert and confident, the naval men accept the great risks of their position. The ship's bows are entered to the water at a steep incline. Every minute the balance is weighing, casting her stern high in the air. The bulkheads are by now taking place of keel and bearing the huge weight of her on the water. At any moment she may go without a warning, to crash into the light hull of the destroyer and bear her down. For all the circling watch of her sister ship, the submarine—if still he lives—may get in a shot at the standing target. It is with a deep relief we signal the captain to bear off. Her decks are jammed to the limit. She can carry no more. *Nemesis* lists heavily under her burdened decks as she goes ahead and clears.

Forty minutes! The zigzag clock in the wheelhouse goes on ringing the angles of time and course as though we were yet under helm and speed. For a short term we have noted that the ship appears to have reached a point of arrest in her foundering droop. She remains upright as she has been since righting herself after the first inrush of water. Like the lady she always was, she has added no fearsome list to the sum of our distress. The familiar bridge, on which so many of our safe sea-days have been spent, is canted at an angle that makes foothold uneasy. She cannot remain for long afloat. The end will come swiftly, without warning—a sudden rupture of the bulkhead that is sustaining her weight. We are not now many left on board. Striving and wrenching to man-handle the only remaining boat—rendered idle for want of the tackles that have parted on service of its twin—we succeed in pointing her outboard, and await a further deepening of the bows ere launching her. Of the military, the officer commanding, some few of his juniors, a group of other ranks, stand by. The senior officers of the ship, a muster of seamen, a few stewards, are banded with us at the last. We expect no further service of the destroyers. The position of the ship is overmenacing to any approach. They have all they can carry. Steaming at a short distance they have the appearance of being heavily overloaded; each has a staggering list and lies low in the water under their deck encumbrance. We have only the hazard of a quick out-throw of the remaining boat and the chances of a grip on floating wreckage to count upon.

On a sudden swift sheer, *Rifleman* takes the risk. Unheeding our warning hail, she steams across the bows and backs at a high speed: her rounded stern jars on our hull plates, a whaler and the davits catch on a projection and give with the ring of buckling steel—she turns on the throw of the propellers and closes aboard with a resounding impact that sets her living deck-load to stagger.

We lose no time. Scrambling down the life-ropes, our small company endeavours to get foothold on her decks. The destroyer widens off at the rebound, but by clutch of friendly hands the men are dragged aboard. One fails to reach safety. A soldier loses grip and goes to the water. The chief officer follows him. Tired and unstrung as he must be by the devoted labours of the last half-hour, he is in no condition to effect a rescue. A sudden deep rumble from within the sinking ship warns the destroyer captain to go ahead. We are given no chance to aid our shipmates: the propellers tear the water in a furious race that sweeps them away, and we draw off swiftly from the side of the ship.

We are little more than clear of the settling fore-end when the last buoyant breath of *Cameronia* is overcome. Nobly she has held afloat to the debarking of the last man. There is no further life in her. Evenly, steadily, as we had seen her leave the launching ways at Meadowside, she goes down.



SALVAGE VESSELS OFF YARMOUTH, ISLE OF WIGHT

XIII

THE SALVAGE SECTION

THE TIDEMASTERS

IF Royal Canute, King of England and Denmark, with his train of servile earls and thanes, could revisit the scene of his famous object-lesson, he would learn a new value in the tide. Suitably, he might improve his homily by presentation of the salvage tidemasters, harnessing the rise and fall of the stubborn element to serve their needs and heave a foundered vessel to sight and service. He would note the cunning guidance of strain and effort, their exact timing of the ruled and ordered habits of the sea. As a moral, he could quote that, if tide may not be ordered to command, it can at least be governed and impressed to performance of a mighty service.

Recovery of ships, their gear and cargo, is no longer wholly an application of practised seamanship. The task is burdened and complicated by powers and conditions that call for auxiliary arts. It is true that the salvage officer's ground, his main asset, is the knowledge and ability to do a seamanlike 'job o' work' when the time and tide are opportune; he must have a seaman's training in the ways of the wind and the sea and be able properly to assess the weather conditions under which alone his precarious work is possible. A scientist of a liberal and versatile type (not perhaps exhaustive in his scope and range), he is able to draw the quantum of his needs from a wide and varied summary. Together with his medical exemplar, he has developed a technique from crude remedies and imperfect diagnoses to application of fine science. He must have a sure knowledge of the anatomy of his great steel patients, be versed in the infinite variety and intricacy of ship construction, and the valves and arteries of their power; be able to pen and plan his formulæ for weight-lifting—the stress and strain of it, down to the calibre of the weakest link. A super-tidesman, he must know to an inch the run of bottom, the swirl and eddy, the value of flood and ebb

and springs, for the tide—Canute's immutable recalcitrant—is his greatest assistant, a familiar *Genius maris* whom he conjures from the deeps of ocean to do his bidding. Shrewd! He is a keen student of the psychology of the distressed mariner; again, like the medical man, he must set himself to extract truth from the tale that is told. His treatment must be prescribed, not to meet a case as presented, but as his skilled knowledge of the probabilities warrants. Tactful, if he is to meet with assistance in his difficult work, he must assume the sympathy of one seaman to another in distress. What, after all, does it matter if he agree heartily that "the touch was very light, we were going dead slow," when, from his divers' reports, he knows that the whole bottom is 'up'?

In the handling of his own men there must be a combination of rigour and reason. Salvage crews are a hardy, tempestuous race who have no ordinary regard for the niceties of law and order; their work is no scheduled and defined occupation with states and margins; they are servants to tide and weather alone; they are embarked on a venture, on a hazard, a lottery. To such men, administering, under his direction, the heroic but destructive remedies of high explosive and compressed air, there cannot be a normal allowance for the economic use of gear and material. He must know the right and judicial discount to be made that will meet the conflicting demands of the expenses department and the results committee. Above all, he must be of an infinite patience, of the mettle that is not readily discouraged. In the great game of seafaring his hand holds the king of disappointment and the knaves of frustration and discouragement. But he has other cards; he holds an ace in stability and determination.

Calm days and smooth seas may lure him to surpassing effort, to work through the tides in feverish energy, making the most of favoured opportunity. The scattered and interrupted work of months has perhaps been geared and bound, the tackle rigged and set for a final dead lift. Buoyancy is figured out and assured; the pumps are in place, throbbing and droning out, throwing steady streams from the weight of water that so long has held the foundered wreck in depth. The work has been long and trying, but an end to difficulty is in sight. Given a day or two of continued fine weather, the sea and the rocks will have to surrender their prisoner.

Comes a darkling to windward and the sea stirs uneasily; jets and spurts of broken water appear over the teeth and spit of rocky ledges. The salvors look around with calculating eyes and note the signs of a weather break. Still, there is no slackening of effort; there may be time to complete the work before the sea rises to interfere; if anything, the omens only call for another spur to the flank, a new sting to the lash.

Beaten to the knees, the gear and tackle swaying perilously in breaking seas, the lifting-barges thundering at their curbs, the pumps groaning and protesting their inability to overcome the lap of blue water, there is no alternative but to abandon the work and return to harbour. From the beach the salvage officer may watch his labour of weeks—or months—savagely undone in an hour or two of storm and fury of the sea!

It is a great catalogue, that schedule of virtues and accomplishments. To it must be added, as a supplement, that he must be a 'made' man—made in a long hard pupilage in a stern school that appraises strictly on results. It is of little use to show that, in theory, a certain course was right and proper, when the broad but damning fact remains that the property is still in Davy Jones his locker, and likely—there to remain. Many are called, but few are chosen. The salvage service has no room for the merely mediocre officer: the right man goes inevitably to his proper place, the wrong one goes back to a junior, and less responsible, post at sea.

It is doubtful if the Naval Service could produce the type required. Their candidate would be, to a degree, inelastic. He would be an excellent theorist, a sound executant, a strict disciplinarian; but his training and ideas would fit ill to the wide range of conflicting interests, and the shutting out of all manœuvre, however skilled and stimulating—but that of securing a maximum of result by a minimum of effort. Perhaps it was for these reasons our salvage services before the war were almost wholly mercantile and commercial. Certainly, most Admiralty efforts in this direction were confined to ports and harbours where method could be ordered and controlled by routine; their more arduous and unmanageable cases on the littoral were frequently handed over to the merchantmen—not seldom after naval efforts had been unavailing. Among the protestations of our good faith to the world in time of peace, it may be cited that we made no serious provision for a succession of maritime casualties; there was no specially

organized and equipped Naval Salvage Service. True, there were the harbour gear, divers, a pump or two, and appliances and craft for attending submarine accidents, but their energies were bent largely to humane purposes—to marine first aid. Of major gear and a trained personnel to control equipment and operation there was not even a nucleus. Salvage was valued at a modest section of the "Manual on Seamanship" (written by a mercantile expert), and a very occasional lecture at the Naval College. At war, and the toll of maritime disaster rising, the need grew quickly for expert and special service. There was no longer a relative and profitable balance to be struck between value of sea-property and cost of salvage operations. A ship had become beyond mere money valuation; as well assess the air we breathe in terms of finance. No cost was high if a keel could be added to our mercantile fleets in one minute less than the time the builders would take to construct a new vessel. The call was for competent ship-surgeons who could front-rank our maritime C Threes. By whatever skill and daring and exercise of seamanship, the wrecks must be returned to service. Happily, there was no necessity to go far afield; the merchants' salvage enterprise, like the merchants' ships and the merchants' men, was ready at hand for adoption.



IN A SALVAGE VESSEL: OVERHAULING THE INSULATION OF THE POWER LEADS

The Salvage Section, Admiralty, is a dignified caption and has an almost imperial address, but, camouflages and all, it is not difficult to see the hem of old sea-worn garments of our mercantile companies peeping out below the gold braid. If in peace-time they did wonders, war has made their greatest and most successful efforts seem but minor actions compared to their present-day victories. The practice and experience gained in quick succession of 'cases' has tuned up their operations to the highest pitch of efficiency. New and more powerful appliances have come to their hands; a skilled and technical directorate has liberated initiative. Strandings, torpedo or mine damage, fire, collisions—frequently a compound of two or three—or all five—provide them with occasion for every shift of ingenuity, every turn of resource. There is no stint to the gear, and no limits to invention, or device, if there is a possibility of a damaged ship being brought to the dry docks. Is it not on record that an obstinate, stranded ship, driven high on the beach, was finally relaunched on the crest of an artificially created 'spring' tide, the wash and suction of a high-speed destroyer, plying and circling in the shallows?

Many new perils are added to the risks and hazards of their normally dangerous work. Casualties that call for their service are rarely located in safe and protected waters; open coast and main channels are the marches of the Salvage Section, where the enemy has a keen and ready eye for a 'potting' shot by which he may prevent succour of a previous victim. The menace of sea-mines is particularly theirs; the run and swirl of Channel tides has strength to weigh a stealthy mooring and carry a power of destruction up stream and down. They have a new and deadly danger to be guarded against in the ammunition and armament of their stricken wards. Many have gone down at 'action stations,' and carry 'hair-sprung' explosive charges, the exact condition and activity of which are usually a matter for conjecture. It calls for a courage of no ordinary measure to grope and stumble under water amid shattered wreckage for the safety-clutch of the charges, or grapple in the mud and litter for torpedo firing-levers. This the pioneer of the divers must do, as the first and most important of his duties.

With skill enhanced by constant and encouraged practice, they set out to bind the wounds and raise our damaged ships to a further lease of sea-activity. So definite and sure are their methods, so skilled and rapid their execution, they steam ahead of reconstruction and crowd the waiting-room at the dry-dock gates. Lined up at the anchorage awaiting their turn, the recovered vessels may be crippled and bent, and showing torsion and distress in the list, and staggering trim with which they swing flood and ebb. They may rest, halting, on the inshore shallow flats, but, laid by for a term of repair, their day is to come again. The Salvage Section has reclaimed their rent and stranded hulls from the misty sea-Front; the Repair Section, working day and night, will hammer and bind and reframe the gaps of their steel; the Sea Section will take them out on the old stormy road, sound and seaworthy, with the flag at the peak once more.

A DAY ON THE SHOALS

THE rigger was engaged at second tucks of a five-inch wire-splicing job, and hardly looked in the direction we indicated. "Them," he said. "Them's crocks wot we don't want nothin' more t' do with! Two on 'em's got frozen mutton. High? Excelsi-bloody-or! . . . an' that feller as is down by th' 'ead—

Gawd! 'e don't 'arf smell 'orrible!" A pause, while he hammered down the strands and found fault with his assistant, gave us time to disentangle the negatives of his opening. "Grain, she 'as—an' of all th' ruddy messes wot I ever see—she gets it! We 'ad four days at 'er—out there 'n th' Padrig Flats, an' she sickened nigh all 'ands! . . . Now we're well quit o' 'er, an' th' longshore gangs is unloadin' th' bulk, in nosebags an' gas 'elmets, t' get 'er a-trim for th' dry dock!"

As we passed alee of the grain-carrier there was no doubt of the truth of the rigger's assurance. Steam-pumps on her fore-deck were forcing a sickly mixture of liquid batter through hoses to a barge alongside, and the overpowering stench of the mess blew down to us and set eyes and noses quickening with instant nausea. The men on the barges were garbed in odd headgear, high cowls with staring circular eyepieces, and each carried a knapsack cylinder on his back. Clouds of high-pressure steam from the winches and pumps threw out in exhaust, and the hooded, ghost-like figures of the labourers passed and repassed in drifts of white vapour. To the hiss and rumble of machines, clamour of block-sheaves and chain and piston joined action to make a setting of *Inferno*, the scene might well be imagery for a stage of unholy rites.

Past her, we turned to the clean salt breeze again and stood on to the open sea. The salvage officer, a Commander, R.N.R., joined us at the rail. "What about that now? Sa—lubrious?" he said.

We wondered how men could be got to work in such an atmosphere, how it was possible to handle such foul-smelling litter in the confined holds.

"Oh! We go through that all right. A bit inconvenient and troublesome, perhaps, working in a restricting gas-rig; but now, the chemists have come to our assistance and we can sweeten things up by a dose of anti-stink. . . . But you won't see that to-day. Our 'bird' has got no cargo, only clean stone ballast—a soft job."

The 'soft job' had had a rough time, a combination and chapter of sea and war hazard. Inward bound from the United States with a big cargo, a German torpedo had found a mark on her. She settled quickly by the stern, but the undamaged engines worked her gallantly into a small seaport where

she brought up with her main deck awash. There she was lightened of her precious load, temporary baulks and patches were clamped and bolted to her riven shell-plate, and she set off again on a short coastwise voyage to the nearest port where definite and satisfactory repair could be effected. Off the Heads, the enemy again got sights on her. Crippled, and steaming at slow speed to ease strain on the bulkheads, she made a 'sitting' target for a second torpedo, that shattered rudder and stern-post and sheared the propellor from the shaft.

"We came on her just before dark," said the commander. . . . "Some of the crew were in the boats, close by, but the captain and a Trinity pilot and others were still aboard. She was down astern to the counter and up forward like a ruddy unicorn. We got fast and started to tow. Tow?—Might as well have taken on the Tower Bridge. There was no way of steering her, and a strong breeze from the south'ard blew her head down against all we could do. . . . Anyway, we hung on, and at daylight in the morning the wind let up on us a bit, and we guided her drift—that's about all we could do—inshore, till she took the bottom on good ground a little north of the Westmark Shoal. We filled her up forrard as the weather was looking bad—a good weight of water to steady her through a gale. She's lain out there for two months now. We've had a turn or two at her occasionally—shoring up the after bulkheads and that, while we had weather chances. *Titan* has been out at her since yesterday morning. . . . It looks good and healthy now." He cast an eye around appreciatively at the calm sea and quiet sky, the gorse-banked cliffs dimmed by a promising summer haze, at seagulls lazily drifting on the tide or becking and bowing in the glassy ripples of our wash. "Good and healthy; I like to see these old 'shellbacks' sitting low and not shrilling overhead with all sail set. . . . If this weather holds I shouldn't wonder if we get the old bus afloat on high tide to-day!"

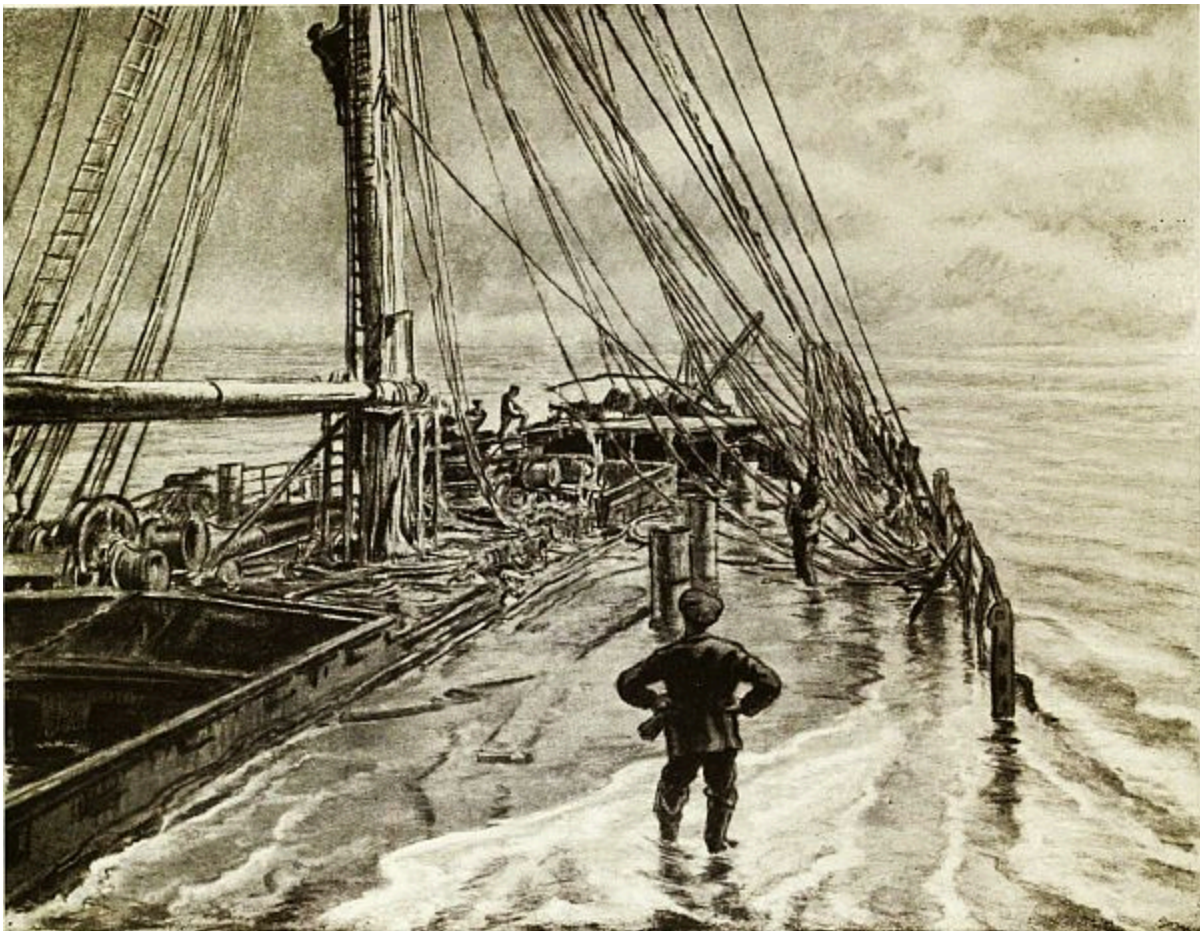
Clear of harbour limits and heading out to the shoals, a brisk rigging of gear and tackle brings action to the decks of the salvage steamer. Already we had thought the narrow confines from bulwark to bulwark congested by the bulk of appliances, but, from hole and corner and cunning stowage, further coils and shoots and lengths of flexible, armoured hose are dragged and placed in readiness for operations. Derricks are topped up and purchases rove for handling the heavy twelve-inch motor-pumps. Hawsers are uncovered and coiled clear, stout fenders thrown over in preparation for

a grind alongside the wreck. Mindful of possibilities, the engineer-lieutenant and his artificers go over the insulation of their power leads in minute search for a leak in the cables that may occasion a short circuit later on. The terminals and couplings are buffed and polished with what seems exaggerated and needless precision—but this is salvage, where sustained effort is only possible in the rare and all-too-brief union of favourable tide and weather conditions. A cessation of the steady throw of the pumps, however instant and skilful the adjustment, may mean the loss of just that finite measure in buoyancy that could spring the weight of thousands in tons. Second chances are rarely given by a grudging and jealous sea; there must be no hitch in the gear, no halt in weighing the mass.

A drift of lazy smoke on the sea-rim ahead marks our rendezvous, where *Titan* and a sisterly tug-boat are already at work on the wreck. A screen of motor-patrols are rounding and lining out in the offing, with a thrust of white foam astern that shows their speed. Coastwise, a convoy of merchant ships zigzag in confusing angles on their way to sea, guarded by spurring destroyers and trawler escort. Seaplanes are out, hawking with swoop and wheel for sight of strange fish. The seascape is busy with a shipping that must remind the coastguard and lightkeepers of old and palmy days when square sail was standard at sea. The Westmark Shoal lies some distance from the normal peace-time track of direct steaming courses. It lies in the bight of a bay, where rarely steamers closed the land. Sailing ships, close-hauled and working a tack inshore, or fisher craft on their grounds, had long been the only keels to sheer water in the deeps, but war practice has renewed our acquaintance with many old sea-routes and by-paths, and we are back now to charts and courses that have long been out of our reckoning.

The tide is at low-water slack, and whirls and eddies mark the run over shallows. At easy speed and handing the lead, we approach the wreck. Her weathered hull, gilt and red-rusted by exposure to sun and wind and sea, stands high and bold against the deep blue of a summer sky. Masts and rigging and cordage are bleached white, like tracery of a phantom ship. The green sea-growth on her underbody fans and waves in the tide, showing long voyaging in the crust and stage of it. She lies well and steadily, with only a slight list to seaward that marks the gradient on which she rests. Through fracture on the stern and counter, the twisted and shattered frames

and beams and angles can be seen plainly. Sunlight, in slanting rays, shines through the rents and fissures of the upper deck, and plays on the free flood that washes in and out of the exposed after hold; seaweed and flotsam surges on the tide, clinging to the jagged, shattered edges of the plating, and breaking away to lap in the dark recesses. To eyes that only know the lines and mould of sightly, seaworthy vessels, she seems a hopeless and distorted mass of standing iron—a sheer hulk, indeed, fit only for a lone sea-perch to gull and gannet and cormorant. It appears idle for the salvors to plan and strive and wrestle for such a prize, but their keen eyes are focused to values not readily apparent. "A fine ship," says the commander, now happily assured that his 'soft job' has suffered no worse than a weathering on the ledge that his skill has secured her. "A job o' work for the repairers, certainly . . . but they will set her up as good as new in a third of the time it would take to build a substitute!"



A TORPEDOED MERCHANTMAN ON THE SHOALS: SALVAGE OFFICERS MAKING A SURVEY

We anchor at a length or two to seaward. There is not yet water alongside for our draught, but *Titan*, drawing less, is berthed at her stern and their men are taking advantage of low water to pin and tomp and strengthen the rearmost bulkhead that must now do duty for the demolished stern section. A boat from *Titan* brings the officer in charge, and he greets his senior with no disguised relief. A serious leak has developed in one of the compartments that they had counted on for buoyancy. . . . "Right under the bilge, and ungetatable, with all that rubble in th' holds. A good job you brought out these extra pumps. We should manage now, all right!"

Technical measures are discussed and a plan of operations agreed. At half-flood there will be water for us alongside, and a 'lift' can be tried. Number one hold is good and tight, but still has a bulk of water to steady her on the ledge; number two is clear and buoyant; three has the obstinate leak; the engine-room is undamaged, but water makes through in moderate quantity. Number four—"the bulkhead is bulged in like the bilge of a cask, but that cement we put down last week has set pretty well, and the struts and braces should hold." Number five? There is no number five, most of it lies on deep bottom off the Heads, some miles away!

With his colleague, the commander puts off to the wreck, to assess the prospects, and we have opportunity to note the inboard trim of her derelict posts and quarters. Davits, swung outboard as when the last of her crew left her, stand up in unfamiliar dejection, the frayed ends and bights of the boat-falls dangling overside and thrumming on the rusty hull. The boat-deck shows haste and urgency in the litter of spars and tackle thrown violently aside: a seaman's bag with sodden pitiful rags of apparel lies awry on the skids, marking some cool and forethinking mariner denied a passage for his goods. Living-rooms and crew quarters show the indications of sudden call, in open desks—a book or two cast side, quick-thrown bedspreads, an array of clothing on a line; the range-guards in the cook's galley have caught the tilt of pots and mess-kits as they slid alee in the grounding. The bridge, with chart and wheelhouse open to the wind and spray, and sea-gear adrift and disordered, strikes the most desolating note in the abandon of it all. Tenantless and quiet, the same scene would be commonplace and understood in dock or harbour, with neighbourly shore structures to point a reason for absence of ship-life, but out here—the clear horizon of an open sea in view around, with vessels passing on their courses, the desertion of

the main post seems final and complete, with no navigator at the guides and no hand at the wheel.

The flood tide making over the shoals sets in with a *thruss* of broken water alee of the wreck. The salvors' cutter, from which the mate is sounding and marking bottom, spins in widening circles in the eddies and shows the strength of early springs. As yet the stream binds the wreck hard to the bank, setting broad on from seaward, but relief will come when the spent water turns east on the last of the flood. Survey completed, the salvage officers clamber to the deck again. The leak in number three is their only concern; if that can be overcome, there seems no bar to a successful programme. The commander questions the mate as to the depth of water alongside, is assured of draught, and signals his vessel to heave up and come on. The strength and onrush of the tidal race makes the manœuvre difficult, and it is on second attempt, with a wide sweep and backing on plane of the current, she drives unhandily to position. The impact of her boarding, for all the guardian fenders, jars and stirs the wreck, but brings a confident look to the salvors' faces; as readily shaken as that, they assure themselves the responding hull will come off with 'a bit of a pinch' on the angle of withdrawal that they have planned on the tidal chart.

With hawsers and warps barely fast, the great pumps are hove up in air and swung over the hatchway of the doubtful hold. But for the general order to carry on, there are few directions and little admonition. Every man of the busy group of mechanics and riggers has 'a brick for the wall,' and the wriggling lengths of armoured hose are coupled and launched over the coamings as quickly as the massive motors are lowered. Foundering with splash and gurgle, like uncouth sea-monsters in their appanage of tortuous rubber tentacles, the sheen of their polished bulk looms through the green translucent flood of solid seawater, the grave and surely augmented tide that they are trimmed to master. Again, the seeming hopelessness of the task, the handicap of man against element, presents a doubt to one's mind. Two shell-like casings of steel, a line of piping and cab-tyre coils for power leads—to compete with the infiltration of an ocean; there are even small fish darting in the flood of it, a radiating Medusa floats in and out the weltering 'tween-decks, waving loathsome feelers as though in mockery of human efforts!

Like a war-whoop to the onslaught the dynamos of the salvage vessel start motion, and hum in *crescendo* to a high tenor tone; the vibrations of their speed and cycle are joined in conduct to the empty hull of the wreck, and she quickens with a throb and stir as of her arteries coursing. There is no preparatory trickle at outboard end of the hose ejections; with a rush and roar, a clean, solid flood pours over, an uninterrupted cascade at seven tons from each per minute!

The carpenter sounds the depth with rod and chalked lanyard, then lowers a tethered float to water-level of the flooded compartment. In this way he sets a starting mark for the competition, a gauge for the throw of the pumps. In interest with the issue, the salvage men gather round the hatchway, and all eyes are turned to the bobbing cork disc to note the progress of the contest. Stirring and drifting to slack of the line, the float seems serenely indifferent to its important motion; wayward and buoyant, it trims, this way and that, then steadies suddenly on a taut restraint; slowly it seems to rise in the water as though drawn by an invisible hand. It spins a little to lay of the cord, then hangs, moisture dropping and forming rings on the glassy surface of the well! By no seeming effort but the pulse-like quiver of the hose, the level falls away. A bolt-head on the plating shows under water, then tips an upper edge above; a minute later the round is exposed and drying in a slant of the sun.

The tense regard with which we have scanned the guide-mark gives way to jest and relief when it is seen that drainage is assured; a facetious mechanic at the hose-end makes motions as of pulling a bar handle to draw a foaming glass. "Sop it up, old sport!" says the rigger, patting the pipes. "Sop it up an' spit! Ol' Neptune ain't arf thusty!"

During our engagement, *Titan* has not been idle. There remains only an hour or two of flood tide and much has to be done. Leaving steam-pumps to cope with the more moderate leakage at the after section, she has hauled forward on the rising tide on the shoal side of the wreck. At the bows she has applied suction to the prisoned water in the fore holds, and a new stream pours overside in foaming ejection. The roar and throb of her power motors adds further volume and vibration to the rousing treatment by which the nerves of the stranded hulk seem braced. Stirred by the new life on her, the old ship may well forget she has no stern and only part a bottom.

Already the decks, gaunt and red-rusted as they are, take on a cheering look of service and animation. The seamen in the rigging and workmen crowded round the hatchways might be the dockers boarded for a day's work on the loading, and only the thunder of the motors and crash of the sluicing torrent remain foreign to a normal ship-day.

The sun has gone west when the tidal current surging past shows a change in direction. We throw sightly flotsam overboard and note the drift that takes the refuse astern. No longer the green slimy plates of the hull show above water, the tide has lapped their sea-growth and ripples high on a cleaner surface. With high water approaching we draw near the point of balance in buoyancy, and the salving tenders tighten up headfasts and stern ropes in readiness for a slip or drag. The sea-tug that has till now been a quiet partner in operations, smokes up and backs in astern to pass a hawser to the wreck. She drops away with a good scope, and lies handy to tow at orders.

Tirelessly, droning and throbbing with insistent monotony, the pumps continue their labour and draw the weight of water that holds the wreck down. At number three hold the flood below is no longer a still and placid well. The penned and mastered water seethes and whirls in impotent fury at the suction that draws and churns only to expel. Some solid matter, seaweed perhaps, has drifted to the leak and stems a volume of the incoming water; there seems a prospect that a single pump may keep the level.

In somewhat tense expectancy, we await a crisis in the operations. There is a feeling that all these masterly movements should lead to a spectacular resurrection—a stir and tremor in the frame of her, reviving sea-throes, a lurch, a list, a mighty heave, and a staggering relaunch to the deeps.

Precise and businesslike, modern salvage avoids such a flourishing end to their labours. As skilful surgeons, they object strongly to excitement. Their frail and tortured sea-patients can rarely stand more than gentle suasion. As surely as the tide they work by, the factors of weight and displacement and trim have been figured and calculated. . . . The commander draws our attention to a quiet and steady rise in the bows, the knightheads perceptibly edging nearer to a wisp of standing cloud. Without

a jar or surge the wreck becomes a floating ship; she lists a little, as the towing hawser creaks and strains, and we draw off gently to seaward.

THE DRY DOCK

A DOWNPOUR of steady, insistent rain makes quagmire of the paths on the dockside, and the half-light of a cheerless early morning gives little guidance to progress among the raffle of discarded ship-gear that lies about the yard. Stumbling over shores and stagings, skirting gaunt mounds of damaged plates and angles, we reach the sea-gate where the ship victims of mine and torpedo are moored in readiness for treatment in the great sea-hospital. In the uncertain light and under wet lowering skies, they make a dismal picture. The symmetry of conventional docking—ships moored in line and heading in the same direction—that is an orderly feature of the harbours, is not possible in the overcrowded basin. There is need to pack the vessels closely. They lie at awkward angles, the stern of one overhanging the bows of another. Masts and funnels and deck erections, upstanding at varied rakes, emphasize the confused berthing and draw the eye to the condition of the mass of damaged shipping. Not all of the vessels are shattered hulks. A number are here for hull-cleaning or overhaul, but their high sides with the rust and barnacles and weedy green scum, make as drab a feature in the combination as the listed hulls of the cripples.



A TORPEDOED SHIP IN DRY DOCK

Though nominally daylight, the arc-lamps of the pier-head still splutter in wet contacts and spread a sickly glow over the oilskin-clad group of dockmen and officials gathered to enter the ships. A chill breeze from the sea blows in and carries reek and cinder of north-country coal to thicken the lash of the rain. The waft comes from heeling dock tugs that strain at their hawsers, spurring the muddy tide to froth in their task of moving the helpless vessels in the basin. The long expanse of flooded dock, brimming to the uppermost ledge, lies open for their entry; the bruised and shattered stern of a large ship is pointed over the sill at an awkward angle that marks an absence of steam-power aboard to control her wayward sheer. The dockmaster, in ill mood with her cantrips, roars admonition and appeal to the smoking tugs to "lie over t' s'uth'ard and right her!" By check, and the powerful heave of a shore capstan, she warps in and straightens to the line of the docks. As she draws on to her berth the high bows of a second cripple swing over from the tiers, and the tugs back out to fasten on and drag her to the gate.

With entry of the ships, the glistening pier-head becomes thronged by tidesmen and their gear; like a drill-yard, with the lusty stamp of the marching lines of dockmen trailing heavy hawsers and handing check and hauling ropes. In an hour or so the gangs of the ship-repair section will be ready to 'turn to' at the new jobs, and the ships must be settled and ready against the wail of the starting 'buzzer.' Shrill whistle signals, orders and hails add to the stir of the labourers, and clatter of the warping capstan joins in with ready chorus. Not least of the medley is the bull roar of the harassed dockmaster, who finds a need in the press for more than one pair of hands at the reins to guide and halt his tandem charges.

The ships are marked in company, to settle bow to stern, with no room to spare, in the length of the dock. Conduct must be ruled in duplicate to exact the full measure of utility from every foot of space. On the last tide a pair of sound ships were floated out to service, braced and bound and refitted for further duty as stout obverse to the 'Sure Shield.' Keel-blocks and beds for the new patients have been set up and rearranged in the brief interval of occupancy, and now, quick on the wash of the outgoers, are new cases for the shearing plate-cutters and the swing of hammers.

Mindful to conserve their precious dry-dock space to the limit of good service, the repair section select the vessels with rare judgment. It is no haphazard turn of the wheel that brings an American freighter, shattered in stern section, to the same operating-table as an east-coast tramp (having her engines in scrap, boilers fractured, and the frames of her midships blown to sea-bottom). The combined measure of their length and the similarity of extent in hull damage has brought them to the one line of blocks. Odd cases, and regular ship-cleaning and minor repairs may be allotted to single-ship dry docks, but here, in sea-hospital with a twin-berth, there is a need for parallel treatment. The two ships must be considered as one, and all efforts be promoted towards refloating them, when hull repairs are completed, on one opening of the sea-gate.

In this, strangely, they are assisted by the enemy. True, his accommodation could well be spared, but it does have an influence on repair procedure. The exact and uniformly graded proportions of the enemy explosive reproduces a correspondingly like extent and nature in ship damage. Location and sea-trim may vary the fractures in proportion to

resistance but, with the vessels on the blocks together, working time may be adjusted to these conditions and a balance be struck that will further a simultaneous completion.

So the dockmaster ranges his pair on the centre line of the keel-blocks, sets tight the hawsers that hold them in position, and bars the sea-entry with a massive caisson. Presently he passes an order to the pumpman, and the power-house echoes to the easy thrust of his giant engine.

The keel-blocks have been set to meet the general lines of the vessels, with only a marginal allowance for the contour of damaged plating. To remedy any error divers, with their gear and escort, are ready on the dockside, and they go below with first fall in the water-level. The carpenters straggle out from sheltered corners and bear a hand. Riggers and dockmen have placed the ships, and it remains for the 'tradesmen' to bed them down and prop against a list by shores and blocks. They are ill content with the vile weather and their job in the open, where the rain lashes down pitilessly, soaking their working clothes. Doubtless they envy the dry divers their suits of proofed rubber, when they are called on to manhandle the heavy timber shores from the mud and litter of the dockside and launch them out towards the steel sides of the settling vessels. There the tide-workers on deck secure them by lanyards, and the spars hang in even order, sighted on doublings of the plates, ready to pin the ships on a steady keel when the water drains away.

With the timbers held in place, the carpenters split up to small parties and stand by to set a further locking strain by prise of block and wedge. The dockmaster blows a whistle signal at the far end of the basin, and casts up his hand as though arresting movement; the thrust of the main pump stills, and he swings his arm. At the sign, the carpenters ram home . . . the thunder of their forehammers on the hardwood wedges rings out in chorus that draws a quavering echo from the empty, hard-pressed hulls.

Settled and bedded and pinned, the ships are left till the water drains away and to await the coming of the shipwrights and repairing gangs. The carpenters shoulder their long-handled top-mauls and scatter to a shelter from the steady, continuous downpour. Up from the floors with their work completed, the divers doff their heavy head-gear and sit a while, *resting*

comfortably under the thrash of the same persistent rain. Anon, their awkward garb discarded, they walk off, striding with a crook at the knees, like farmer folk on ploughed land. The great pumps now pulsate at full speed, drawing water to their sluices in an eddying current that spins the flotsam and bares ledge after ledge of the solid dock masonry. From gaping wounds of the crippled vessels a full tide of seawater gushes and spurts to join the troubled wash below. The beams and side-planking, and temporary measures of the salvage section, uncover and come to sight, showing with what patience and laborious care the divers have striven to stem an inrush.

On the second ship the receding water-line exposes the damage to her engine- and boiler-rooms. A litter of coal and oily scum showers from angles of the wrecked bunker and stokehold to the floor of the dock, and leaves the fractured beams and tubes to stand out in gaunt twist and deformity. Through the breaches the shattered cylinders and broken columns of the engines lie distorted in a piled raffle of wrenched pipe sections, valves and levers, footplates, skeleton ladders, and shafting. The mass of distorted metal has still a shine and token of polish, and these signs of late care and attention only serve to make the ruin seem the more complete and irremediable.

An hour later a strident power syren sounds out from roof of the repair 'shops.' The workmen, hurrying to 'check in' at the gates, scarcely glance at their new jobs on the blocks of the dry-dock. To them it seems quite a commonplace that the round of their industry should suffer no halt, that the two seaworthy ships they completed yesterday should be so quickly replaced by the same type of casualty for their attention. The magnitude of the task—the vast extent of plating to be sheared and rebuilt, the beams to be withdrawn or straightened in place, the litter to be cleared—holds no misgivings. Short on the stroke of 'turn to' they straggle down the dockside to start the round anew. With critical eye, foremen and surveyors chalk off the cypher of their verdicts on the rusted displaced remnants; the gangs apportion and assemble with tools and gear; the huge travelling cranes rumble along on their railways, and lower slings and hooks in readiness for a load of damaged steel.

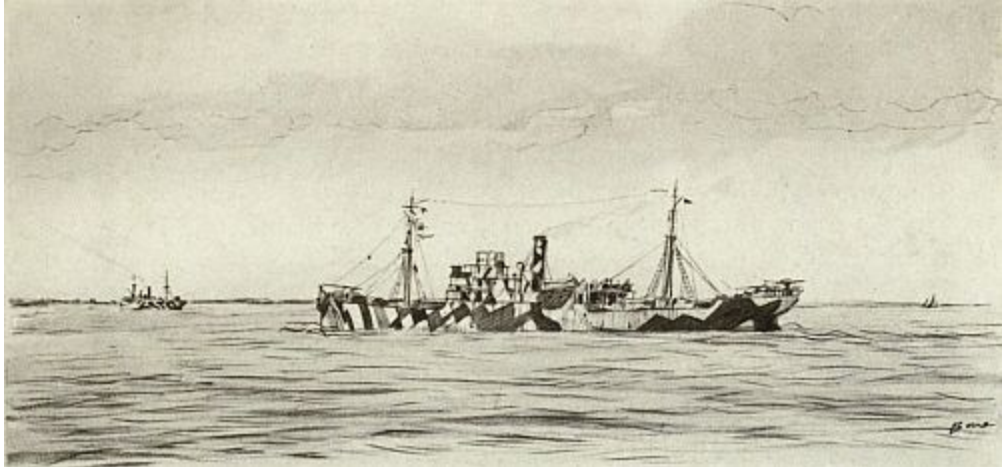
With the men lined out to the gangways and filing down the dock steps, chain linking in trial over the crane sheaves, and the bustle of preparation

on ship and shore, everything seems set for an instant beginning—but no hammer falls as yet. There is, first, a sad freight to be discharged; not all the crew of the ship with the wrecked engines have gone to the pay-table. Three sombre closed wagons are waiting by the dockside, and towards them down the long gangways from the ship, the bodies of an engineer and some of the stokehold crew are being carried. The weltering flood that held them has drained to the dock, and busy hands have searched in the wreckage where they died at their post.

We have no flags to honour, no processional march to accompany our dead. Their poor bodies, dripping and fouled, are draped in a simple coarse shroud that hardly conceals the line of their mangled limbs. Awkwardly the carriers stumble on the sodden planking and rest arms and knees on the guiding hand-lines. The workmen pause on the ship and gangways and look respectfully, if curiously, at the limp burdens as they are carried by.

Here and there a man speaks of the dead, but the most are silent, with lowering looks, set teeth—a sharp intake of the breath. . . . Who knows? Perhaps the spirits of the murdered seamen may come by a payment at the hands of the shipwright gangs. The best monument to their memory will stand as another keel on the deep—a quick ripost to the enemy, in his victim repaired and strengthened and returned to sea.

Lowering looks, set teeth, a hissing intake of the breath are the right accompaniment to a blow struck hard home; the thunder of hammers and drills, the hiss and sparkle of shearing cutters, that breaks out when the wagons have gone, marks a start to their monument!



DAZZLE

XIV

ON CAMOUFLAGE—AND SHIPS' NAMES

EARLY in the war the rappel of 'Business as usual' was as deadly at sea as elsewhere. Arrogant and super-confident in our pride of sea-place, we made little effort to trim and adapt our practice to rapidly altering conditions; there were few visible signs to disquiet us, we hardly deviated from our peaceful sea-path, and had no concern for interference. We carried our lights ablaze, advertised our doings in plain wireless, announced our sailings and arrivals, and even devoted more than usual attention to keeping our ships as span in brave new paint and glistening varnish as the hearts of impressionable passengers could desire.

We had difficulties with our manning. The seamen were off, at first tuck of drum, to what they reckoned a more active part in the great game of war—the strictly Naval Service—and we were left with weak crews of new and raw hands to carry on the sea-trade. So, from the very first of it, we engaged in a moral camouflage in our efforts to keep up appearances, and show the neutrals with whom we did business that such a thing as war could hardly disturb the smooth running of our master machine—the Merchants' Service!

Some there were among us who saw the peril in such prominence, and took modest (and somewhat hesitating) steps to keep out of the limelight, by setting lonely courses on the sea, restraining the comradely gossip of wireless operators, and toning down appearances from brilliant polish to the more sombre part suiting a sea in war-time. Deck lights were painted over and obscured, funnel and masts were allowed to grey to neutral tints, the brown ash that discomposes fine paint at sea was looked upon with a new and friendly eye. The bias of chief mates (in a service where promotion is the due for a clean and tidy ship) was, with difficulty, overcome, and a new era of keen look-out and sea-trim started.

There was but moderate support for these bold iconoclasts who dared thus to affront our high fetish. Ship painting and decoration and upkeep

were sacrosanct rites that even masters must conform to; the enactments of the Medes and Persians were but idle rules, mere by-laws, compared to the formulæ and prescriptions that governed the tone of our pantry cupboards and the shades of cunning grain-work. We were peaceful merchantmen; what was the use of our dressing up like a parish-rigged man-o'-war? As to the lights—darkening ship would upset the passengers; there would be rumours and apprehension. They would travel in less 'nervous' vessels!

The mine that shattered *Manchester Commerce* stirred the base of our happy conventions; the cruise of the *Emden* set it swaying perilously; the torpedoes that sank *Falaba* and *Lusitania* blew the whole sham edifice to the winds, and we began to think of our ships in other terms than those of freight and passenger rates. Our conceptions of peaceful merchantmen were not the enemy's!

We set about to make our vessels less conspicuous. Grey! We painted our hulls and funnels grey. In many colours of grey. The nuances of our coatings were accidental. Poor quality paint and variable untimely mixings contributed, but it was mainly by crew troubles (deficiency and incapacity) that we came by our first camouflage. As needs must, we painted sections at a time—a patch here, a plate or two there—laid on in the way that real sailors would call 'inside-out'! We sported suits of many colours, an infinite variety of shades. Quite suddenly we realized that grey, in such an ample range—red-greys, blue-greys, brown-greys, green-greys—intermixed on our hulls, gave an excellent low-visibility colour that blended into the misty northern landscape.

Bolshevik now in our methods, we worked on other schemes to trick the murderer's eye. Convention again beset our path. The great god Symmetry—whom we had worshipped to our undoing—was torn from his high place. The glamour of Balances, that we had thought so fine and shipshape, fell from our eyes, and we saw treachery in every regular disposition. Pairs—in masts, ventilators, rails and stanchions, boat-groupings, samson posts, even in the shrouds and rigging—were spies to the enemy, and we rearranged and screened and altered as best we could, in every way that would serve to give a false indication of our course and speed. Freighters and colliers (that we had scorned because of ugly forward rake of mast and funnel) became the leaders of our fashion. We wedged our masts forward (where we could)

and slung a gaff on the fore side of the foremast; we planked the funnel to look more or less upright; we painted a curling bow wash over the propellor and a black elaborate stern on the bows. We trimmed our ships by the head, and flattered ourselves that, Janus-like, we were heading all ways!

Few, including the enemy, were greatly deceived. At that point where alterations of apparent course were important—to put the putting Fritz off his stroke—the deck-houses and erections with their beamwise fronts or ends would be plainly noted, and a true line of course be readily deduced. With all our new zeal, we stopped short of altering standing structures, but we could paint, and we made efforts to shield our weakness by varied applications. Our device was old enough, a return to the chequer of ancient sea-forts and the line of painted gun-ports with which we used to decorate our clipper sailing ships. (That also was a camouflage of its day—an effort to overawe Chinese and Malay pirates by the painted resemblance to the gun-deck of a frigate.) We saw the eye-disturbing value of a bold criss-cross, and those of us who had paint to spare made a 'Hobson-jobson' of awning spars and transverse bulkheads.

These were our sea-efforts—rude trials effected with great difficulty in the stress of the new sea-warfare. We could only see ourselves from a surface point of view, and, in our empirics, we had no official assistance. During our brief stay in port it was impossible to procure day-labouring gangs—even the 'gulls' of the dockside were busy at sea. On a voyage, gun crews and extra look-outs left few hands of the watch available for experiments; in any case, our rationed paint covered little more than would keep the rust in check. We were relieved when new stars of marine coloration arose, competent shore concerns that, on Government instruction, arrayed us in a novel war paint. Our rough and amateurish tricks gave way to the ordered schemes of the dockyard; our ships were armed for us in a protective coat of many colours.

Upon us like an avalanche came this real camouflage. Somewhere behind it all a genius of pantomimic transformation blazed his rainbow wand and fixed us. As we came in from sea, dazzle-painters swarmed on us, bespattered creatures with no bowels of compassion, who painted over our cherished glass and teakwood and brass port-rims—the last lingering evidences of our gentility. Hourly we watched our trim ships take on the

hues of a swingman's roundabouts. We learned of fancy colours known only in high art—alizarin and grey-pink, purple-lake and Hooker's green. The designs of our mantling held us in a maze of expectation. Bends and ecartelés, indents and rayons, gyrony and counter-flory, appeared on our topsides; curves and arrow-heads were figured on boats and davits and deck fittings; apparently senseless dabs and patches were measured and imprinted on funnel curve and rounding of the ventilators; inboard and outboard we were streaked and crossed and curved.

With our arming of guns there was need for instruction in their service and maintenance; artificial smoke-screens required that we should be efficient in their use; our Otters called for some measure of seamanship in adjustment and control. So far all governmental appliances for our defence relied on our understanding and operation, but this new protective coloration, held aloof from our confidence, it was quite self-contained, there was no rule to be learnt; we were to be shipmates with a new contrivance, to the operation of which we had no control. For want of point in discussion, we criticized freely. We surpassed ourselves in adjectival review; we stared in horror and amazement as each newly bedizened vessel passed down the river. In comparison and simile we racked memory for text to the gaudy creations. "Water running under a bridge." . . . "Forced draught on a woolly sheep's back." . . . "Mural decoration in a busy butcher's shop." . . . "Strike *me* a rosy bloody pink!" said one of the hands, "if this 'ere don't remind me o' jaundice an' malaria an' a touch o' th' sun, an' me in a perishin' dago 'orspittel!"

While naming the new riot of colour grotesque—a monstrosity, an outrage, myopic madness—we were ready enough to grasp at anything that might help us in the fight at sea. We scanned our ships from all points and angles to unveil the hidden imposition. Fervently we hoped that there would be more in it than met our eye—that our preposterous livery was not only an effort to make Gargantuan faces at the Boche! Only the most splendid results could justify our bewilderment.

Out on the sea we came to a better estimate of the value of our novel war-paint. In certain lights and positions we seemed to be steering odd courses—it was very difficult to tell accurately the line of a vessel's progress. The low visibility that we seamen had sought was sacrificed to

enhance a bold disruption of perspective. While our efforts at deception, based more or less on a one-colour scheme of greys, may have rendered our ships less visible against certain favouring backgrounds of sea and sky, there were other weather conditions in which we would stand out sharply revealed. Abandoning the effort to cloak a stealthy sea-passage, our newly constituted Department of Marine Camouflage decked us out in a bold pattern, skilfully arranged to disrupt our perspective, and give a false impression of our line of course. With a torpedo travelling to the limit of its run—striking anything that may lie in its course, range is of little account. Deflection, on the other hand, is everything in the torpedo-man's problem—the correct estimation of a point of contact of two rapidly moving bodies. He relies for a solution on an accurate judgment of his target's course; it became the business of the dazzle-painters to complicate his working by a feint in colour and design. The new camouflage has so distorted our sheer and disrupted the colour in the mass as to make our vessels less easy to hit. If not invisible against average backgrounds, the dazzlers have done their work so well that we are at least partially lost in every elongation.

The mystery withheld from us—the system of our decoration—has done much to ease the rigours of our war-time sea-life. In argument and discussion on its origin and purpose we have found a topic, almost as unfailling in its interest as the record day's run of the old sailing ships. We are agreed that it is a brave martial coat we wear, but are divided in our theories of production. How is it done? By what shrewd system are we controlled that no two ships are quite alike in their splendour? We know that instructions come from a department of the Admiralty to the dockyard painters, in many cases by telegraph. Is there a system of abbreviations, a colourist's shorthand, or are there maritime Heralds in Whitehall who blazon our arms for the guidance of the rude dockside painters? It can be worked out in fine and sonorous proportions:

For s.s. CORNCRIX

Party per pale, a pale; first, gules, a fesse dancette, sable; second, vert, bendy, lozengy, purpure cottised with nodules of the first; third, sable, three billets bendwise in fesse, or: sur tout de tout, a barber's pole cockbilled on a sinking gasometer, all proper. For motto: "Doing them in the eye."

One wonders if our old conservatism, our clinging to the past, shall persist long after the time of strife has gone; if, in the years when war is a memory and the time comes to deck our ships in pre-war symmetry and grace of black hulls and white-painted deck-work and red funnels and all the gallant show of it, some old masters among us may object to the change.

"Well, have it as you like," they may say. "I was brought up in the good old-fashioned cubist system o' ship painting—fine patterns o' reds an' greens an' Ricketts' blue, an' brandy-ball stripes an' that! None o' your damned newfangled ideas of one-colour sections for me! . . . *Huh!* . . . And black hulls, too! . . . Black! A funeral outfit! . . . No, sir! I may be wrong, but anyway, I'm too old now to chop and change about!"

If we have become reconciled to the weird patterns of our war-paint, every instinct of seafaring that is in us rebels against the new naming of our ships. Is it but another form of camouflage—like the loving Indian mother abusing her dear children for deception of a malicious listening Djinn? *War Cowslip, War Dance, War Dreamer!* War Hell! Are our new standard ships being thus badly named, that the enemy may look upon them as pariahs, unworthy of shell or torpedo? Perhaps, as a thoughtful war measure, it may be chargeful of pregnant meaning; our new war names for the ships may be germane to some distant world movement, the first tender shoot of which we cannot yet recognize! More than likely, it is the result of the fine war-time frolic of fitting the cubest of square pegs in the roundest of holes. How is it done? Is there, in the hutments of St. James's Park, an otherwise estimable and blameless greengrocer, officially charged with the task of finding names for vessels, 015537-68 inclusive, presently on the Controller's lists and due to be launched?

We sailors are jealous for our vessels. Abuse us if you will, but have a care for what you may say of our ships. We alone are entitled to call them bitches, wet brutes, stubborn craft, but we will stand for no such liberties from the beach; strikes have occurred on very much less sufficient ground. Ridicule in the naming of our ships is intolerable. If *War* is to be the prefix, why cannot our greengrocer find suitable words in the chronicles of strife? Can there be anything less martial than the *War Rambler, War Linnet, War Titmouse, War Gossamer*? Why not the *War Teashop, the War Picture*

House, the—the—the *War Lollipop*? Are we rationed in ships' names? Is there a Controller of Marine Nomenclature? The thing is absurd!

If our controllers had sense they would see the danger in thus flouting our sentiment; they would value the recruiting agency of a good name; they would recognize that the naming of a ship should be done with as great care as that of an heir to an earldom. Is the torpedoed bos'n of the *Eumaeus* going to boast of a new post on the *War Bandbox*? What are the feelings of the captain of a *Ruritania* when he goes to the yards to take over a *War Whistler*? Why *War*? If sober, businesslike argument be needed, it is confusing; it introduces a repetition of initial syllable that makes for dangerous tangles in the scheme of direction and control.

It is all quite unnecessary. There are names and enough. Fine names! Seamanlike names! Good names! Names that any sailor would be proud to have on his worsted jersey! Names that he would shout out in the market-place! Names that the enemy would read as monuments to his infamy! Names of ships that we knew and loved and stood by to the bitter end.

XV

FLAGS AND BROTHERHOOD OF THE SEA

UNLIKE the marches of the land, with guard and counterguard, we had no frontiers on the sea. There were no bounds to the nations and their continents outside of seven or ten fathoms of blue water. We all travelled on the one highway that had few by-paths on which trespassers might be prosecuted. And our highway was no primrose path, swept and garnished and safeguarded; it had perils enough in gale and tempest, fog, ice, blinding snow, dark moonless nights, rock and shoal and sandbar. Remote from ordered assistance in our necessity, we relied on favour of a chance passer-by, on a fallible sea-wanderer like ourselves. So, for our needs, we formed a sea-bond, an International Alliance against our common hazards of wind and sea and fire, an assurance of succour and support in emergency and distress. Out of our hunger for sea-companionship grew a union that had few rules or written compacts, and no bounds to action other than the simply humane traditions and customs of the sea. There were no statutory penalties for infringement of the rules unwritten; we could not, as true seamen, conceive so black a case. We had no Articles of our Association, no charters, no covenants; our only documents were the International Code of Signals and the Rule of the Road at Sea. With these we were content; we understood faith and a blood-bond as brother seamen, and we put out on our adventures, stoutly warranted against what might come.

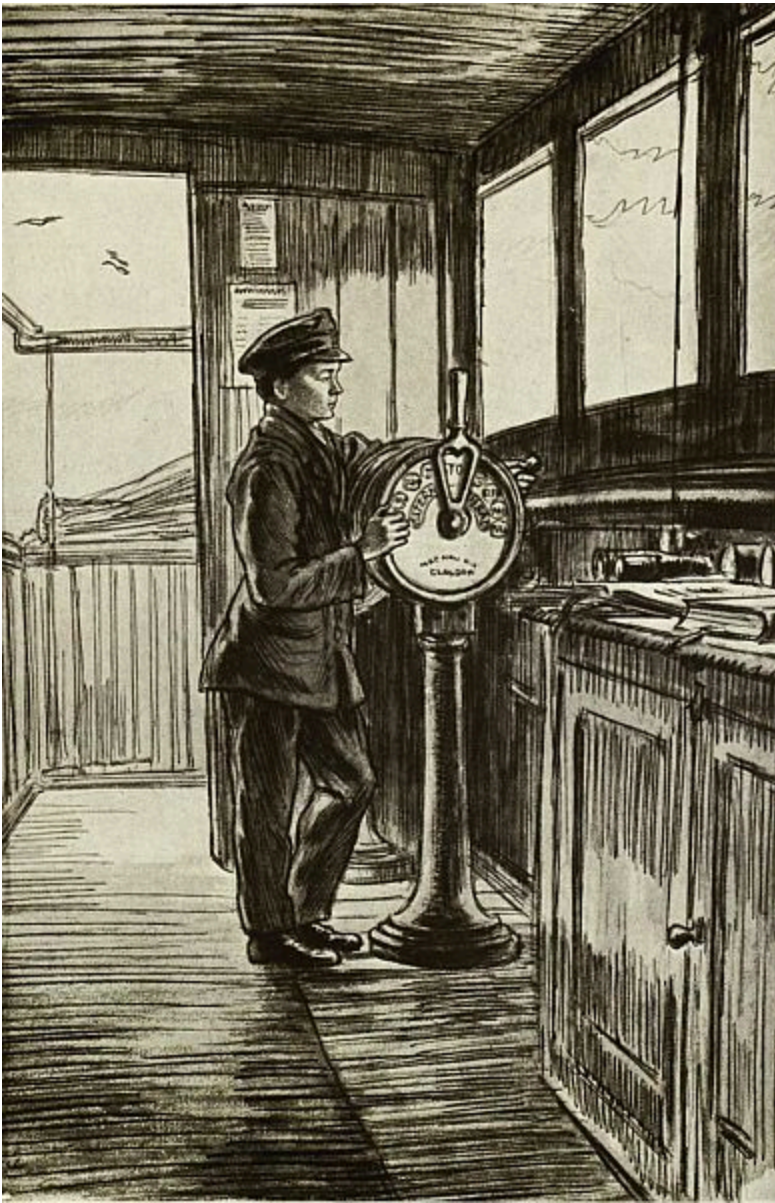
In the Code of Signals we had a language of our own, more immediate and attractive than Volapük or Esperanto. The dire fate of the builders of the Tower held no terror for us, for our intercourse was that of sight and recognition, not of speech. Our code was one of bright colours and bold striking design—flags and pendants fluttering pleasantly in the wind or, in calmer weather, drooping at the halyards with a lift for closer recognition. The symbol of our masonry was a bold red pendant with two vertical bars of white upon it. We had fine hoists for hail and farewell; tragic turn of the

colours for a serious emergency, hurried two-flag sets for urgent calls, leisurely symbols of three for finished periods.

'*Can you*' required three flags to itself; *me* or *I* or *it* came all within our range. We told our names and those of our ports by a long charge of four; we could cross our *t*'s and dot our *i*'s by beckon of a single square. We lowered slowly and rehoisted ('knuckles to the staff, you young fool!') our National Ensign, as we would raise our hat ashore. It was all an easy, courteous and graceful mode of converse, linguistically and grammatically correct, for we had no concern with accent or composition, taking our polished phrases from the book. It suited well the great family of the sea, for, were we a Turk of Galatz and you an Iceland brigantine, we could pass the time of day or tell one another, simply and intelligibly, the details of our ports and ladings. Distance, within broad limits, was small hindrance to our gossip; there were few eyes on the round of the sea, to read into our confidences. We could put a hail ashore, too. Passing within sight of San Miguel, we could have a message on the home doorsteps on the morrow, by hoisting our 'numbers'; the naked lightkeeper on the *Dædalus* could tell us of the northern winds by a string of colours thrown out from the upper gallery.

Good news, bad news, reports, ice, weather, our food-supply, the wages of our seamen, the whereabouts of pirates and cannibals, the bank rate, high politics (we had S.L.R. for Nuncio)—we had them all grouped and classed and ready for instant reference. Medicine, stocks, the law (G.F.H., King's Bench; these sharps who never will take a plain seaman's clear word on salvage or the weather, or the way the fog-whistle was duly and properly sounded!) Figures! We could measure and weigh and divide and subtract; we could turn your Greek *Daktylas* into a Japanese *Cho* or *Tcho*, or Turkish *Parmaks* into the *Draas* of Tripoli! Some few world measures had to be appended; a *Doppelzentner* was Z.N.L. What is a *Doppelzentner*?

As evidence of our brotherly regard, our peaceful intent, we had few warlike phrases. True, we had hoists to warn of pirates, and we could beg a loan, by signal, of powder and cannon-balls—to supplement our four rusty Snyders, with which we could defend our property, but there was no group in our international vocabulary that could read, "I am torpedoing you without warning!" Seamanlike and simple, we saw only one form of



AN APPRENTICE IN THE MERCHANTS' SERVICE

warfare at sea, and based our signals on that. "Keep courage! I am coming to your assistance at utmost speed!" . . . "I shall stand by during the night!" . . . "Water is gaining on me! I am sinking!" . . . "Boat is approaching your quarter!" These, and others alike, were our war signals, framed to meet our ideas of the greatest peril we might encounter in our conflict with the elements.

Of all this we write in a sad past tense. Our sea-bond is shattered. There is no longer a brotherhood on the sea. The latest of our recruits has betrayed us. The old book is useless, for it contains no reading of the German's avowal, "Come on the deck of my submarine. I am

about to submerge!" . . . "Stand by, you helpless swine in the boats, while I shell you and scatter your silly blood and brains!"

No longer will the receipt of a call of distress be the instant signal (whatever the weather or your own plight) for putting the helm over. We have shut the book! We are grown hardened and distrustful. S.O.S. may be the fiend who has just torpedoed a crowded Red Cross, and endeavours by his lying wireless to lure a Samaritan to the net. A heaving boat, or a lone

raft with a staff and a scrap, may only be closed with fearful caution; they may be magnets for a minefield.

". . . still he called aloud, for he was in the track of steamers. And presently he saw a steamer. She carried no lights, but he described her form, a darker shape upon the sea and sky, and saw the sparks volley from her funnel.

"He shrieked till his voice broke, but the steamer went on and vanished. The Irishman was furiously enraged, but it was of no use to be angry. He went on calling. So did the other four castaways, but their cries were growing fainter and less frequent.

"Then there loomed another steamer, and she, too, went on. By this time, perhaps, an hour had gone by, and the Arab firemen had fallen silent. The Irishman could see them no longer. He never saw them again. A third steamer hove in sight, and she, too, went on. The Irishman cursed her with the passionate intensity peculiar to the seaman, and went on calling. It was a desperate business. . . ."

The shame of it!

Lusitania, Coquet, Serapis, Thracia, Mariston, The Belgian Prince, Umaria . . .

". . . The commanding officer of the submarine, leaning on the rail of the conning-tower, looked down upon his victims.

"Crouched upon the thwarts in the sunlight, up to their knees in water, which, stained crimson, was flowing through the shell-holes in the planking, soaked with blood, holding their wounds, staring with hunted eyes, was the heap of stricken men.

"The German ordered the boat away. The shore was fifteen miles distant. . . ."

He ordered the boat away! The shame of it! The abasing, dishonouring shame of it!

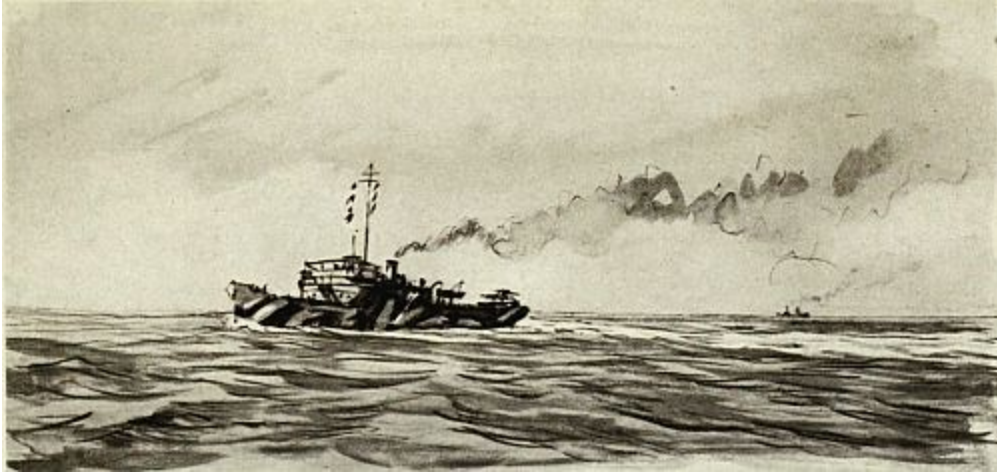
Bitterly, tarnished—we realize our portion in the guilt, our share in this black infamy—that seamen should do this thing!

What of the future? What will be the position of the German on the sea when peace returns, let the settlement by catholic conclave be what it may?

Sailorfolk have long memories! Living a life apart from their land-fellows, they have but scant regard for the round of events that, on the shore, would be canvassed and discussed, consented—and forgotten. There is no busy competing commercial intrigue, no fickle market, no grudging dalliance on the sea. We stand fast to our own old sea-justice; we have no shades of mercy or condonation, no degrees of tolerance for this bastard betrayer of our unwritten sea-laws. No brotherhood of the sea can be conceived to which he may be re-admitted. Not even the dethronement of the Hohenzollern can purge the deeds of his marine Satraps, for their crimes are individual and personal and professional.

In the League of Nations a purged and democratic Germany may have a station, but there is no redemption for a Judas on the sea. There, by every nation, every seafarer, he will remain a shunned and abhorred Ishmael for all time.

PART III



A STANDARD SHIP AT SEA

XVI

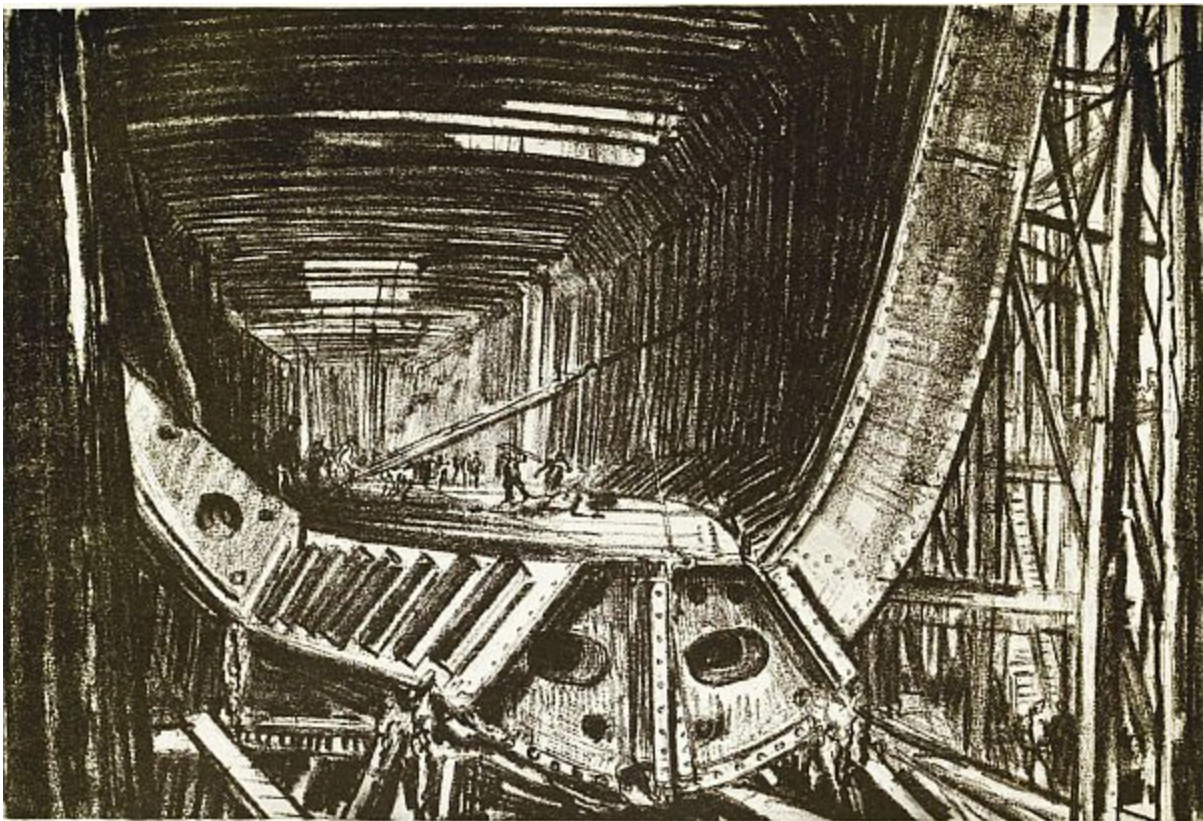
THE CONVOY SYSTEM

EARLY in 1917 the losses of the merchants' ships and men had assumed a proportion that called for a radical revision of the systems of naval protection. Concentrating their energies on but one specific form of sea offence, the enemy had developed their submarine arm to a high point of efficiency. Speed and power and lengthy sea-keeping qualities were attained. To all intents and purposes the U-boats had become surface destroyers with the added conveniency of being able to disappear at sight. They conducted their operations at long distance from the land and from their bases. The immense areas of the high seas offered a peculiar facility for 'cut-and-run' tactics: the system of independent sailings of the merchantmen provided them with a succession of victims, timed in a progression that allowed of solitary disposal. Notwithstanding the matured experience of submarine methods gained by masters, the rapid evolution of counter-measures by the Royal Navy, the courage and determination of all classes of seafarers, our shipping and that of our Allies and the neutral nations was being destroyed at a rate that foreshadowed disaster.

Schemes of rapid ship construction were advanced, lavish expenditure incurred, plans and occupation designed—all to ensure a replacement of tonnage at a future date. More material in point of prompt effect were the efforts of the newly formed Ministry of Shipping to conserve existing tonnage by judicious and closely controlled employment. All but sternly necessary sea-traffic was eliminated: harbour work in loading and unloading was expedited: the virtues of a single control enhanced the active agency of the merchants' ships—now devoted wholly to State service. Joined to the provisional and economic measures of the bureaux, Admiralty reorganized their methods of patrol and sea-supervision of the ships. The entry of the United States into the world war provided a considerable increase of naval strength to the Allied fleets. Convoy measures, that before had been deemed impracticable, were now possible. Destroyers and sloops could be released from fleet duties and were available as escorts. American

flotillas crossed the Atlantic to protect the sea-routes: Japanese war craft assisted us in the Mediterranean.

In the adoption of the convoy system the Royal Navy was embarking on no new venture. Modern ships and weapons may have brought a novel complication to this old form of sea-guardianship, but there is little in seafaring for which the traditions of the Naval Service cannot offer text and precedent. The constant of protection by convoy has remained unaltered by the advance of armament and the evolution of strange war craft: the high spirit of self-sacrifice is unchanged. When, in October 1917, the destroyers *Strongbow* and *Mary Rose* accepted action and faced three German cruisers, their commanders—undismayed by the tremendous odds—reacted the parts of the common sea-dramas of the Napoleonic wars. The same obstinate courage and unconquerable sea-pride forbade them to desert their convoy of merchantmen and seek the safety that their speed could offer. H.M.S. *Calgarian*, torpedoed and sinking, had yet thought for the convoy she escorted. Her last official signal directed the ships to turn away from the danger.



BUILDING A STANDARD SHIP

The convoy system did not spring fully served and equipped from the earlier and less exacting control. Tentative measures had to be devised and approved, a large staff to be recruited and trained. The clerical work of administration was not confined to the home ports; similar adjustment and preparation had to be conducted in friendly ports abroad. As naval services were adapted to the new control, the system was extended. The comparatively simple procedure of sending destroyer escorts to meet homeward-bound convoys became involved with the timing and dispatch of a mercantile fleet sailing from a home port. The escorts were ordered out on a time-table that admitted of little derangement. Sailing from a British port with a convoy of outward-bound vessels, the destroyers accompanied that fleet to a point in the Atlantic. There the convoy was dispersed, and the destroyers swung off to rendezvous with a similar convoy of inward-bound vessels. While the outgoing merchantmen were allowed to proceed independently after passing through the most dangerous area, the homeward-bound vessels were grouped to sail in company from their port abroad. An ocean escort was provided—usually a cruiser of the older class—and there was opportunity in the longer voyage for the senior officer to drill the convoy to some unity and precision in manœuvre.

The commander of the ocean escort had no easy task in keeping his charges together. The age-old difficulty of grouping the ships in the order of their sailing (now steaming) powers has not diminished since Lord Cochrane, in command of H.M.S. *Speedy*, complained of the 'fourteen sail of merchantmen' he convoyed from Cagliari to Leghorn. In the first enthusiasm of a new routine, masters were over-sanguine in estimation of the speed of their ships. The average of former passages offered a misleading guide. While it was possible to average ten and a half knots on a voyage from Cardiff to the Plate, proceeding at a speed that varied with the weather (and the coal), station could not easily be kept in a ten-knot convoy when—at the cleaning of the fires—the steam went 'back.' Swinging to the other extreme (after experience of the guide-ship's angry signals), we erred in reserving a margin that retarded the full efficiency of a convoy. Our commodores had no small difficulty in conforming to the date of their convoy's arrival at a rendezvous. The 'cruising speed' of ten knots, that we had so blithely taken up when sailing from an oversea port, frequently toned down to an average of eight—with all the consequent derangement of the

destroyers' programme at the home end; a declared nine-knot convoy would romp home at ten, to find no escort at the rendezvous.

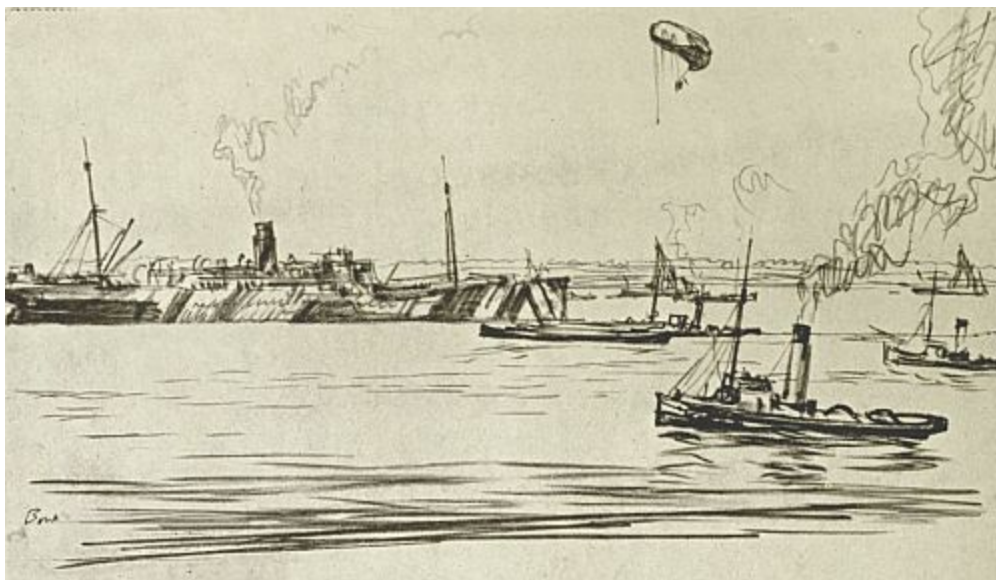
In time, we adjusted our estimate to meet the new demands. Efforts of the Ministry of Shipping to evolve an order in our voyaging that would reduce irregularities had good results. The skilfully thought-out appointment of the ships to suitable routes and trades had effect in producing a homogeneity that furthered the employment of our resources to the full. The whole conduct of our seafaring speedily came within the range of governmental control, as affecting the timely dispatch and arrival of the convoys. The quality of our fuel, the state of the hull, competence of seamen, formed subject for close investigation. The rate of loading or discharge, the urgency of repairs and refitment, were no longer judged on the note of our single needs; like the states of the weather and the tide, they were weighed and assessed in the formula that governed our new fleet movements.

The system of convoy protection had instant effect in curbing the activities of the U-boats. They could no longer work at sea on the lines that had proved so safe for them and disastrous for us. To get at the ships they had now to come within range of the destroyers' armament. Hydrophones and depth-charges reduced their vantage of submersion. The risks of sudden rupture of their plating by the swiftly moving keel of an escorting vessel did not tend to facilitate the working of their torpedo problem. In the coastal areas aircraft patrolled overhead the convoys, to add their hawk-sight to the ready swerve of the destroyers. The chances of successful attack diminished as the hazard of discovery and destruction increased. Still, they were no fainthearts. The German submarine commanders, brutal and hell-nurtured, are no cowards. The temptation of a massed target attracted them, and they sought, in the confusion of the startled ships, a means of escape from the destroyers when their shot into the 'brown' had run true.

Convoy has added many new duties to the sum of our activities when at sea. Signals have assumed an importance in the navigation. The flutter of a single flag may set us off on a new course at any minute of the day. Failure to read a hoist correctly may result in instant collision with a sister ship. We have need of all eyes on the bridge to keep apace with the orders of the commodore. In station-keeping we are brought to the practice of a branch of

seamanship with which not many of us were familiar. Steaming independently, we had only one order for the engineer when we had dropped the pilot. 'Full speed ahead,' we said, and rang a triple jangle of the telegraph to let the engineer on watch know that there would be no more 'backing and filling'—and that he could now nip into the stokehold to see to the state of the fires. Gone—our easy ways! We have now to keep close watch on the guide-ship and fret the engineer to adjustments of the speed that keep him permanently at the levers. The fires may clag and grey down through unskilful stoking—the steam go 'back' without warning: ever and on, he has to jump to the gaping mouth of the voice-tube: "Whit? Two revolutions? Ach! Ah cannae gi' her ony mair!"—but he does. Slowly perhaps, but surely, as he coaxes steam from the errant stokers, we draw ahead and regain our place in the line. No small measure of the success of convoy is built up in the engine-rooms of our mercantile fleets.

Steaming in formation at night without lights adds to our 'grey heires.' The menace of collision is ever present. Frequently, in the darkness, we have no guide-ship in plain sight to regulate our progress. The adjustments of speed, that in the daytime kept us moderately well in station, cannot be made. It is best to turn steadily to the average revolutions of a former period, and keep a good look-out for the broken water of a sister ship. On occasion there is the exciting medley of encountering a convoy bound the opposite way. In the confusion of wide dispersal and independent alterations of course to avert collision, there is latitude for the most extraordinary situations. An incident in the Mediterranean deserves imperishable record: "We left Malta, going east, and that night it was inky dark and we ran clean through a west-bound convoy. How there wasn't an accident, God only knows. We had to go full astern to clear one ship. She afterwards sidled up alongside of us and steamed east for an hour and a half. Then she hailed us through a megaphone: 'Steamer ahoy! Hallo! Where are you bound to?' 'Salonika,' we said. 'God Almighty,' he says. 'I'm bound to Gibraltar. Where the hell's *my* convoy?'"



THE THAMES ESTUARY IN WAR-TIME

XVII

OUTWARD BOUND

CUSTOMS clerks--may their name be blessed--are worth much more than their mere weight in gold. We do not mean the civil servants at the Custom House, who listen somewhat boredly to our solemn Oath and Compearance. Doubtless they, too, are of value, but our concern is with the owner's shipping clerk who attends our hesitating footsteps in the walk of ships' business when we come on shore. He greets us on arrival from overseas, bearing our precious letters and the news of the firm: he has the devious paths of our entry-day's course mapped out, down to the train we may catch for home. As an oracle of the port, there is nothing he does not know: the trains, the week's bill at the 'Olympeambra,' the quickest and cheapest way to send packages to Backanford, suitable lodging in an outport, the standing of the ship laundries, the merits of the hotels--he has information about them all. During our stay in port he attends to our legal business. He speeds us off to the sea again, with all our many folios in order.

In peace, we had a settled round that embraced the Custom House for entry, the Board of Trade for crew affairs, the Notary for 'Protest.' ("... and experienced the usual heavy weather!") War has added to our visiting-list. We must make acquaintance with the many naval authorities who control our movements; the Consuls of the countries we propose to visit must see us in person; it would be discourteous to set sail without a p.p.c. on the Dam-ship and Otter officers. Ever and on, a new bureau is licensed to put a finger in our pie: we spend the hours of sailing-day in a round of call and counter-call. The Consul wishes to *visé* our Articles—the Articles may not be handed over till we produce a slip from the Consul, the Consul will grant no slip till we have seen the S.I.O. "Have we identity papers for every member of the crew, with photograph duly authenticated?"—"We are instructed not to grant passports!" Back and forward we trudge while the customs clerk at our side tells cheerfully of the very much more trying time that fell to Captain Blank.

By wile and industry and pertinacity he unwinds the tangle of our longshore connections. He reconciles the enmity of the bureaux, pleads for us, apologizes for us, fights for us, engages for us. All we have to do is to sign, and look as though the commercial world stood still, awaiting the grant of that particular certificate. Undoubtedly the customs clerk is worth his weight in red, red gold!

On a bright summer afternoon we emerge from the Custom House. We have completed the round. In the case which the clerk carries we have authority to proceed on our lawful occasions. Customs have granted clearance; our manifests are stamped and ordered; the Articles of Agreement and the ship's Register are in our hands. The health of our port of departure is guaranteed by an imposing document. Undocking permit, vouchers for pilotage and light dues, discharge books, sea-brief, passports, and store-sheets, are all there for lawful scrutiny. In personal safe-keeping, we have our sea-route ordered and planned. The hard work is done. There is no more *business*—nothing to do but to go on board and await the rise of tide that shall float us through the river channels to sea.

Cargo is stowed and completed; the stevedores are unrigging their gear when we reach the ship. Our coming is noted, and the hatch foremen (in anticipation of a 'blessing') rouse the dockside echoes with carefully phrased orders to their gangs: "T' hell wit' yes, now! Didn't Oi tell ye, Danny Kilgallen, that *th'* *Cyaptin* wants thim tarpolyan sames turned fore an' aff!" (A shilling or two for him!)"—"Beggin' yer pardon, sir—I don't see th' mate about—will we put them fenders below *for ye* before we close th' hatch?" (Another *pourboire!*)—Number three has finished his hatchway, but his smiling regard calls for suitable acknowledgment. (After all, we shall have no use for British small coinage out West!) The head foreman, dear old John, is less ambitious. All he wants is our understanding that he has stowed her tight—and a shake of the hand for good luck. Firmly we believe in the good luck that lies in the hand of an old friend. "Bye, John!"

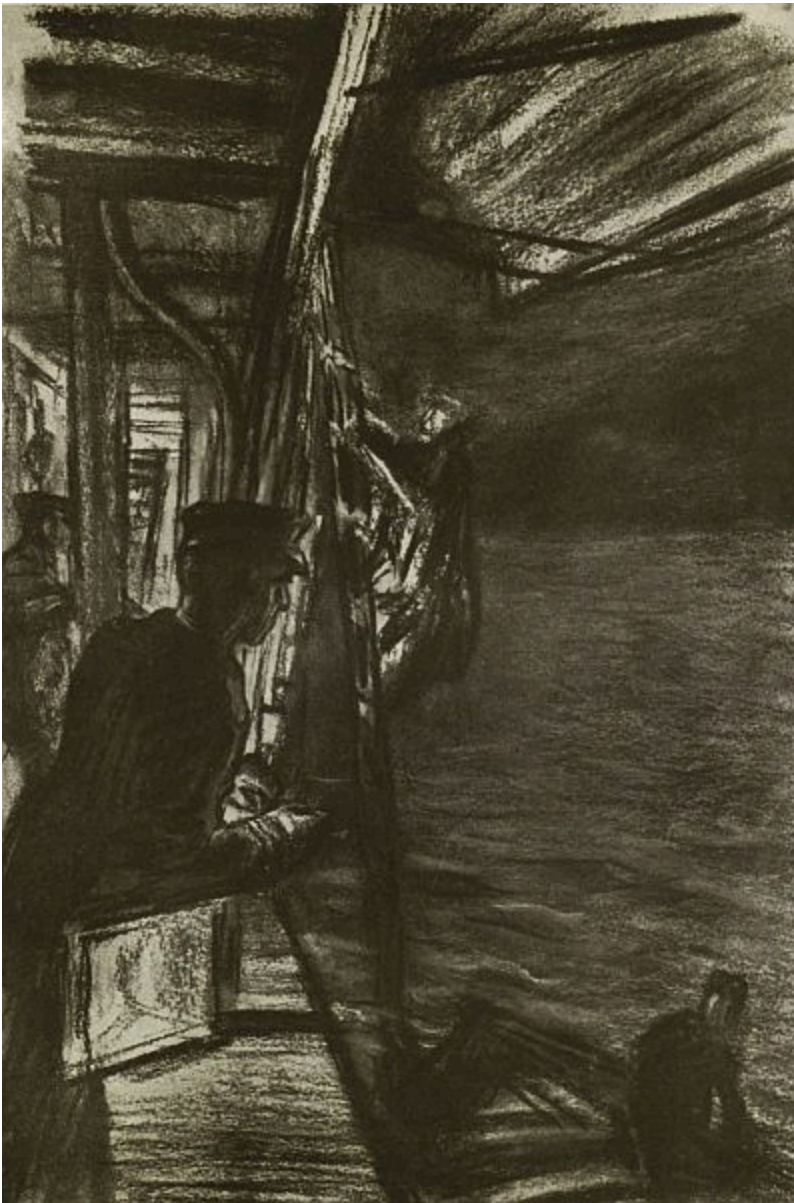
In groups, as their work is finished, the dockers go on shore, and leave to the crew the nowise easy task of clearing up the raffle, lashing down, and getting the lumbered decks in something approaching sea-trim. Fortunately, there is time for preparation. Usually, we are dragged to the dock gates with the hatches uncovered, the derricks aloft, and the stowers still busy blocking

off the last slings of the cargo. This time there will be no hurried (and improper) finish—the stevedores hurling their gear ashore at the last minute, slipping down the fender lanyards, scurrying to a 'pier-head jump,' with the ship moving through the lock! Some happy chance has brought completion within an hour or two of tide-time. The mate has opportunity to clear ship effectively, and we have leisure to plot and plan our sea-route (in anticipation of hasty chart glances when we get outside) before the pier-master hails us—"Coom along wi' t' *Massilia!* "

Tugs drag us through the inner gates, pinch and angle our heavy hull in the basin, and enter us into the locks. The massive gates are swung across, the sluices at the river-end eased to an outflow and, slowly, the great lock drains to the river level. The wires of our quay-fasts tauten and ring out to the tension of the outdraft, as we surge in the pent water-space and drop with the falling level. Our high bridge view over the docks and the river is pared in inches by our gradual descent; the deck falls away under cope of the rough masonry; our outlook is turned upwards to where the dockmaster signals his orders. The ship seems suddenly to assume the proportions of a canal-boat in her contrast with the sea-scarred granite walls and the bulk of the towering gates.

At level with the flood, the piermen heave the outer lock-gates open for our passage. We back out into the river, bring up, then come ahead, canting to a rudder pressure that sheers us into the fairway. The river is thronged by vessels at anchor or under way, docking and undocking on the top of the tide, and their manœuvres make work for our pilot. At easy speed we work a traverse through the press at the dock entrances and head out to seaward.

Evening is drawing on as we enter the sea-channels—a quiet close to a fine summer day. Out on the estuary it is hard to think of war at sea. Shrimpers are drifting up on the tide, the vivid glow of their tanned canvas standing over a mirrored reflection in the flood. The deep of the fairway is scored by passage of coasting steamers, an unending procession that joins lightship to lightship in a chain of transport. The sea-reaches look in no way different from the peaceful channels we have known so long, the buoys and the beacons we pass in our courses seem absurdly tranquil, as though lacking any knowledge that they are signposts to a newly treacherous sea. Only from the land may one draw a note of warning—on shore there are



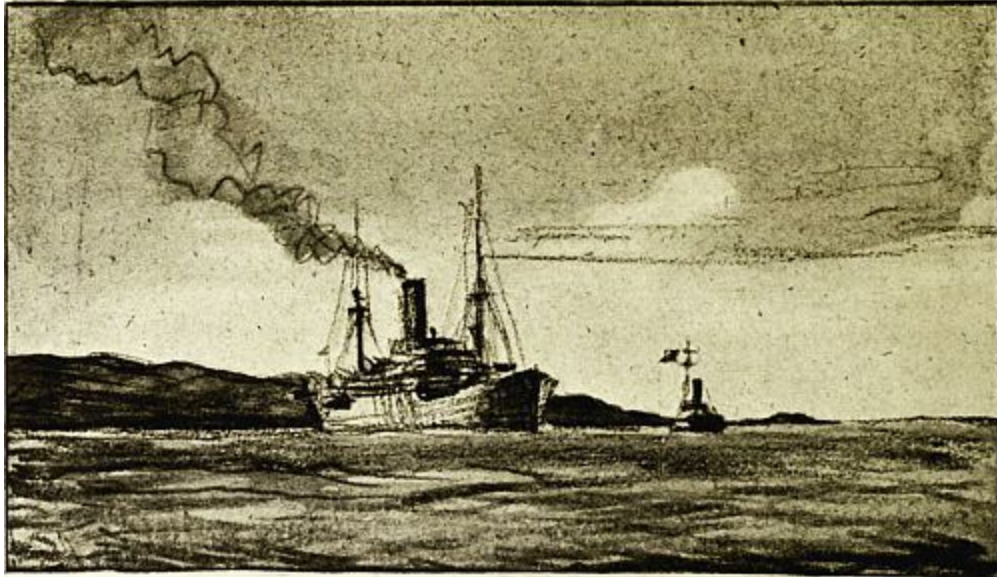
DROPPING THE PILOT

buoy and reduce speed to drop our pilot. The night is windless and a calm sea gives promise of a good passage. We bring up close to the cutter, and, shortly, with a stout 'Good-bye,' the pilot swings overside and clambers down the long side-ladder to his boat. We shut off all lights and steer into the protecting gloom of the night.

visible signs of warfare. The searchlights of the forts, wheeling over the surface of the channels, turn on us and steady for a time in inspection. Farther inland, ghostly shafts and lances are sweeping overhead, in ceaseless scrutiny of the quiet sky.

At a bend in the fairway we close and speak the channel patrol steamer and draw no disquieting impression from her answer to our hail. The port is still open and we may proceed on our passage to join convoy at ——. An escort will meet us in 1235 and conduct us to 5678. 'Carry on!'

It is quite dark when we round the outer



**EXAMINATION SERVICE PATROL BOARDING AN INCOMING
STEAMER**

XVIII

RENDEZVOUS

ALMOST hourly they round the Point, turning in from seaward with a fine swing and thrash of propellers to steer a careful course through the boom defences. Screaming gulls wheel and poise and dive around them, exulting to welcome the new-comers in, and the musical clank and rattle of anchor cables, as the ships bring up in the Roads, mark emphatic periods to this--the short coasting section of the voyage.

"Safe here!" sing the chains, as they link out over the open hawse. "Thus far, anyway, in spite of fog and coast danger, of mine and submarine," and the brown hill-side joins echo to the clamour of the wheeling gulls, letting all know the ships have come in to join the convoy.

The bay, that but a day ago lay broad and silent and empty, now seems to narrow its proportions as each high-sided merchantman comes in; the hills draw nearer with every broad hull that anchors, wind-rode, in the blue of the bay. As if in key with the illusion, the broad expanse of shallow, inshore water, that before gave distance to the hills, now sheds its power, cut and furrowed as it becomes by thrash and wake of tugs and launches all making out to serve the larger vessels.

On the high mound of the harbour-master's look-out, keen eyes note all movements in the bay. The signal-mast and yard bear a gay setting of flags and symbols, and rapid changes and successions show the yeoman of signals and his mates at work, recording and replying, taking mark and tally of the ships as they arrive. Up and down goes the red-and-white-barred answering pendant to say that it is duly noted—"War *Trident*, *Marmion*, and *Pearl Shell* report arrival"—or the semaphore arms, swinging smartly, tell H.M.S. 03xyz that permission to enter harbour (she having safely escorted the trio to port) is approved.

Out near the entrance to the bay, where the 'gateships' of the boom defences show clear water, the patrol steamer of the Examination Service

lays-to, challenging each incoming vessel to state her name and particulars. These, in turn, are signalled to the shore and the yeoman writes: "Begins war trident for norfolk va. speed nine knots is ready for sea stop marmion for Bahia reports steering engine broken down will require ten hours complete repairs stop pearl shell nine and half short-handed one fireman two trimmers report agents stop ends."

If room is scanty, the convoy office has at least an atmosphere in keeping with its mission. Nestling close under the steep brow of the harbour-master's look-out, it was, in happier days, the life-boat coxswain's dwelling, and a constant reminder of sea-menace and emergency almost blocks the door—the long boat-house and launch-ways of the life-boat. Four square and solid, the little house only has windows overlooking the bay, as if attending strictly to affairs at sea and having no eyes for landward doings; the peering eaves face straight out towards the 'gateships' as though even the stone and lime were intent on the sailing of the convoys, whose order and formation are arranged within their walls. The upper room has a desk or two, a telephone, a chart table, and a typewriter, and here the port convoy officer and his assistants trim and index and arrange the ships in order of their sailing. At the window a seaman-writer is typing out 'pictures' for the next sailing—signal tables, formation and dispersal diagrams, call signs, zigzags, constantly impressing that Greenwich Mean Time is the thing (no Summer Time at sea), and that courses are True, *not* Magnetic. The clack and release of his machine seem quite a part of conversation between the convoy officer and his lieutenant; the whole is so apparently disjointed in references to this ship and that, to repairs and tides, and shortage of 'hands' and water-supply and turns in the hawse, and even Spanish influenza! To one accustomed to single-ship work the whole is mildly bewildering, and one readily understands that sailing a merchant convoy calls for more than the simple word of command.

"*War Trident*, nine knots," reads the junior, from a signal slip. "*Marmion*, a doubtful starter—steering-gear disabled. *Pearl Shell*, three stokehold hands short."

"*Trident* only nine! That be damned for a yarn!" says his senior, reaching for the slip. "Nine will reduce the speed of the whole convoy a knot. She must be good for more—new ship, isn't she?"

"Yes. One of these new standards—built for eleven knots and chocked up afterwards with fancy gear and 'gadgets' to rob the boilers."

"Lemme see—nine knots"—turning to the pages of a tide-book, the convoy officer makes a rough sum of it. "High water at Oysterpool—so—arrived here—distance—and seventy-one. Why, he's come on from Oysterpool at ten, no less, and that's not allowing for the zigzag either!"

The lieutenant looks round for his cap. Clearly there is a definite 'drill' for captains who come on from Oysterpool at ten and declare their speed as nine, and he is ready when the P.C.O. passes orders. "All right. You go off and see the captain. Try to get him to spring at least half a knot. I expect he's allowing a bit for 'coming up,' and going easy till he knows his new ship. . . . I'll 'phone *Pearl Shell's* agents and warn 'em to hustle round for firemen. *Marmion*? Yes. Board *Marmion* on your way back. Wants ten hours—she should be able to keep her sailing." A year ago there would have been but moderate and passive interest in the varying troubles of the ships and their crews, but much water has flowed over the Red Ensign since then, and we are learning.

The convoy lieutenant goes down a winding path to the boat-slip and boards his launch to set off for the Roads. The morning, that broke fair and unclouded, has turned grey; a damp sea-mist is wandering over the bay in thin wraiths and feathers, but sunlight on the brown of the distant hills promises a clearing as the day draws on. Fishing-smacks, delayed by want of wind, are creeping in to the market steps under sweep of their long oars, and their lazy canvas rustles, and the booms and sheet-blocks creak as the wash of the picket-launch sets them swaying. In from the sea channels, with their sweeps still wet and glistening, come the *Agnes Whitwell*, *Fortuna*, the *Dieudonné*, and *Brother Fred*, each with a White Ensign aloft and a naked grey gun on their high bows. They are late in their return, and one can guess at deadly iron spheres stirred from the depths of the fairways, thrown buoyant in the wash astern, and destroyed by crack of gunfire. The commodore of the sisterly pairs, a young lieutenant of Reserve, waves a cheery greeting as we pass.



DAWN: CONVOY PREPARING TO PUT TO SEA

And now the Roads, windless and misty, the anchored merchantmen swung at different angles, in their gay fantasy of dazzle-paint, borrowing further motley from the mist, and leering grotesquely through the thin vapours. But for her lines, undeniably fine and graceful, *War Trident* is the standardest of standards. Dazzle-painters have slapped their spite at her in lurid swathes and, not content, have draped her sheer in harlequin crenellations. Her low pipe-funnel upstands in rigid perpendicular. ("Chief! Pit yer haun' up an' feel if th' kettle's bilin'!") No masts break the long length of her, saving only a midship signal-pole that serves her wireless aerials and affords a hod-like perch for the look-out aloft. She is stark new, smooth of plating, and showing even the hammer-strokes on her rivets. Through the thin paint on her sides, marks and symbols of construction appear, the letters of her strakes painted in firm white, with here and there an unofficial shipyard embellishment—"Good old Jeemy Quin," or "Tae hell wi' the Kiser!" She is ready for sea, and life-boats and davits, swung outboard, tower overhead as the picket-launch draws up at her gaunt side. She is in ballast trim, and it is evident that her standard carpenters hold strictly to a rule that ignores a varying freeboard—the side ladder is short by eight feet, and only by middling the rungs (a leap at the bottom, a long swaying climb, and a drag at the top) are we able to clamber on board.

A special 'drill' for conducting affairs with masters of brand-new ships should be devised immediately by Admiralty, and the mildest of Low-Church curates (trimmed by previous dire tortures to the utter limit of exasperation) be provided, on whom officials may be well practised. Usually the master has been hurried out of port by the last rivet driven home, with strange officers and the very weakest of new crews, in a ship jam-full of the newest 'gadgets,' and the least possible reserve of gear to work them. Quickly and bitterly the fourth sentence of Confession at Morning Prayer is recalled to him—the things undone crowd round, and there is nothing in the bare hull to serve as a makeshift. The engines and *auxiliaries* (that, with a builder's man at every bearing, worked well on trials) now develop tricks and turns to keep the chief engineer and his fledgling juniors on the run; the mate cries "Kamerad" to all suggestions, pointing to his hopeless watch of one. (Eight deck: four in a watch, less one helmsman and two look-outs, equals one.) Add to the sum of difficulties that the captain has probably been ashore since he lost his last ship, and finds the new tactics and signals and zigzags unfamiliar; through it all the want of familiar little trifles and fixings (that go so far to help a ready action), sustains a feeling of irritation.

It is little wonder that the convoy lieutenant goes warily, and, indeed, but for the brilliant inspiration of using the 'last ship,' it seems probable that the convoy will have to proceed at *Trident's* modest nine knots. Bluntly, the captain is in undisguised ill-humour. He has been on deck practically since leaving the builder's yard, and his weary eyes suggest a need for prompt sleep. His room, still reeking of new paint and varnish, is in some disorder, and shows traces of an anxious passage along the coast. 'Notices to Mariners' lie open at the minefield sketches, with a half-smoked pipe atop to keep the pages open; chart upon chart is piled (for want of a rack) on bed and couch; oilskins, crumpled as when drawn off, hang over the edge of a door—not a peg to hang them on; an open sextant case, jammed secure by pillows, lies on the washstand lid; books of sailing directions, a taffrail log, some red socket-flares, are heaped awry in a corner of the room; the whole an evidence that lockers and minor ship conveniences are not yet standardized. Pray goodness he may have a stout honest thief of a chief mate, able and willing to find a baulk or two of timber, and a few nails and brass screws and copper tacks and a curtain-rod or two and a bolt of canvas!

The convoy lieutenant, unheeding a somewhat surly return of his greeting, produces Convoy Form No. AX, and starts in cheerfully to fill the vacant columns. "Tonnage, captain?—register will do. Crew? Guns? Coal?—consumpt. at speeds. Revolutions per half-knot?" The form completed, he hands it over for signature, thus tactfully drawing the captain's attention to the secretarial work he has done for him. "What's the speed? Nine and a half?" "Speed!" answers the Old Man. "Hell! This bunch of hair-springs can't keep out of her own way! Speed? The damned funnel's so low we can't get draught to burn a cigarette-paper; and these new pumps they've given her! . . . Well, we might do nine, but only in fine weather, mind you. Nine knots!"

"You'll have to do better for this convoy, captain. There's not a ship under nine and a half; but there may be a bunch of eight-knotters going out in five days."

"Nothing under nine and a half! What? Why, there's *Pearl Shell* came in with us. She hasn't a kick above nine. When I was in the old *Collonia*, we. . . ."

"The *Collonia*? A fine ship, Gad! Were you in her, captain, when she was strafed? Let's see—Mediterranean, wasn't it?" The captain nods pleasantly, as if accepting a compliment.

"Umm! Mediterranean—troops—a hell of a job to get them off. Lost some, though"—regretfully.

The convoy lieutenant turns a good card. "Must be a change to come down to ten knots, captain, after a crack ship like *Collonia*. What could she do? Sixteen?"

"Oh no. We could get an eighteen-knot clip out of her—more, if we wanted!" (If *War Trident*'s speed be low and doubtful, the Old Man can safely pile the knots on his stricken favourite.) "*She* was a ship, not a damned parish-rigged barge like this—a poverty-stricken hulk that. . . ."

"Yes. I heard about her from Benson, of *War Trumpet*. He sailed in last convoy. Said he was glad he wasn't appointed here."

"Wasn't appointed here, be damned! Didn't have the chance. Why, that ship of his isn't in the same class at all. The *Trident* can steer, anyway, and when we get things fixed up. . . . She has the hull of a fine ship. If only we could get a decent funnel on her. . . . Here, I'll try her at your nine and a half knots! I'll bet *War Trumpet* can't do a kick above nine!"

Be it noted that the convoy officers have the wavy gold lace of the R.N.R. for their rank stripes; plain half-inch ones of the Royal Navy might have had to let the convoy sail at nine, after all—not knowing the 'grip' of the 'last ship.'



EVENING: PLYMOUTH HOE

XIX

CONFERENCE

"**A** LAUNCH will be sent off at 3 p.m., S.T., to bring masters on shore for conference. You are requested to bring"--etc. So reads the notice, and p.m. finds the coxswain of the convoy office picket-boat steaming and backing from ship to ship, and making no secret of his disapproval of a scheme of things that keeps him waiting (tootling, perhaps, an impatient blast), while leisurely shipmasters give final orders to their mates at the gangways. ("That damned ship's cat in the chart-room again, sir!")

More ships have come in since the clearing of the morning mist, and calm weather and vagaries of the tide have combined to crowd the ships in the anchorage into uncomfortably close quarters; perhaps, after all, it would be rather the counter-swing of that River Plate boat, anchoring close abeam ("Given me a foul berth, damn him!"), than the insanitary ways of the ship's cat that kept the captain, one leg over the rail, so long in talk with his mate.

Never, since the days of sailing ships and the leisurely deep-sea parliaments in the ship-chandler's back room, have we been brought so much together. The bustle and dispatch of steamer work, in pre-war days, kept us apart from our sea-fellows; there were few forgatherings where we could exchange views and experiences and abuse 'square-heads' and damn the Board of Trade. Now, the run of German torpedoes has banded us together again, and in convoy and their conferences, we are coming to know one another as never before. At first we were rather reserved, shy perhaps, and diffident, one to another. Careless, in a way, of longshore criticism and opinion, we were somewhat concerned that conduct among our peers should be dignified and seaworthy; then, the fine shades of precedence—largely a matter of the relative speeds of our commands—had to average out before the 'master' of an east-coast tramp and the 'captain' of an R.M.S. found joint and proper equality. In this again, the enemy torpedo served a turn, and we are not now surprised to learn that the 'captain' of a

modest nine-knot freighter had been (till she went down with the colours apeak) 'master' of His Majesty's Transport of 16,000 tons.

So we crowd up together in the convoy launch, and introduce ourselves, and talk a while of our ships and crews till stoppage of the engines and clatter of hardwood side-ladders mark another recruit, sprawling his way down the high wall-side of a ballasted ship. The coxswain sighs relief as he pockets his list—the names all now ticked off in order of their boarding—and puts his helm over to swing inshore. "A job o' work," he says. "Like 'unt th' slipper, this 'ere! 'Ow can I tell wot ships they is, names all painted hover; an' them as does show their names is only damn numbers!"

In pairs, colloquing as we go, we mount the jetty steps and find a way to the conference-room. We make a varied gathering. Some few are in their company's service uniform, but most of us, misliking an array but grudgingly tolerated in naval company, wear longshore clothes and, in our style, affect soft felt hats and rainproof overcoats. Not very gallant raiment, it is true, but since brave tall hats and plain brass buttons and fancy waistcoats and Wellingtons went out with the lowering of the last single topsail, we have had no convention in our attire. In conference we come by better looks—bareheaded, and in stout blue serge, we sit a-row facing the blackboard on which our 'drills' are chalked. Many find a need for eyeglasses, the better to read the small typescript (uniformly bad) handed round to us, that sets forth our stations and the order of our sailing, and one wonders if the new look-out has brought us at last to the hands of the opticians; certainly, our eyes are 'giving' under the strain.

Of all the novel routine that war has brought to seafaring, convoy work is, perhaps, the most apart from our normal practice. We have now to think of concerted action, outboard the limits of our own bulwark; we have become subject to restriction in our sailing; we conform to movements whose purpose may not, perhaps, be plainly apparent. Trained and accustomed to single and undisputed command, it was not easy to alter the habits of a lifetime at sea. We were autocrats in our small sea-world, bound only by our owner's instruction to proceed with prudence and dispatch. We had no super-captain on the sea to rule our lines and set our courses and define our speeds. We made 'eight bells!'

But the 'bells' we made and the courses we steered and the rate we sped could not bring all of us safely to port. They gave us guns—and we used them passing well—but guns could not, at that date, deflect torpedoes, and ships went down. Then came convoy and its success, and we had to pocket our declarations of independence, and steer in fleets and company; and gladly enough, too, we availed ourselves of a union in strength, though it took time to custom us to a new order at sea.

At first we were resentful of what, ill-judging, we deemed interference. Were we not master mariners, skilled seamen, able to trim and handle our ships in any state or case? And if, on our side, the great new machine revolved a turn or two uneasily, it is true that the naval spur-wheel was not itself entirely free of grit. The naval officers, who drilled us down, were at first distant and superior; masters were a class, forgotten since sail went out, who had now no prototype in His Majesty's Service; there was no guide to the standard of association. Having little, if any, knowledge of merchant-ship practice, naval officers expected the same many-handed efficiency as in their own service. Crew troubles were practically unknown in their experience; all coal was 'Best Welsh Navigation'; all ships, whatever their lading, turned, under helm, apace! Gradually we learned—as they did. We saw, in practice, that team work and not individual smartness was what counted in convoy; that, be our understanding of a signal as definite and clear as the loom of the Craig, it was imperative, for our own safety, that the reading of out-wing and more distant ships should be as ready and accurate. In this, our convoy education, the chief among our teachers were the commodores, R.N. and R.N.R., who came to sea with us, blest, by a happy star, with TACT!



A CONVOY CONFERENCE

So, we learned, and now sit to listen, attentively and with respect, to what the King's Harbour Master has to say about our due and timely movements in forming up in convoy. On him, also, the happy star has shone, and we are conscious of an undernote that admits we are all good men and true and know our work. One among us, a junior by his looks, dissents on a movement, and not all-friendly eyes we turn on him; but he is right, all the same, and the point he raises is worthy the discussion that clears it. Our ranks are evidence of a world-wide league of seafarers against German brutality. While his frightfulness has barred the enemy for ever from sea-brotherhood, it has had effect in banding the world's seamen in a closer union. We are not alone belligerents devising measures of warfare; in our international gathering we represent a greater movement than a council of arms. British in majority, with Americans, Frenchmen, a Japanese, a Brazilian—we are at war and ruling our conduct to the sea-menace, but among us there are neutrals come to join our convoy; peaceful seamen seeking a place with us in fair trade on the free seas. Two Scandinavian masters and a Spaniard listen with intent preoccupation to the lecture—a recital in English, familiar to them as the Esperanto of the sea.

The K.H.M.'s careful and detailed routine has a significance not entirely connected with our sailing of the morrow; in a way it impresses one with

the extent of our sea-empire. Most of us have taken station as he orders, have all the manœuvres by rote, but even at this late date, there are those among us, called from distant seas, to whom the instructions are novel. For them, we say, the emphasis on clearing hawse overnight, the definition of G.M.T., the exactitude of zigzag, and the necessity of ready answer to signals. We are old stagers now, we know all these drills, we— Damn! We, too, are becoming superior! In turn, the commodore who is to sail with us has his say. Signals and look-out, the cables of our distance, wireless calls, action guns and smoke-screen, the rubbish-heap, darkening ship, fog-buoys and hydroplanes, he deals with in a fine, confident, deep sea-voice. Only on question of the hearing of sound-signals in fog do we throw our weight about, and we make reminiscent tangents not wholly connected with the point at issue. Yarn-spinners, courteously recalled from their digressions, wind up somewhat lamely, and commodore goes on to deal with late encounters with the enemy in which a chink in our armour was bared. Methods approved to meet such emergencies are explained, and his part is closed by attention to orders detailed for convoy dispersal. The commander of the destroyer escort has a few words for us; a brief detail of the power of his under-water armament, a request for a 'fair field in action.' Conference comes to an end when the shipping intelligence officer has explained his routes and given us our sailing orders.

Till now we have been actually an hour and a half without smoking, and our need is great. As one man we fumble for pipes and tobacco (a few lordly East-Indiamen flaunt cheroots), and in the fumes and at our ease arrange, in unofficial ways, the small brotherly measures that may help us at sea.

"Oh yes, *Chelmsford*, you're my next ahead. Well, say, old man, if it comes fog, give me your brightest cargo 'cluster' to shine astern—daytime, too—found it a good——" "Fog, egad! What about fog when we are forming up? Looked none too clear t' the south'ard as we came ashore!"

Somewhat late, we realize that not a great deal has been said about weather conditions for the start-off. The port convoy officer is still about, but all he can offer is a pious hope and the promise that he will have tugs on hand to help us out. "No use 'making almanacks' till the time comes," says our Nestor (a stout old greybeard who has been twice torpedoed). "We shall

snake into column all right, and, anyhow, we're all bound the same way!" "What about towing one another out?" suggests a junior, and, the matter having been brought to jest, we leave it at that.

The caretaker jangles his keys and, collecting our 'pictures,' we go out to the quayside, where thin rain and a mist shroud the harbour basin, and the dock warehouses loom up like tall clippers under sail. The coxswain comes, clamping in heavy sea-boots and an oilskin, to tell that the launch is at the steps, ready to take us off. Two of us have business to conclude with our agent, and remain on the jetty to see our fellows crowd into shelter of the hood and the launch back out. We call cheerfully, one to another, that we shall meet at Bahia or New York or Calcutta or Miramichi, and the mist takes them.

Up the ancient cobbled street we come on an old church and, the rain increasing to a torrent, we shelter at the porch. Who knows, curiosity perhaps, urges us farther and we step quietly down-level to the old stone-flagged nave. The light is failing, and the tombs and monuments are dim and austere, the inscriptions faint and difficult to read. A line of Drakes lie buried here, and tablets to the memory of old sea-captains (whose bones may lie where tide is) are on the walls. A sculptured medallion of ships on the sea draws our attention and we read, with difficulty, for the stone is old and the lines faint and worn.

". . . INTERRED YE BODY OF EDMOND LEC——, FORMERLY COMMANDER OF HER MAJ—— SHIP YE *LINN FRIGOT*, 17— . . . A FRENCH CORVAT FROM WHOM HE PROTECTED A LARGE FLEET OF MERCHANT SHIPS ALL INTO SAFETY. . . . AND BRAVELY HE GAVE YE ENEMY BATTEL AND FORCED HIM TO BEAR AWAY WITH MUCH DAMMAGE. . . ."

We looked at one another. A good charge to take to sea in 1918! Quietly we closed the door and came away.



THE OLD HARBOUR, PLYMOUTH

XX

THE SAILING

FOG, AND THE TURN OF THE TIDE

RAINY weather overnight has turned to fog, and the lighthouse on the Point greets breaking dawn with raucous half-minute bellows. Less regular and insistent, comes a jangle of anchor-bells, breaking in from time to time, ship after ship repeating, then subsiding a while until the syren of a moving tugboat—as if giving time and chorus to the din—sounds a blast, and sets the look-outs on the anchored ships to their clangour again. From the open sea distant reedy notes tell that the minesweeping flotilla is out and at work, clearing the course for draught of the out-bound convoy, and searching the misty sea-channels for all the enemy may have moored there. The 'gateships' of the boom defences rasp out jarring discords to warn mariners of their bobbling floats and nets. Inshore the one sustained and solemn toll of bell at the pier-head measures out time to the sum of a dismal dayspring.

By all the sound of it, it is ill weather for the sailing of a convoy. In time of peace there would not be a keel moving within harbour limits through such a pall. "Call me when the weather clears," would be the easy order, and we would turn the more cosily to blanket-bay, while the anchor-watch would pace athwart overhead, in good content, to await the raising of the curtain. Still and all, it is yet early to assess the rigour of the fog. Sound-signals, started late in the coming of it, became routine and mechanical, and persist—through clearing—till their need is more than over. The half-light of breaking day has still to brighten and diffuse; who knows; perhaps, after all, this may be only that dear and fond premise of hopeful sailormen—the pride o' the morning!

The elder fishermen (the lads are out after the mines) have no such optimism. Roused by the habits of half a century, they turn out for a pipe

and, from window and doorway, assure one another that their idle 'stand-by' decreed by harbour-master for outgoing of the convoy, is little hardship on a morning like this. "'Ark t' them bells," they say, thumb over shoulder. "All 'ung up. Thick as an 'edge out there, an' no room t' back an' fill. There won't be no move i' th' Bay till 'arf-ebb, my oath!"

But they are wrong in that, if right in their estimation of the weather and congestion in the roads, for we are at war, and the port convoy officer, hurrying to his launch, is already sniffing for the bearings of the leader of the line. Prudently he has mapped their berths as they came in to anchor, and has, at least, a serviceable, if rough, chart to guide him on his rounds.



CONVOY SAILING FROM PLYMOUTH SOUND

So far there are no reports from the sea-patrols that would call for an instant alteration of the routes, and for that the P.C.O. has a thankful heart. A 'hurrah's nest,' a panic on Exchange, a block at the Bank crossing, would be feeble comparison to the confusion he might look for in a combination of dense fog, counter-mandates, and a congested roadstead, for, even now, the ships to form up the next convoy are thrashing their way down the coast and (Article XVI of the Rule of the Road being lightly held by in war-time) may be expected off the 'gateships' before long. To them, as yet, the port is 'closed,' but every distant wail from seaward sets him anxiously wondering whether it be a minesweeper signalling a turn to his twin or a distant deep-waterman, early on the tide, standing in for the land. The sailor's morning litany—"Who wouldn't sell a farm and go to sea"—is near to him as he turns up the collar of his oilskin and gives a rough course to his coxswain. "South, s'west, and ease her when you hear th' Bell buoy. *British Standard*

first—she's lying close south of it." Turning out, the picket-boat sets her bows to the grey wall of mist and her wash and roundel of the screws (that on a clear busy day would scarce be noted) sound loud and important in the silence of the bay. The coxswain, cunning tidesman, steers a good course and reduces speed with the first toll of the buoy. The clamour of its iron tongue seems out of all relation to the calm sea and the cause is soon revealed. Silently, closely in line ahead, four grey destroyers break the mist, fleet swiftly across the arc of vision ahead, and disappear. "Near it," says the coxswain (and now sounds a blast of *his* whistle). "Them fellers ain't 'arf goin' it!" Cautiously he rounds the buoy, noting the gaslight crown shining yet, though pale and sickly in the growing day. Out now, in seven fathoms, the lingering inshore fog has given place to a mist, through which the ships loom up in sombre grey silhouette. Full speed for a turn or two brings the launch abeam of a huge oil-tanker that, sharp to the tick of Greenwich Mean Time, already has her Convoy Distinguishing Flags hoisted and the windlass panting white steam to raise anchor. A small flag in the rigging assures the P.C.O. that the pilots have boarded in good time, and it is with somewhat of growing satisfaction that he hails the bridge and asks the captain to 'carry on!'

Doubts and hesitancies that may have lingered in the prudent captain's mind are dispelled by the P.C.O.'s appearance. "It is decided, then, that the orders stand," and there is at least a certain relief in his tone as he orders, "Weigh anchor!"

The *British Standard* is deep-loaded, in contrast to the usual empty war-time outward bound, but her lading is clean salt water, no less, run into her compartments on the sound theory that Fritz, by a strafe, may only 'change the water in the tanks.' Homeward, from the west, there will be no such fine assurance, for a torpedo may well set her ablaze from stem to stern, and the enemy takes keen and peculiar delight in such *Schrecklichkeit*. Still, there is little thought to that; *British Standard* is to lead the line, and her anchor comes to the hawse and she backs, then comes ahead again, swinging slowly under helm towards the sound of 'gateships' hand-horns. High on the stern emplacement her men are uncovering her gun and clearing the ranges, and the long grey barrel is trained out to what will be the sun-glare side of the first tangent of her sea-course. Close astern of her comes *War Ordnance*, her pushful young captain having taken heed of the sounds of

Standard's weighing. "Good work," says the P.C.O. cheerfully, and cons his rough chart for the whereabouts of Number Three.

As though the devil in the wind had heard him, down comes the fog again, dense this time, a thick blanket-curtain of it that shuts off the misty stage on which the prompter had hoped, passably, to complete his dispatch of the fleet.

The compass again. "East 'll do," and the launch slips through the grey of it. All around in the roadstead the clank of cable linking over the spurs, and hiss and thrust of power windlasses are indication that *British Standard's* movement has given signal to weigh, that it is plain to the others—"Convoy will proceed in execution of previous orders." A propellor, thrashing awash in trial, looms up through the fog ahead, but 'East' has brought the launch wide of her mark, and *Massilia* is answer to the P.C.O.'s hail. *Massilia* is Number Four, but needs must when the fog drives, so he advises the captain to get under way and head out.

Number Three has stalled badly and is hot in a burst of graceless profanity from bridge to forecastle-head, and (increasing in volume and blood-red emphasis) from there to the chain-locker. There is a foul stow. Her nip-cheese builders have pared the locker-space to the mathematical limit (to swell her carrying tonnage), and the small crew that her nip-cheese owners have put on her are unable to range the tiers. Twenty fathoms of chain remain yet under water, the locker is jammed, and the mate, roughed (and through a megaphone, too), from the bridge, is calling on strange deities to take note that, 'of all the damn ships he ever sailed in. . . .' The pilot calls out from the bridge that they are going to pay out and restow, and the convoy officer, blessing the forethought that had bade him send off Number Four, swings off to speed the succession.

High water has made and the tide ebbs, swinging the ships yet anchored till they head inshore, and adding to the pilots' worry of narrowed vision the need to turn short round in crowded waters. For this the tugs have been sent out in readiness, and the convoy launch has a busy mission in casting about to find and set them to the task of towing the laggards round. It is nothing easy, in the fog and confusion of moving ships, to back the *Seahorse* in and harness her by warp and hawser, but with every vessel, canted, that

straightens to her course, the press is lightened by so much sea-room cleared. Gradually the hail and counter-hail, hoarse order and repeat, whistle-signals, protest of straining tow-ropes, die away with the lessening note of each sea-going propeller.

To Number Three again, last of the line and out of her station, the convoy officer seeks to return. The fog is denser than ever, and the echoes of the bay, now transferred to seaward, augment the uneasy short-blast mutterings where the ships, closed up at the narrow 'gateway,' are slowing and backing to drop their pilots. In his traverse of the anchorage the coxswain has lost bearing of the *Cinderella* and steers a zigzag course through the murk. The sun has risen, brightening the overhead but proving (in sea glare and misty daze) an ally to the veil. No sound of heaving cable or thunder of escaping steam that would mark a vessel hurrying to get her anchor and make up for time lost is to be heard. Frankly puzzled, the coxswain stops his engines. "Must 'a sailed, sir," he says at length. "There ain't nothin' movin' this end o' th' bay."

The convoy officer nods. "*Mmm!* She may have gone on, while we were dragging *Marmion* clear of th' stern of that 'blue funnel' boat. A good job. Well, carry on! Head in—think that was th' pier-head bell we heard abeam!"

At easy speed the launch turns and coxswain bends to peer at the swinging compass-card. As one who has held out to a job o' work completed, the P.C.O. stretches his arms and yawns audibly and whole-hearted. "A good bath now and a bite o' breakfast and— Oh, hell! What's that astern?"

The turn in the wake has drawn his eye to a grey blur in the glare of the mist. An anchored ship!

Keeping the helm over, the coxswain swings a wide circle and steadies on the mark. "Damn if it ain't her!" he says, as the launch draws on.

The *Cinderella* lies quiet with easy harbour smoke rising straight up from her funnel and no windlass party grouped on the forecastle-head; quiet, as if fog and convoy and the distant reverberations of her sister ships held no concern for her. To the P.C.O.'s surprised and somewhat indignant hail there is returned a short-phrased assurance that the ruddy anchor is

down—and is going to remain down! "Think I'm going out in this to hunt my place in the pack? No damn fear!" says the captain. "Why, I can scarce see who's hailing me, less a line o' ships barging along!"

The pilot, in a tone that suggests he has already 'put out an oar'—with little effect—joins in to reassure. "Clearin' outside now, captain. I haven't heard th' lighthouse syren for twenty minutes or more! The fog'll be hangin' here in harbour a bit."

"Aye, aye! But it's here we are, pilot—not outside yet. A clearing out there doesn't show us th' leading marks, and I'll not risk it. I've no fancy for nosing into th' nets and booms. I know where I am here, and I won't stir a turn—unless"—bending over the light screen towards the launch—"unless you lead ahead!"

The convoy officer is somewhat embarrassed. Certainly the weather is as thick as a hedge; there is no 'drill' of convoy practice that empowers him to order risks to be taken—navigation of the ships is not his province. It is enough for him to arrange and advise and assist. If he leads out and anything *does* happen?

Still, it is maddening to think of one hitch in a good programme—'almost a record, too!' He looks at his watch and notes that only fifty minutes have elapsed since *British Standard* weighed.

"Oh, hell! Right, captain," he says. "Heave up and I'll give you a lead out to clear weather!"

'IN EXECUTION OF PREVIOUS ORDERS'

WE are Number Four in the line; *Vick—beer—code* is our address, and we steam somewhat faster than the fog warrants to keep touch with our next ahead. She, in turn, is packing close up on the leader, and if, in the strict ruling of a 'line ahead,' we are stepping out a trifle wide, at least we keep in company. The farthest we can see is the thrash of foam, white in the grey, of *War Ordnance's* propeller—a good moving mark, that, though faint, draws the eye by the lead of broken water. Nearer, we have a steering-guide in her hydroplane, cutting and dancing under the bows and throwing a slightly

feather of spray. The sea is flat calm, save for our leader's wake—a broad ribbon of troubled water through which we steer. Our eyes, now limited in range by the fog, seem to focus readily on trifles; for want of major objects, roving glances take in driftwood and ship-litter, and turn on minute patches of seaweed with an interest that a wider range would dissipate. Spurring, black-crested puffins come at us from under the misty pall, floating still, as if set in glass, till our bow wash plays out and sets them, squawking in distress, to an ungainly splutter on the surface, or dipping swiftly to show white under-feathers and the widening rings of their dive.

Astern of us, a medley of sound and steering-signals marks the gateway of the harbour where our followers are striving to drop their pilots and join in convoy; one loud trumpeter is drawing up at speed and showing, by the frequency of her whistle-blasts, anxiety to sight our wake. The lighthouse syren roars a warning of shoal-water out on the landward beam, a raucous discord of two weird notes. These, with the rare mournful wail of our leader, are our guiding sounds, but we have sight now and then of the destroyer escort passing and turning mistily on the rim of our narrowed vision, like swift sheep-dogs folding the stragglers of a scattered flock.

The fog, that settled dense and deep as we got under way, shows a little sign and promise of thinning, a small portent that draws our eyes to the lift above the funnel. There is no wind, but our smoke-wrack, after curving with our speed to masthead height, seems turned by light upper draughts to the eastward. The sun has risen and peers mistily over the top of the grey curtain that surrounds us. The day is warming up. Pray fortune, a stout west wind may come out of it all, to clear the muck and give us one good honest look at one another, when we are due for that 'six-point' turn to the south'ard!

To keep in station on our pacemaker, we call for constant alterations in the speed—a range of revolutions that rattles up scale and down, like first lessons on the piano, and sets the engineers below to a plaintive verge of tears. The junior officer at the voice-pipe looks reflective, after each order he passes, as though comparing the quality of the reply with the last sulphurous rejoinder. The fog has added to our starting vagaries and postponed a happy understanding, but we shall do better later on when we

have gauged and discovered—and pitied—the tiresome vacillations of the *other* ships!

Meantime, as best we can, we chase the sheering hydroplane ahead that seems endowed with every chameleon gift of the classic gods. It vanishes, invisible, in a drift of fog, and though we con a course as steady as a cat on eggs, a clearing comes to show us its white feather broad on the bows and edging off at an angle to dip under the thick of the mist! It drops down to us; we sheer aside and slow a pace, and it lingers and dallies sportively abeam. It slips suddenly ahead, with a rush and a rip, as though, like a child among the daisies, it recalls a parent in advance.

The trumpeter astern has come up and sighted our wake and fog-buoy, and the clamour of her questing syren is stilled. She looms up close on our quarter, a huge menacing bulk of sheering steel with the foam thundering under her bows and curling and shattering on her grey hull. *They* have great difficulty in adjusting to our speed. She slows and fades back into the mist, grows again from gloomy shadow to threatening detail, steadies at a point for a few minutes, and resumes the round of her previous motions in irritating cycle. "Whatever can be the matter with them?" (We take the stout point of a position as steady as the Rock, and grow scornful of their clumsy efforts to keep station.) "*Huh!* These gold-laced London men! Why can't they steady up a bit? Why can't they——" We note that our steering-mark and the wash of *War Ordnance's* propeller are no longer in sight ahead, and set in to count the beats of the screw. ". . . t'-one, t'-two, t'-three, t'— *Hell!* Didn't we order seventy? Go full speed!" Jumping to the tube, the junior attends. "*I* said seven-owe, sir, but he thought I said six-four! Says th' bl—, th' engines working, sir—can't hear properly!"

Grudgingly, as though loath to give us our sight again, the fog clears. The first of the tantalizing rift in the curtain is signalled by the high lookout, who calls that he can see the topmasts of our near neighbours piercing the low-lying vapours. The sun shines through, showing now and then a clear-cut limb in place of the luminous misshapen brightening that has been with us since sunrise. In fits and starts the fog thins, and thickens again, at the will of wandering airs.

A west wind comes away, freshens, and stirs the vapour till it whips close overhead in wraiths and streamers, raises here and there a fold on the distant horizon, then dies again. Growing in vigour, the breeze returns; a gallant breath that ruffles the smooth of the sea and sweeps the round of it, routing the lingering flurries that settle, dust-like, when the mass is cleared.

The clearing of our outlook produces a curious confusion to the eye. We have become accustomed to a limited range in sight, and the sudden change to distant vision, in which there is no standard of position, no mark to judge by, effects an illusion as of a photographer's plate developing. Fragments, wisps, and sections of the sea-rim appear, breaking through as the fog lifts, and seeming strangely high and foreign in position. Topmasts and a funnel-wreath of black smoke loom up almost in mid-air; the water-line of a ship's hull grows to sight, low in the plane as though dangerously close. Distant, obscure, and blurred formations sharpen suddenly to detail and show our destroyer escort as almost suspended in mirage, floating in air. Piece by piece, the plate develops in sensible gradation, fitting and joining with exactitude; the ships ahead take up their true proportions, the sea-horizon runs to a definite hard line. Mast and funnel and spar stand out against the piled and shattered fog-bank, whose rear-guard lingers, sinking but slowly and sullenly, on the rim of the eastern horizon.

The fog cleared, and a busy seascape in sight, we shake ourselves together and take heed of appearances. Our convoy signal hangs damp and twisted on the halyards, and needs to be cleared to blow out for recognition; the mirrored arc-lamp that we turned astern to aid the trumpeter is switched out. With the fog-buoy we are less urgent; it will be time enough to haul it aboard when we are assured the new-born breeze is healthy and likely to remain with us. The press of work about the decks has lessened with the hawsers and docking gear stowed away. Sea-trim is the order now—a war sea-trim, in which the boats, swung outboard and ready for instant use, rafts tilted to a launching angle, hoses rigged to lead water, and crew at the guns, form a constant reminder (if that be needed) of lurking under-water peril. In marked contrast to less exciting days, when we could afford to disregard whatever might go on behind us, we place look-outs to face all ways. The enemy may gamble on our occupation with the view ahead, but, with a new war wariness, we have grown eyes to search the sea astern.

In the clearing weather we become sensitive to the strict and proper reading of our sailing orders. There must be no more faults in the voice-tube to let us down from confidence in our right to a sudden sense of guilt. We adjust our station in the line by sextant angles of the leader, measuring his height to fractions, and set an ear to the note of our engine-beats to ensure a steady gait.

Clearing our motes, we turn a purged and critical eye on our fellows, now all clear of the mist, and steaming in sight. To far astern, where the land lies and the sun plays on wet roof and flashing window-pane, a long line of ships snakes out in procession, their smoke blowing and curling merrily alee to join the cumulus of the foundering fog-banks. There are gaps and kinks in our formation that would, perhaps, call for angry signals in a line of battle, but the laggards are closing up in hasty order to right the wayward tricks of sound and distance in the fog. If not quite ruled and ordered to figures of our text, at least we conform to the spirit, and are all at sea together, steering out on our ventures.

Our distance run, *British Standard* puts her helm over and turns out. Forewarned, all eyes have been focused on the line of her masts, and her sheer gives signal for a general cut and shuffle. We change partners. Curtsying to full rudder pressure, we join the dance, and swing to her measure, adjusting speed to mark time while other important leaders of columns draw up abeam. The flat bright sea is cut and curved by thrashing wakes as the convoy turns south. Ahead and abeam, round and about, the destroyers wheel and turn, fan in graceful formation and swerve quickly on their patrolling courses.

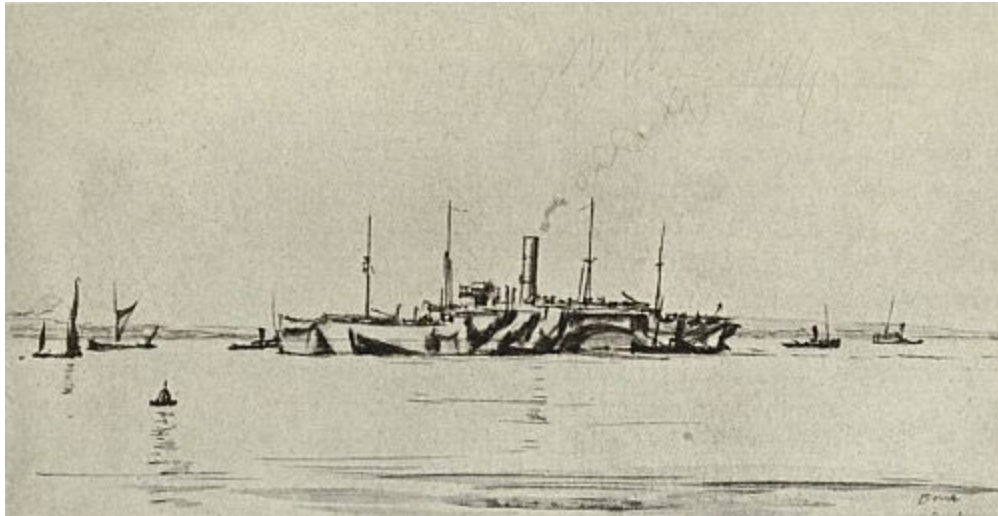
We are less expert in the figures of our cotillion. It cannot be pretended that we slip into our convoy stations with anything approaching their speed and precision. We are too varied in our types, in turning periods, in the range of our dead-weight, to manœuvre alike. Most of us have but a slender margin of speed to draw on, and, 'all bound the same way,' the spurt to an assigned position proves the stern a long chase. The fog, at starting, has thrown many of us out of our proper turn, and we zigzag, unofficially, this way and that, to gain our stations without reduction of speed. In the confusion to our surface eyes, there is this consoling thought—that the same perplexing evolutions (calling for frequent appeals to the high gods

for enlightenment as to the 'capers' of the *other* fellows) have, at least, no better meaning in the reflected angles of a periscope.

Now the hum and drone that has puzzled us in the fog reveals itself as the note of a covey of seaplanes searching the waters ahead. They have come out at first sign of a clearing, and now fly low, trimming and banking in their flight like gannets at the fishing. A winking electric helio on one of them spits out a message to the leader of the destroyers, and she flashes answer and acknowledgment as readily as though the seaplane were a sister craft. A huge coastal airship thunders out across the land to join our forces. She grows to the eye as though expanding visibly, and noses down to almost masthead height in a sharp and steady-governed decline; abeam, she turns broad on, manœuvring with ease and grace, and the sunlight on her silvered sides glints and sparkles purely, as though to shame the motley camouflage of the ships below.

The commodore poises the baton as his ship draws up to her station. Till now we have steamed and steered 'in execution of previous orders' and, considering the dense fog and the press of ships at the anchorage and pilot-grounds, we have not been idle or neglectful. Now we are in sea order, and, with the ships closing up in formation, we attend our senior officer's signals as to course and speed. A string of flags goes up, fluttering to the yard of his ship, and we fret at the clumsy fingers that cannot get a similar hoist as quickly to ours. Anon, on all the ships, a gay setting of flags repeats the message, and we stand by to take measure and sheer of a tricky zigzag, at tap of the baton.

The line of colour droops and fades quickly to the signalman's gathering; the convoy turns and swings into the silver-foil of the sun-ray.



INWARD BOUND

XXI

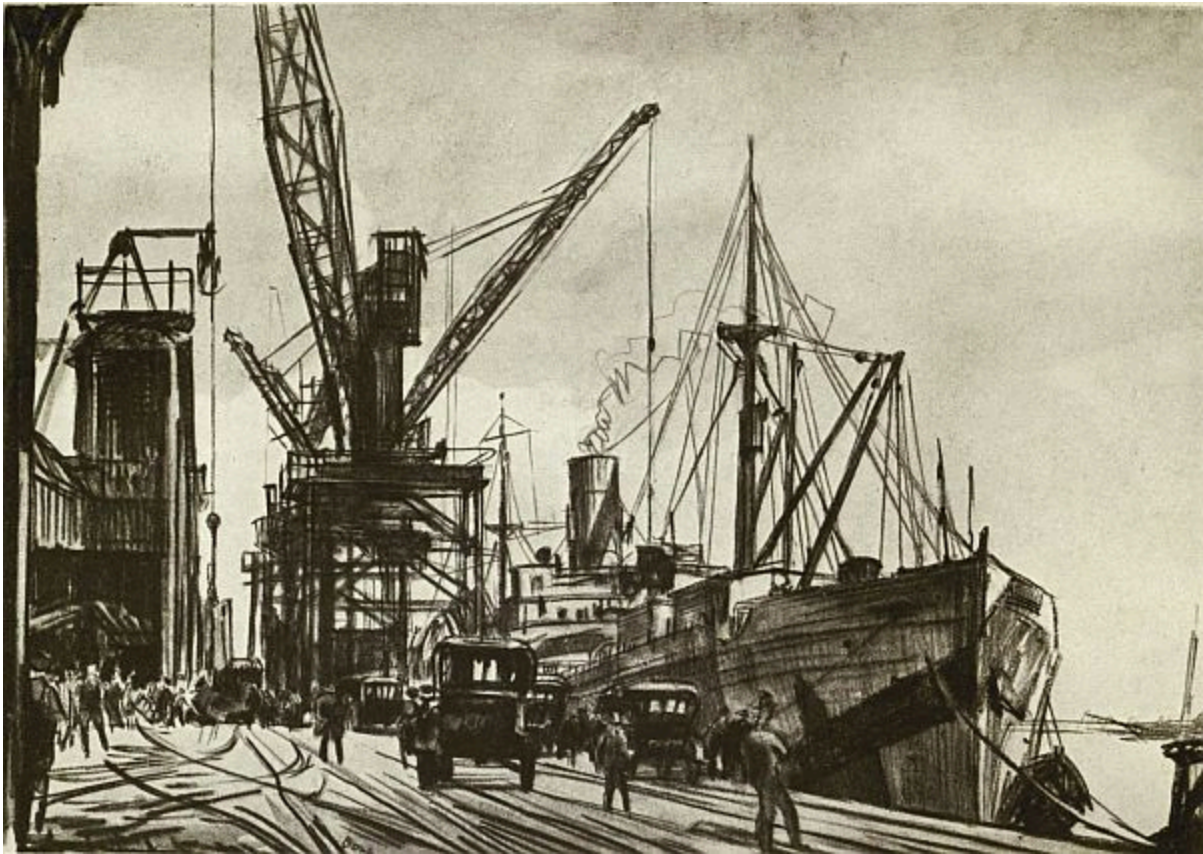
THE NORTH RIVER

THE broad surface of the Hudson is scored by passage of craft of all trades and industries. Tugs and barges crowd the waterway in unending succession, threading their courses in a maze of harbour traffic; high-sided ferry-boats surge out from their slips and angle across the tide—crab-wise—towards the New Jersey shore; laden ocean steamers hold to the deeps of the fairway on their passage to the sea. Up stream and down, back and across, sheering in to the piers and wharves, the harbour traffic seems constantly to be scourged and hurried by the lash of an unseen taskmaster. The swift outrunning current adds a movement to the busy plying of the small craft—a hastening sweep to their progress, that suggests a driving power below the yellow tide. The stir of it! The thrash of screw and lapping of discoloured water, the shriek of impatient whistle-blasts, the thunder of escaping steam!

As we approach from seaward, there is need for caution. The railway tugmen—who live by claims for damages from ocean steamers—are alert and determined that we shall not pass without a suitable parting of their hawsers, damage to barges, strain to engines and towing appliances. Off the Battery, they sidle to us in coy appeal, but we carry bare steerageway. As the pilot says: "Thar ain't nothin' doin'!" We disengage their ardent approach, and make a slow progress against the tide to our loading-berth. There, we drop in towards the pier-head and angle our bows alongside the guarding fenders. A flotilla of panting tugboats takes up station on our inshore side and 'punches' into us—head on—to shove our stern round against the full pressure of the strong ebb tide. The little vessels seem absurdly small for their task. They 'gittagoin',' as instructed by the pilot, and wake the dockside echoes with the strain of their energy. White steam spurts from the exhausts with every thrust of their power. The ferry-boats turning in to their slips come through the run of a combined stern wash that sets them on the boarding with a heavy impact. Power tells. Our stern wavers, then we commence to bear up-stream in a perceptible measure. The Hudson

throws a curl of eddying water to bar our progress, but we pass up—marking our progress by the water-side of the west shore. Anon, the thunder of the tugs' pulsations eases, then stops: they back away, turn, and speed off on a quest for other employment—while we move ahead, out of the run of the tide, and make fast at the pier.

Our ship is keenly in demand. The dockers are there, ready with gear and tackle to board and commence work. The wharf superintendent hails us from the dockside before the warps are fast. He is anxious to know the amount of ballast coal to be shifted from the holds before he can commence loading. "Toosday morning, capt'n," he adds, as reason for his anxiety—"Toosday morning—an' she's gotta go!" Tuesday, eh! And this is Saturday morning! They will have to hustle to do it.



A TRANSPORT LOADING

'Hustle'—as once he told us—is the superintendent's maiden name. Already the narrow water-space between us and our neighbour is jammed tight by laden barges, brought in to await our coming. Billets of steel, rough-cast shells, copper ingots, bars of lead and zinc are piled ready for

acceptance. The shed on our inshore tide is packed by lighter and more perishable cargo, all standing to hand for shipment. Preparation for our rapid dispatch is manifest and complete. Before the pilot is off the ship with his docket signed, the blocks of our derricks are rattling and the stevedores are setting up their gear for an immediate start. Barred, on the sea-passage, from communication by wireless, we have been unable to give a timely advice of our condition to the dock. The factor of the coal to be shifted—till now unknown to them—is the first of many difficulties. We have no cargo to discharge (having crossed in ballast trim), but—the storms of the North Atlantic calling for a weight to make us seaworthy—we have a lading of coal sufficient to steam us back to our home port. This has all to be raised from the holds and stowed in the bunker spaces: the holds must be cleaned for food-stuffs: for grain in bulk there is carpenter-work in fitting the midship boards to ensure that our cargo shall not shift. Tuesday morning seems absurdly near!

With a thud and jar to clear the stiffening of a voyage's inaction, our deck winches start in to their long heave that shall only end with the closing of the hatches on a laden cargo. The barges haul alongside at the holds that are ready for stowage and loading begins. The slings of heavy billets pass regularly across the deck and disappear into the void of the open hatchways. In the swing and steady progression there seems an assurance that we shall keep the sailing date, but our energy is measured by the capacity of the larger holds. In them there is the bulk of fuel to be handled. The superintendent concentrates the efforts of his gangs on this main issue: the loading of the smaller compartments is only useful in relieving the congestion of the barges overside.

Under his direction the coalmen set to work at their hoists and stages and soon have the baskets swinging with loads from the open hatchways. The coal thunders down the chutes to the waiting barges, and raises a smother of choking dust. The language of South Italy rings out in the din and clatter. "Veera, veera," roars the stageman (not knowing that he is passing an ancient order on a British ship). It is a fine start. Antonio and Pasquali and their mates are fresh: they curse and praise one another alternately and impartially: they seem in a fair way to earn their tonnage bonus by having the holds cleared before the morning.

It is almost like an engagement in arms. Good leadership is needed. There are grades and classes in the army of dockers; groups as clearly specialized in their work as the varied units that form an army corps. Italian labourers handle the coal; coloured men are employed for the heavy and rough cargo work; the Irish are set to fine stowage. There is little infringement of the others' work. Artillery and infantry are not more set apart in their special duties than the grades of the dockers. Certainly there is a rivalry between the coloured men and the Irish—the line that divides the cargo is perhaps lightly drawn. "Hey! You nigger! You gitta hell out o' this," says Mike. The coloured man bides his time. The thunder of the winches pauses for an instant—he shouts down the hatchway: "Mike! Ho, Mike!" An answering bellow sounds from below. "Ah say, Mike! When yo' gwine back hom' t' fight fo' King Gawge?"

Sunday morning, the 'macaroni' gangs knock off work for a term. The holds are cleared, but our fuel has again to be hove up from the barges and stowed in the bunkers. That can be done while loading is in progress. Meantime—red-eyed and exhausted—the coalmen troop ashore and leave the ship to one solitary hour of Sunday quiet. At seven the turmoil of what the superintendent calls a 'fair start' begins. Overnight a floating-tower barge for grain elevation has joined the waiting list of our attendant lighters. She warps alongside and turns her long-beaked delivery-pipes on board; yellow grain pours through and spreads evenly over the floor-space of our gaping holds. Fore and aft we break into a full measure of activity. The loading of the cargo is not our only preparation for the voyage. The fittings of the 'tween-decks, thrown about in disorder by the coal-gangs, have to be reconstructed and the decks made ready for troops. Cleaning and refitting operations go on in the confusion of cargo work: conflicting interests have to be reconciled—the more important issues expedited—the fret of interfering actions turned to other channels. At the shore end of the gangways there is riot among the workers. Stores and provisions are delivered by the truckmen with an utter disregard for any convenience but their own. The narrow roadway through the shed is blocked and jammed by horse and motor wagons that, their load delivered, can find no way of egress. Cargo work on the quayside comes to a halt for want of service. The dockers roar abuse at the truckmen, the truckmen—in intervals of argument with their fellows—return the dockers' obloquy with added embellishment.

The 'house-that-Jack-built' situation is cleared by the harassed pier-foreman. The shed gates are drawn across: outside the waiting charioteers stand by, their line extended to a block on the Twenty-Third Street cars.

The roar and thrust and rattle of the straining winches ceases on Monday evening. We are fully stowed: even our double-bottom tanks—intended for water-ballast alone—carry a load of fuel oil to help out the difficulties of transport. The superintendent goes around with his chest thrown out and draws our attention to the state of affairs—the ship drawing but eighteen inches short of her maximum draught, and the 'tween-decks cleared and fitted. "Fifty-four working hours, capt'n," he says proudly. It is no mean work!

The silence of the ship, after the din and uproar of our busy week-end, seems uncanny. The dock is cleared of all our attendant craft, and the still backwater is markedly in contrast to the churned and troubled basin that we had known. From outside the dock a distant subdued murmur of traffic on the streets comes to us. Cross-river ferries cant into a neighbouring slip, and the glow of their brilliant lights sets a reflection on the high facades of the water-front buildings. Overhead, the sky is alight with the warm irradiance of the great city. Ship-life has become quiescent since the seamen bundled and put away their gear after washing decks. Only the dynamos purr steadily, and an occasional tattoo on the stokehold plates tells of the firemen on duty to raise steam. In the unfamiliar quiet of the night and absence of movement in the dock there is countenance to a mood of expectancy. It seems unreasonable that we should so lie idle after the past days of strenuous exertion in preparing for sea. The flood in the North River, dancing under the waterside lights, invites us out to begin the homeward voyage. Why wait?

We are not yet ready. In our lading we have store of necessities to carry across the sea. Food, munitions and furniture of war, copper, arms, are packed tightly in the holds: power-fuel for our warships lies in our tanks. There is still a further burthen to be embarked—we wait a cargo of clear-

headed, strong-limbed, young citizens bound east to bear arms in the Crusade.

They come after midnight. There are no shouts and hurrahs and flag-waving. A high ferry-boat crosses from the west shore and cants into the berth alongside of us. The dock shed, now clear of goods, is used for a final muster. Encumbered by their heavy packs, they line out to the gangways and march purposely on board. The high-strung mimicry of jest and light heart that one would have looked for is absent. There is no boyish call and counter-call to cloak the tension of the moment. Stolidly they hitch their burdens to an easier posture, say 'yep' to the call of their company officer, and embark.

The troops on board, we lose no time in getting under way. Orders are definite that we should pass through the booms of the Narrows at daybreak, and join convoy in the Lower Bay with the utmost dispatch. We back out into the North River, turn to meet the flood-tide, and steer past the high crown of Manhattan.



A CONVOY IN THE ATLANTIC

XXII

HOMEWARDS

THE ARGONAUTS

THE boat guard (one post, section A) stir and grow restive as the hour of their relief draws on. Till now they have accepted wet quarters, the reeling ship, black dark night with fierce squalls of rain and sleet, as all a part of the unalterable purgatory of an oversea voyage. With a prospect of an end to two hours' spell of acute discomfort, of hot 'kawfee,' dry clothes, and a snug warm bunk, their spirits rise, and they show some liveliness. Muffled to the ear-tips in woollens and heavy sodden greatcoats, their rifles slung awkwardly across the bulge of ill-fitting cork life-belts, they shift in lumbering movement from foot to foot, or pace—two steps and a turn—between the boat-chocks of their post. A thunder of shattering salt spray lashes over from break of a sea on the foredeck, and they dodge and dive for such poor shelter as the wing of the bridge affords.

Scraps of their protest to the fates carry to our post in breaks of the wind "Aw, you guys! Say! Wisha was back 'n li'l old N'yok, ringin' th' dial 'n a Twanny-Thoid Street car!" "Whaddya mean—a Scotch highball? Gee! I gotta thoist f'r all th' wet we soak!" "Bettcha Heinie's goin'a pay *me* cents an' dallers f'r this!" ". . . an' a job claenin' me roifle. . . th' sargint, be damn but, he . . ."

"Cut it! Less talk 'round there!" orders their duty officer from somewhere in the darkness; the talk ceases, though stamp and bustle of expectant relief persist, and we are recalled to survey and reflection on the gloom ahead.

Midnight now, and no sign of a change! Anxiously we scan sea and sky for hope or a promise—not a token! A squall of driving sleet has passed over, and has left the outlook moderately clear, but a quick-rising bank of

hard clouds in the nor'east threatens another, and a heavier, by the look, soon to follow. A moonless night, not a star shines through the sullen upper clouds to mark even a flying break in the lift of it. A hopeless turn for midnight, showing no relief, no prospect!

Ahead, the dark bulk of our column leader sways and thrashes through the spiteful easterly sea, throwing the wash broad out and taking the spray high over bow and funnel. In turn, we lurch and drive at the same sea that has stirred her, and find it with strength enough to lash over and fill the fore-deck abrim. Weighed down forward, we throw our stern high, and the mad propeller thrashes in air, jarring every bolt and rivet in her. We cant to windward, joggling in an uneasy lurch, then throw swiftly on a sudden list that frees the decks of the encumbering water. We ease a pace or two as the propeller finds solid sea to churn, steady, then gather way to meet the next green wall. With it the squall breaks and lashes furiously over us, driving the icy slants of hard sleet to our face, cutting at our eyes in vicious persistence. Joined to the wind-burst, a heavy sea shatters on fore-end of the bridge, and ring of the steel bulkhead sounds in with the crash of broken water that floods on us.

In this succession the day and half the night have passed. No 'let-up' in the round of it. Furious wind-bursts marking time on the face of a steady gale. Rain—and now sleet. Sleet! Who ever heard of icy sleet in North Atlantic, this time of the year? Gad! Every cursed thing seems to weigh in against us on this voyage! The weather seems in league with the enemy to baulk our passage. Every cursed thing! Head winds and heavy seas all the way. Fog! These horse transports having to heave-to, and forcing the rest of the convoy to head up and mark their damned time! And now this, just when we were looking for a 'slant' to make the land! Maddening!

The bridge is astir with the change of the watch. A fine job they make of it! Like a burst of damned schoolboys! Oilskin-clad clumsy ruffians barging up the ladders, trampling and stumbling in their heavy sea-boots, across and about, peering to find their mates! Are they all blind? Why can't they arrange set posts for eight bells? Why can't they look where—"Th' light, damn you! Dowse that light! *Huh!* Some blasted idiot foul of that binnacle-screen again! Th' way things are done on this ship! Egad! Would think we were safe in th' Ship Canal, instead of dodging submar—" A slat of

driving spray cuts over and we dip quickly under edge of the weather-screen.

The second officer arrives to stand his watch, and the Third, who goes below, is as damnably cheerful and annoying as the other is dour. "North, —ty-four east, th' course. She's turning seven-six just now, but you'll have to reduce shortly—drawing up on our next ahead. Seven-three or four sh'd keep her in station. *Neleus* ahead there, two cables. Rotten weather all th' watch. Squalls, my hat! There's another big 'un making up now! Th' Old Man over there—like a bear with a sore—raisin' hell 'bout——"

"Oh, a—ll right! Needn't make a song and dance of it! North, —ty-four east? Right!" Picking up binoculars, the Second scans the black of it ahead, as though now definitely set for business.

The watch is taken over and all seems settled, but the Third is not yet completely happy. He gloats a while over the Second's gloomy outlook, and yawns in that irritating *arpeggio*, the foretaste of a good sound sleep. "Oh, d'ya read in orders 'bout th' zigzag for th' morning watch?—a new stunt, fours and sixes; start in at——"

"Oh, g'rr out! How can a man keep a watch, you chewin' th' rag? Yes, I —read—the orders!" *S-snap!*

"*Huh!* A pair of them!" It comes to us that something will have to be said about the way the damned bridge is relieved in this ship!

Into the chart-room, to fumble awkwardly for light ('*T'tt!* That switch out of order again!') and search for a portent in the jeering glassy face of the aneroid. *Tip, tip, whap!* The cursed thing is falling still. 'Twenty-nine owe two—half an inch since ten o'clock! Whatever can be behind all this? That damn glass was never right, anyway!'

Drumming of the wireless-cabin telephone sounds out, and we listen to a brief account of Poldhu's war warning. An S.O.S. has been heard, but a shore station has accepted it. (They can identify the ship—might be the harping of a Fritz.) There is a long code message through, and the quartermaster brings it—a jumble of helplessly ugly consonants that looks as though the German Fleet, at last, is out—but resolves (after a wearisome cryptic wrestle) to back-chat that has little of interest for us. Poldhu has the

reports of the day—mines and derelicts, wreckage, the patrols, and enemy submarines in the channels. Chart work for a while. The wrecks and the derelicts are figured and placed, and we dally with the subs, plotting and measuring to find a clue to their movements. 'Fifteen hours at six, and ten to come or go! *Mmm!* That 'll be the same swine working to the nor'east. Hope he makes a good course into the minefield! This one is solo—and that! A ghastly bunch, anyway!' We project a line of our course, but hesitate at position. 'Not one decent observation in the last three days. Only a muggy guess at a horizon. Dead-reckoning? Of course, there is our dead-reckoning, but—but—wonder where the commodore got his position from? Must have added on th' day of th' month, or fingers and toes or something! Damned if we can see how, at twelve knots, we could be where——'

The outspread chart, glaring white under the electric light, with a maze of heights and soundings, grows strangely indistinct, and it calls for an effort to set the counts and figures in their places. We realize that wandering thought and a warm chart-room are not the combination for wakefulness. So, on deck again, to steady up at the doorway and wonder why the night has become suddenly as hellish black as the pit!



**THE BOWS OF THE KASHMIR DAMAGED
BY COLLISION**

The second officer has found his composure at the bottom of a cup of steaming coffee, and seems mildly astonished that we are unable to pick up *Neleus* in the darkness ahead. "Quite plain, sir, when these squalls pass. A bit murky while they blow over, but—see her clear enough, sir. Reduced two revolutions, and keeping good station on her at that!" Somewhat slowly (for we have been afoot since six yesterday morning) our eyes focus to the gloom and line out the sea and sky in their shaded proportions. *Neleus* grows out of the sombre opacous curtain—a definite guide with the sea breaking white in her wake. Dark patches of smoke-wrack, around and about, mark bearings on the sea-line where our sisters of the convoy are forging through. The next astern has dropped badly in cleaning fires, and is now throwing a whirl of green smoke in the effort to regain her station. The sea seems to have lessened since last we viewed it. Our hot coffee may have had effect in producing a more impressionable frame of mind, but certainly the weather is no worse. The rain and sleet have beaten out a measure of the toppling sea-crests. We see the fore-castle-head, black and upstanding, for longer periods, and only broken spray flies over, where, but a little ago, were green whelming seas. A sign of modest content comes from the boat-deck, where the guards are humming, "*Over there, over there, over there! Th' Yanks are coming!*"

The duty officer (troops) comes to us to pass the time of the morning. He salutes with punctilio. (He has not yet learned that we are only a damn civilian, camouflaged, and not entitled to such respect.) It is reported to him that one of the ship's boats had been badly damaged by a sea during the night. "In event of—of an accident, is it in orders that the troops allocated [his word] to that boat shall not go in any other?"

Good lad! For all that darkness and the gale, he looks very fine and bold, standing stiffly, if somewhat unsteadily, demanding detail of the Birkenhead Drill! We assure him that there will be no immediate need for regrouping the men, that measures have already been taken to repair the damaged planking, that half an hour of daylight will serve us—and turn the talk to less disquieting affairs. He is very keen. Till now he has never been farther out to sea than the Iron Steamboat Company would take him—to Coney Island or the more subdued delights of the Hook. A New-Yorker, he tempers quite natural vaunts to be the more in keeping with the great and impending trial that awaits. For all that, he is gravely concerned that we should

recognize his men as good and true—"the best ever, yessa!" With a good experience of their conduct, under trying conditions, we assent.

". . . They kin number us up all they wanna, but we're the—th N' Yok National Guard—a right good team! Down there on th' Mexican barder, we sure got trimmed, good and planny! Hot! My! Saay, cap'n, I guess— Ah well, a' course you've been through some heat, too—but it was sure some warm hell down there! Yes—sir!" A bright lad!

His words recall to us a windy afternoon on Fifth Avenue, in the days when our Uncle Sam was dispassionate and neutral. Flags whipping noisily in the high breeze, the crowds, the bands, and the long khaki column in fours winding towards the North River ferries to embark for Mexico, on a task that called for inhuman restraint. Newsboys were shouting aloud the peril of Verdun, and the thought came to us then—"Will that stream of manhood ever march east?" And now, under our feet and in our charge, fourteen hundred—"the best ever, yessa!"—are bound east by every thrust of the screw, and out on the heaving waste of water around us are fifteen thousand more; and the source is sure, and the stream, as yet, is but trickling.

ON OCEAN PASSAGE

THE weather has certainly moderated. In but an hour the sea has gone down considerably. There is no longer height enough in the tumble of it to throw us about like a Deal lugger. We steam on a more even keel; the jar and racket of the racing propeller has altered to a steady rhythmic pulse-beat that thrusts our length steadily through the water. At times the rain lashes over and shuts out sight of our neighbours, but we have opportunity to regulate our station in the lengthening intervals between the squalls. Improvement in the wind and sea has brought our somewhat scattered fleet into better and closer order. The rear horse-transport have come up astern and seem to have got over the steering difficulties that their high topsides and small rudder-immersion effected in the heavier sea. Only the barometer shows no inclination to move, in keeping with the better conditions—the rain, perhaps, is keeping the mercury low.

It seems plain sailing for a while. The Second can look out for her; no use having too many good men on the bridge. We are only in the way out here, stamping and turning on the wet foot-spars, or throwing bowlines in the 'dodger' stops to pass the night. Four bells—two a.m.—the time goes slowly! We are somewhat footsore. Perhaps, sea-boots off, a seat for a minute or two in the chart-room may ease our limbs for the long day that lies before us.

A long day, and the best part of another long day before we reach port! A wearisome stretch of it! We ought to have some system of relief. Why not? Why not take a relief? The chief officer is as good a man as the master. Why not let him run the bus for a spell? Oh, just—just—just a rotten way we have of doing! In the Navy they make no bones about turning over to their juniors; why should we make it so hard for our— "*Says it is hazy, sir! Told me to let you know he hasn't seen any of the ships for over an hour!*"

Whatever is the man talking about! "*Ships?*" What ships? "*An hour?*"

The quartermaster, in storm-rig of dripping oilskin, stands sheepish in the doorway. "Aff-past-three, sir," he says.

"*Htt!*" In drowsy mood we don oilskin and sea-boots. Overhead the rain is drumming, heavy and persistent, on the deck. A glance at the barometer shows an upward spring. *Tip, tip, tip*—a good glass, that! Well-balanced! The Second is apologetic, almost as though his was the hand that had accidentally turned the tap. "Been like this for over an hour, sir! Was always hoping it would pass off, but there has been no sign of clearing. Would have called you sooner, but thought it would lift. I've kept her steady at average revolutions for the last eight hours' run—seven-three. Haven't seen a thing since shortly after you went below." A query brings answer that the fog-buoy has been streamed and gun's crew cautioned to a sharp look-out astern. Not that there is great need; our sailing experience has been that A ——— will drop astern when 'the gas is turned down!'

The wind has fallen and has hauled to south. It is black dark, with a heavy continuous downpour of rain. The air is milder, and the sea around has a glow of luminous milky patches. So, it is to be southerly weatherly for making the land! It might be worse! At least, this thrash of heavy rain will 'batten hatches' on a rise of the sea, and make a good parade-ground for our

destroyer escort when they join company. We should be able to shove along at better speed when daylight comes. The mist or the haze or whatever combination it may be, is puzzling. From the outlook it is not easy to gauge the range of our vision. Near us the wash from our bows is sharply defined by phosphorescence in the broken water, a white scum churns and curls alongside, brightening suddenly in patches as though our passage had set spark to the fringe. Outboard the open sea merges away into the gloomy sky with no horizon, no ruling of a division. We seem to be steaming into a vertical face of vapour. There is no sound from the ships around us, not a light glimmers in the darkness. The eerie atmosphere through which we pass has effect on the night-life of the ship. On deck there is an inclination to move quietly, to preserve a silence in keeping with the weird spell that seems to environ us. There is no longer chatter and small talk among the duty troops; they sit about, huddled in glistening *ponchos*, peering out at the ghostly glow on the water. From far down in the bowels of the ship the rattle of a stoker's shovel on the plates rings out in startling clamour, and rouses an instant desire to suppress the jarring note. It seems impossible that there can be ships in our company—vessels moving with us through mystic seas. We peer around, on all the bearings, but see nothing on our encircling wall. Smell? We nose at the air, seeking a waft of coal-smoke, but the rain is beating straight down, basting the funnel-wraiths on the flat of the sea.

An average of eight hours' steaming, seven-three revolutions, may be no good guide, considering the racing and the plunging we have gone through. In proper station we ought to see the loom of *Neleus* ahead, or, at least, the wash of her fog-buoy. It is important that we should be in good touch at daybreak. We go full speed for a turn or two and post an officer in the bows to scan for our leader.

New and vexing problems come at us as time draws on. We are due to start a zigzag, 'in execution of previous orders,' before the day breaks. We see a royal 'hurrah's nest'—a rough house—before us if we lay off without a proper sight of our fellows. So far there has come no negative to our orders; we are somewhat concerned. A message cannot have been missed, surely! "Nothing through yet, sir," is the wakeful assurance from the wireless operator. "X's fierce with this rain, but should get any near message all right."

At eight bells we come in sight of one unit of the convoy. She shows up, broad off on our lee bow, in a position we had hardly looked for. There is little to see. A darkling patch, a blurred shadow, in the face of sea and sky, with a luminous curl of broken water astern. We cannot identify her in the darkness; flashing signals are barred in the submarine areas; we must wait daylight for recognition. She should be *Neleus*, but a hair-line on our steering-card may have brought us to the leader of the outside column. In any case we are in touch, and it is with some relief we ease speed to a close approximation of hers. Anon, our anxiety about the zigzag is dispelled by a message from the commodore, cancelling former orders. He has sat tight on it to nearly the last minute, hoping for a clearance.



THE MAYFLOWER QUAY, THE BARBICAN, PLYMOUTH

With the coming of the chief officer's watch we feel that the 'day' is beginning. Twelve to four are unholy hours that belong to no proper order of our reckoning. They are past the night, and have no kinship with the day: bitter, tedious, helpless spaces of time that ought only to be passed in slumber and oblivion. By five, and the lift greying, there is something in the movement about the decks that suggests an awakening of the ship to busy life and action, after the sullen torpor of an uneasy night. The troop 'fatigue

men' turn out to their duties, and traffic to the cooking-galleys goes on, even under the unceasing downpour that falls on us. The guard get busy on their rounds, challenging the men as they step out of the companionways, to show their life-belts in order and properly adjusted. Complaint and discussion are frequent, but the guard are firm in their insistence. "I should worry!" is the strange request, appeal, exhortation, demand, reply, aside, that punctuates each meeting on the decks below. In nowise influenced by the sinister import of the questioning, the duty troops on the boat-deck waken up. The spirit of matutinal expression descends on them, despite the rain, and they whistle cheerful 'harmonic discords,' till barked to silence by Sergeant 'Jawn.'

The watch on deck trail hoses and deck-scrubbers from the racks and set about preparations for washing down, bent earnestly on their standard rites though the heavens fall! The carpenter and his mate are assembling their gear and tools, awaiting better daylight to get on with their repairs to the damaged lifeboats. On the bridge we seem congested. Extra 'day' look-outs obstruct our confined gangways and the bulk of their weather harness, plus life-belts and megaphones, restricts a ready movement. In preparation for busy daylight, the signalmen put out their bunting on the lettered hooks, and ease off the halyards that are set 'bar-tight' by the soaking rain. There is, withal, an air of freshness in the morning bustle that comes in company with the dawn.

With gloom sufficient for our signal needs (and light enough for protection) we flash a message to our consort. She is *Neleus*, and answers that she has other vessels of the convoy in sight to leeward. We sheer into our proper position astern of her and find the outer column showing through the mist in good station. On our report that we had no others in sight, *Neleus* alters course perceptibly to converge on the commodore, and daylight coming in finds us steaming in misty but visible touch with the other columns. The horse transports have dropped astern, and one is bellowing for position. She gets a word or two on the 'buzzer,' comes ahead, and lets go the whistle lanyard.

If commodore's reckoning is right, we should now be on the destroyer rendezvous, but our wireless operator, who has been listening to the twitter of the birds, assures us that they are yet some distance off. We hope for a

clearing to enable them to meet us without undue search; it will not be a simple matter to join company in the prevailing weather conditions, particularly as we are working on four days of dead-reckoning. By seven o'clock there is no sign of the small craft, and we note our ocean escort closing in to engage the commodore with signals. The rain lessens and turns to a deep Scotch mist, our range of vision is narrowed to a length or two. Anon, our advance guardship sets her syren sounding dismal wails at long intervals, as she swings over from wing to wing of the convoy.

By what mysterious channel does information get about a ship? Is there a voice in the aerials? Are ears tuned to the many-tongued whisperings of rivet and shell-plate, that all hands have an inkling of events? The rendezvous is an official secret; the coming of the destroyers is supposedly unknown to all but the master, the navigators, and the wireless operator, but it is not difficult to see a knowing expectancy in the ranks of our company. Despite the wet and clammy mist, ignoring the dry comforts of the "tween-decks," the troops crowd the upper passages and hang long over the rails and bulwarks, pointing and shouting surmise and conjecture to their mates. The crew are equally sensitive. Never were engine-room and stokehold ventilators so tirelessly trimmed to the wind. At frequent intervals, one or other of the grimy firemen ascends to the upper gratings, cranks the cowls an inch or two this way or that, then stands around peering out through the mist for first sight of a welcome addition to our numbers. The official ship look-outs are infected by a new keenness, and every vagary in the wind that exposes a glimpse of our neighbours is greeted by instant hails from the crow's nest.

Eight bells again! The watch is changed and, with new faces on the bridge, the length of our long spell is painfully recalled. With something of envy we note the posts relieved and the men gone below to their hours of rest. "What a life!" The wail of the guardship's syren fits in to our mood—*Wh-o-o-owe!*

Quick on the dying note a new syren throws out a powerful reedy blast, sounding from astern. Thus far on the voyage, with fog so long our portion, we have come to know the exact whistle-notes of our neighbours, down to the cough and steam splutter of the older ships. This is new—a stranger—a musical chime that recalls the powerful tug-boats on the Hudson. Our New-

Yorker troops are quick to recognize the homely note. "Aw! Saay!" is the chorus. "Lissen! Th' *Robert E. Lee!* "

The rear ships of the convoy now give tongue—a medley of confused reverberations. No reply comes to their tumult, but a line of American destroyers emerges from the mist astern and steams swiftly between the centre columns. There is still a long swell on the sea and they lie over to it, showing a broad strake of composition. They are bedizened in gaudy dazzle schemes, and the mist adds to the weird effect. The Stars and Stripes flies at each peak, standing out, board-like, from the speed of their carriers. As they pass, in line ahead, a wild tumult of enthusiasm breaks out among the troops. They join in a full-voiced anthem, carried on from ship to ship, "The Star-spangled Banner!"

'ONE LIGHT ON ALL FACES'

A SLIGHT lift in the mist, edging from sou'west in a freshening of the wind, extends our horizon to include all ships of the convoy. With this modest clearing, the shield of vapour that has cloaked us from observation since early morning is withdrawn. Although still hazy, there is sight enough for torpedo range through a periscope, and the long-delayed zigzag is signalled by the commodore.

There is no time lost in settling to the crazy courses. At rise of the mist we are steaming through the flat grey sea in parallel columns, our lines ruled for us by the wakes of our leaders. The contrasts of build and tonnage, the variegations of our camouflage, are dulled to a drab uniformity by the lingering mist, and we make a formal set-piece in the seascape, spaced and ordered and defined. The angle of the zigzag disturbs our symmetry. As one movement, on the tick of time, we swing over into an apparent confusion, like the flush of a startled covey. We make a pattern on the smooth sea with our stern wash. Wave counters wave and sets up a running break on the surface that draws the eye by its similarity to a sheering periscope; not for the first time we turn our glasses on the ripples, and scan the spurt of broken water in apprehension.

Our escort is now joined by British sloops returning from their deep-sea patrols. The faster American destroyers spur out on the wings and far ahead, leaving the less active warships to trudge and turn in rear of the convoy. With our new additions, ship by ship steering to the east, we make a formidable international gathering on the high seas, a powerful fleet bringing the Pilgrim sons back over the weary sea-route of their fathers' *Mayflower!*

Having far-flung scouts to safeguard our passage, there seems no reason for concern about our navigation, but the habits of a sea-routine urge us to establish a position—to right the uncertainty of four days' dead-reckoning. The mist still hangs persistently about us, but there is a prospect that the sun may break through. The strength of the wind keeps the upper vapours moving, but ever there are new banks to close up where a glimpse of clear vision shows a 'pocket' in the clouds. The westering sun brightens the lift and plays hide-and-seek behind the filmy strata. Time and again we stand by for an observation, but, should a nebulous limb of the sun shine through, the horizon is obscured—when the sea-line clears to a passable mark, the sun has gone! A vexing round of trial after trial! We put away the sextant, vowing that no tantalizing promise shall tempt us. "Bother the sun! 'We should worry!' We have got an approximation by soundings, we can do without—we— *Look out, there!*"—we are hurrying for the instrument again and tapping 'stand by' to the marksman at the chronometer!

At length a useful combination of a clean lower limb and a definite horizon gives opportunity for contact, and it is with a measure of satisfaction we figure the result on the chart, and work back to earlier soundings for a clue to the latitude. Busied with pencil and dividers, our findings are disturbed by gunfire—the whine of a slow-travelling shell is stifled by a dull explosion that jars the ship!

On deck again; the men on the bridge have eyes turned to the inner column. The rearmost transport of that line has a high upheaval of debris and broken water suspended over her; it settles as we watch, and leaves only a wreath of lingering dust over the after part of the ship; she falls out of line, listing heavily; puffs of steam on her whistle preface the signal-blasts that indicate the direction from which the blow was struck. From a point astern of us a ruled line of disturbed water extends to the torpedoed

ship—the settling wake of the missile! The smack and whine of our bomb-thrower speaks out a second time, joined by other vessels opening fire.

Events have brought our ship's company quickly to their stations. The chief officer stands, step on the ladder, awaiting orders. "Right! Lay aft! Cease fire, unless you have a sure target! Look out for the destroyers blanking the range!" He runs along, struggling through the mass of troops. The men are strangely quiet; perhaps the steady beat of our engines measures out assurance to them—as it does to us. Their white-haired colonel has come to the bridge, and stands about quietly. Other officers are pushing along to their stations. There is not more than subdued and controlled excitement in a low murmur. The men below crowd up the companionways from the troop-decks. In group and mass, the ship seems packed to overflowing by a drab khaki swarm; the light on all faces turned on the one cant, arms pointing in one direction, rouses a haunting disquiet. However gallant and high of heart, they are standing on unfamiliar ground—at sea, in a ship, caged! If—

Two destroyers converge on us at frantic speed, tearing through the flat sea with a froth in their teeth. As the nearest thunders past, her commander yells a message through his megaphone. We cannot understand. Busied with manœuvres of the convoy, with the commodore's signal for a four-point turn, we miss the hail, and can only take the swing and wave of his arms as a signal to get ahead—"Go full speed!" The jangle of the telegraph is still sounding, when we reel to a violent shock. The ship lists heavily, every plate and frame of her ringing out in clamour with the impact of a vicious sudden blow. She vibrates in passionate convulsion on recovery, masts oscillate like the spring of a whip-shaft, the rigging jars and rattles at the bolts, a crash of broken glass showers from the bridge to the deck below!

The murmur among the troops swells to a higher note, there is a crowding mass-movement towards the boats. The guard is turned to face inboard. The colonel is impassive; only his eyes wander over the restless men and note the post of his officers. He turns towards us, inquiringly. What is it to be? His orderly bugler is standing by with arm crooked and trumpet half raised.

Our lips are framing an order, when a second thundering shock jars the ship, not less in violence and shattering impact than the first. A high hurtling column of water shoots up skyward close astern of the ship. We suppress the order that is all but spoken, stifle the words in our throat. We are not torpedoed! Depth-charges! The destroyers' work! At a sign, the bugler sounds out "*Still!*" and slowly the tumult on deck is arrested.

The commodore's *half-right* has been instantly acted on, and we are steadied on a new course, bearing away at full speed, with the torpedoed horse transport and the racing, circling destroyers astern. Suddenly our bows begin to swing off to port, falling over towards the outer column. The helmsman has the wheel hard over against the sheer; we realize that our steering-gear has gone; the second depth-charge has put us out of control. We swing on the curve of a gathering impetus—it is evident that the rudder is held to port; converging on us at full speed, the rear ship of the outer column steams into the arc of our disorder!

The signalman is instant with his 'not under command' hoist, the crew are scattered to throw in emergency gear, but there is no time to arrest the sheer. The first impulse is to stop and go astern. If we arrest the way of the ship, a collision is inevitably assured, but the impact may be lessened to a side boarding, to damage that would not be vital; if we swing as now, we may clear—our eye insists we should clear. If our tired eyes prove false, if the strain of a long look-out has dulled perception, our stem will go clean into her—we shall cut her down! Reason and impulse make a riot of our brain. The instinct to haul back on the reins, to go full astern on the engines, is maddening. Our hand curves over the brass hood of the telegraph, fingers tighten vice-like on the lever; with every nerve in tension, we fight the insane desire to ring up and end the torturing conflict in our mind!

A confusion of minor issues comes crowding for settlement, small stabs to jar and goad in their trifling. There is a call to carry on side-actions. Every bell on the bridge clamours for attention. The engine-room rings up, the chief officer telephones from aft that the starboard chain has parted, the rudder jammed hard to port. From the upper spars, the signalman calls out a message from an approaching destroyer—"What is the matter? Are you torpedoed?" Through all, we swing out—swiftly, inexorably!

Troops and look-outs scurry off the forecastle-head, in anticipation of a wrecking blow. On the other ship, there is outcry and excitement. She has altered course and her stern throws round towards us, further encroaching on the arc of our manœuvre. So near we are, we look almost into the eyes of her captain as we head for the bridge. Troops, the boat-guard, are scrambling aboard from the out-swung lifeboats, their rifles held high. On her gun-platform the gunners slam open their breech, withdraw the charge, and hurry forward to join the mass of men amidships. All eyes are centred on the narrowing space of clear water that separates us, on our high sheering stem that cuts through her out-flung side-wash.

Strangely the movement seems to be all in our sweeping bow. The other vessel appears stationary, inert—set motionless against the flat background of misty cloud; our swinging head passes point upon point of the chequered camouflage on her broadside; subconsciously we mark the colours of her scheme—red and green and grey. We clear her line of boats, and sway through the length of her after-deck—waver at the stern-house, then cover the grey mounting of her gun-emplacement. In inches we measure the rails and stanchions on her quarter, as our upstanding bow drives on. Tensely expectant, our mind trembles on the crash that seems inevitable.

It does not come. Our eye was right—we clear her counter! With some fathoms to spare we sheer over the thrash of her propellers, the horizon runs a line across our stem, we have clear yielding blue water under the bows!

The illusion of our sole movement is reversed as the mass of the other vessel bears away from us. The unbroken sea-line offers no further mark to judge our swing; we seem to have become suddenly as immobile as a pier-head, while our neighbour starts from our forefoot in an apparent outrush, closing and opening the line of her masts and funnels like shutting and throwing wide the panels of a door.

With no indecision now we pull the lever over hood of the telegraph. One case is cleared; there still remains the peril of the lurking submarine. The destroyers are busy on the chase, manœuvring at utmost speed and exploding depth-charges in the area. We are now some distance from them but the crash of their explosion sends an under-running shock to us still. Our sheer has brought us broadside on to the position from which the

enemy loosed off his torpedo. At full astern we bring up and swing over towards the receding convoy. If we are barred from carrying on a zigzag by the mishap to our helm, we can still put a crazy gait on her by using the engines. Backing and coming ahead, we make little progress, but at least we present no sitting target.

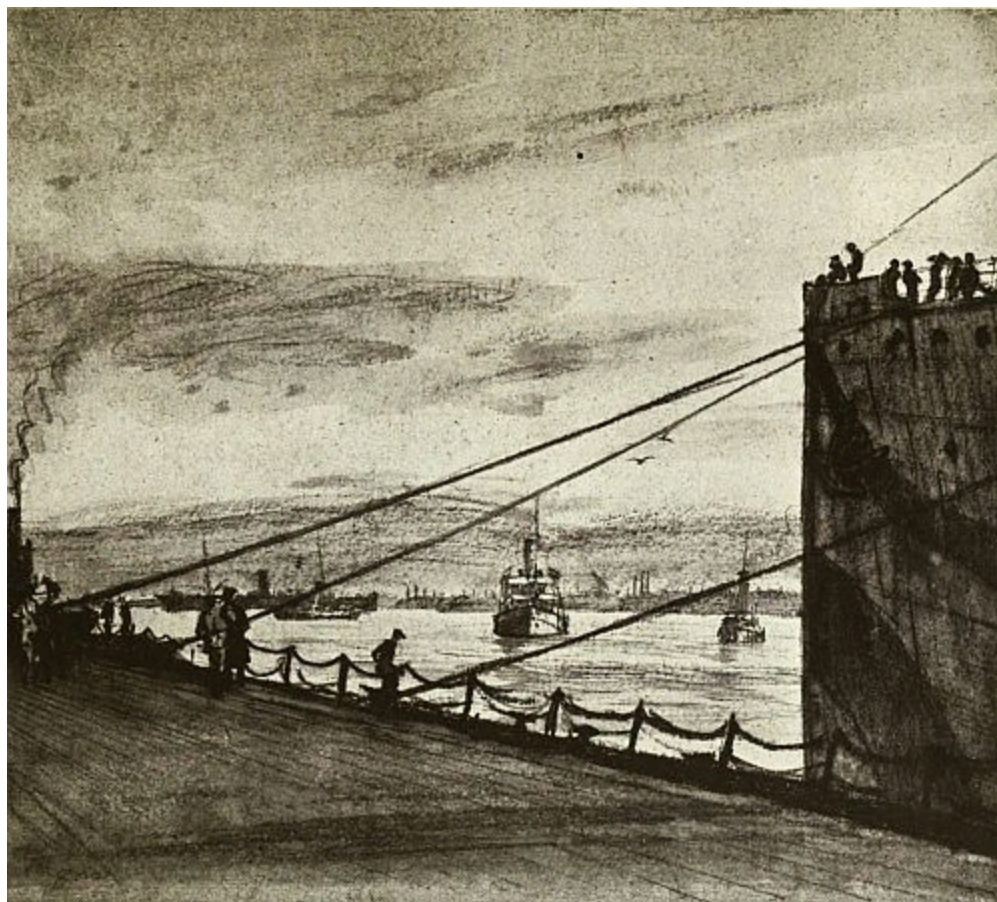
Reports come through from aft that the broken chain, springing from a fractured link, has jammed hard under the quadrant; the engineers are at work, jacking up to release the links; they will be cleared in ten minutes! The chief asks for the engines to be stopped; sternway is putting purchase on the binding pressure of the rudder. Reluctantly we bring up and lie-to. In no mood to advertise our distress, we lower the 'not under command' signals, and summon what patience may be left to us to await completion of repairs.

A long 'ten minutes!' Every second's tick seems fraught with a new anxiety. Fearfully we scan the sea around, probing the line of each chance ripple for sight of an upstanding pin-point. Anon, steam pressure rises and thunders through the exhaust, throwing a battery of spurting white vapour to the sky, and letting even the sea-birds know we are crippled and helpless.

The torpedoed ship still floats, though with a dangerous list and her stern low in the water. A sloop is taking her in tow, and we gather assurance of her state in the transport's boats still hanging from the davits; they have not abandoned. She falters at the end of the long tow-rope and sheers wildly in the wake of her salvor. The convoy has vanished into the grey of the east, and only a lingering smoke-wreath marks the bearing where they have entered the mist. The sun has gone, leaving but little afterglow to lengthen twilight; it will soon be dark. Apparently satisfied with their work the destroyers cease fire; whether there is oil on their troubled waters we cannot see. They linger a while, turning, then go on in the wake of the convoy. One turns north towards us, with a busy windmill of a signalman a-top the bridge-house. "*What is the matter? Do you wish to be towed?*" We explain our case, and receive an answer that she will stand by, "*but use utmost dispatch effect repair.*"

'Use utmost dispatch!' With every minute, as the time passes, goes our chance of regaining our station in the convoy; we are in ill content to linger!

We have a liking for our chief engineer—a respect, an admiration—but never such a love as when he comes to the bridge-ladder, grimy, and handling his scrap of waste. "They're coupling up now! A job we had! Chain jammed and packed under th' quadrant, like it had been set by a hydraulic ram! If that one landed near Fritz, he'll trouble us no more!"

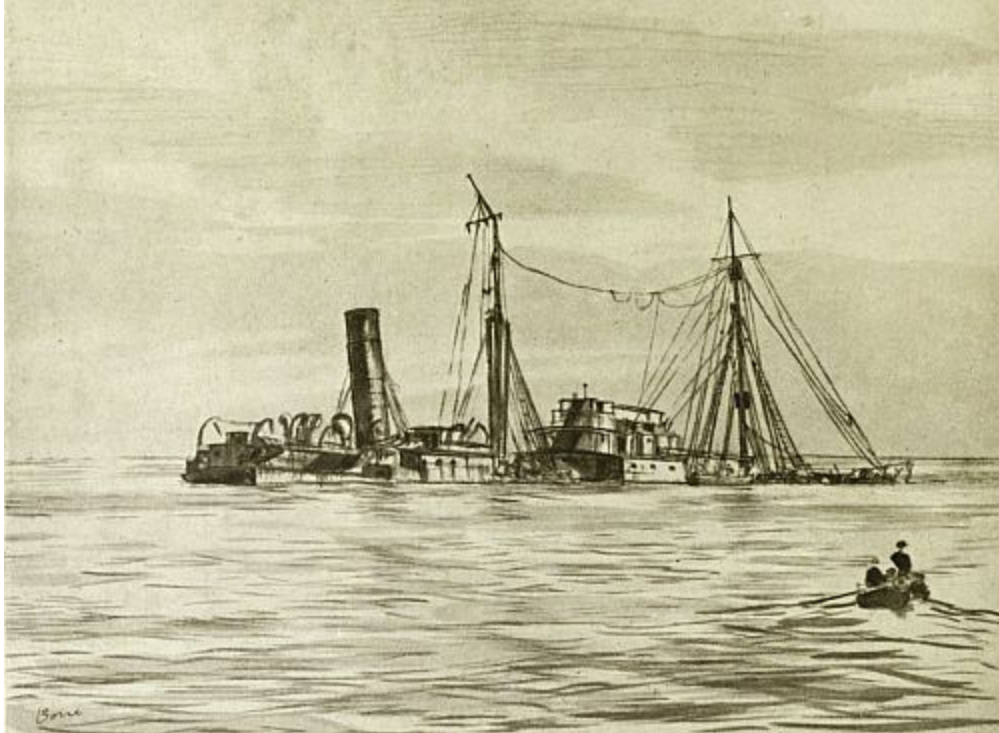


EVENING: THE MERSEY FROM THE LANDING-STAGE

With the engines turning merrily, and helm governance under our hand, we regain composure. Our task is yet none too easy. Even at our utmost speed we cannot now rejoin the convoy before nightfall; snaking through the ships in the dark to take up station offers another harassing night out! Still, it might be worse—much worse! We think of the torpedoed ship towing so slowly abeam—of the khaki swarm on our decks, 'the light on all faces turned on one cant.' Surely our luck is in! The infection of the measured beat in our progress recalls a job unfinished; we step into the chart-room and take up pencil and dividers.



THE STEERSMAN



THE WORK OF A TORPEDO

XXIII

'DELIVERING THE GOODS'

OCTOBER on the Mersey is properly a month of hazy autumn weather, but the few clear days seem to gain an added brilliance from their rarity, and present the wide estuary in a vivid, clear-cut definition. The distant hills of North Wales draw nearer to the city, and stand over the slated roofs of the Cheshire shore as though their bases were set in the peninsula. Seaward the channel buoys and the nearer lightships are sharply distinct, cutting the distant sea-line like the topmast spars of ships hull down. Every ripple and swirl of the tide is exaggerated by the lens of a rare atmosphere; the bow wash of incoming vessels is thrown upward as by mirage.

On such a day a convoy bears in from the sea, rounding the lightships under columns of drifting smoke. Heading the merchantmen, the destroyers and sloops of the escort steam quickly between the channel buoys and pass in by New Brighton at a clip that shows their eagerness to complete the voyage. A sloop detaches from the flotilla and rounds-to off the landing-stage. Her decks are crowded by men not of her crew. Merchant seamen are grouped together at the stern, and a small body of Uncle Sam's coloured troops line the bulwarks in attitudes of ease and comfort. They are a happy crowd, and roar jest and catchword to the passengers on the crossing ferries. The merchantmen are less boisterous. They watch the preparations of the bluejackets for mooring at the stage with a detached professional interest; some of them gaze out to the nor'ard where the transports of the convoy are approaching. Doubtless their thoughts are with the one ship missing in the fleet—their ship. The sloop hauls alongside the stage and a gangway is passed aboard. Naval transport officers and a major of the U.S. Army staff are waiting, and engage the commander of the man-of-war in short conversation. The men are disembarked and stand about in straggling groups. There is little to be said by the sloop's commander. "A horse transport torpedoed yesterday. No! No losses. Tried to tow her for a bit, but had to cast off. She went down by the stern."



TRANSPORTS DISCHARGING IN LIVERPOOL DOCKS

The trooper horse-tenders are marshalled in some order and pass over to the waiting-rooms under charge of the American officer. With a word or two and a firm handshake to the sloop's commander, the master of the torpedoed ship comes ashore and joins his men. No word of command! He jerks his head in the direction of the Liver Buildings and strides off. The seamen pick up their few bundles of sodden clothing and make after him, walking in independent and disordered groups. As they straggle along the planking of the stage, a

military band—in full array—comes marching down from the street-way. They step out in fine swing, carrying their glittering brasses. "Here, Bill," says one of the seamen, hitching his shoulder towards the burdened drummers, "who said we was too late for th' music!"

The transports have come into the river. Every passing tug and ferry-boat gives *rrr—oot* on her steam-whistle to welcome them as they round-to off

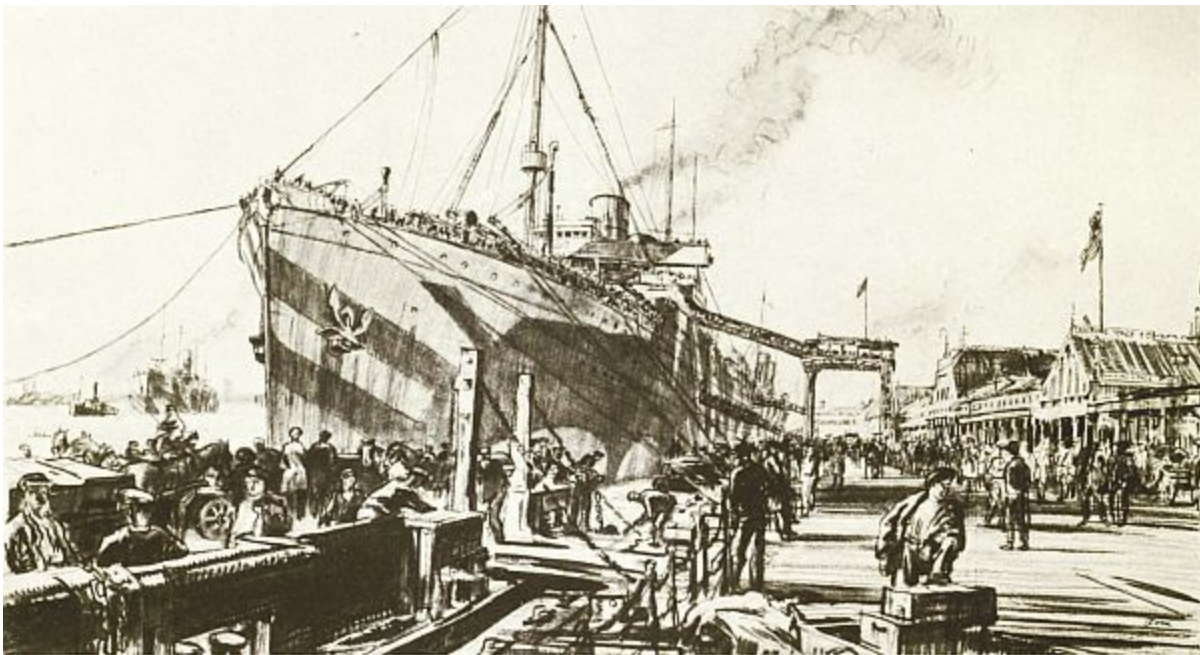
the docks and landing-stage. Loud bursts of cheer and answering cheer sound over the water. The wide river, so lately clear of shipping, seems now narrowed to the breadth of a canal by the huge proportions of the liners bringing up in the tideway. The bizarre stripes and curves and the contrasted colours of their dazzle schemes stand out oddly against the background of the Cheshire shore. It is not easy to disentangle the lines of the ships in the massed grouping of funnel and spar and high topsides. They are merged into a bewildering composition with only the mastheads and the flags flying at the trucks to guide the eye in attempting a count. Fifteen large ships, brimming at the bulwarks with a packed mass of troops, all at a deep draught that marks their load below decks of food and stores and munitions.

The landing-stage becomes rapidly crowded by disembarkation officers and their staffs. Transport wagons and cars arrive at the south end and run quietly on the smooth boarding to their allotted stands. A medical unit, gagged with fearsome disinfectant pads, musters outside their temporary quarters. Most prominent of all, tall men in their silver and blue, a sergeant and two constables of the City police stand by—the official embodiment of law and order.

A flag is posted by the stage-men at the north end, and its flutter calls an answering whistle-blast from the nearest transport. Steadily she disengages from the press of ships and closes in towards the shore. The tugs guiding her sheer strain at the hawsers and lie over in a cant that shows the tremendous weight of their charge. A row-boat dances in the wash of their screws as it is backed in to the liner's bows to pass a hawser to the stage. Sharp, short blasts indicate the pilot's orders from the bridge: the stage-master keeps up a commentary on the manœuvres through a huge megaphone. Stir and bustle and high-spirited movement! The troops that pack the liner's inshore rails give tongue to excited gaiety. A milkgirl (slouch hat, trousers and gaiters complete) passes along the stage on her way to the restaurant and is greeted with acclaim, "Thatta gel—thatta goil—oh, you kid!" The policemen come in for it: "Aw, say! Looka th' guys 'n tha lodge trimmings. What's th' secret sign, anyway!" An embarrassed and red-faced junior of the Transport Service is forced to tip it and accept three cheers for "th' Brissh Navy!"

The opening bars of 'The Star-spangled Banner' brings an instant stop to their clamour. The troops spring to attention in a way that we had not observed before in their own land. The spirit of patriotism, pronounced in war! 'God Save the King' keeps them still at attention. As strong as war and patriotism—the spirit of a new brotherhood in arms!

The transport makes fast and high gantries are linked to a position on the stage and their extensions passed on board. The stage-men make up their heaving-lines and move off to berth a second vessel at the south end. The tide is making swiftly in the river, and there must be no delay if the troops are to be disembarked and the ships cast off in time to dock before high water has passed.



TROOP TRANSPORTS DISEMBARKING AT THE LANDING-STAGE, LIVERPOOL

Viewed from the low tidal stage, almost at a level with the water, the ship—that had appeared so delicate of line in the river—assumes a new and stronger character at close hand. The massive bulk of her, towering almost overhead, dwarfs the surrounding structures. The shear that gave her beauty at a distance is lost in the rapid foreshortening of her length: her weathered plating, strake upon strake bound by a pattern of close rivet-work, attracts the eye and imposes an instant impression of strength and seaworthiness. On her high superstructure the figures of men seem absurdly diminished. The sense of their control of such a vessel is difficult of realization. Pouring

from her in an apparently endless stream of khaki, her living cargason passes over the gangways.

They move rapidly from the ship to the shore. Waiting-sheds and the upper platforms are soon littered by their packs and equipment, and the troops squat on the roadway to await formation of their group. Large bodies are marched directly to the riverside station to entrain for camp, but the assortment and enumeration of most of the companies and detachments is carried through on the broad planking of the stage. In and out the mustered files of men, transport cars make a noisy trumpeting progress, piled high with baggage and stores, and each crowned by a waving party of high-spirited soldiers. A second transport is brought in at the other end of the stage, and adds her men to the throng of troops at the water-side. The disembarkation staff have work with the sheep and the goats. There is the natural desire to learn how 'th' fellers' got on in the other ship, and the two ships' complements are mixed in a fellowship that makes a tangle of the 'nominal rolls' and drives the harassed officers to an outburst of profanity. Ever and on, a block occurs on the gangways where the inevitable 'forgetters' are struggling back through the press of landing men, to search for the trifles of their kit.

A prolonged blast of her siren warns the military officers that the first transport is about to cast off, and the movement of the troops is accelerated to a hurried rush and the withdrawal of the gangways. The waiting tugs drag the ship from the stage, and she moves slowly down-stream to dock at the Sandon entrance, there to discharge the burden of her packed holds. Another huge vessel takes her place, canting in at the north end, and shortly sending out more men to the already congested landing. She carries two full battalions, and they are disembarked with less confusion than the former varied details. Forming fours, and headed by their own band, they march off up the long bridgeway to the city streets.

The tide is approaching high water and the pilots are growing anxious lest they should lose opportunity of docking on the tide. Already the dock gates are open, and the smaller vessels of the convoy have dropped out of the river into the basins. With three ships disembarked and a fourth drawing alongside, the Naval Transport officers decide that they can handle no more men on the stage, and send the remaining steamers to land their men in

dock. There, with the troops away, an army of dockers can get to work to unload the store of their carriage from overseas.



'M N'

CONCLUSION

'M N'

S HIMMERING in gilt sunlit threads, the grey North Sea lay calm and placid, at peace with the whip of the winds after days of storm and heavy weather. The sun had come up to peer over a low curtain of vapour that hung in the east. Past the meridian, the moon stood clear-cut in the motionless upper sky. The ring of quiet sea accepted the presence of the waiting ships as of friendly incomers, familiar to the round of the misty horizon. Two British destroyers, a flotilla of motor-vessels, drifters—the brown sails of Thames barges appearing, then vanishing, in the wisps of fickle vapour. A breathless dawn. Sun, the silver moon, the grey flat sea bearing motionless ships, were witness to the drama—the giving up of the murder craft, the end of piracy.

Growing out of the mist, a squadron of British light cruisers and their convoy approached the rendezvous where the destroyers lay in readiness to take over charge of the German submarines. Two enemy transports under their commercial flags, headed the line of the water-snakes. Aircraft circled overhead and turned and returned on the line of progress. The leading ships swung out on approaching the destroyers and engaged them by signal. The destroyers weighed anchor and proceeded to carry out their orders. Each carried a number of officers and men to be placed aboard the submarines, to accept their surrender, to direct their further passage to within the booms at Harwich.

The commander of *Melampus* focused his glasses on the eleventh submarine of the long straggling line. The U-boat had a wash over his screws and was apparently steaming ahead to overtake his fellows, now fading into the mist in the direction of their prison gates.

"Our group," he said: then, to the signalman, "Tell him to stop instantly!"

The bluejacket stood out on the sparring of the bridge and signalled with his hand-flags. The submarine still moved ahead at speed, his exhaust

panting at pressure. The German commander could not (or would not) understand, and it was necessary to hoist 'M N' of the International Code. The two flags were sufficient: he threw his engines astern and brought up to await further orders. His followers arrived on the station. Some cast anchor, others slowed and stopped. All took note of the flags—St. Andrew's cross over blue and white checquers, hoisted at the destroyer's yard-arm—and obeyed the summary signal.

'M N!' International Code! The old flags of the days when there was peace on the sea, when the German commercial ensign was known and familiar and respected in the seaports of the world!

How many of the Germans would understand the full significance of the hoist that brought them to a standstill—the import of the flags drooping in the windless air—the beckoning of the coloured fabric that ended their murder trade. The day had long passed since they had used this warning signal for a procedure in law and order. No 'M N' to *Lusitania* before littering the Irish Sea with wreckage and the pitiful bodies of women and small children: no signal to *Arabic* or *Persia*: no warning to *Belgian Prince*, to *California*, to all the long and ghastly list: no summons to the hospital ships—alight and blazoned to advertise their humane mission. And now—their ensign dishonoured, their name as seamen condemned to the everlasting tale of infamy, their proud commercial seafaring destroyed—to come in with the blood on their hands, and render and submit to the mandate of a two-flag hoist!

'M N!' The Code of the Nations! The summons to peaceful seafarers! 'Stop instantly!' Disobey at your peril! At last, at long last, the Freedom of the Seas—the security of the ships—the safety of all who pass on their lawful occasions—completely re-established by the flaunt of the old flags!

APPENDIX

COMPELLED by the nature of their work to be long absent from home ports, seamen are frequently in ignorance of the current of longshore opinion. Newspapers do not reach out to the sea-routes (as yet), and the media of Guild Gazettes and Association Reporters come somewhat late on the tide of an appreciation. The tremendous historical importance of the Nation's Thanks to its Fighting Forces (in which the Merchants' Service was included) has not adequately been realized by the merchantmen. Some do not even know of it. For these reasons—not in a spirit of 'pride above desert'—the writer quotes the following:

The Resolution of Parliament of October 29, 1917, placed upon record—

"That the thanks of this House be accorded to the officers and men of the Mercantile Marine for the devotion to duty with which they have continued to carry the vital supplies to the Allies through seas infested with deadly perils."

A year later, an equally generous appreciation of the work of the Merchants' Service was issued by the Board of Admiralty.

"On the occasion of the first Meeting of the Board of Admiralty after the signing of the German Armistice, their Lordships desire, on behalf of the Royal Navy, to express their admiration and thanks to the Owners, Masters, Officers, and Crews of the British Mercantile Marine, and to those engaged in the Fishing Industry, for the incomparable services which they have rendered during the War, making possible and complete the Victory which is now being celebrated.

"The work of the Mercantile Marine has been inseparably connected with that of the Royal Navy, and without the loyal co-operation of the former, the enemy's Submarine Campaign must inevitably have achieved its object. The Mercantile Marine from the beginning met this unprecedented form of warfare with indomitable courage, magnificent endurance, and a

total disregard of danger and death, factors which the enemy had failed to take into account and which went far towards defeating his object.

"In no small measure also has the success achieved against the submarine been due to the interest taken by Owners in the defensive equipment of their ships, and to the ability, loyalty, and technical skill displayed by Masters and Officers in carrying out Admiralty regulations which, though tending to the safety of the vessels from submarine risks, enormously increased the strain and anxiety of navigation. The loyal observance of these precautions has been the more commendable since the need for absolute secrecy, on which safety largely depended, has prevented the reasons for their adoption being in all cases disclosed.

"Further, the Convoy System, which has played such an important part in frustrating the designs of the enemy and securing the safe passage of the United States Army, could never have attained its success but for the ability and endurance displayed by Masters, Officers, and crews of the Merchant Service forming these Convoys. This system has called for the learning and practising of a new science—that of station-keeping—the accuracy of which has depended in no small measure on the adaptability and skill of the Engineers and their Departments.

"Their Lordships also desire to acknowledge the ready response of Owners to the heavy calls made on the Merchant Service for Officers and men to meet the increasing requirement of the Navy. On board our ships of every type, from the largest Dreadnought down to the smallest Patrol Boat are to be found Officers and men of the Merchant Navy who have combined with those of the Royal Navy in fighting the enemy and defeating his nefarious methods of warfare at sea.

"The Merchant Service and the Royal Navy have never been so closely brought together as during this War. In the interests of our glorious Empire this connection must prove a lasting one."

The Resolution of Parliament of August 6, 1919, placed upon record—

"That the thanks of this House be accorded to the officers and men of the Mercantile Marine for the fine and fearless seamanship by which our people have been preserved from want and our cause from disaster."

INDEX

ABERDEEN, [82](#)

Admiralty, [xiii](#), [11](#), [32](#), [35-37](#), [41](#), [42](#), [64](#), [110](#), [113](#), [144](#), [178](#), [195](#), [255](#)

Adriatic, [25](#), [99](#)

Agnes Whitwell, [192](#)

Aleppo, [4](#)

"Allo," [118](#), [119](#)

Amadas, [6](#)

Anglo-Californian, [18](#)

Antwerp, [45](#), [78](#)

Aquitania, [13](#), [100](#)

Arabic, [253](#)

Arctic Ocean, [25](#)

Arklow, [82](#)

Arlington, [108](#)

Armada, The Great, [4](#), [6](#)

Atalanta, [82](#)

Atlantic, [107](#), [117](#), [128](#), [178](#), [218](#), [224](#), [225](#)

Atlantic City, [104](#), [109](#)

Augustine, [62](#), [63](#), [66](#)

Australia, [126](#)

Austrian Navy, [99](#)

Avocet, [85](#)

BABYLON, [4](#)

Backhouse, John, [5](#)

Baffin, [6](#)

Bahia, [191](#), [204](#)

Balsara, [4](#)

Barlow, [6](#)

Barnegat Lighthouse, [109](#)

Beaumaris, [78](#)

Beaverbrook, Lord, [xiii](#)

Belgian Prince, [173](#), [253](#)

Bell, Captain, [79](#)

Bengal, [9](#)

Bennett, Arnold, [xiii](#)

Bermuda, [107](#), [108](#)

Berry, [11](#)

Biggatt, William, [5](#)

Birchgrove, [85](#)

Black Middens, [78](#)

Blake, [10](#)

Board of Trade, [61-66](#), [121](#), [131](#), [185](#)

Boer War, [22](#)

Boom defences, [100](#), [191](#), [206](#)

Bordeaux, [55](#)

Boston, [108](#)

Boy Ernie, [101](#)

Boy Jacob, [78](#)

Bremen Hansa Line, [31](#)

Brennell, Captain, [85](#)

Bridgwater, [10](#)

Bristol Channel, [80](#), [116](#)

Britannic, [78](#)

British Standard, [209](#), [210](#), [211](#), [215](#)

Brixham, [101](#)

Brother Fred, [192](#)

Brownrigg, Rear-Admiral Sir Douglas, [xiii](#)

Brussels, [79](#)

Cabotia, [114](#)

Cagliari, [181](#)

Calcutta, [204](#)

Calgarian, [178](#)

California, [253](#)

Cameronia, [140](#)

Canada, [126](#), [128](#)

Canute, [141](#), [142](#)

Cape Cod, [108](#)

Cardiff, [181](#)

Caspian Sea, [4](#)

Cats, [198-99](#)

Cavendish, Thomas, [6](#)

Cerigo Channel, [114](#), [132](#)

Channel, The, [88](#), [89](#), [95](#), [96](#), [131](#), [147](#)

Charles II, [57](#)

Chaucer, [55](#)

Chelmsford, [204](#)

Cheshire, [244](#), [247](#)

China, [22](#), [165](#)

Cinderella, [210](#), [211](#)

Clyde, [xi](#), [100](#), [116](#)

Cochrane, Lord, [181](#)

Collins, Captain Greenville, [57](#), [58](#), [64](#)

Collonia, [196](#)

Commissioner, [82](#)

Coney Island, [230](#)

Constantinople, [4](#)

Contalmaison, [51](#)

Cook, James, [10](#)

Copenhagen, [11](#)

Coquet, [173](#)

Cork, [85](#)

Cornford, L. Cope, [xiii](#)

Costello (boatswain of *Gull*, trawler), [99](#)

Cottingham, [80](#), [81](#)

Coverley, Captain, [114](#)

Crane, [99](#)

Crimea, [22](#)

Crystal Palace, [122](#)

Cunard Line, [118](#), [119](#)

Custom House, [184](#), [185](#)

DÆDALUS Light, [170](#)

Dampier, [10](#)

Davis, [6](#)

Deal, [77](#), [89](#), [230](#)

Deptford, [56](#), [58](#)

Deutschland, [104](#)

Devonshire (East Indiaman), [9](#)

Dieppe, [31](#)

Dieudonné, [192](#)

Dixon, W. Macneile, [xiii](#)

Dogger Bank, [99](#)

Doiran, [133](#)

Downs, The, [9](#), [88-90](#), [96](#)

Drake, [6](#)

Drei Geschwister, [85](#)

Dublin, [15](#)

EAST India Company, [9](#)

Eddystone Lighthouse, [91](#)

Egypt, [22](#), [126](#)

Elbow Buoy, [97](#)

Emden, [164](#)

Empress of Fort William, [95](#)

Esperanto, [169](#), [203](#)

FAIR Head, [117](#)

Falaba, [164](#)

Fermo, [101](#)

Fishermen, [98-102](#), [206](#), [255](#)

Flanders, [51](#), [79](#), [126](#)

Floandi, [99](#)

Foley, [11](#)

Forbes, Captain, [63](#)

Foreign consuls, [185](#)

Formidable, [101](#)

Fortuna, [192](#)

France, [125](#), [126](#)

Franz Fischer, [85](#)

Frobisher, [6](#)

Fryatt, Captain, [79](#)

Fürst, [27](#), [28](#)

GALATZ, [170](#)

Gallipoli, [127](#)

"Gamecock" Fleet, [99](#)

Garron Head, [116](#)

German Navy, [17](#), [24](#), [26](#), [28](#), [41](#), [253](#)

Crimes on the sea, [170-74](#), [253](#)

Fishing-boats, [79](#), [100](#), [101](#)

Hospital ships, [92](#), [173](#), [253](#)

Lightships, [92](#)

Merchantmen, [17](#), [81](#), [82](#), [85](#), [114](#), [164](#), [173](#)

Mines, [92](#), [95](#)

Rafts and open boats, [92](#)

Submarine minelayers, [80](#), [92](#)

See under Merchants' Service: [German Schrecklichkeit](#), and [Submarine piracy](#)

Submarines, [17-20](#), [37](#), [41-43](#), [79-86](#), [95](#), [96](#), [107-9](#), [113-15](#), [118-19](#), [173](#), [177](#), [181](#), [182](#), [237-40](#), [252-53](#), [255-56](#)

Gibbs, Richard, [5](#)

Gibraltar, [126](#), [183](#)

Glasgow, [33](#), [39](#), [78](#), [116](#)

Goodwin Sands, [89](#), [96](#)

Gowan Lea, [99](#), [100](#)

Grand Banks, [103](#)

Gravesend, [3](#), [59](#)

Greece, King of, [63](#)

Greenwich Mean Time, [191](#)

Gulflight, [109](#)

Gull (trawler), [99](#)

HAKLUYT, Richard, [4](#), [5](#), [9](#)

Halifax, [108](#)

Hardy, [11](#)

Harwich, xi, [90](#), [253](#)

Hawkins, Sir John, [6](#)

Henderson, Algernon C. F., [v](#)

Henry VIII, [6](#)

Hohenzollern, [174](#)

Hollesay Bay, [90](#)

Holy Land, [54](#)

Horn, Cape, [77](#)

Hudson, [6](#)

Hudson Bay, [86](#)

Hudson River, [217](#), [218](#)

Hull, [90](#)

ICELAND, [170](#)

Imperial War Museum, [xiii](#)

India, [22](#), [126](#)

International Code of Signals, [169](#)

Islay, [117](#)

Isle of Man, [32](#)

Istria, [55](#)

Jane Williamson, [82](#)

Japan, [178](#)

Java, [13](#)

Johnson, Dr., [47-48](#)

Justitia, [78](#)

Karlsruhe, [104](#)

Kashmir, [227](#)

Keith, Captain, [131](#)

Kiel, [24](#)

King John, [55](#)

Kingsdown, [89](#)

King's Harbour Master, [203](#)

Kingsway, [124](#)

Kölnische Zeitung, [81](#)

Kronprinz Wilhelm, [104](#)

Lady of the Lake, [79](#)

Lamport and Holt Line, [109](#)

Langton, Stephen, Archbishop of Canterbury, [55](#)

Leggatt (of *Crane*), [99](#)

Leghorn, [181](#)

Leviathan, [135](#)

Leyland Line, [117](#), [118](#), [119](#)

Liège, [27](#)

Lightships, [57](#), [91-96](#)

Gull Lightship, [53](#), [89](#), [93](#), [96](#)

Ostend Lightship, [92](#)

Royal Sovereign Lightship, [95](#)

Shambles Lightship, [95](#)

South Goodwin Lightship, [95](#)

Linn ("frigot"), [204](#)

Liver Buildings, [49](#), [247](#)

Liverpool, [62](#), [69](#), [78](#), [103](#), [116](#), [135](#), [245](#), [249](#)

Lobelia, [100](#)

London, [89](#), [213](#)

Louis XIV, of France, [91](#)

Lowestoft, [90](#)

Lusitania, [164](#), [173](#), [253](#)

MALAY pirates, [165](#)

Maloja, [95](#)

Malta, [126](#), [128](#), [132](#), [182](#)

Manchester, [124](#)

Manchester Commerce, [164](#)

Manhattan, [223](#)

"Manual on Seamanship," [144](#)

Marconi, [123](#)

Marconi Company, [110](#)

Maréchal de Villars, [82](#)

Margaret, [15](#)

Mariston, [173](#)

Maritime Code, [54](#) seq.

Marmion, [191](#), [192](#), [211](#)

Marseilles, [128](#), [131](#), [132](#)

Mary Rose, [178](#)

Massilia, [186](#), [210](#)

Mayflower, [236](#)

Mayo, Walter H., [xiii](#)

Meadowside, [140](#)

Mediterranean, [30](#), [114](#), [127](#), [128](#), [178](#), [182](#), [196](#)

Melampus, [253](#)

Merchant Adventurers, [4](#), [5](#)

Merchants' Service:

Growth, [3-5](#);

parent of Navy, [6](#), [18](#);

Imperial significance, xii, [10](#), [12](#), [14](#), [15](#), [126](#);

unrecognized work, [10](#), [46 seq.](#);

educational function, [11](#);

introduction of steamships, [11](#), [12](#), [13](#);

international supremacy, [12](#), [13](#);

outbreak of Great War, [14](#), [24](#), [163](#);

submarine piracy, [17](#), [18 seq.](#), [51](#), [177](#), *passim*;

arming of, [19](#), [37 seq.](#), [113-5](#);

differences with Navy, [7](#), [21-25](#);

reconciliation, [25 seq.](#);

liaison with Navy, [14](#), [18-19](#), [21-43](#), [120-3](#), [134](#), [191](#), [197](#), [199-203](#), [255-6](#) *passim*;

commerce-raiders, [28](#), [51](#);

Naval War Staff, [30 seq.](#);

transporting of troops, [14](#), [30 seq.](#), [44](#), [45](#), [125-40](#), [223](#), [224-51](#);

popular recognition of, and the longshore view, [46 seq.](#), [255](#);

discipline, [64](#), [68-71](#);

wanted, a Ministry of Marine, [66](#);
manning, [68-73](#);
German *Schrecklichkeit*, [72](#), [73](#), [80-85](#), [92](#), [173](#), [203](#), [209](#);
coastal Services, [77-86](#);
war-time navigation, [85-6](#), [87-95](#);
traditions, [4](#), [5](#), [9-13](#), [23](#), [98](#), [99-100](#);
signals and wireless, [110-3](#), [120-4](#), [169-70](#), [191](#);
destroyer escort, [29-30](#), [126-8](#), [178](#), [203](#), [235](#);
torpedoing of a transport, [137-40](#), [237-40](#);
camouflage and dazzle, [163-7](#);
naming of standard ships, [168](#);
owners' customs clerks, [184](#), [185](#);
clearing for sea, [184-9](#);
convoy conference, [198-204](#);
putting to sea, [205-16](#);
unloading and loading, [217-223](#), [247-51](#). *See under* [Navy](#).

Merchant Shipping Act, [67](#)

Mersey, [49](#), [80](#), [107](#), [241](#), [244](#)

Mexico, [230](#)

Middleton, [6](#)

Minesweeping, [25](#), [88](#), [89](#), [91](#), [95](#), [97-99](#), [102](#), [205](#)

Ministry of Information, [xiii](#)

Ministry of Shipping, [178](#), [181](#)

Miramichi, [204](#)

'M N', [252-53](#)

Mobbs, Engineman, [99](#)

Mons, [126](#)

Monson, [6](#)

Muscovy Company, [5](#)

NANTUCKET Lightship, [104](#), [107](#), [109](#)

Nautical Magazine, [xiii](#)

Navesink, [109](#)

Navy:

Offshoot from Merchants' Service, [6](#), [18](#);

press-gangs, [9](#);

naval science, [11](#);

arming of merchantmen, [19](#), [37](#);

War Staff, [30](#) seq., [128](#);

Naval Transport Officer, [32](#) seq., [131](#), [247](#), [251](#);

Shipping Intelligence Officer, [36](#) seq., [185](#);

D.A.M.S., [37-8](#), [185](#);

'Otters,' [38](#), [185](#);

convoys, [41](#), [110](#), [121](#), [178-83](#), [189-97](#), [198-204](#), [205-12](#), [255](#);

anti-submarine measures, [41-3](#), [95-6](#), [110](#), [113](#), [203](#), [255](#);

'Q. ships,' [42](#);

gunnery, [43](#);

wireless on sea-going merchantmen, [110-3](#);

Transport Department, [126](#), [127](#);

Salvage Section, [144-7](#);

Examination Service, [190](#). See under [Merchants' Service](#)

Nebraskan, [109](#)

Neleus, [226](#), [229](#), [232](#), [233](#), [234](#)

Nelson, [11](#), [22](#)

Nelson, [79](#)

Nemesis, [139](#)

Neutral shipping, [17](#), [91](#), [177](#), [203](#)

New Brighton, [247](#)

Newhaven, [31](#)

New Jersey, [217](#)

Newport, U.S.A., [107](#)

New York, [27](#), [103](#), [109](#), [204](#)

New York Herald, [109](#)

Nichols, Skipper, [99](#)

Nile, [11](#)

Nore, [97](#)

Norfolk, Va., [104](#), [191](#)

North River, [217-23](#), [230](#)

North Sea, [85](#), [88](#), [252](#)

Olympic, [131](#)

'Otters,' 38, [80](#), [166](#)

Oversay, [117](#)

PADRIG Flats, [148](#)

Palermo, [81](#)

Palm Branch, [18](#), [19](#), [20](#)

Patrols, [28](#), [114](#), [117](#), [128](#), [150](#), [189](#), [190](#), [236](#)

Pearl Shell, [191](#), [192](#)

Persia, [253](#)

Philanthropic Seamen's Societies, [65](#)

Pilots, [59](#), [87-91](#), [98](#), [149](#), [187](#), [189](#), [210](#), [218](#), [251](#)

Plymouth, [205](#), [233](#)

Hoe, [198](#)

Sound, [207](#)

Poldhu, [226](#)

Portliskey, [81](#)

Present Help, [97](#), [98](#)

Prince Line, [109](#)

Provident, [101](#)

Prussian Guard, [51](#)

Psiloriti, [132](#)

Purchas, [5](#)

Queen Alexandra, [131](#)

Quetta Staff College, [30](#)

RAMSGATE, [56](#), [98](#)

Rate of Exchange, [103](#), [109](#)

Rathbone, Master John, [11](#)

Rathlin Island, [116](#)

Rathlin Sound, [116](#)

Rea (of *Crane*), [99](#)

Redcap, [82](#)

Richard Cœur de Lion, [54](#), [55](#)

Rifleman, [139](#)

River Plate, [4](#), [13](#), [181](#), [199](#)

Rôles d'Oléron, [54](#) *seq.*

Rowlatt, Mr. Justice, [62](#)

Royal Edward, [127](#)

Royal Naval Reserve, [23-25](#), [36-37](#), [41](#), [64](#), [122](#), [197](#)

Royal Naval Reserve (*Temporary*), [18](#), [25](#), [26](#)

Royal Naval Volunteer Reserve, [122](#)

Rozhdestvensky, [98](#)

Rue Point, [117](#)

ST. HELENS, [131](#)

St. Paul, [108](#)

Salonika, [128](#), [132](#), [133](#), [183](#)

Salvage, [141-62](#)

Salving a merchantman, [147-56](#)

Repairing in dry dock, [156-62](#)

Sandon, [251](#)

Sandy Hook, [230](#)

San Miguel, [170](#)

Sarah Pritchard, [78](#)

Seahorse, [210](#)

Seaplanes, [85](#), [92](#), [150](#), [182](#), [215](#)

Sea-slang, [26](#)

Selsey Bill, [95](#)

Serapis, [173](#)

Shanghai, [42](#)

Sir Ralph the Rover, [91](#)

Skullmartin, [116](#)

Smeaton, [91](#)

Smith (of *Crane*), [99](#)

South Africa, [126](#)

Southampton, [65](#), [67](#), [129](#), [131](#)

Southampton Water, [131](#)

Speedy, [181](#)

Stevedores, [185](#), [186](#), [218](#)

Straits of Dover, [85](#)

Strand, [124](#)

Strongbow, [178](#)

Suda Bay, [128](#), [132](#)

Suez Canal, [137](#)

Sutherland, [132](#)

THAMES, [78](#), [79](#), [86](#), [184](#), [252](#)

Thordis, [79](#)

Thracia, [173](#)

Titan, [149-55](#)

Titanic, [123](#)

Tor Point, [116](#)

Trafalgar, [11](#)

Trinity Bay, [89](#)

Trinity House, [56-61](#), [88](#), [95](#), [149](#)

Tripolis, [4](#)

Turkey Company, [9](#)

Tuscania, [15](#)

Tyne, [78](#), [86](#)

Umaria, [173](#)

United States, [72](#), [89](#), [103](#), [128](#), [178](#)

United States Seamen's Act, [72](#)

U 53, [109](#)

VALENCIA, [13](#)

Valparaiso, [42](#)

Vanilla, [101](#)

Verdun, [230](#)

Virginia, [104](#)

Virginia, [114](#)

Volapük, [169](#)

Volturno, [123](#)

Vosges, [17](#), [18](#)

WALMER, [89](#)

Wandle, [79](#)

'War Channel,' [86](#), [88](#)

War Ordnance, [209](#), [211](#), [213](#)

War Risks Associations, [36](#)

War Trident, [191](#), [192](#), [195-97](#)

Waterford, [30](#)

Watt, Skipper Joseph, [99](#)

Westmark Shoal, [149](#), [150](#)

White Star Line, [109](#)

Whymper, F., [xiii](#)

Wilhelmshaven, [24](#), [100](#)

William, [82](#)

William (East Indiaman), [9](#)

Wood, Walter, [xiii](#)

YARMOUTH, [90](#)

Yarmouth, I. of W., [141](#)

ZEPPELINS, [85](#), [92](#)

PRINTED AT
THE COMPLETE PRESS
WEST NORWOOD
LONDON

Transcriber's Notes:

Obvious punctuation errors repaired.

Text uses both propeller and propellor. Varied hyphenation where a majority usage could not be confirmed within the text was retained. For example: three uses of seamen and two of sea-men.

Page 259, the reference to page 136 was removed as this is a blank page following an illustration. The original read (coastal Services, 77-86, 136;)

The remaining corrections made are indicated by dotted lines under the corrections. Scroll the mouse over the word and the original text will appear.

*** END OF THE PROJECT GUTENBERG EBOOK MERCHANTMEN-
AT-ARMS : THE BRITISH MERCHANTS' SERVICE IN THE WAR ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.